1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%.
2. ALL CAPACITANCE VALUES ARE IN MICROFARADS.
3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.

REV	ECN	DESCRIPTION OF REVISION	CK APPD DATE
A	0000892544	PRODUCTION RELEASED	2010-04-15

K75F MLB_MEMSWAP

LAST_MODIFIED=Thu Apr 15 11:21:16 2010

DRAWING
TITLE=K22
ABBREV=DRAWING
LAST_MODIFIED=Thu Apr 15 11:21:16 2010

DRAWING TITLE
SCH,K75F,MLB_MEMSWAP

Apple Inc.

DRAWING NUMBER 051-8600 SIZE D
REVISION A.0.0
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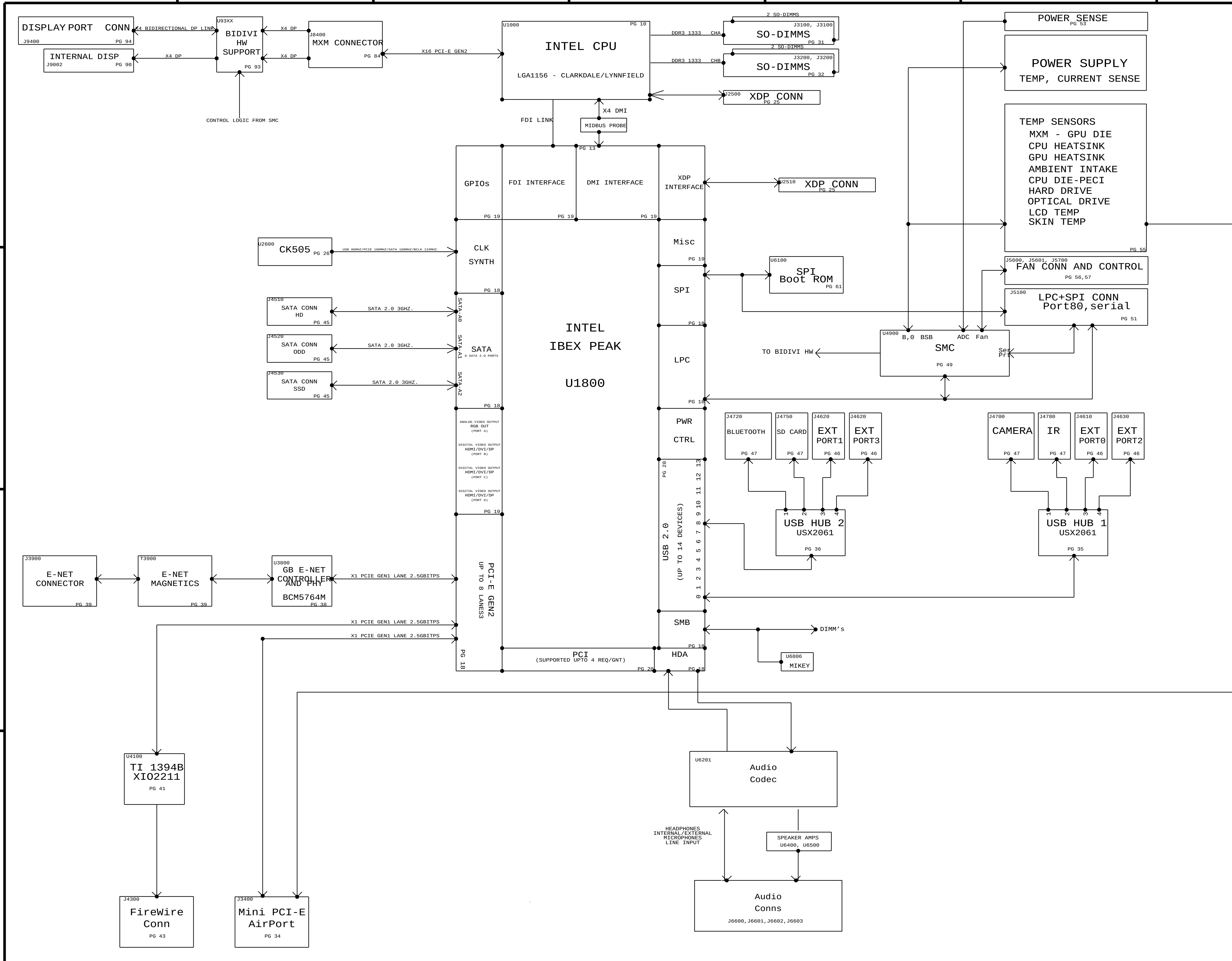

SYNC_MASTER=K75F_MLB SYNC_DATE=04/14/2010

PAGE TITLE

System Block Diagram

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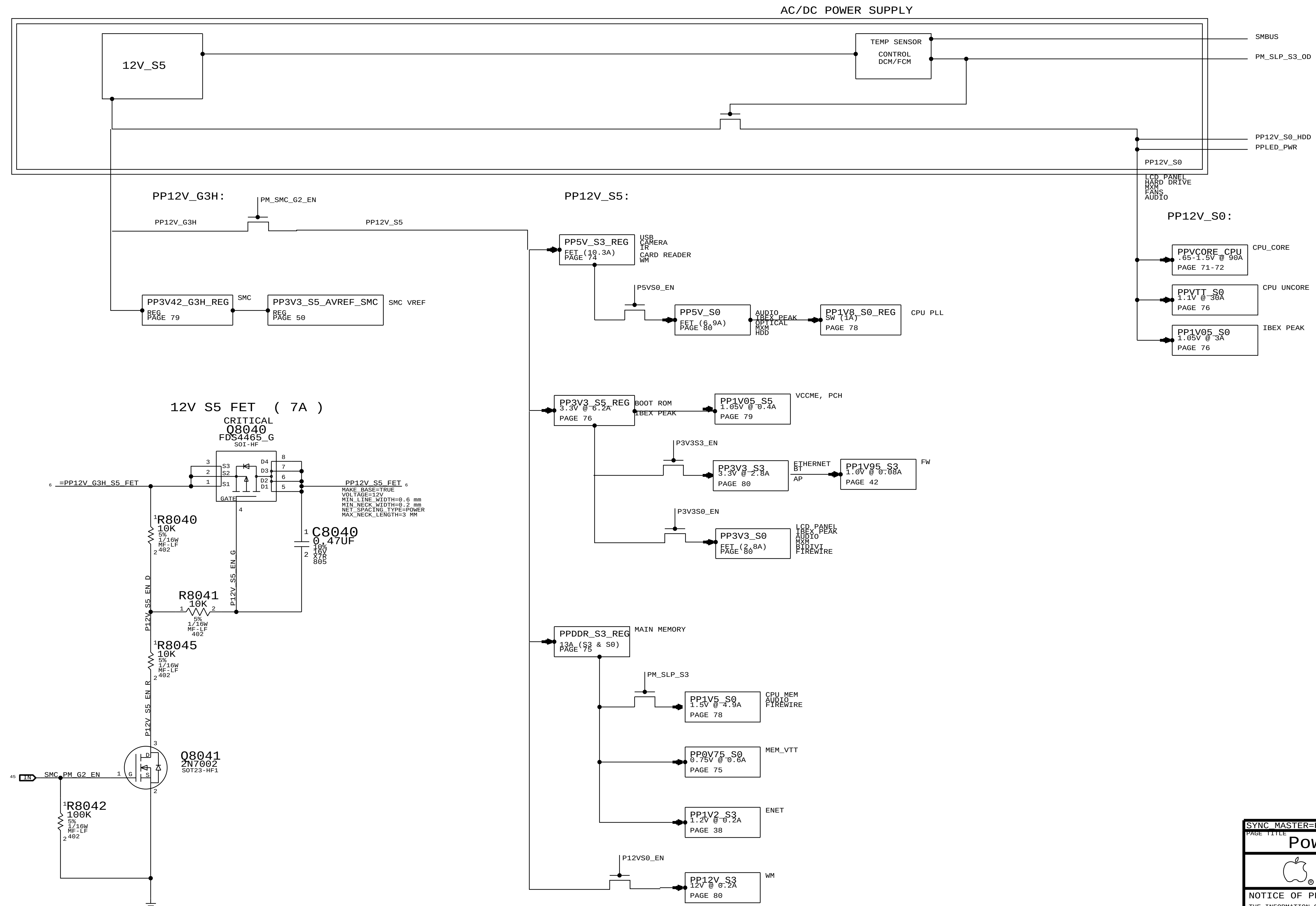

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PAGE TITLE

Power Block Diagram

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BOM Variants

BOM NUMBER	BOM NAME	BOM OPTIONS
085-1609	PCBA,MLB,DEV,K75F	DEVELOPMENT,DEV_GROUP
639-1103	PCBA,MLB,K75F,3.20GHZ,CKD	K75F,3P20GHZ_CKD_CPU,BASIC,CPUPOC_IMAX_100_120
639-1105	PCBA,MLB,K75F,3.60GHZ,CKD	K75F,3P60GHZ_CKD_CPU,BASIC,CPUPOC_IMAX_100_120
639-1104	PCBA,MLB,K75F,2.66GHZ,LFD	K75F,2P66GHZ_LFD_CPU,BASIC,CPUPOC_IMAX_100_120
639-1106	PCBA,MLB,K75F,2.93GHZ,LFD	K75F,2P93GHZ_LFD_CPU,BASIC,CPUPOC_IMAX_100_120

BOM GROUPS

BOM GROUP	BOM OPTIONS
BASIC	COMMON,ALTERNATE,XDP,BETTER,MXM,XDP_CPU_BPM,INT_VREF,PCH_VRM,BUF_CLK,PRODUCTION
DEV_GROUP	XDP_CONN,LPCPLUS,MOJOMUX,CPU_TDIODE

CPU SOCKET & ILM SUB-BOMS

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
511S0063	1	SOCKET,LGA1156,CPU-LF	U1000	CRITICAL	MOLEX_SOCKET
604-0942	1	ASSY,PURCHASED,ILM,MOLEX,K75	ILM	CRITICAL	MOLEX_SOCKET
511S0069	1	SOCKET,LGA1156,CPU-LF	U1000	CRITICAL	FOXCONN_SOCKET
604-0988	1	ASSY,PURCHASED,ILM,FOXCONN,K75	ILM	CRITICAL	FOXCONN_SOCKET

ALTERNATE SOCKET VENDORS MUST USE MATCHING ILM

BOM NUMBER	BOM NAME	BOM OPTIONS
607-6876	SUB ASSY,CPU SOCKET,K75F,MOLEX	MOLEX_SOCKET
607-6877	SUB ASSY,CPU SOCKET,K75F,FOXCONN	FOXCONN_SOCKET

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
607-6876	1	MOLEX CPU SOCKET AND ILM	SKT_ILM	CRITICAL	

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
607-6877	607-6876		SKT_ILM	FOXCONN ALTERNATIVE

BOARD STACK-UP

TOP	SIGNAL
2	GROUND
3	SIGNAL
4	POWER
5	POWER
6	SIGNAL
7	GROUND
BOTTOM	SIGNAL

COMMON

	PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
	337S3828	1	IC,IBEX PEAK B3 PRQ,DESKTOP,FCBGA,P425	U1800	CRITICAL	
	359S0157	1	IC,SLG2AP108,CLK GEN,CK505,QFN3	U2600	CRITICAL	BUF_CLK
	341T0230	1	IC,EFI BOOTROM,K74/K75	U6100	CRITICAL	
	338S0765	1	IC,XIO2211ZAY,1394B_PCIE,PHY/LINK	U4100	CRITICAL	
	343S0485	1	IC,BCM5764M,68PIN QFN	U3700	CRITICAL	
RAW: 335S0663	341T0269	1	ENET 1MBIT FLASH,CII,K74/K75F	U3701	CRITICAL	
	825-7122	1	MLB LABEL,48.0X4.8	X14	CRITICAL	

CPUS

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
337S3911	1	CKD,SLBUD,PRQ,3.20,73W,1333,K0,4M,LGA	CPU	CRITICAL	3P20GHZ_CKD_CPU
337S3910	1	CKD,SLBTM,PRQ,3.60,73W,1333,K0,4M,LGA	CPU	CRITICAL	3P60GHZ_CKD_CPU
337S3810	1	LFD,SLBLC,PRQ,2.66,95W,1333,B1,8M,LGA	CPU	CRITICAL	2P66GHZ_LFD_CPU
337S3861	1	LFD,SLBJG,PRQ,2.93,95W,1333,B1,8M,LGA	CPU	CRITICAL	2P93GHZ_LFD_CPU

ALTERNATES

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
128S0298	128S0293		C7460,C7461,C7462,C7463	
518S0811	518S0685		J9002	

K75F PARTS

	PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
	051-8600	1	SCH,K75F,MLB_MEMSWAP	SCH1		K75F
	820-2901	1	PCBF,K75F,MLB_MEMSWAP	MLB1		K75F
(338S0489 - BLNK)	341T0273	1	IC,SMC,K75F	U4900	CRITICAL	K75F

SYNC_MASTER=MASTER SYNC_DATE=N/A

PAGE TITLE
BOM Configuration

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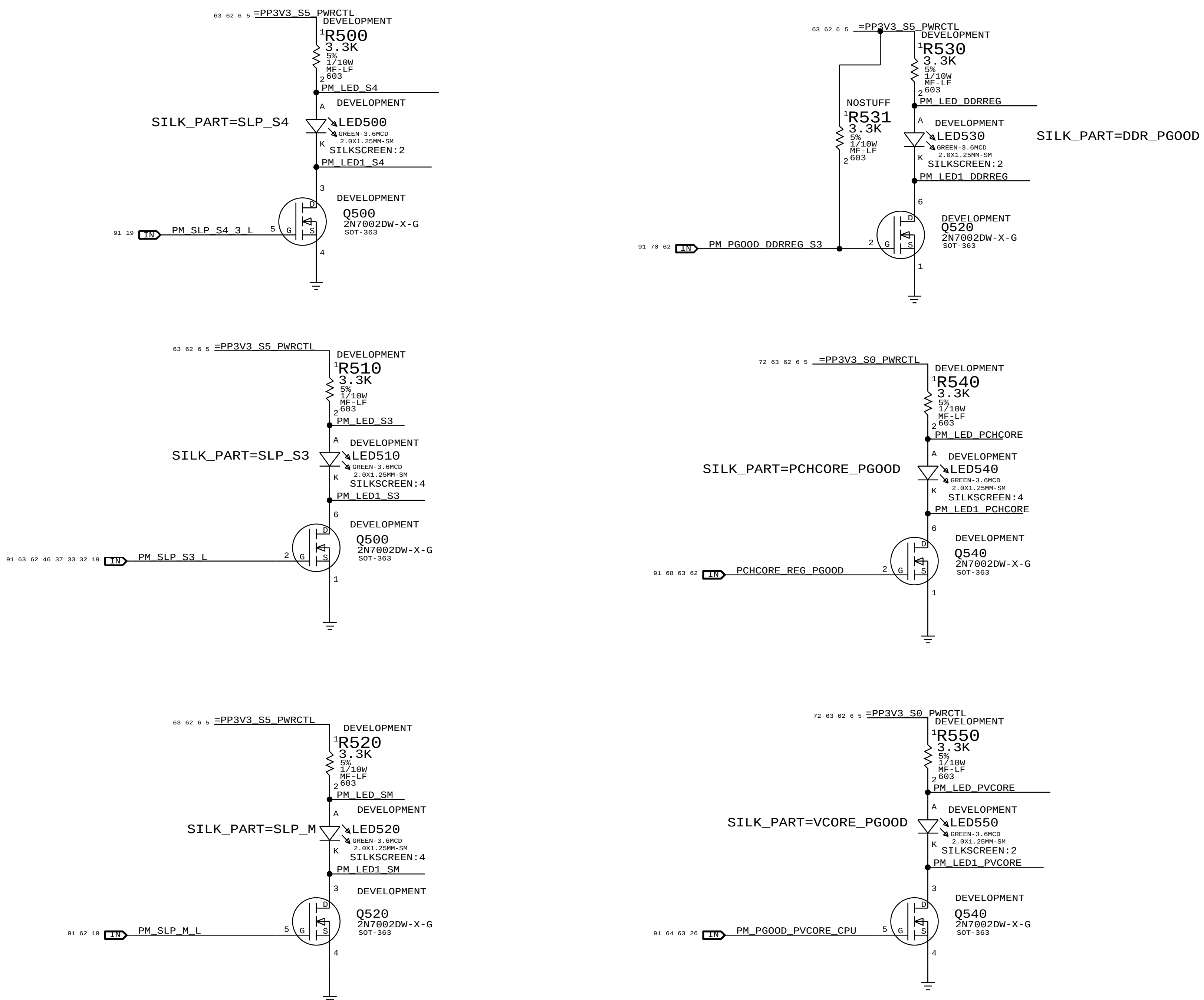
CAN DELETE THESE FOR PROTO1 AND BEYOND



State	Manageability	SMC_PM_G2_ENABLE	PM_S4_STATE_L	PM_SLP_S3_L	PM_SLP_S4_L	PM_SLP_M_L
Run (S0/M0)	N/A	1	1	1	1	1
Sleep (S3/M1)	On	1	1	0	1	1
Soft-Off (S5/M1)	On	1	0	0	1	1
Sleep (S3/M-Off)	Off	1	1	0	1	0
Soft-Off (S5/M-Off)	Off	1	0	0	0	0
Battery Off (G3Hot)	N/A	0	0	0	0	0

SYNC_MASTER=K75F_MLB SYNC_DATE=04/14/2010

PAGE TITLE
PROTO 0 DEBUG LEDS

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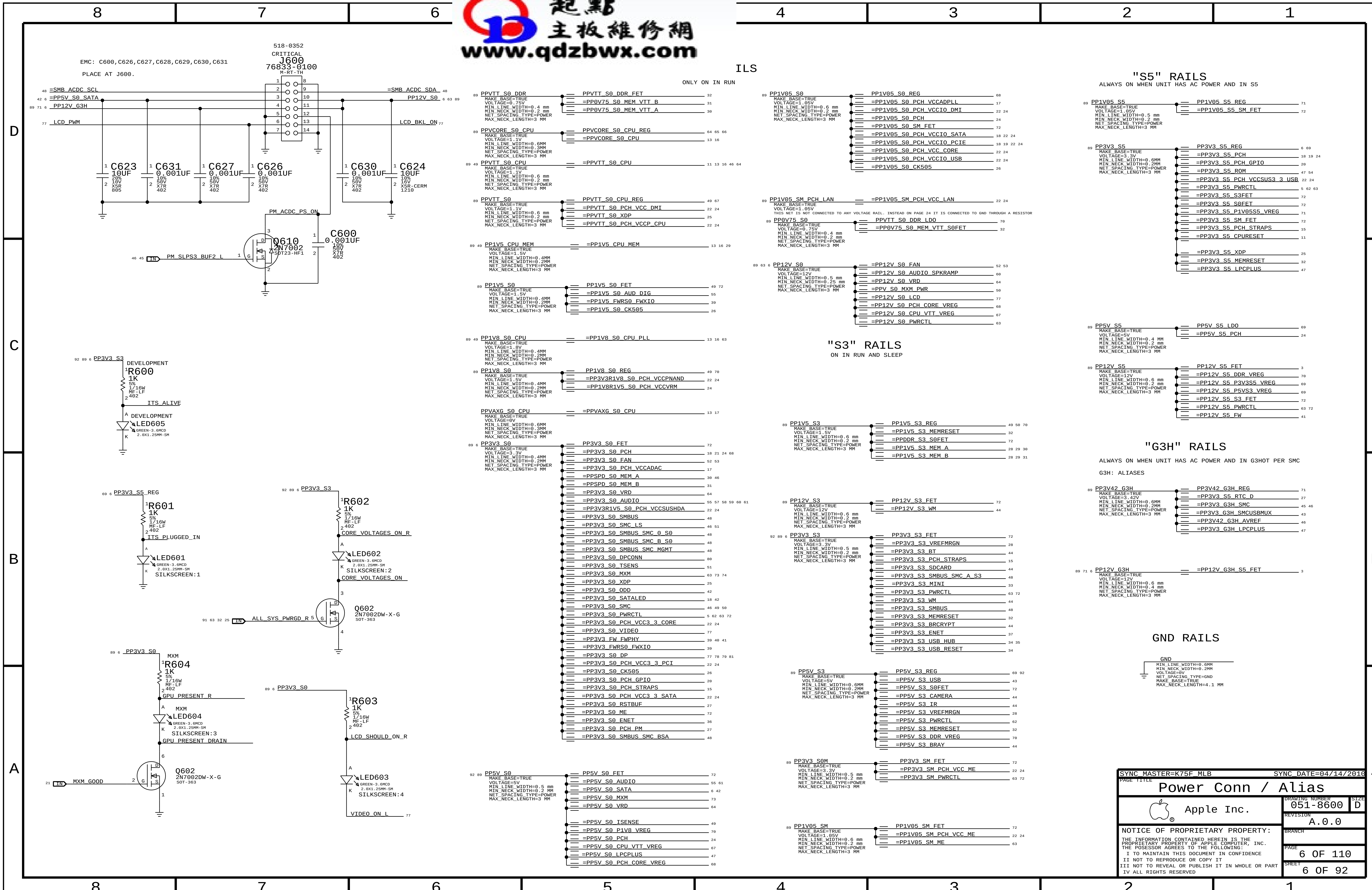

EMC: C600,C626,C627,C628,C629,C630,C631
PLACE AT J600.
518-0352
CRITICAL
J600
76833-0100
M-RT-TH
SMB_ACDC_SCL
PP5V_S0_SATA
PP12V_G3H
LCD_PWM
SMB_ACDC_SDA
PP12V_S0
LCD_BKL_ON
PM_ACDC_PS_ON
PM_SLPS3_BUF2_L
Q610
2N7002
C600
ONLY ON IN RUN
"S5" RAILS
ALWAYS ON WHEN UNIT HAS AC POWER AND IN S5
"S3" RAILS
ON IN RUN AND SLEEP
"G3H" RAILS
ALWAYS ON WHEN UNIT HAS AC POWER AND IN G3HOT PER SMC
G3H: ALIASES
GND RAILS
DEVELOPMENT
R600
ITS_ALIVE
LED605
R601
ITS_PLUGGED_IN
LED601
SILKSCREEN:1
R602
CORE_VOLTAGES_ON_R
LED602
SILKSCREEN:2
CORE_VOLTAGES_ON
Q602
2N7002DW-X-G
ALL_SYS_PWRGD_R
MXM
R604
GPU_PRESENT_R
LED604
SILKSCREEN:3
GPU_PRESENT_DRAIN
MXM_GOOD
R603
LCD_SHOULD_ON_R
LED603
SILKSCREEN:4
VIDEO_ON_L
SYNC_MASTER=K75F_MLB
SYNC_DATE=04/14/2010
PAGE TITLE
Power Conn / Alias
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051-8600
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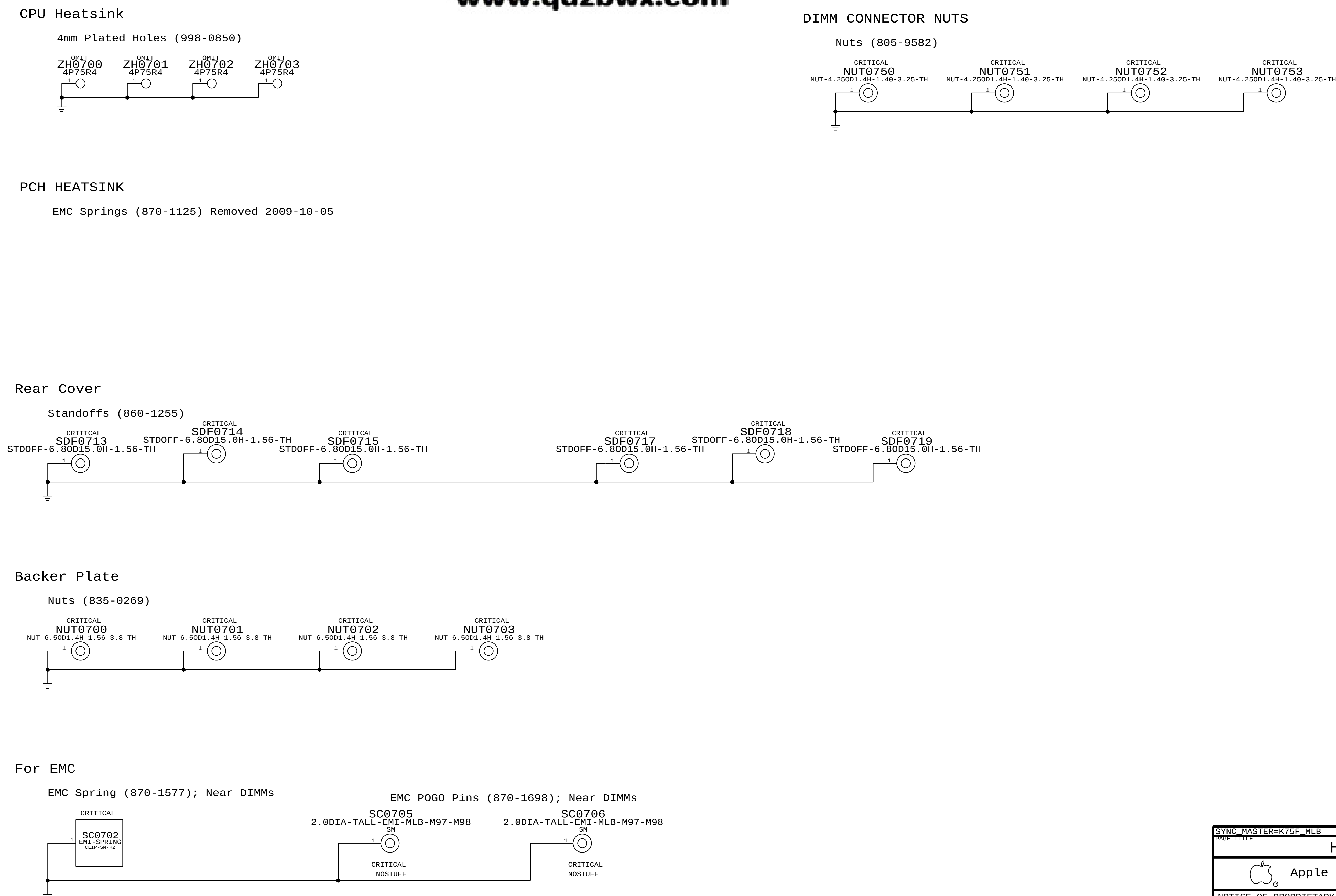

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UNUSED CPU SIGNALS

10 TP_CPU_RSVD<41..29> = NC_CPU_RSVD<41..29> MAKE_BASE=TRUE NO_TEST=TRUE
10 TP_CPU_RSVD<26..1> = NC_CPU_RSVD<26..1> MAKE_BASE=TRUE NO_TEST=TRUE
13 TP_CPU_FC_AE38 = NC_CPU_FC_AE38 MAKE_BASE=TRUE NO_TEST=TRUE
13 TP_CPU_FC_AG40 = NC_CPU_FC_AG40 MAKE_BASE=TRUE NO_TEST=TRUE

NC ON UNUSED PCI ALIASES

20 TP_PCI_AD<31..0> = NC_PCI_AD<31..0> MAKE_BASE=TRUE NO_TEST=TRUE
20 TP_PCI_C_BE_L<3..0> = NC_PCI_C_BE_L<3..0> MAKE_BASE=TRUE NO_TEST=TRUE

20 TP_PCI_PAR = NC_PCI_PAR MAKE_BASE=TRUE NO_TEST=TRUE
20 TP_PCI_RESET_L = NC_PCI_RESET_L MAKE_BASE=TRUE NO_TEST=TRUE
21 TP_PCIE_CLK100M_XDPP = NC_PCIE_CLK100M_XDPP MAKE_BASE=TRUE NO_TEST=TRUE
21 TP_PCIE_CLK100M_XDPN = NC_PCIE_CLK100M_XDPN MAKE_BASE=TRUE NO_TEST=TRUE
21 TP_DMI_CLK100M_LAP = NC_DMI_CLK100M_LAP MAKE_BASE=TRUE NO_TEST=TRUE
21 TP_DMI_CLK100M_LAN = NC_DMI_CLK100M_LAN MAKE_BASE=TRUE NO_TEST=TRUE
18 TP_LPC_DREQ1_L = NC_LPC_DREQ1_L MAKE_BASE=TRUE NO_TEST=TRUE
18 TP_LPC_DREQ0_L = NC_LPC_DREQ0_L MAKE_BASE=TRUE NO_TEST=TRUE

NC ON UNUSED NAND ALIASES

20 TP_NV_CE_L<3..0> = NC_NV_CE_L<3..0> MAKE_BASE=TRUE NO_TEST=TRUE
20 TP_NV_DQS<1..0> = NC_NV_DQS<1..0> MAKE_BASE=TRUE NO_TEST=TRUE
20 TP_NV_DQ<15..0> = NC_NV_DQ<15..0> MAKE_BASE=TRUE NO_TEST=TRUE

20 TP_NV_RB_L = NC_NV_RB_L MAKE_BASE=TRUE NO_TEST=TRUE
20 TP_NV_WR_RE_L<1..0> = NC_NV_WR_RE_L<1..0> MAKE_BASE=TRUE NO_TEST=TRUE
20 TP_NV_WE_CK_L<1..0> = NC_NV_WE_CK_L<1..0> MAKE_BASE=TRUE NO_TEST=TRUE

NC ON UNUSED MISC ALIASES

25 TP_JTAG_XDP_TRST_L = NC_JTAG_XDP_TRST_L MAKE_BASE=TRUE NO_TEST=TRUE
21 TP_PCH_PWM0 = NC_PCH_PWM0 MAKE_BASE=TRUE NO_TEST=TRUE
21 TP_PCH_PWM1 = NC_PCH_PWM1 MAKE_BASE=TRUE NO_TEST=TRUE
21 TP_PCH_PWM2 = NC_PCH_PWM2 MAKE_BASE=TRUE NO_TEST=TRUE
21 TP_PCH_PWM3 = NC_PCH_PWM3 MAKE_BASE=TRUE NO_TEST=TRUE
21 TP_PCH_SST = NC_PCH_SST MAKE_BASE=TRUE NO_TEST=TRUE

NC ON UNUSED MEM ALIASES

12 TP_MEM_A_CS_L<7..4> = NC_MEM_A_CS_L<7..4> MAKE_BASE=TRUE NO_TEST=TRUE
12 TP_MEM_A_DQ_CB<7..0> = NC_MEM_A_DQ_CB<7..0> MAKE_BASE=TRUE NO_TEST=TRUE
83 12 TP_MEM_A_DQS_N<8> = NC_MEM_A_DQS_N<8> MAKE_BASE=TRUE NO_TEST=TRUE
83 12 TP_MEM_A_DQS_P<8> = NC_MEM_A_DQS_P<8> MAKE_BASE=TRUE NO_TEST=TRUE
12 TP_MEM_B_CS_L<7..4> = NC_MEM_B_CS_L<7..4> MAKE_BASE=TRUE NO_TEST=TRUE
12 TP_MEM_B_DQ_CB<7..0> = NC_MEM_B_DQ_CB<7..0> MAKE_BASE=TRUE NO_TEST=TRUE
83 12 TP_MEM_B_DQS_N<8> = NC_MEM_B_DQS_N<8> MAKE_BASE=TRUE NO_TEST=TRUE
83 12 TP_MEM_B_DQS_P<8> = NC_MEM_B_DQS_P<8> MAKE_BASE=TRUE NO_TEST=TRUE

TP_PCI_T28_D2R_N<3..0> = NC_PCI_T28_D2R_N<3..0> MAKE_BASE=TRUE NO_TEST=TRUE
TP_PCI_T28_D2R_P<3..0> = NC_PCI_T28_D2R_P<3..0> MAKE_BASE=TRUE NO_TEST=TRUE
TP_PCI_T28_R2D_C_N<3..0> = NC_PCI_T28_R2D_C_N<3..0> MAKE_BASE=TRUE NO_TEST=TRUE
TP_PCI_T28_R2D_C_P<3..0> = NC_PCI_T28_R2D_C_P<3..0> MAKE_BASE=TRUE NO_TEST=TRUE

18 TP_HDA_SDIN1 = NC_HDA_SDIN1 MAKE_BASE=TRUE NO_TEST=TRUE
18 TP_HDA_SDIN2 = NC_HDA_SDIN2 MAKE_BASE=TRUE NO_TEST=TRUE
18 TP_HDA_SDIN3 = NC_HDA_SDIN3 MAKE_BASE=TRUE NO_TEST=TRUE
18 TP_PCIE_CLK100M_PE5P = NC_PCIE_CLK100M_PE5P MAKE_BASE=TRUE NO_TEST=TRUE
18 TP_PCIE_CLK100M_PE5N = NC_PCIE_CLK100M_PE5N MAKE_BASE=TRUE NO_TEST=TRUE
20 TP_NV_RCOMP = NC_NV_RCOMP MAKE_BASE=TRUE NO_TEST=TRUE
18 PCIE_CLK100M_EXCARD_P = NC_PCIE_CLK100M_EXCARD_P MAKE_BASE=TRUE NO_TEST=TRUE
18 PCIE_CLK100M_EXCARD_N = NC_PCIE_CLK100M_EXCARD_N MAKE_BASE=TRUE NO_TEST=TRUE

20 TP_USB_7N = NC_USB_7N MAKE_BASE=TRUE NO_TEST=TRUE
20 TP_USB_7P = NC_USB_7P MAKE_BASE=TRUE NO_TEST=TRUE
20 TP_USB_6N = NC_USB_6N MAKE_BASE=TRUE NO_TEST=TRUE
20 TP_USB_6P = NC_USB_6P MAKE_BASE=TRUE NO_TEST=TRUE
20 TP_USB_1N = NC_USB_1N MAKE_BASE=TRUE NO_TEST=TRUE
20 TP_USB_1P = NC_USB_1P MAKE_BASE=TRUE NO_TEST=TRUE
20 TP_USB_3N = NC_USB_3N MAKE_BASE=TRUE NO_TEST=TRUE
20 TP_USB_3P = NC_USB_3P MAKE_BASE=TRUE NO_TEST=TRUE
20 TP_USB_5N = NC_USB_5N MAKE_BASE=TRUE NO_TEST=TRUE
20 TP_USB_5P = NC_USB_5P MAKE_BASE=TRUE NO_TEST=TRUE
20 TP_USB_9N = NC_USB_9N MAKE_BASE=TRUE NO_TEST=TRUE
20 TP_USB_9P = NC_USB_9P MAKE_BASE=TRUE NO_TEST=TRUE
20 TP_USB_10N = NC_USB_10N MAKE_BASE=TRUE NO_TEST=TRUE
20 TP_USB_10P = NC_USB_10P MAKE_BASE=TRUE NO_TEST=TRUE
20 TP_USB_11N = NC_USB_11N MAKE_BASE=TRUE NO_TEST=TRUE
20 TP_USB_11P = NC_USB_11P MAKE_BASE=TRUE NO_TEST=TRUE
20 TP_USB_12N = NC_USB_12N MAKE_BASE=TRUE NO_TEST=TRUE
20 TP_USB_12P = NC_USB_12P MAKE_BASE=TRUE NO_TEST=TRUE
20 TP_USB_13N = NC_USB_13N MAKE_BASE=TRUE NO_TEST=TRUE
20 TP_USB_13P = NC_USB_13P MAKE_BASE=TRUE NO_TEST=TRUE

18 PCIE_EXCARD_D2R_P = NC_PCIE_EXCARD_D2R_P MAKE_BASE=TRUE NO_TEST=TRUE
18 PCIE_EXCARD_D2R_N = NC_PCIE_EXCARD_D2R_N MAKE_BASE=TRUE NO_TEST=TRUE
18 PCIE_EXCARD_R2D_C_P = NC_PCIE_EXCARD_R2D_C_P MAKE_BASE=TRUE NO_TEST=TRUE
18 PCIE_EXCARD_R2D_C_N = NC_PCIE_EXCARD_R2D_C_N MAKE_BASE=TRUE NO_TEST=TRUE
19 TP_SDVO_TVCLKINN = NC_SDVO_TVCLKINN MAKE_BASE=TRUE NO_TEST=TRUE
19 TP_SDVO_TVCLKINP = NC_SDVO_TVCLKINP MAKE_BASE=TRUE NO_TEST=TRUE
19 TP_SDVO_STALLN = NC_SDVO_STALLN MAKE_BASE=TRUE NO_TEST=TRUE
19 TP_SDVO_STALLP = NC_SDVO_STALLP MAKE_BASE=TRUE NO_TEST=TRUE
19 TP_SDVO_INTN = NC_SDVO_INTN MAKE_BASE=TRUE NO_TEST=TRUE
19 TP_SDVO_INTP = NC_SDVO_INTP MAKE_BASE=TRUE NO_TEST=TRUE

NC ON UNUSED SATA ALIASES

16 TP_SATA_D_D2RN = NC_SATA_D_D2RN MAKE_BASE=TRUE NO_TEST=TRUE
16 TP_SATA_D_D2RP = NC_SATA_D_D2RP MAKE_BASE=TRUE NO_TEST=TRUE
16 TP_SATA_D_R2D_CN = NC_SATA_D_R2D_CN MAKE_BASE=TRUE NO_TEST=TRUE
16 TP_SATA_D_R2D_CP = NC_SATA_D_R2D_CP MAKE_BASE=TRUE NO_TEST=TRUE
16 TP_SATA_E_D2RN = NC_SATA_E_D2RN MAKE_BASE=TRUE NO_TEST=TRUE
16 TP_SATA_E_D2RP = NC_SATA_E_D2RP MAKE_BASE=TRUE NO_TEST=TRUE
16 TP_SATA_E_R2D_CN = NC_SATA_E_R2D_CN MAKE_BASE=TRUE NO_TEST=TRUE
16 TP_SATA_E_R2D_CP = NC_SATA_E_R2D_CP MAKE_BASE=TRUE NO_TEST=TRUE
16 TP_SATA_F_D2RN = NC_SATA_F_D2RN MAKE_BASE=TRUE NO_TEST=TRUE
16 TP_SATA_F_D2RP = NC_SATA_F_D2RP MAKE_BASE=TRUE NO_TEST=TRUE
16 TP_SATA_F_R2D_CN = NC_SATA_F_R2D_CN MAKE_BASE=TRUE NO_TEST=TRUE
16 TP_SATA_F_R2D_CP = NC_SATA_F_R2D_CP MAKE_BASE=TRUE NO_TEST=TRUE

NC ON UNUSED DISPLAY ALIASES

19 TP_CRT_IG_DDC_CLK = NC_CRT_IG_DDC_CLK MAKE_BASE=TRUE NO_TEST=TRUE
19 TP_CRT_IG_DDC_DATA = NC_CRT_IG_DDC_DATA MAKE_BASE=TRUE NO_TEST=TRUE
19 TP_CRT_IG_RED = NC_CRT_IG_RED MAKE_BASE=TRUE NO_TEST=TRUE
19 TP_CRT_IG_GREEN = NC_CRT_IG_GREEN MAKE_BASE=TRUE NO_TEST=TRUE
19 TP_CRT_IG_BLUE = NC_CRT_IG_BLUE MAKE_BASE=TRUE NO_TEST=TRUE
19 TP_CRT_IG_HSYNC = NC_CRT_IG_HSYNC MAKE_BASE=TRUE NO_TEST=TRUE
19 TP_CRT_IG_VSYNC = NC_CRT_IG_VSYNC MAKE_BASE=TRUE NO_TEST=TRUE

19 TP_DP_IG_B_MLN<3..0> = NC_DP_IG_B_MLN<3..0> MAKE_BASE=TRUE NO_TEST=TRUE
19 TP_DP_IG_B_MLP<3..0> = NC_DP_IG_B_MLP<3..0> MAKE_BASE=TRUE NO_TEST=TRUE
19 TP_DP_IG_B_AUX_N = NC_DP_IG_B_AUXN MAKE_BASE=TRUE NO_TEST=TRUE
19 TP_DP_IG_B_AUX_P = NC_DP_IG_B_AUXP MAKE_BASE=TRUE NO_TEST=TRUE
19 TP_DP_IG_B_HPD = NC_DP_IG_B_HPD MAKE_BASE=TRUE NO_TEST=TRUE
19 TP_DP_IG_B_DDC_CLK = NC_DP_IG_B_CTRL_CLK MAKE_BASE=TRUE NO_TEST=TRUE
19 TP_DP_IG_B_DDC_DATA = NC_DP_IG_B_CTRL_DATA MAKE_BASE=TRUE NO_TEST=TRUE

19 TP_DP_IG_C_MLN<3..0> = NC_DP_IG_C_MLN<3..0> MAKE_BASE=TRUE NO_TEST=TRUE
19 TP_DP_IG_C_MLP<3..0> = NC_DP_IG_C_MLP<3..0> MAKE_BASE=TRUE NO_TEST=TRUE
19 TP_DP_IG_C_AUX_N = NC_DP_IG_C_AUXN MAKE_BASE=TRUE NO_TEST=TRUE
19 TP_DP_IG_C_AUX_P = NC_DP_IG_C_AUXP MAKE_BASE=TRUE NO_TEST=TRUE
19 TP_DP_IG_C_HPD = NC_DP_IG_C_HPD MAKE_BASE=TRUE NO_TEST=TRUE
19 TP_DP_IG_C_CTRL_CLK = NC_DP_IG_C_CTRL_CLK MAKE_BASE=TRUE NO_TEST=TRUE
19 TP_DP_IG_C_CTRL_DATA = NC_DP_IG_C_CTRL_DATA MAKE_BASE=TRUE NO_TEST=TRUE

19 TP_DP_IG_D_MLN<3..0> = NC_DP_IG_D_MLN<3..0> MAKE_BASE=TRUE NO_TEST=TRUE
19 TP_DP_IG_D_MLP<3..0> = NC_DP_IG_D_MLP<3..0> MAKE_BASE=TRUE NO_TEST=TRUE
19 TP_DP_IG_D_AUXN = NC_DP_IG_D_AUXN MAKE_BASE=TRUE NO_TEST=TRUE
19 TP_DP_IG_D_AUXP = NC_DP_IG_D_AUXP MAKE_BASE=TRUE NO_TEST=TRUE
19 TP_DP_IG_D_HPD = NC_DP_IG_D_HPD MAKE_BASE=TRUE NO_TEST=TRUE
19 TP_DP_IG_D_CTRL_CLK = NC_DP_IG_D_CTRL_CLK MAKE_BASE=TRUE NO_TEST=TRUE
19 TP_DP_IG_D_CTRL_DATA = NC_DP_IG_D_CTRL_DATA MAKE_BASE=TRUE NO_TEST=TRUE

13 TP_GFX_VID<0..6> = NC_GFX_VID<0..6> MAKE_BASE=TRUE NO_TEST=TRUE
13 TP_GFX_VSENSE_N = NC_GFX_VSENSE_N MAKE_BASE=TRUE NO_TEST=TRUE
13 TP_GFX_VSENSE_P = NC_GFX_VSENSE_P MAKE_BASE=TRUE NO_TEST=TRUE

SYNC_MASTER=K75F_MLB SYNC_DATE=04/14/2010
PAGE TITLE
UNUSED SIGNAL ALIAS

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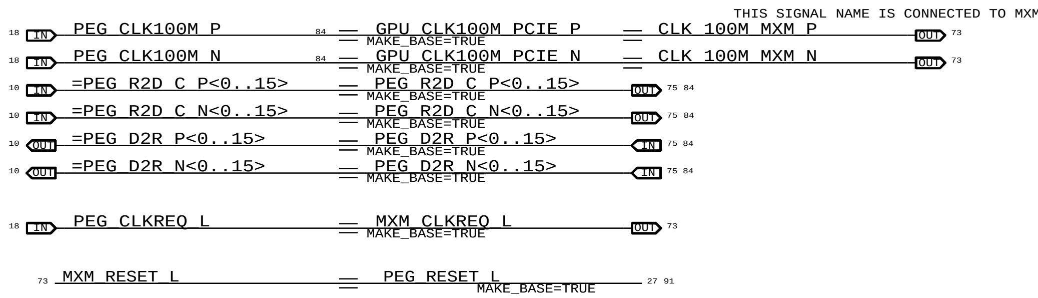
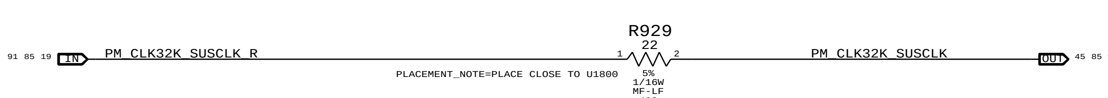
PEG Slot Support





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Signal Aliases

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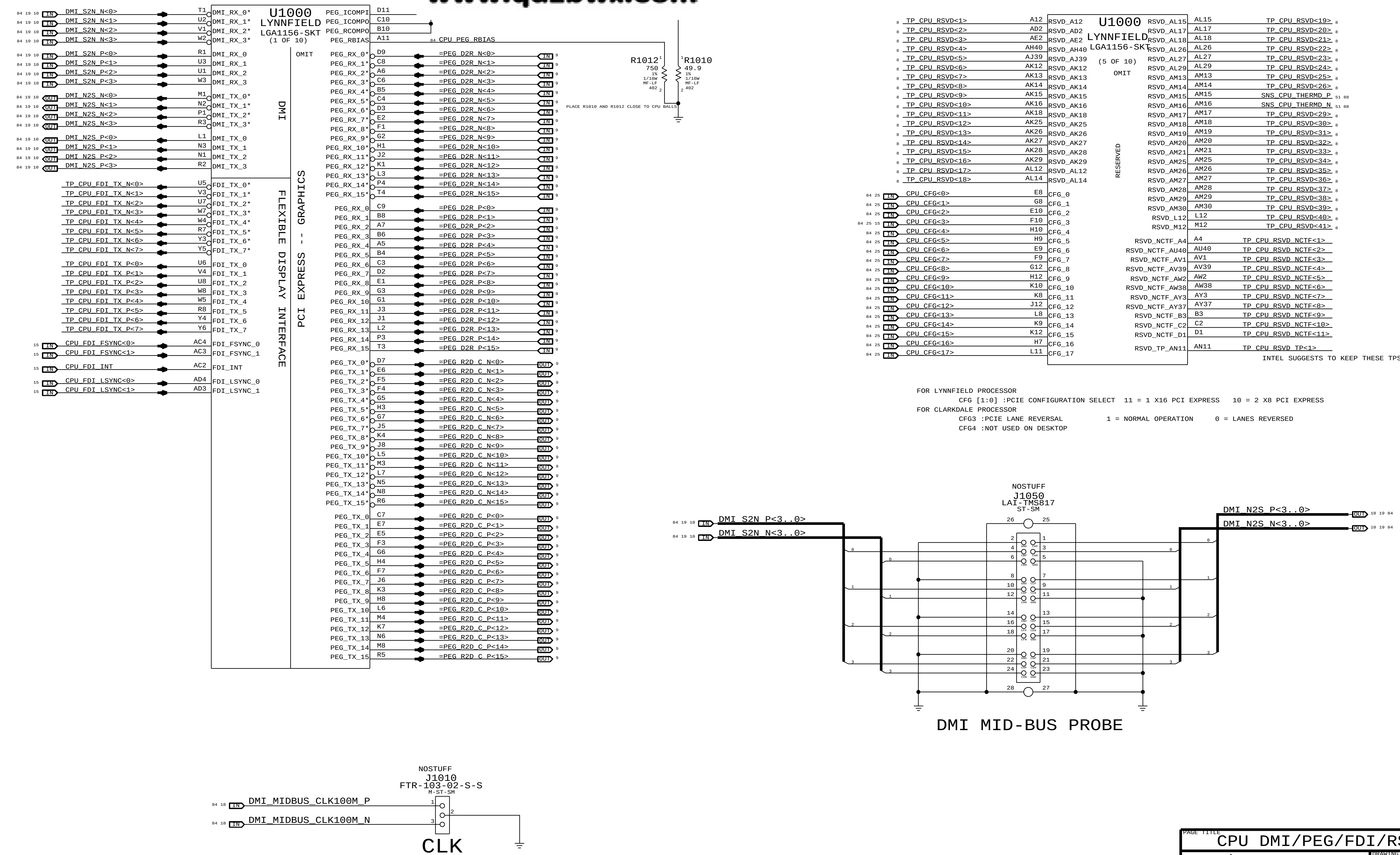
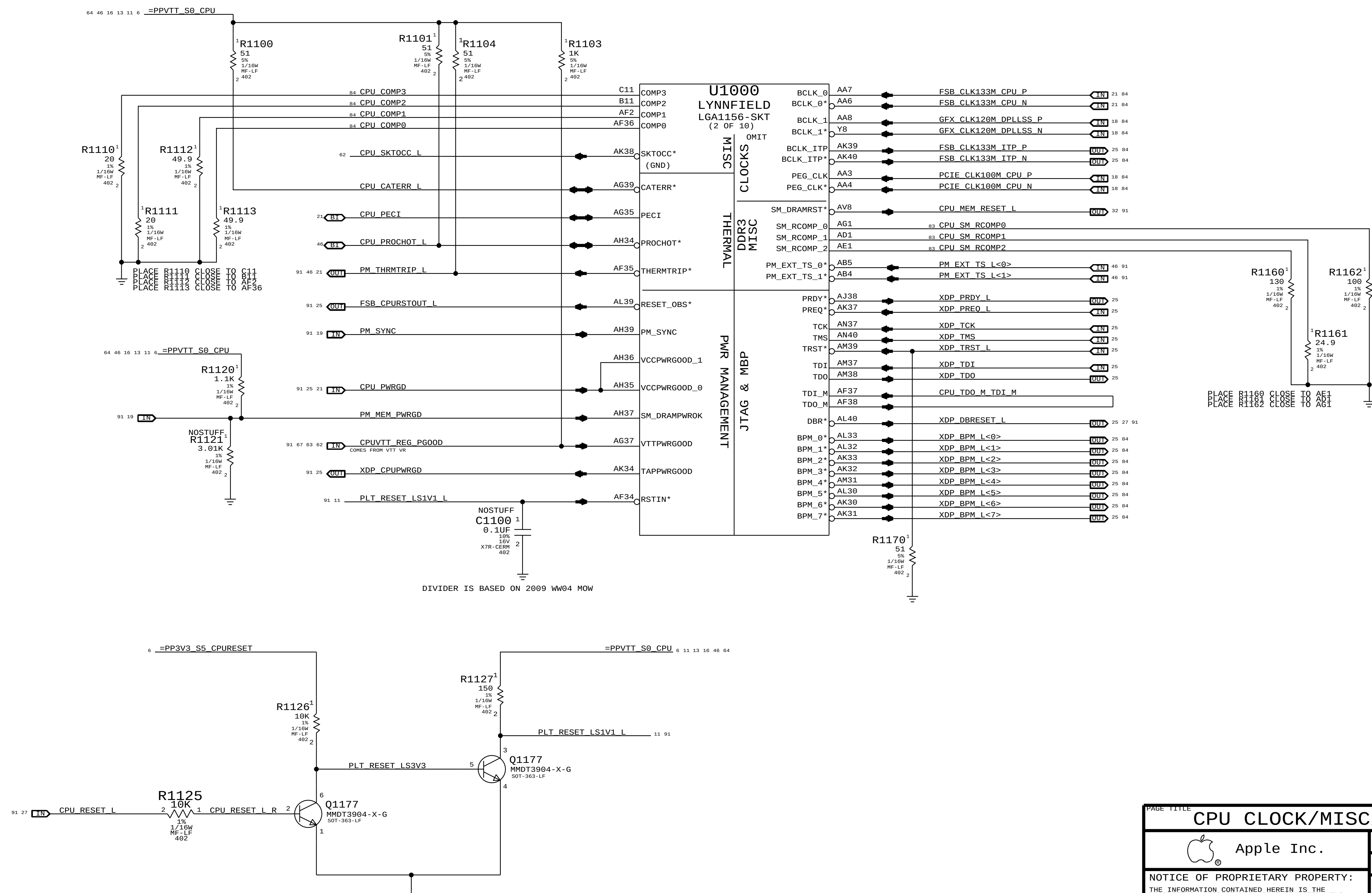


PAGE TITLE: CPU DMI/PEG/FDI/RSVD	DRAWING NUMBER: 051-8600	SIZE: D
Apple Inc.	REVISION: A.0.0	
NOTICE OF PROPRIETARY PROPERTY:	BRANCH	
	PAGE: 10 OF 110	
	SHEET: 10 OF 92	



CPU CLOCK/MISC/JTAG

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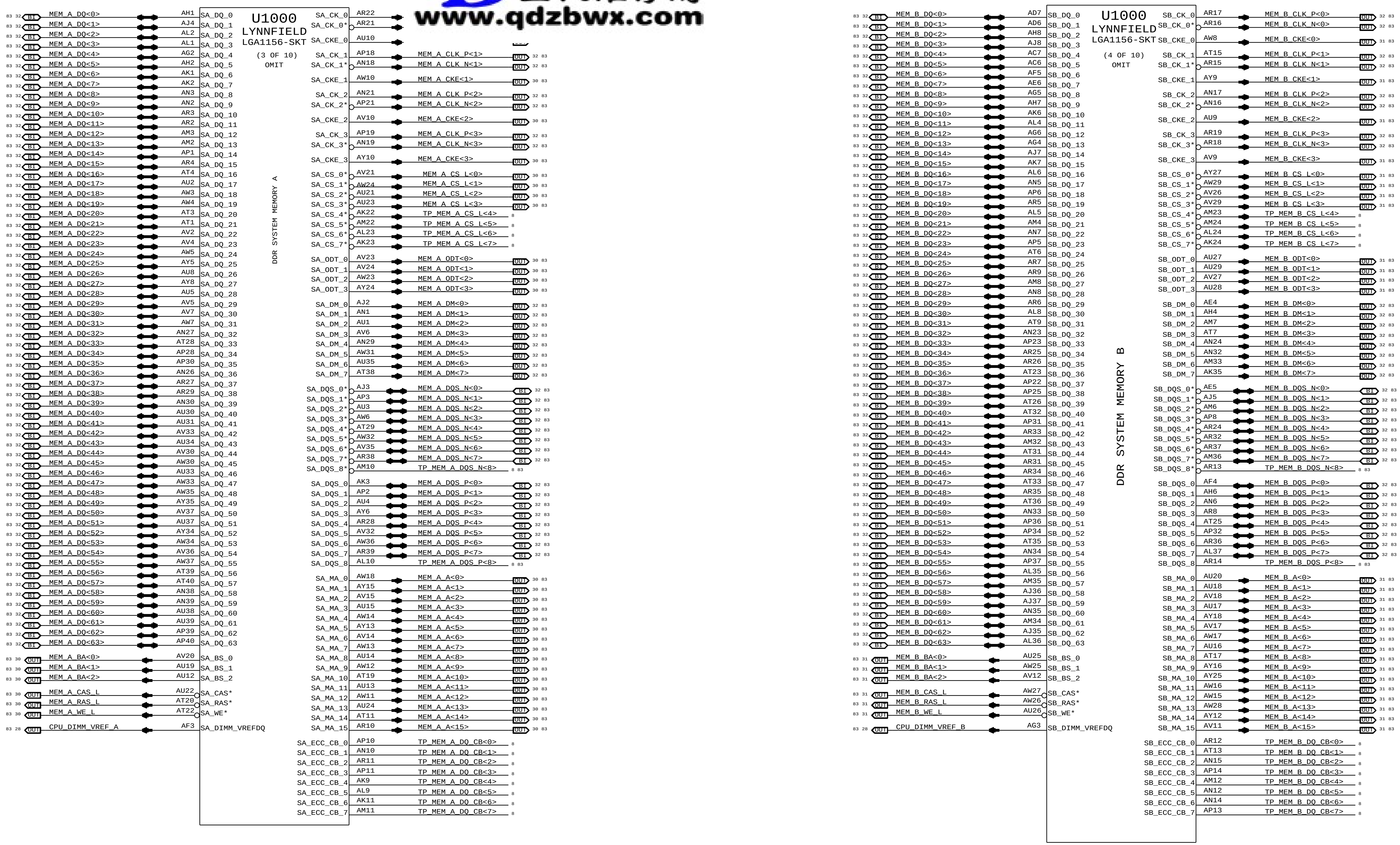

SYNC_MASTER=K75F_MLB SYNC_DATE=04/14/2010

PAGE TITLE

CPU DDR3 INTERFACES

Apple Inc.

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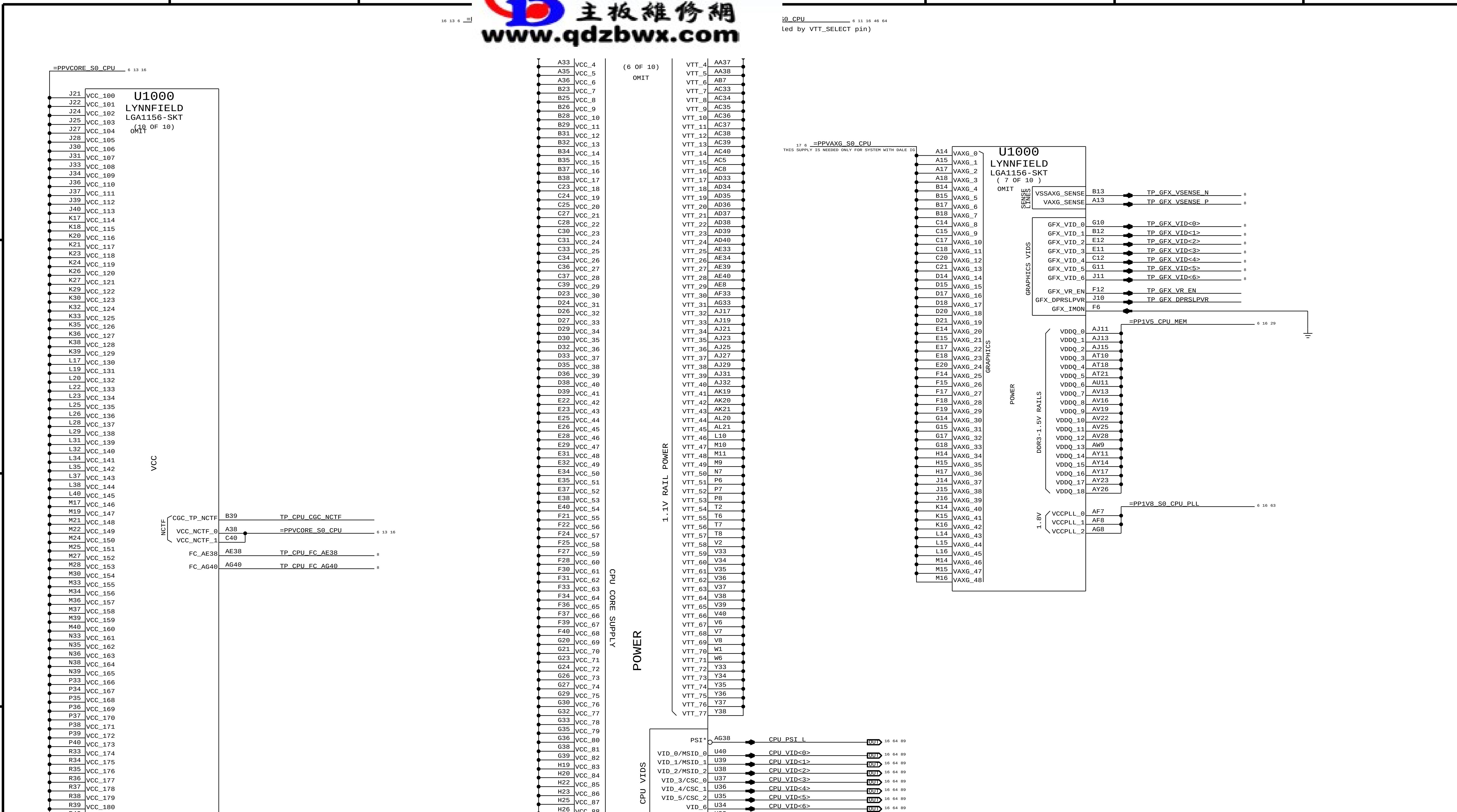

SYNC_MASTER=K75F_MLB SYNC_DATE=04/14/2010

PAGE TITLE

CPU POWER

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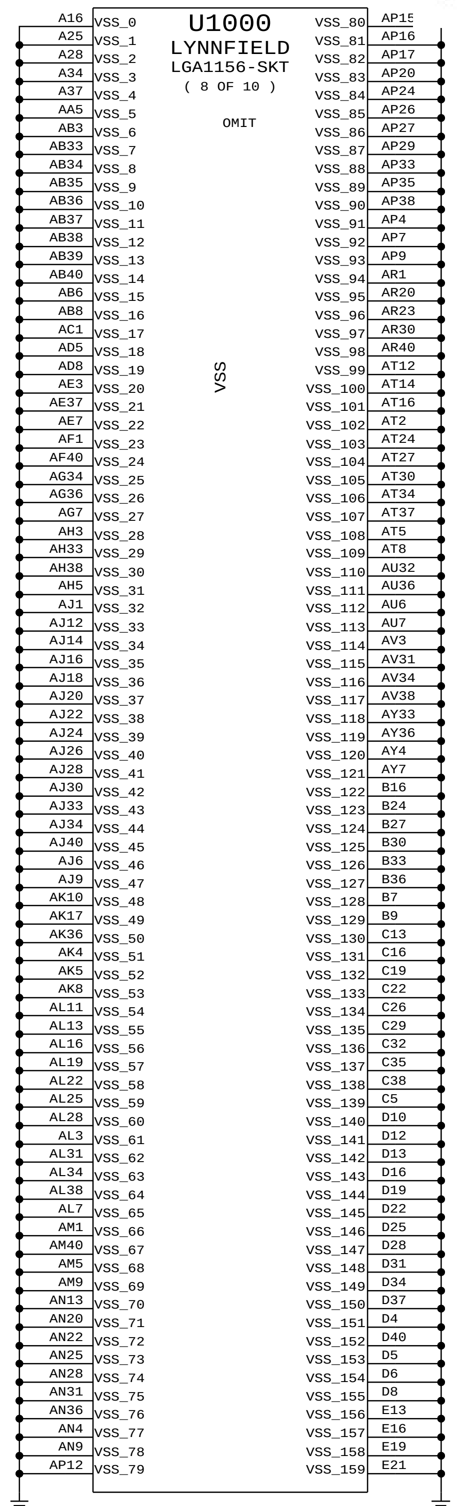
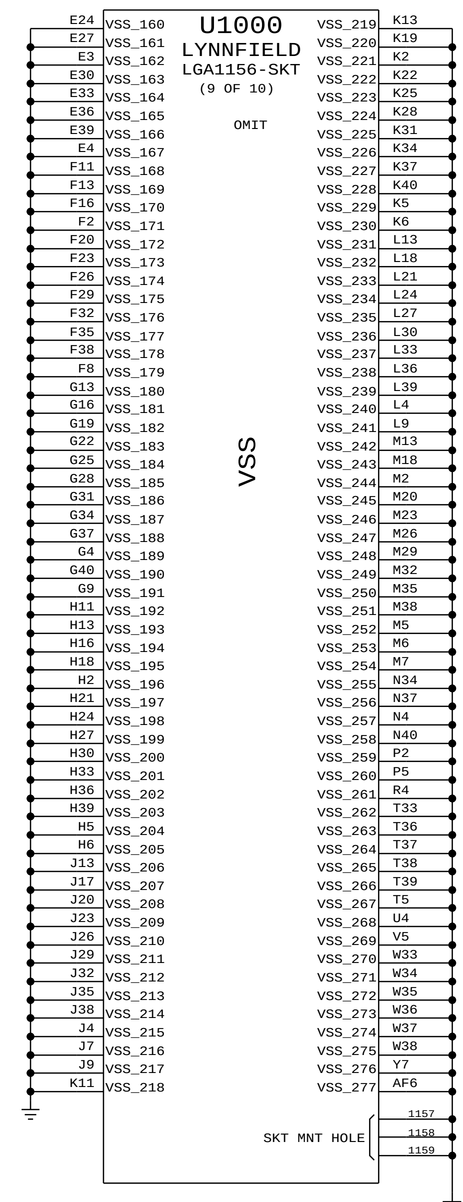
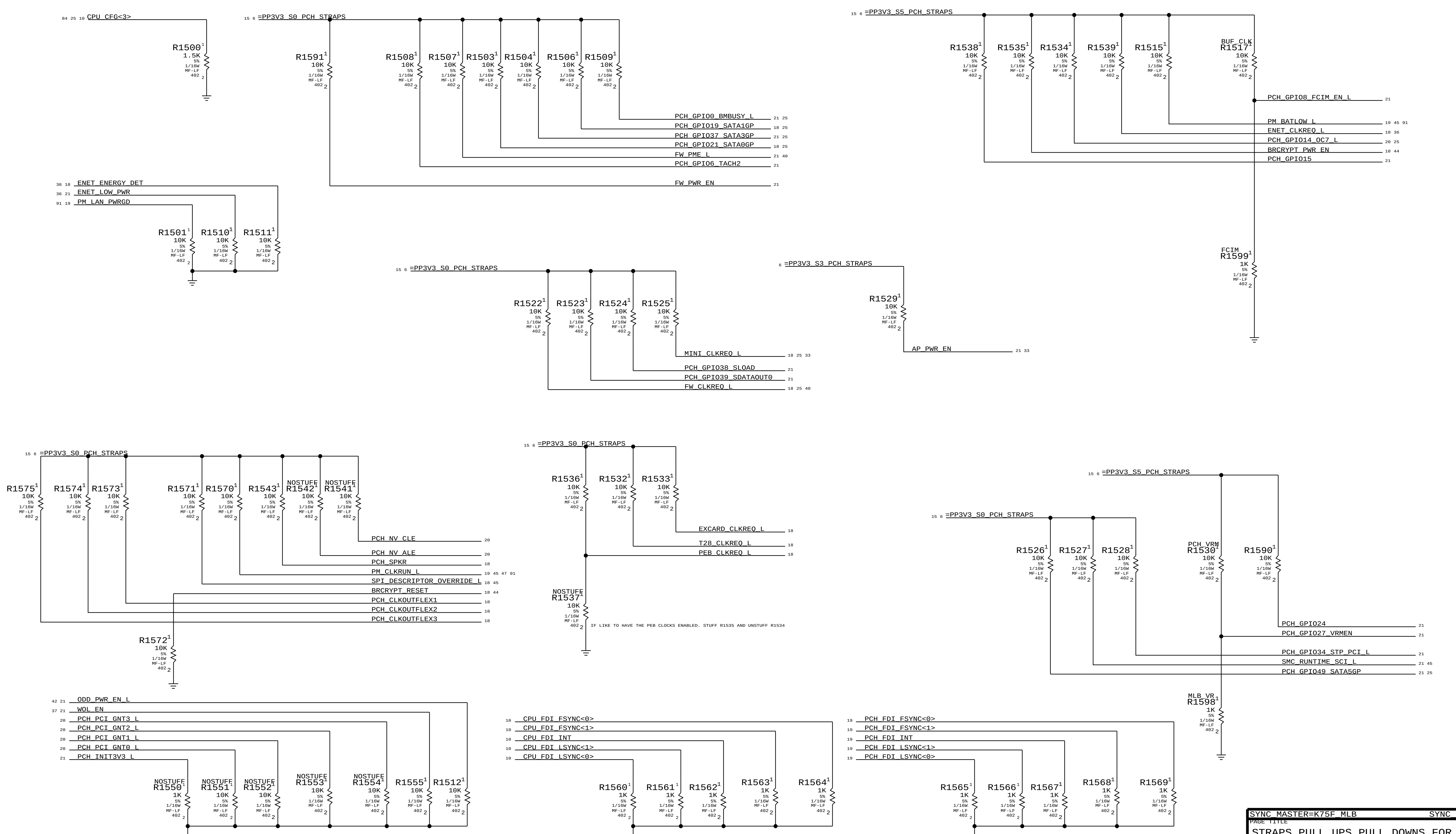




PAGE TITLE
CPU GROUNDS

Apple Inc.

DRAWING NUMBER	051-8600
SIZE	D
REVISION	A.0.0
BRANCH	
PAGE	14 OF 110
SHEET	14 OF 92



SYNC_MASTER=K75F_MLB SYNC_DATE=04/14/2010

PAGE TITLE

STRAPS,PULL UPS,PULL DOWNS FOR PCH AND CPU

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DRAWING NUMBER 051-8600 SIZE D

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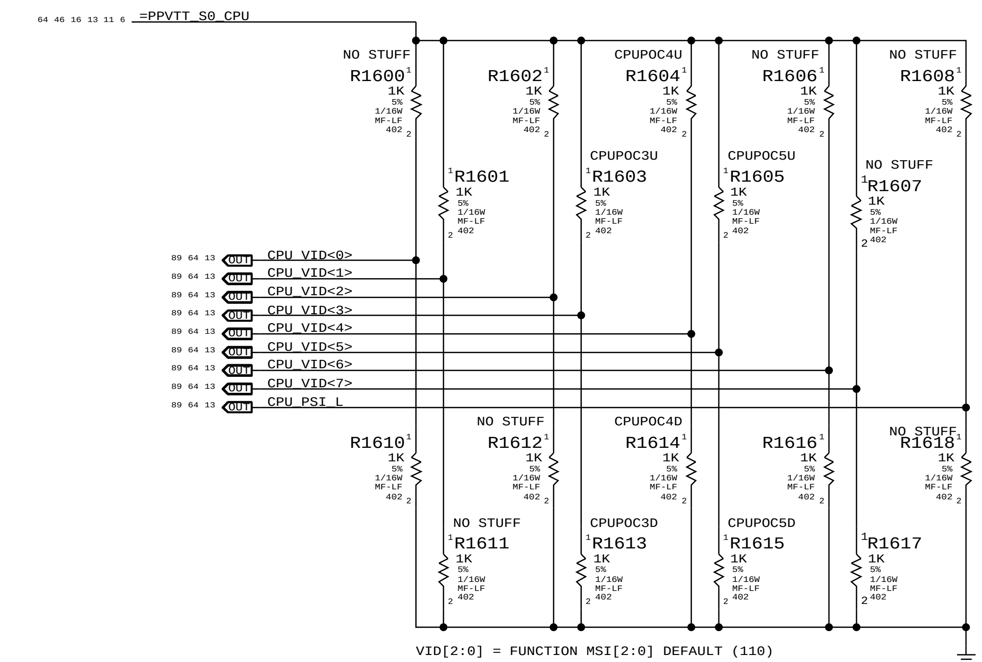
CPU Power On Configuration (POC) Straps

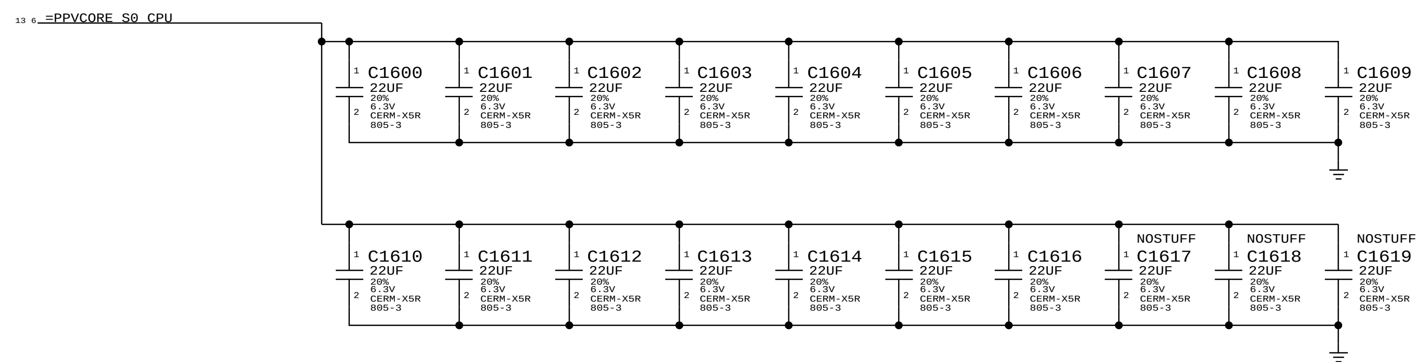
Intel recommends all option straps should be provided in layout



VID[2:0] = FUNCTION MSI[2:0] DEFAULT (110)
VID[5:3] = IMON CONFIG DEFAULT (101)
VID[6] = Reserved (0)
VID[7] = VRD SELECT (0)
PSI# = Reserved (0)

CPU VCORE DECOUPLING

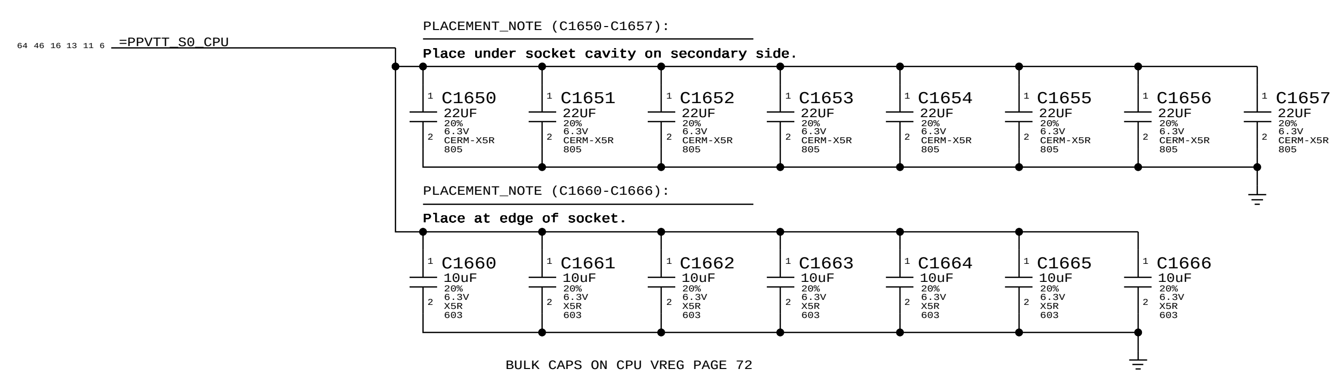
INTEL RECOMMENDATION 17X 22UF 0805



BULK CAPS ON CPU VREG PAGE 74

VTT (CPU Uncore) DECOUPLING

8X 22UF 0805, 7X 10UF 0805 INTEL RECOMMENDATION 9X22UF 0805



BULK CAPS ON CPU VREG PAGE 72

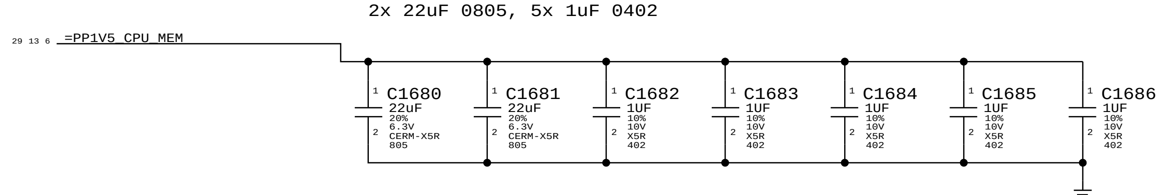
MSI - MARKET SEGMENT IDENTIFICATION
PREVENTS THE PLATFORM BOOTING USING A HIGHER POWERED CPU

BOM GROUP	IMAX @ 900mV	CPU Gain Setting	BOM OPTIONS	Equivalent Gain
CPUPOC_IMAX_DIS		000	CPUPOC3D,CPUPOC4D,CPUPOC5D	
CPUPOC_IMAX_0_40	40A	001	CPUPOC3D,CPUPOC4D,CPUPOC5U	45
CPUPOC_IMAX_40_60	60A	010	CPUPOC3D,CPUPOC4U,CPUPOC5D	30
CPUPOC_IMAX_60_80	80A	011	CPUPOC3D,CPUPOC4U,CPUPOC5U	22.5
CPUPOC_IMAX_80_100	100A	100	CPUPOC3U,CPUPOC4D,CPUPOC5D	18
CPUPOC_IMAX_100_120	120A	101	CPUPOC3U,CPUPOC4D,CPUPOC5U	15
CPUPOC_IMAX_120_140	140A	110	CPUPOC3U,CPUPOC4U,CPUPOC5D	12.857
CPUPOC_IMAX_140_180	180A	111	CPUPOC3U,CPUPOC4U,CPUPOC5U	10

NOTE: BOM Configurations should not call out CPUPOCnU/D BOMOPTIONs directly.
Instead call out appropriate BOM GROUP defined in tables above.

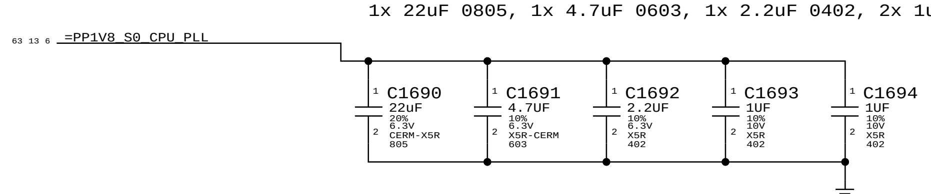
Memory (CPU VCCDDR) DECOUPLING

2x 22uF 0805, 5x 1uF 0402



PLL (CPU VCCSFR) DECOUPLING

1x 22uF 0805, 1x 4.7uF 0603, 1x 2.2uF 0402, 2x 1uF 0402



SYNC_MASTER=K75F_MLB SYNC_DATE=04/14/2010

PAGE TITLE: CPU NON-GFX DECOUPLING

Apple Inc.

DRAWING NUMBER: 051-8600 SIZE: D
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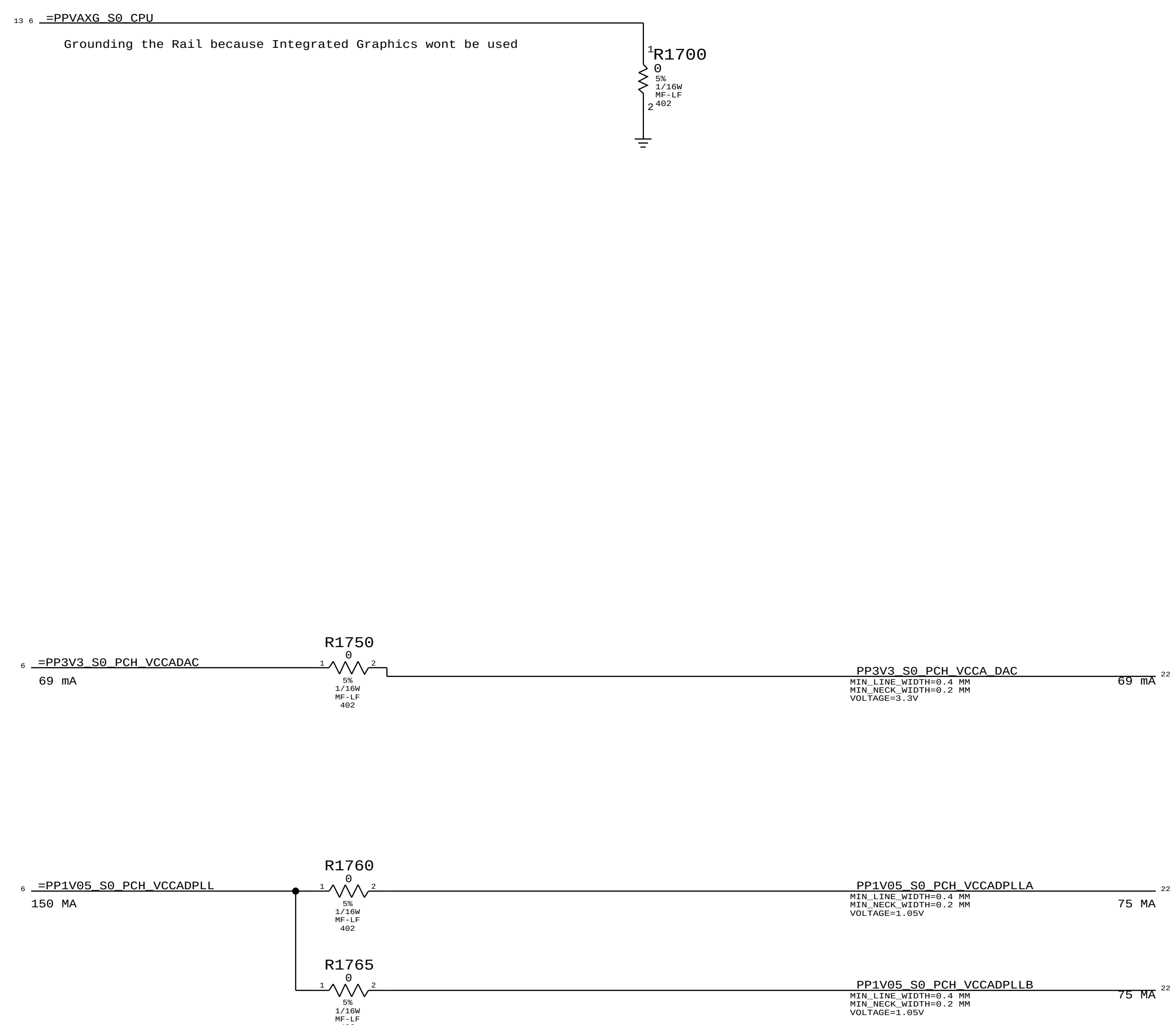

SYNC_MASTER=K75F_MLB SYNC_DATE=04/14/2010

PAGE TITLE: CPU/PCH GFX DECOUPLING	
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	REVISION: A.0.0
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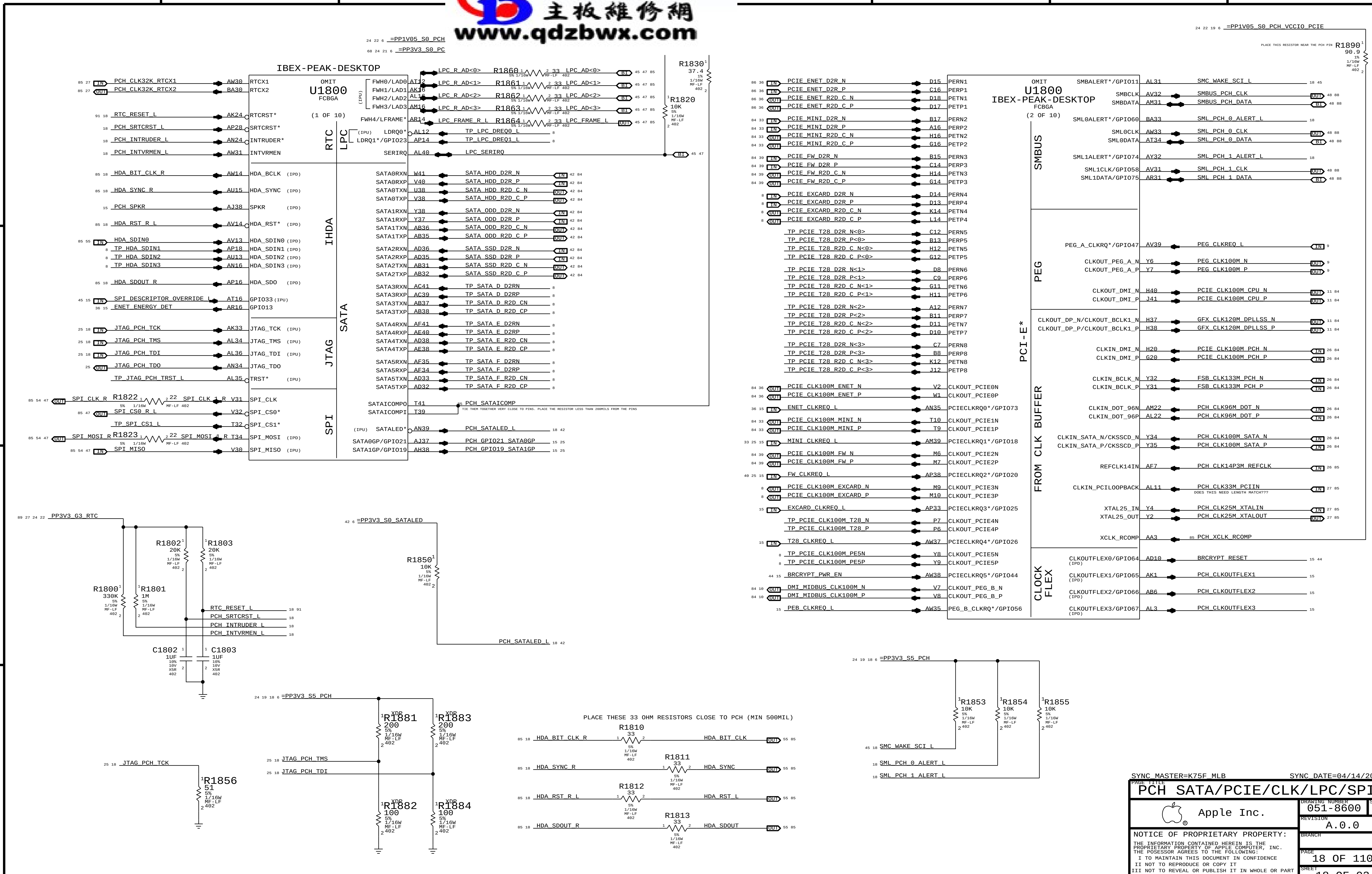
	SHEET: 17 OF 92



SYNC_MASTER=K75F_MLB SYNC_DATE=04/14/2010

PAGE TITLE

PCH SATA/PCIE/CLK/LPC/SPI

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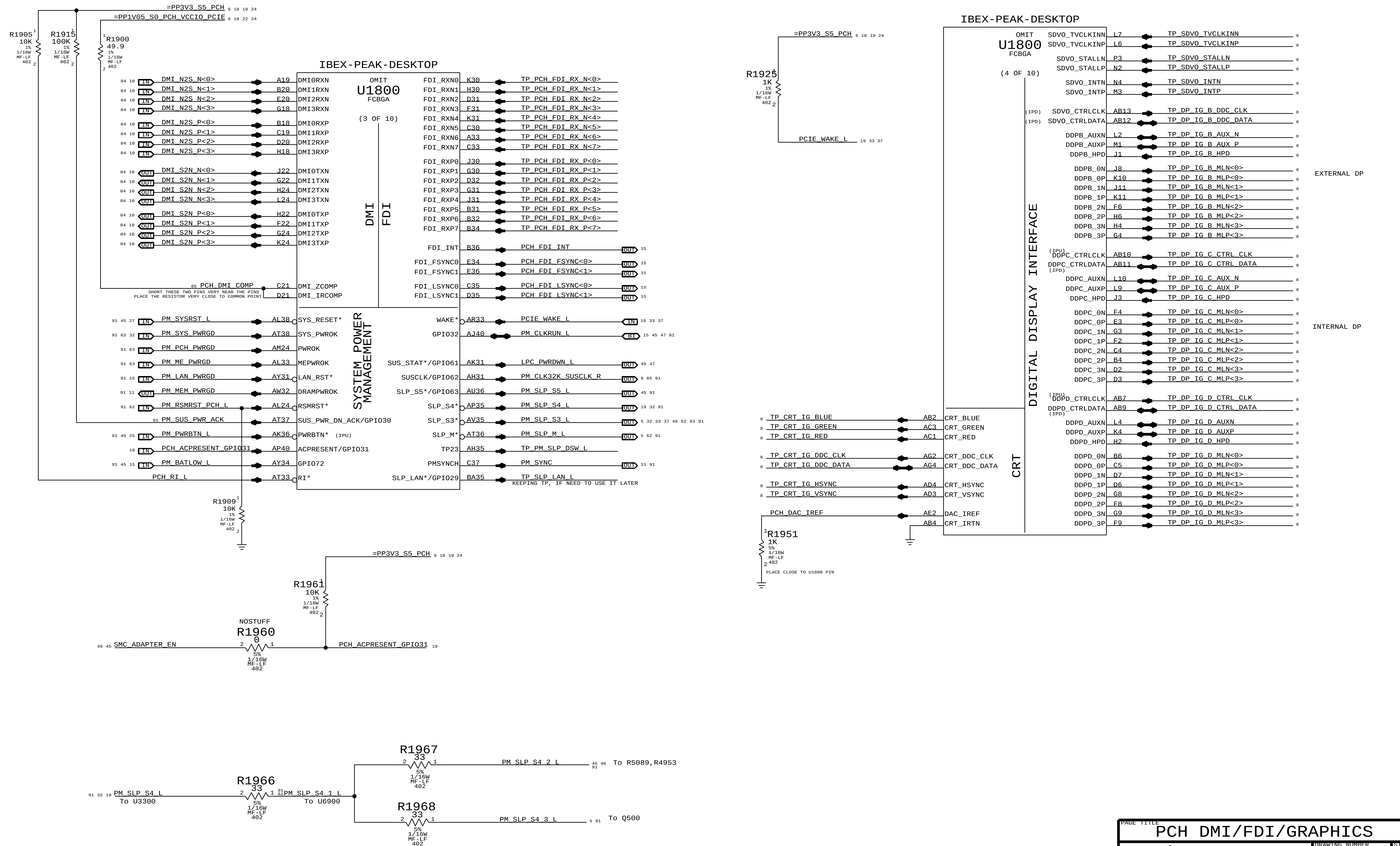

PAGE TITLE

PCH DMI/FDI/GRAPHICS

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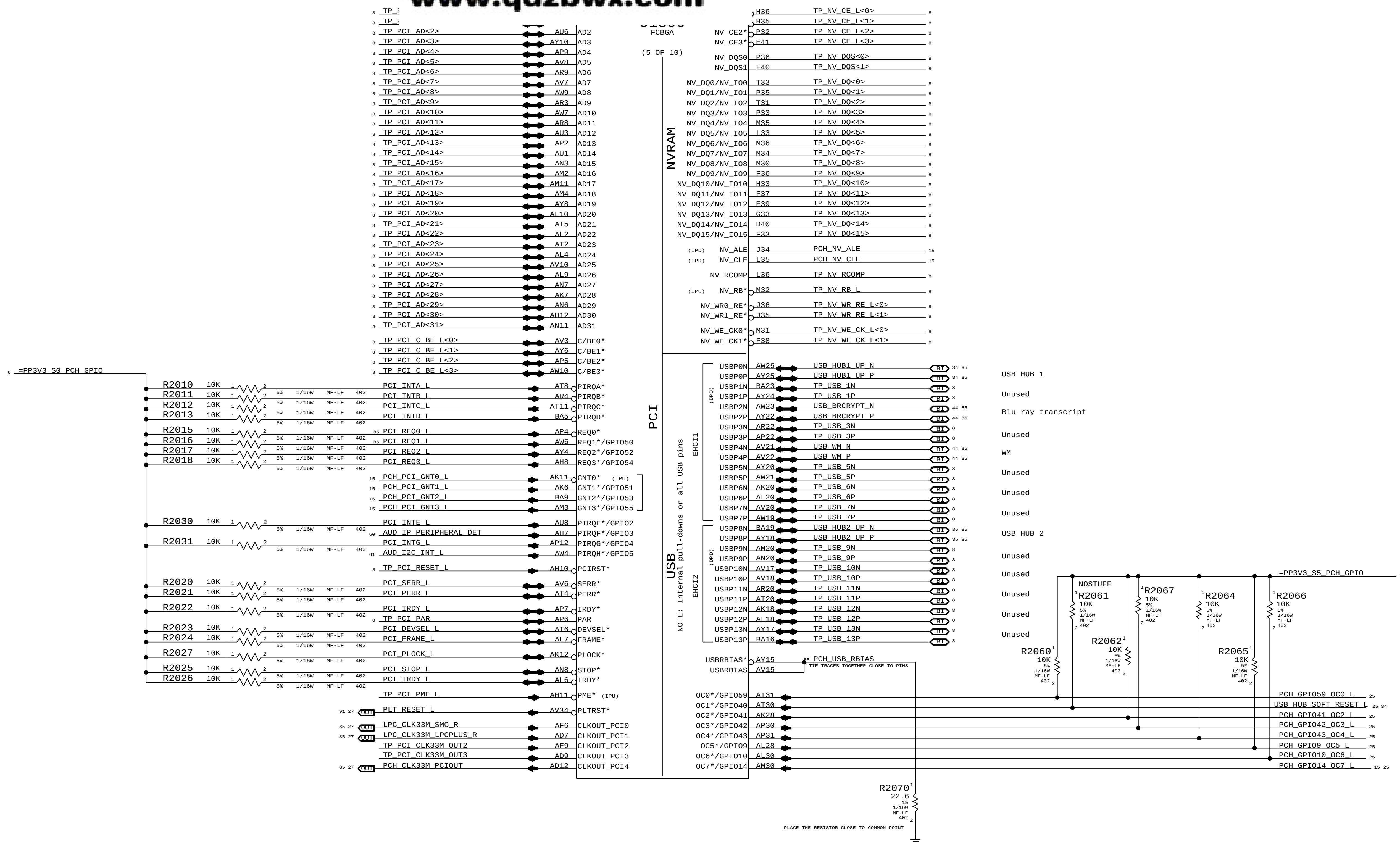

SYNC_MASTER=K75F_MLB SYNC_DATE=04/14/2010

PAGE TITLE

PCH PCI/FLASHCACHE/USB

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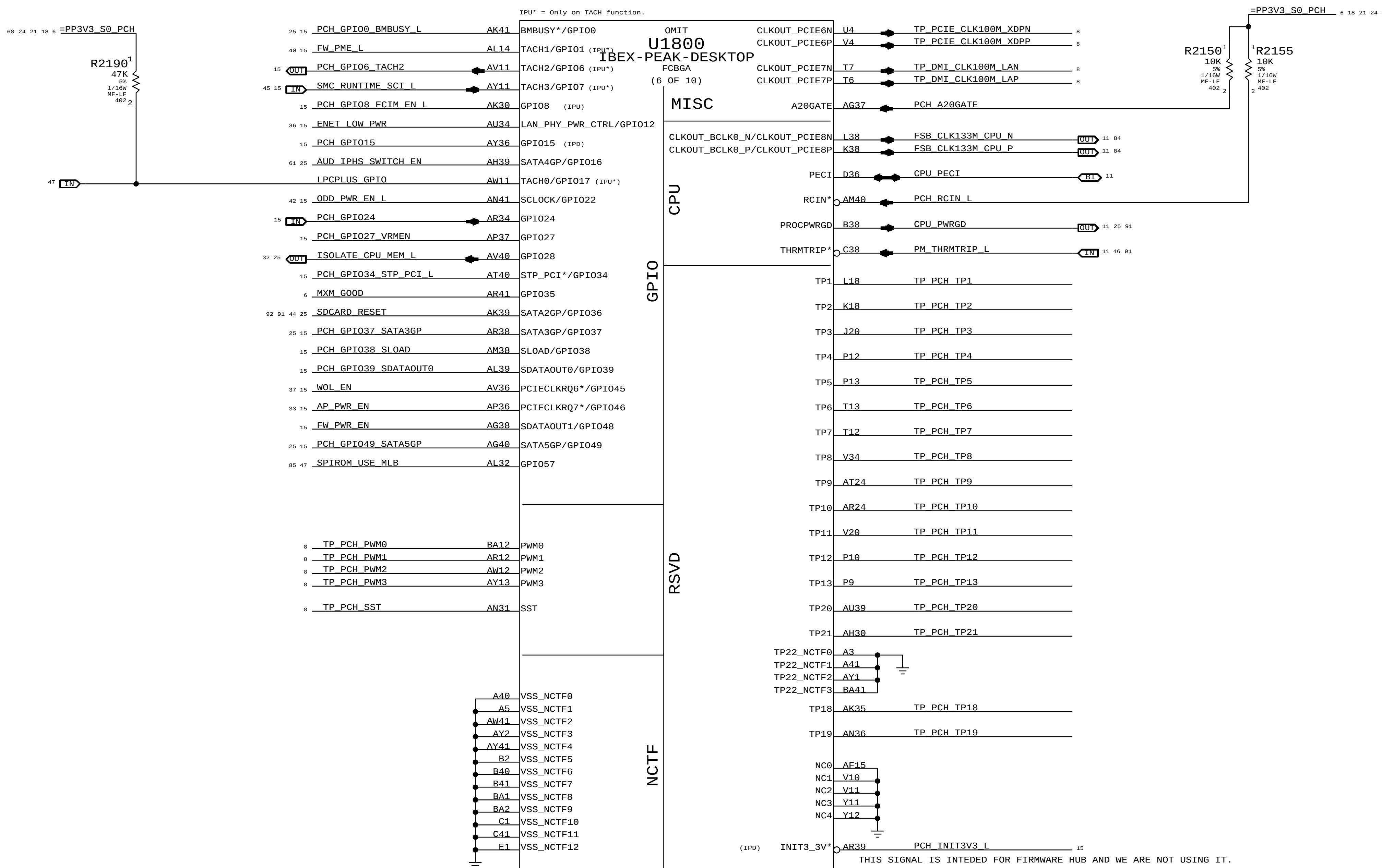

SYNC_MASTER=K75F_MLB SYNC_DATE=04/14/2010

PAGE TITLE
PCH MISC

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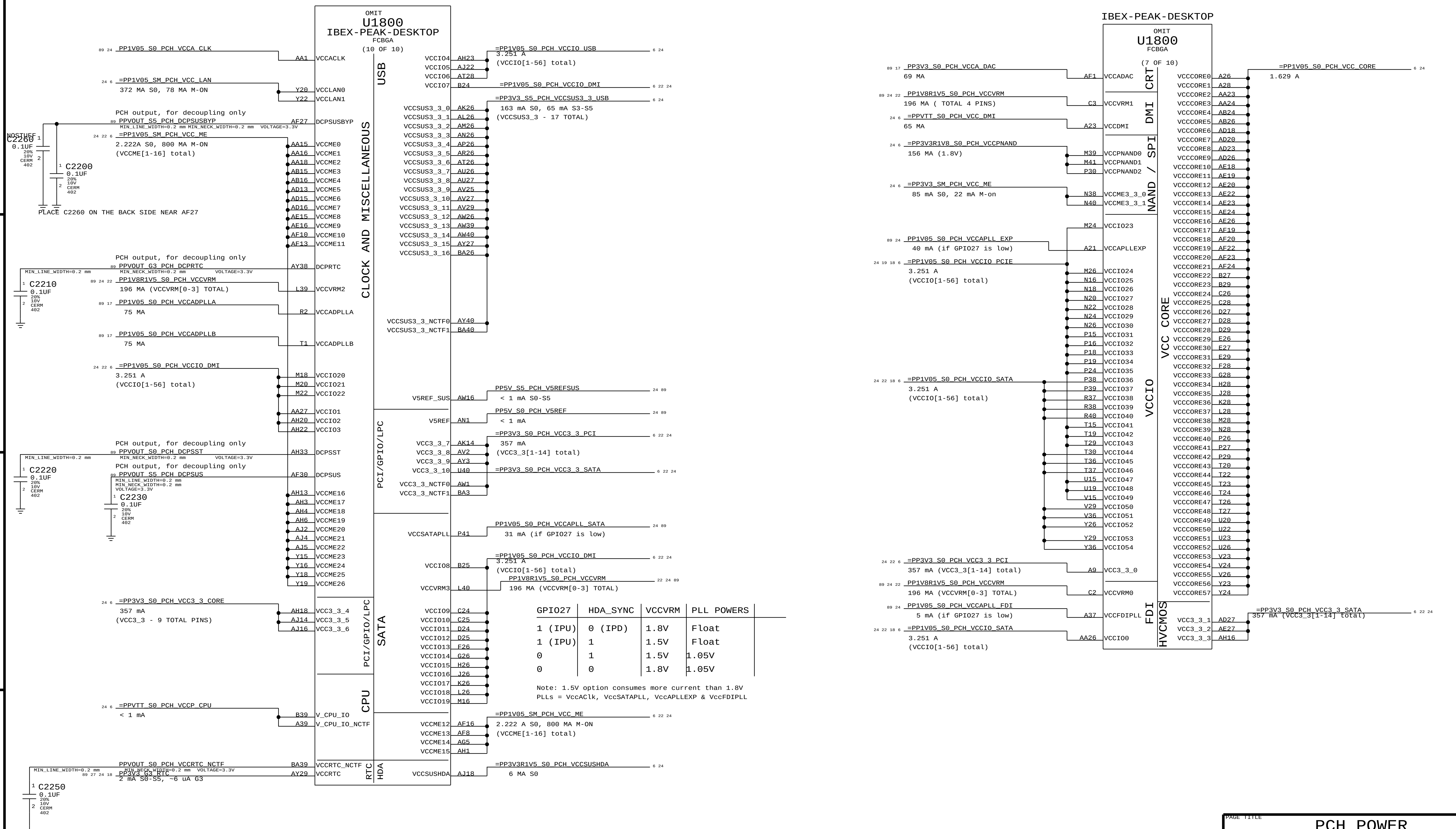

GPIO27	HDA_SYNC	VCCVRM	PLL POWERS
1 (IPU)	0 (IPD)	1.8V	Float
1 (IPU)	1	1.5V	Float
0	1	1.5V	1.05V
0	0	1.8V	1.05V

Note: 1.5V option consumes more current than 1.8V
PLLs = VccAClk, VccSATAPLL, VccAPLLEXP & VccFDIPLL

PAGE TITLE
PCH POWER

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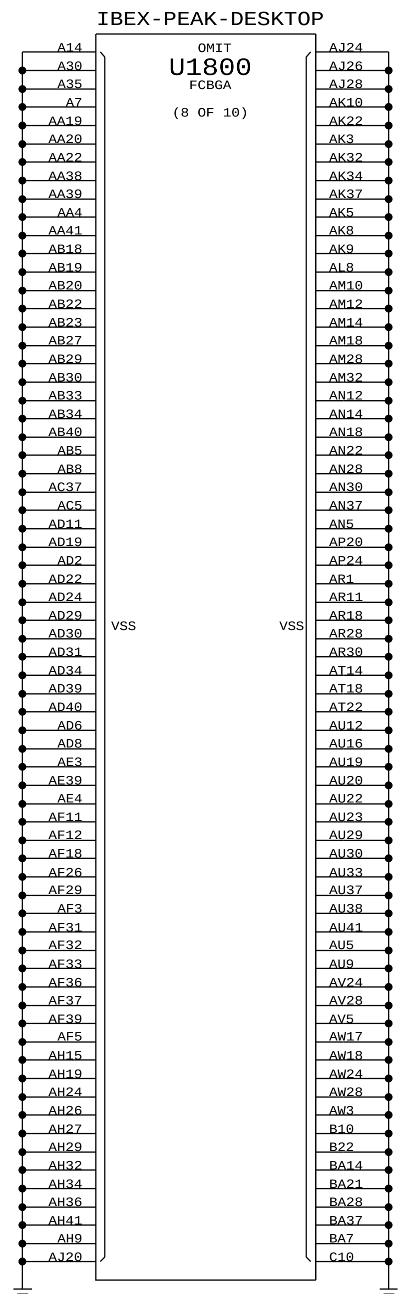
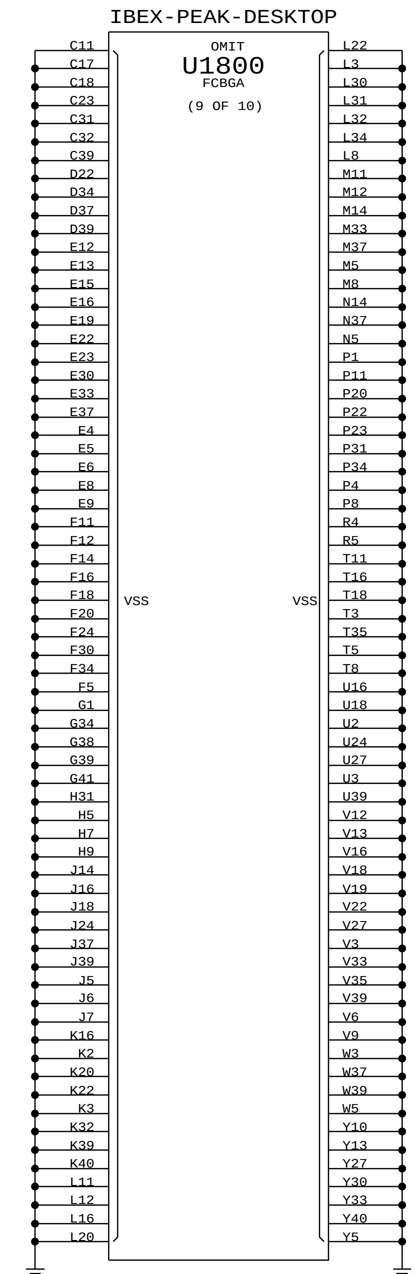
PAGE 22 OF 110
SHEET 22 OF 92





SYNC_MASTER=K75F_MLB SYNC_DATE=04/14/2010

PAGE TITLE
PCH GROUNDS

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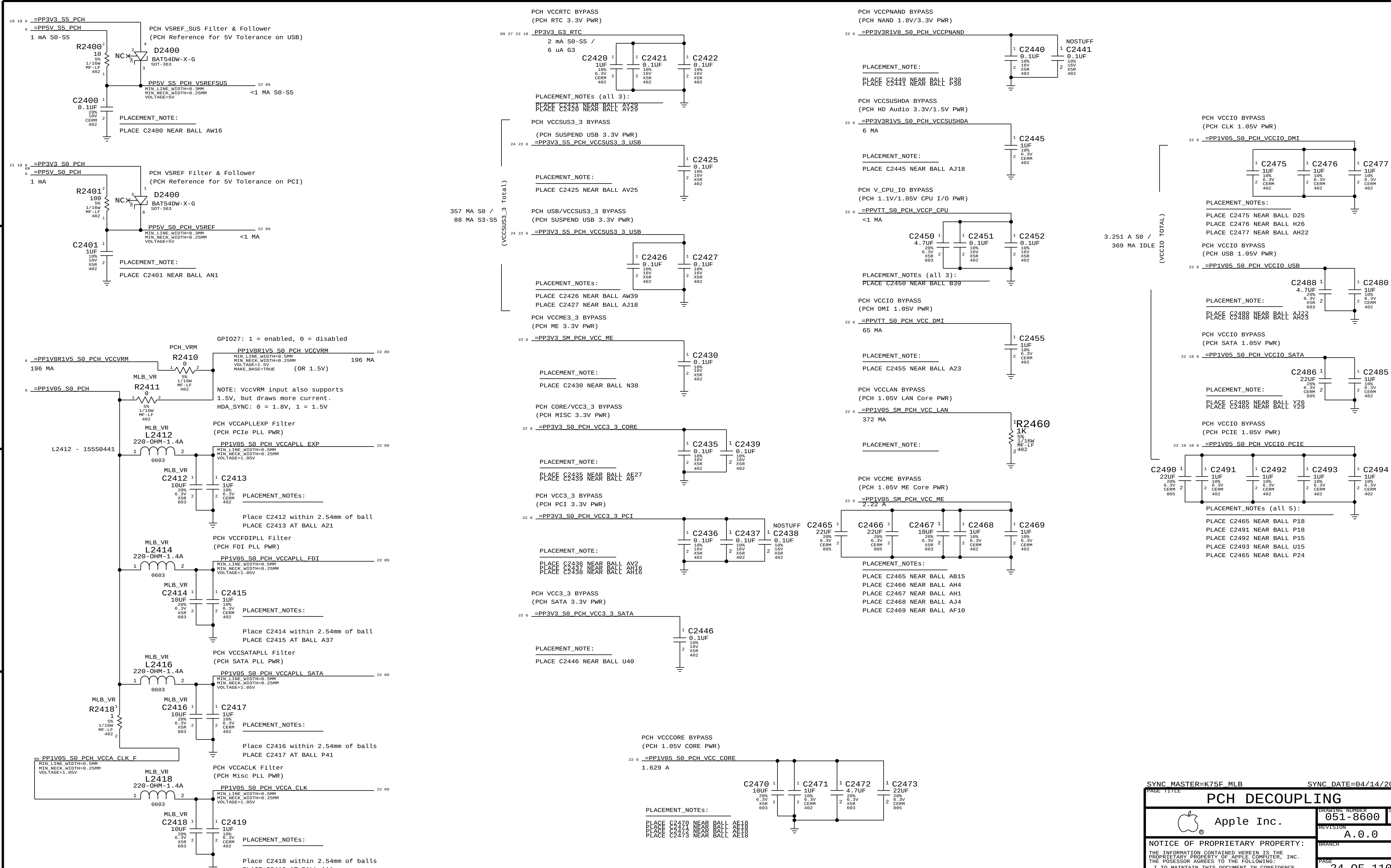

Current numbers from Ibex Peak EDS Spec Update rev 0.71, doc #386904 (Table 8-3). Pre-Silicon Mobile Estimates.

SYNC_MASTER=K75F_MLB SYNC_DATE=04/14/2010

PAGE TITLE
PCH DECOUPLING

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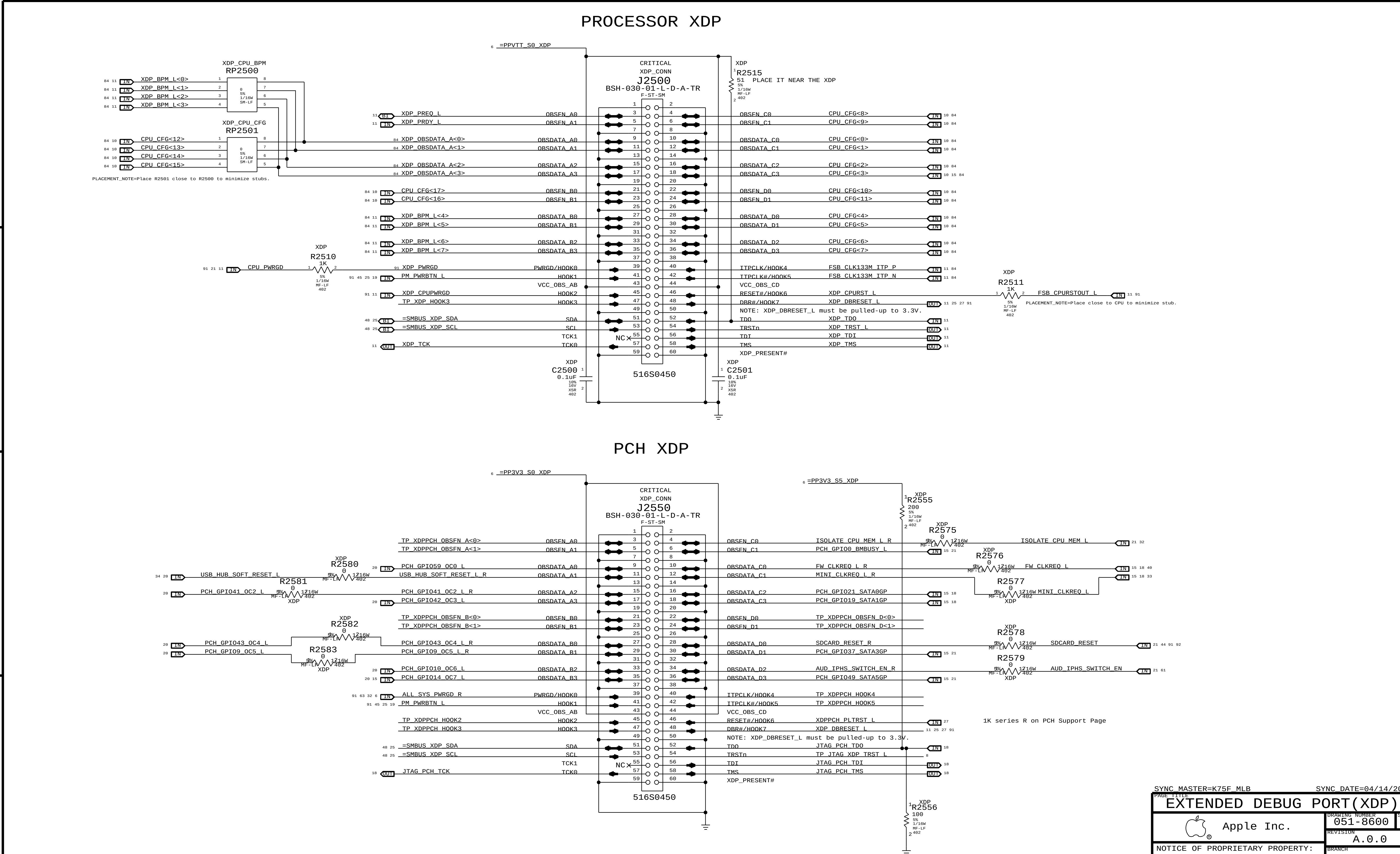

SYNC_MASTER=K75F_MLB SYNC_DATE=04/14/2010

PAGE TITLE

EXTENDED DEBUG PORT(XDP)

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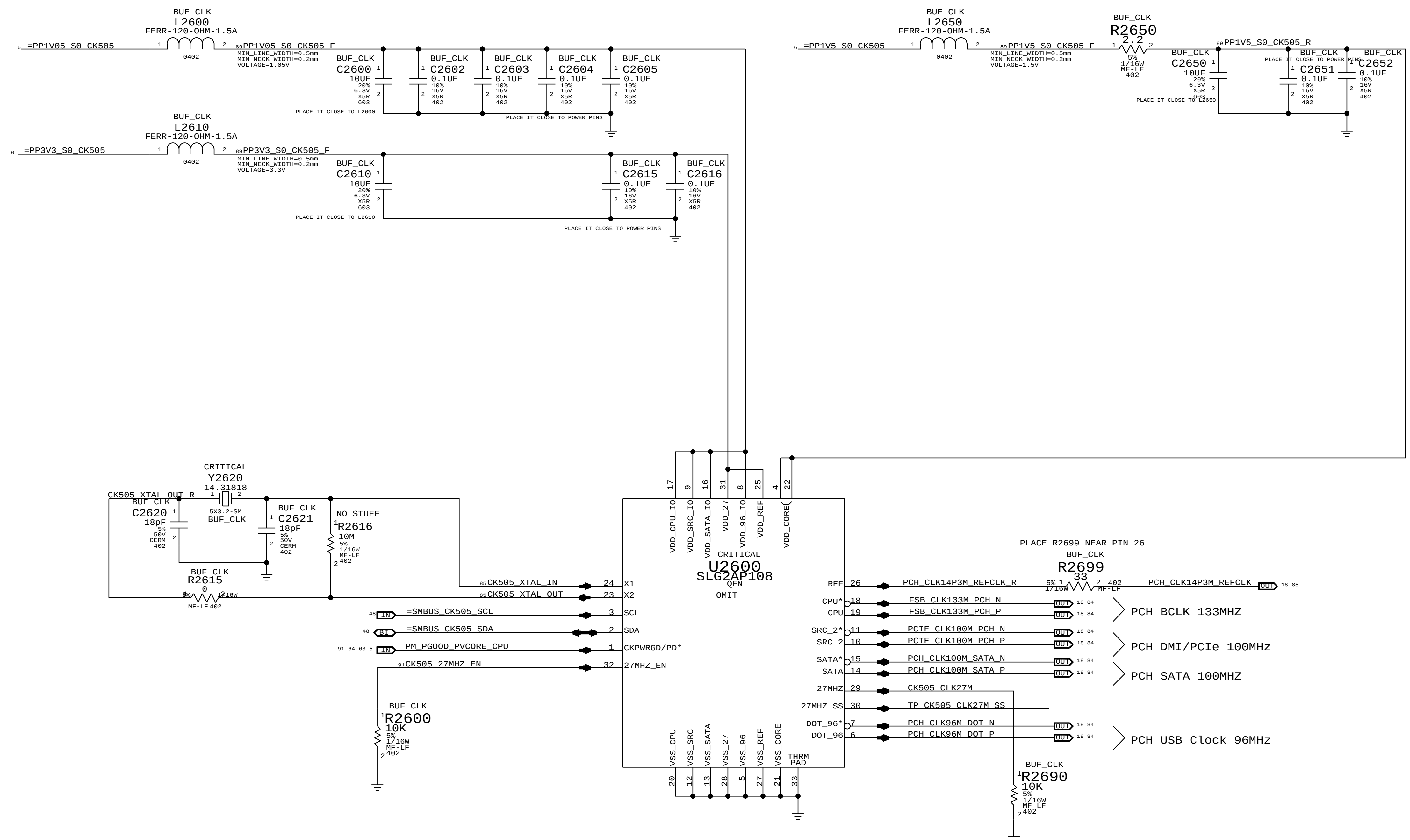

SYNC_MASTER=K75F_MLB SYNC_DATE=04/14/2010

PAGE TITLE

CLOCK (CK505)

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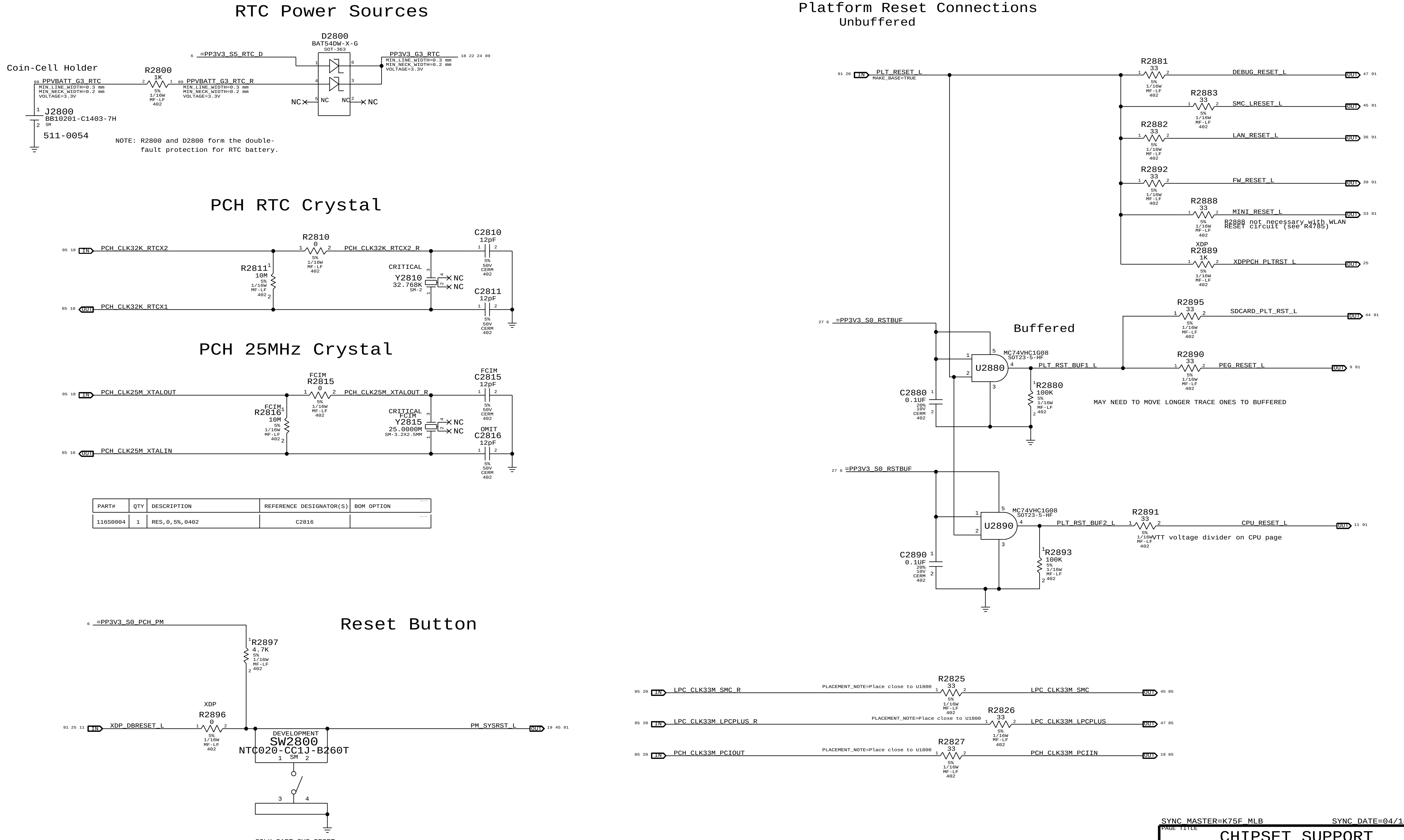

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
116S0004	1	RES,0,5%,0402	C2816	

SYNC_MASTER=K75F_MLB SYNC_DATE=04/14/2010

PAGE TITLE
CHIPSET SUPPORT

Apple Inc.

DRAWING NUMBER 051-8600 SIZE D

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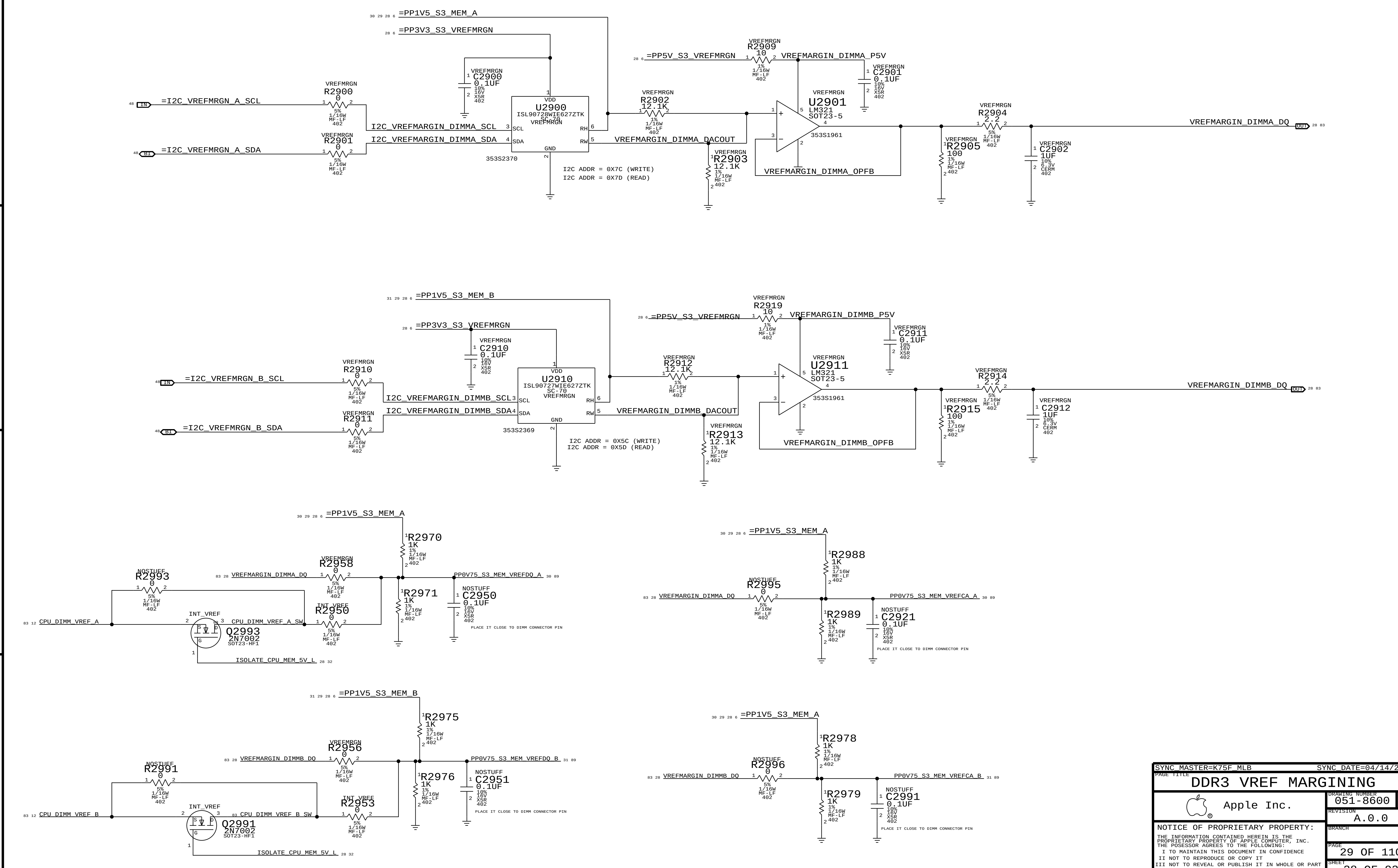

SYNC_MASTER=K75F_MLB SYNC_DATE=04/14/2010

PAGE TITLE

DDR3 VREF MARGINING

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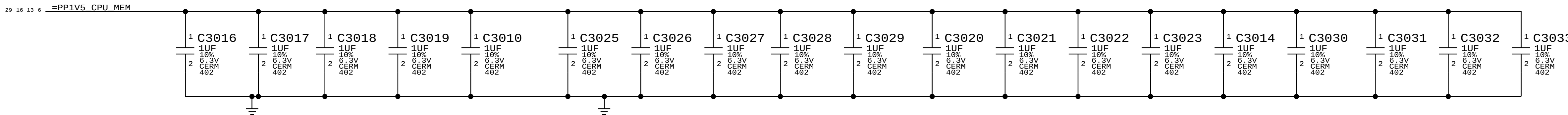
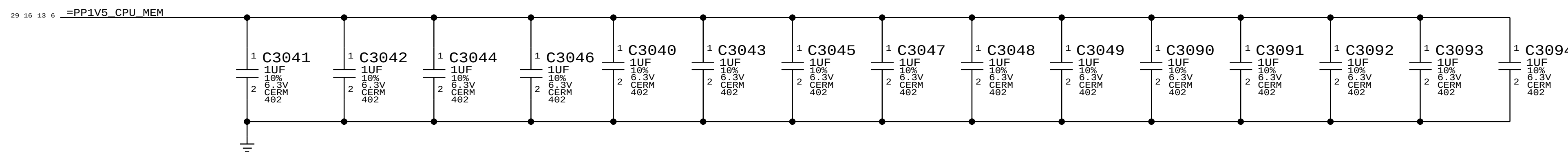
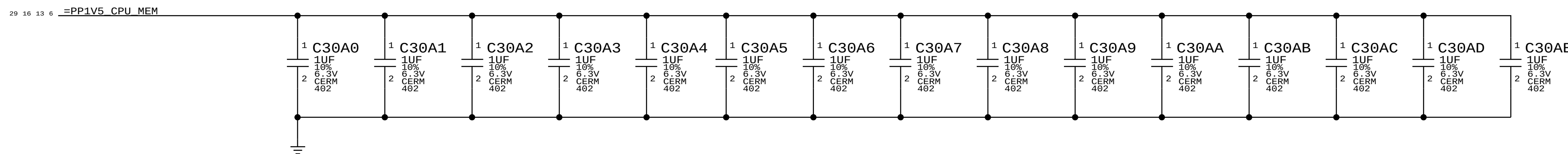
DIMM A (FURTHER FROM CPU)   CAPS TO COUPLE CPU 1V5_MEM   DIMM B (CLOSER TO CPU)

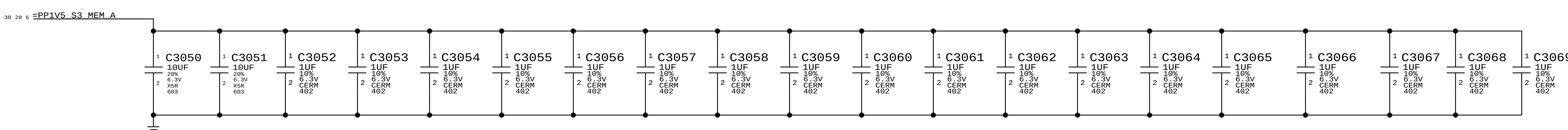


EXTRA DECOUPLING CAPS FOR CPU MEM RAIL

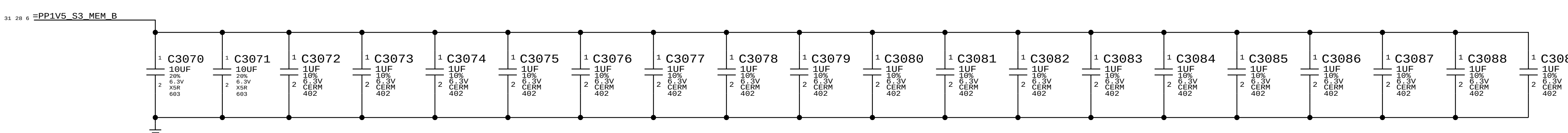




DECOUPLING CAPS FOR DIMM ON CHANNEL A - AT CONNECTOR



DECOUPLING CAPS FOR DIMM ON CHANNEL B - AT CONNECTOR



SYNC_MASTER=K75F_MLB	SYNC_DATE=04/14/2010
PAGE TITLE	MEMORY CAPS
Apple Inc.	DRAWING NUMBER 051-8600 SIZE D
	REVISION A.0.0
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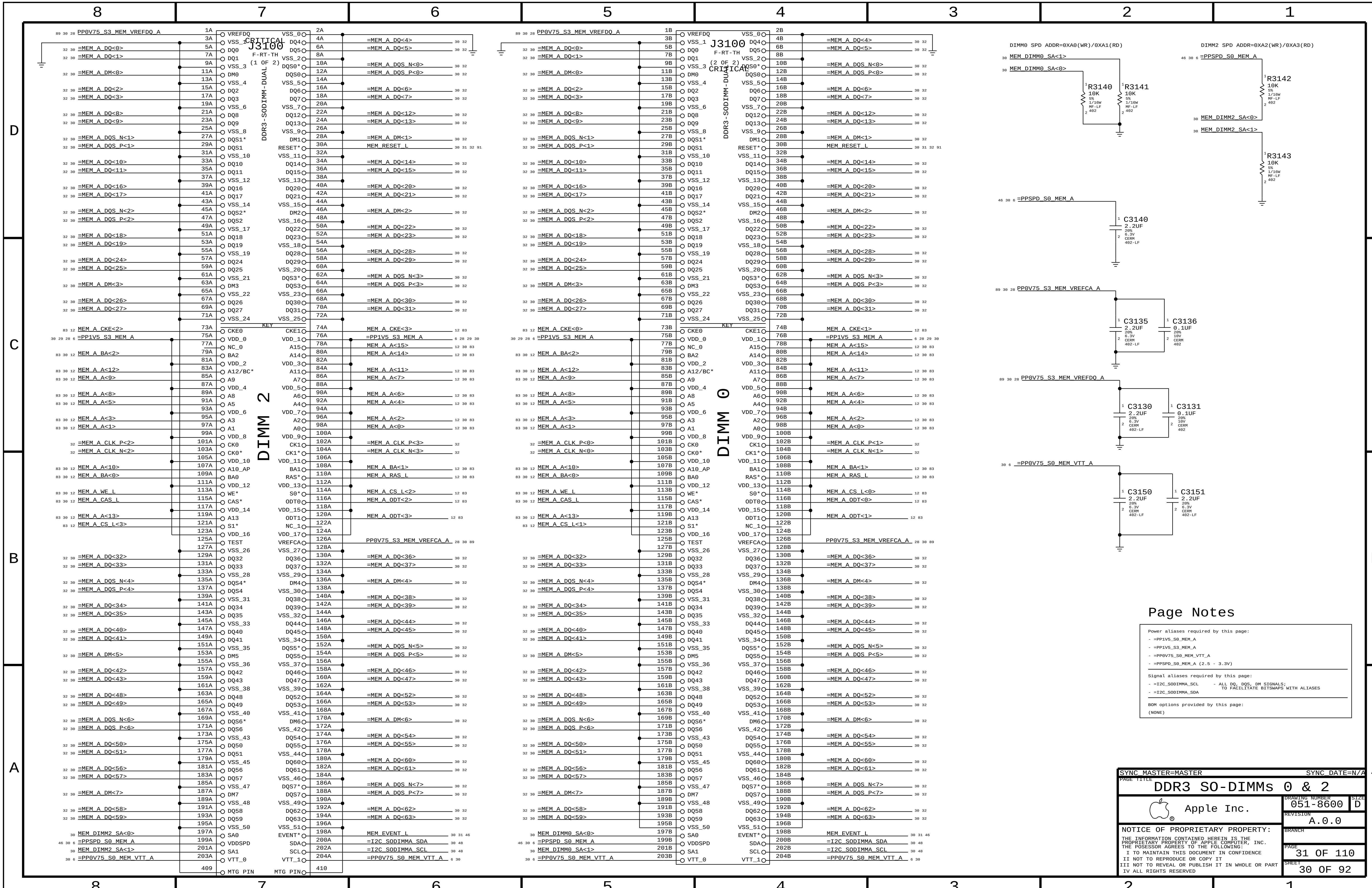

J3100
F-RT-TH
(1 OF 2)
CRITICAL
DDR3-SODIMM-DUAL
DIMM 2
J3100
F-RT-TH
(2 OF 2)
CRITICAL
DDR3-SODIMM-DUAL
DIMM 0
KEY
DIMM0 SPD ADDR=0XA0(WR)/0XA1(RD)
DIMM2 SPD ADDR=0XA2(WR)/0XA3(RD)
MEM_DIMM0_SA<1>
MEM_DIMM0_SA<0>
=PPSPD_S0_MEM_A
MEM_DIMM2_SA<0>
MEM_DIMM2_SA<1>
R3140 10K 5% 1/16W MF-LF 402
R3141 10K 5% 1/16W MF-LF 402
R3142 10K 5% 1/16W MF-LF 402
R3143 10K 5% 1/16W MF-LF 402
C3140 2.2UF 20% 6.3V CERM 402-LF
PP0V75_S3_MEM_VREFCA_A
C3135 2.2UF 20% 6.3V CERM 402-LF
C3136 0.1UF 20% 10V CERM 402
PP0V75_S3_MEM_VREFDQ_A
C3130 2.2UF 20% 6.3V CERM 402-LF
C3131 0.1UF 20% 10V CERM 402
=PP0V75_S0_MEM_VTT_A
C3150 2.2UF 20% 6.3V CERM 402-LF
C3151 2.2UF 20% 6.3V CERM 402-LF
MEM_RESET_L
MEM_EVENT_L
=I2C_SODIMMA_SDA
=I2C_SODIMMA_SCL
=PP1V5_S3_MEM_A
Page Notes
Power aliases required by this page:
- =PP1V5_S0_MEM_A
- =PP1V5_S3_MEM_A
- =PP0V75_S0_MEM_VTT_A
- =PPSPD_S0_MEM_A (2.5 - 3.3V)
Signal aliases required by this page:
- =I2C_SODIMMA_SCL - ALL DQ, DQS, DM SIGNALS; TO FACILITATE BITSWAPS WITH ALIASES
- =I2C_SODIMMA_SDA
BOM options provided by this page:
(NONE)
SYNC_MASTER=MASTER
SYNC_DATE=N/A
PAGE TITLE
DDR3 SO-DIMMs 0 & 2
Apple Inc.
DRAWING NUMBER
051-8600
SIZE
D
REVISION
A.0.0
BRANCH
PAGE
31 OF 110
SHEET
30 OF 92
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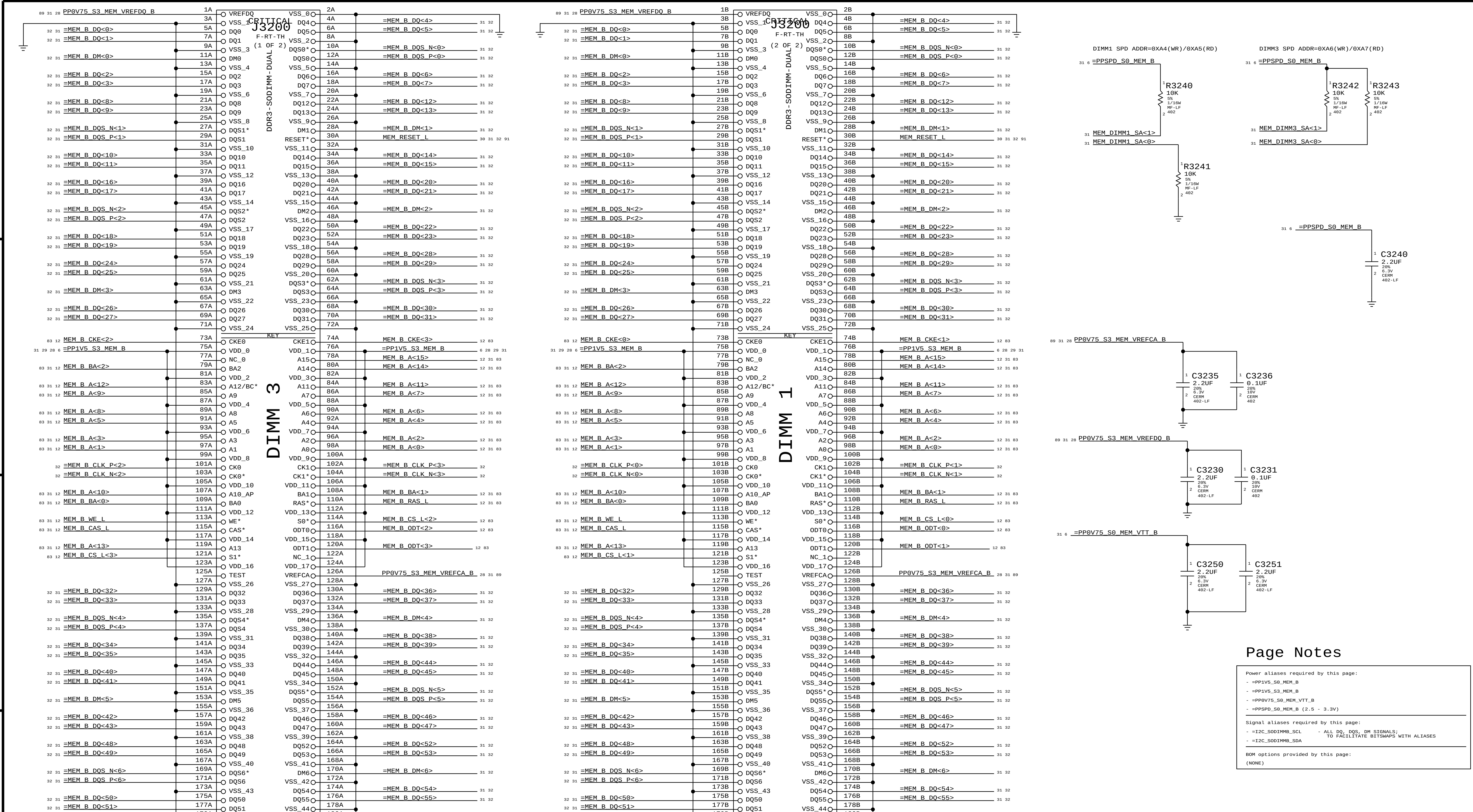

Page Notes

Power aliases required by this page:
- =PP1V5_S0_MEM_B
- =PP1V5_S3_MEM_B
- =PP0V75_S0_MEM_VTT_B
- =PPSPD_S0_MEM_B (2.5 - 3.3V)

Signal aliases required by this page:
- =I2C_SODIMMB_SCL — ALL DQ, DQS, DM SIGNALS; TO FACILITATE BITSWAPS WITH ALIASES
- =I2C_SODIMMB_SDA

BOM options provided by this page:
(NONE)

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PAGE TITLE
DDR3 SO-DIMM CONNECTOR B

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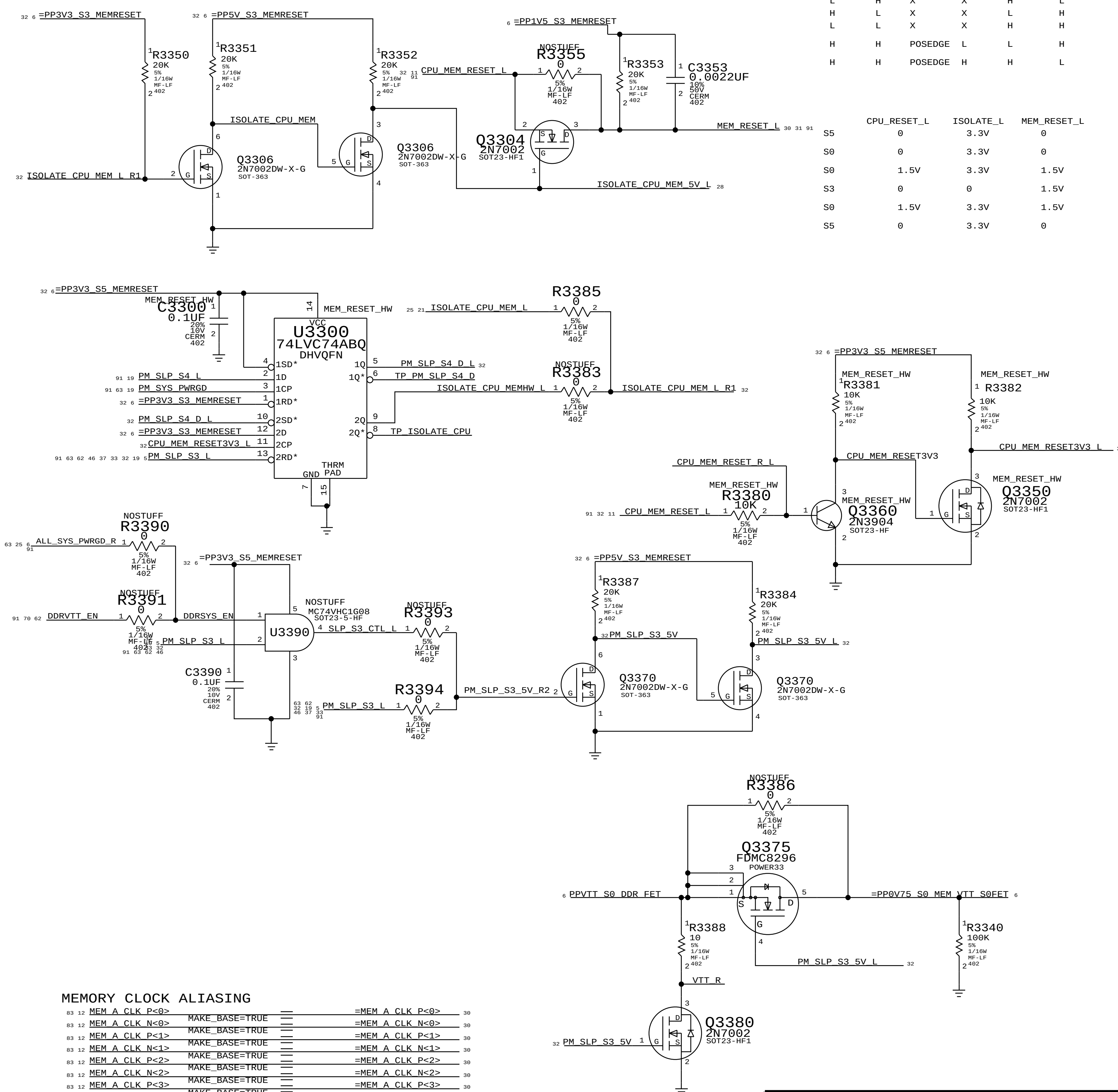
DDR3 RESET Support

LFD CANNOT CONTROL THIS SIGNAL DIRECTLY SINCE IT MUST BE HIGH IN SLEEP AND CPU MEM RAILS ARE NOT POWERED IN SLEEP.



S	R	CLK	D	Q	QB
L	H	X	X	H	L
H	L	X	X	L	H
L	L	X	X	H	H
H	H	POSEDGE	L	L	H
H	H	POSEDGE	H	H	L

	CPU_RESET_L	ISOLATE_L	MEM_RESET_L
S5	0	3.3V	0
S0	0	3.3V	0
S0	1.5V	3.3V	1.5V
S3	0	0	1.5V
S0	1.5V	3.3V	1.5V
S5	0	3.3V	0

CPU CHANNEL A DQS 0 -> DIMM A DQS 7

83 12	MEM_A_DQS_N<0>	MAKE_BASE=TRUE	=MEM_A_DQS_N<7>	30
83 12	MEM_A_DQS_P<0>	MAKE_BASE=TRUE	=MEM_A_DQS_P<7>	30
83 12	MEM_A_DM<0>	MAKE_BASE=TRUE	=MEM_A_DM<7>	30
83 12	MEM_A_DQ<7>	MAKE_BASE=TRUE	=MEM_A_DQ<57>	30
83 12	MEM_A_DQ<6>	MAKE_BASE=TRUE	=MEM_A_DQ<56>	30
83 12	MEM_A_DQ<5>	MAKE_BASE=TRUE	=MEM_A_DQ<58>	30
83 12	MEM_A_DQ<4>	MAKE_BASE=TRUE	=MEM_A_DQ<59>	30
83 12	MEM_A_DQ<3>	MAKE_BASE=TRUE	=MEM_A_DQ<60>	30
83 12	MEM_A_DQ<2>	MAKE_BASE=TRUE	=MEM_A_DQ<61>	30
83 12	MEM_A_DQ<1>	MAKE_BASE=TRUE	=MEM_A_DQ<62>	30
83 12	MEM_A_DQ<0>	MAKE_BASE=TRUE	=MEM_A_DQ<63>	30

CPU CHANNEL A DQS 1 -> DIMM A DQS 6

83 12	MEM_A_DQS_N<1>	MAKE_BASE=TRUE	=MEM_A_DQS_N<6>	30
83 12	MEM_A_DQS_P<1>	MAKE_BASE=TRUE	=MEM_A_DQS_P<6>	30
83 12	MEM_A_DM<1>	MAKE_BASE=TRUE	=MEM_A_DM<6>	30
83 12	MEM_A_DQ<15>	MAKE_BASE=TRUE	=MEM_A_DQ<49>	30
83 12	MEM_A_DQ<14>	MAKE_BASE=TRUE	=MEM_A_DQ<52>	30
83 12	MEM_A_DQ<13>	MAKE_BASE=TRUE	=MEM_A_DQ<51>	30
83 12	MEM_A_DQ<12>	MAKE_BASE=TRUE	=MEM_A_DQ<50>	30
83 12	MEM_A_DQ<11>	MAKE_BASE=TRUE	=MEM_A_DQ<48>	30
83 12	MEM_A_DQ<10>	MAKE_BASE=TRUE	=MEM_A_DQ<53>	30
83 12	MEM_A_DQ<9>	MAKE_BASE=TRUE	=MEM_A_DQ<55>	30
83 12	MEM_A_DQ<8>	MAKE_BASE=TRUE	=MEM_A_DQ<54>	30

CPU CHANNEL A DQS 2 -> DIMM A DQS 5

83 12	MEM_A_DQS_N<2>	MAKE_BASE=TRUE	=MEM_A_DQS_N<5>	30
83 12	MEM_A_DQS_P<2>	MAKE_BASE=TRUE	=MEM_A_DQS_P<5>	30
83 12	MEM_A_DM<2>	MAKE_BASE=TRUE	=MEM_A_DM<5>	30
83 12	MEM_A_DQ<23>	MAKE_BASE=TRUE	=MEM_A_DQ<40>	30
83 12	MEM_A_DQ<22>	MAKE_BASE=TRUE	=MEM_A_DQ<45>	30
83 12	MEM_A_DQ<21>	MAKE_BASE=TRUE	=MEM_A_DQ<46>	30
83 12	MEM_A_DQ<20>	MAKE_BASE=TRUE	=MEM_A_DQ<47>	30
83 12	MEM_A_DQ<19>	MAKE_BASE=TRUE	=MEM_A_DQ<41>	30
83 12	MEM_A_DQ<18>	MAKE_BASE=TRUE	=MEM_A_DQ<44>	30
83 12	MEM_A_DQ<17>	MAKE_BASE=TRUE	=MEM_A_DQ<43>	30
83 12	MEM_A_DQ<16>	MAKE_BASE=TRUE	=MEM_A_DQ<42>	30

CPU CHANNEL A DQS 3 -> DIMM A DQS 4

83 12	MEM_A_DQS_N<3>	MAKE_BASE=TRUE	=MEM_A_DQS_N<4>	30
83 12	MEM_A_DQS_P<3>	MAKE_BASE=TRUE	=MEM_A_DQS_P<4>	30
83 12	MEM_A_DM<3>	MAKE_BASE=TRUE	=MEM_A_DM<4>	30
83 12	MEM_A_DQ<31>	MAKE_BASE=TRUE	=MEM_A_DQ<37>	30
83 12	MEM_A_DQ<30>	MAKE_BASE=TRUE	=MEM_A_DQ<33>	30
83 12	MEM_A_DQ<29>	MAKE_BASE=TRUE	=MEM_A_DQ<35>	30
83 12	MEM_A_DQ<28>	MAKE_BASE=TRUE	=MEM_A_DQ<34>	30
83 12	MEM_A_DQ<27>	MAKE_BASE=TRUE	=MEM_A_DQ<36>	30
83 12	MEM_A_DQ<26>	MAKE_BASE=TRUE	=MEM_A_DQ<32>	30
83 12	MEM_A_DQ<25>	MAKE_BASE=TRUE	=MEM_A_DQ<38>	30
83 12	MEM_A_DQ<24>	MAKE_BASE=TRUE	=MEM_A_DQ<39>	30

CPU CHANNEL A DQS 4 -> DIMM A DQS 3

83 12	MEM_A_DQS_N<4>	MAKE_BASE=TRUE	=MEM_A_DQS_N<3>	30
83 12	MEM_A_DQS_P<4>	MAKE_BASE=TRUE	=MEM_A_DQS_P<3>	30
83 12	MEM_A_DM<4>	MAKE_BASE=TRUE	=MEM_A_DM<3>	30
83 12	MEM_A_DQ<39>	MAKE_BASE=TRUE	=MEM_A_DQ<28>	30
83 12	MEM_A_DQ<38>	MAKE_BASE=TRUE	=MEM_A_DQ<29>	30
83 12	MEM_A_DQ<37>	MAKE_BASE=TRUE	=MEM_A_DQ<27>	30
83 12	MEM_A_DQ<36>	MAKE_BASE=TRUE	=MEM_A_DQ<31>	30
83 12	MEM_A_DQ<35>	MAKE_BASE=TRUE	=MEM_A_DQ<25>	30
83 12	MEM_A_DQ<34>	MAKE_BASE=TRUE	=MEM_A_DQ<24>	30
83 12	MEM_A_DQ<33>	MAKE_BASE=TRUE	=MEM_A_DQ<26>	30
83 12	MEM_A_DQ<32>	MAKE_BASE=TRUE	=MEM_A_DQ<30>	30

CPU CHANNEL A DQS 5 -> DIMM A DQS 2

83 12	MEM_A_DQS_N<5>	MAKE_BASE=TRUE	=MEM_A_DQS_N<2>	30
83 12	MEM_A_DQS_P<5>	MAKE_BASE=TRUE	=MEM_A_DQS_P<2>	30
83 12	MEM_A_DM<5>	MAKE_BASE=TRUE	=MEM_A_DM<2>	30
83 12	MEM_A_DQ<47>	MAKE_BASE=TRUE	=MEM_A_DQ<17>	30
83 12	MEM_A_DQ<46>	MAKE_BASE=TRUE	=MEM_A_DQ<16>	30
83 12	MEM_A_DQ<45>	MAKE_BASE=TRUE	=MEM_A_DQ<22>	30
83 12	MEM_A_DQ<44>	MAKE_BASE=TRUE	=MEM_A_DQ<23>	30
83 12	MEM_A_DQ<43>	MAKE_BASE=TRUE	=MEM_A_DQ<21>	30
83 12	MEM_A_DQ<42>	MAKE_BASE=TRUE	=MEM_A_DQ<20>	30
83 12	MEM_A_DQ<41>	MAKE_BASE=TRUE	=MEM_A_DQ<18>	30
83 12	MEM_A_DQ<40>	MAKE_BASE=TRUE	=MEM_A_DQ<19>	30

CPU CHANNEL A DQS 6 -> DIMM A DQS 1

83 12	MEM_A_DQS_N<6>	MAKE_BASE=TRUE	=MEM_A_DQS_N<1>	30
83 12	MEM_A_DQS_P<6>	MAKE_BASE=TRUE	=MEM_A_DQS_P<1>	30
83 12	MEM_A_DM<6>	MAKE_BASE=TRUE	=MEM_A_DM<1>	30
83 12	MEM_A_DQ<55>	MAKE_BASE=TRUE	=MEM_A_DQ<12>	30
83 12	MEM_A_DQ<54>	MAKE_BASE=TRUE	=MEM_A_DQ<13>	30
83 12	MEM_A_DQ<53>	MAKE_BASE=TRUE	=MEM_A_DQ<10>	30
83 12	MEM_A_DQ<52>	MAKE_BASE=TRUE	=MEM_A_DQ<15>	30
83 12	MEM_A_DQ<51>	MAKE_BASE=TRUE	=MEM_A_DQ<8>	30
83 12	MEM_A_DQ<50>	MAKE_BASE=TRUE	=MEM_A_DQ<9>	30
83 12	MEM_A_DQ<49>	MAKE_BASE=TRUE	=MEM_A_DQ<11>	30
83 12	MEM_A_DQ<48>	MAKE_BASE=TRUE	=MEM_A_DQ<14>	30

CPU CHANNEL A DQS 7 -> DIMM A DQS 0

83 12	MEM_A_DQS_N<7>	MAKE_BASE=TRUE	=MEM_A_DQS_N<0>	30
83 12	MEM_A_DQS_P<7>	MAKE_BASE=TRUE	=MEM_A_DQS_P<0>	30
83 12	MEM_A_DM<7>	MAKE_BASE=TRUE	=MEM_A_DM<0>	30
83 12	MEM_A_DQ<63>	MAKE_BASE=TRUE	=MEM_A_DQ<5>	30
83 12	MEM_A_DQ<62>	MAKE_BASE=TRUE	=MEM_A_DQ<4>	30
83 12	MEM_A_DQ<61>	MAKE_BASE=TRUE	=MEM_A_DQ<3>	30
83 12	MEM_A_DQ<60>	MAKE_BASE=TRUE	=MEM_A_DQ<2>	30
83 12	MEM_A_DQ<59>	MAKE_BASE=TRUE	=MEM_A_DQ<1>	30
83 12	MEM_A_DQ<58>	MAKE_BASE=TRUE	=MEM_A_DQ<0>	30
83 12	MEM_A_DQ<57>	MAKE_BASE=TRUE	=MEM_A_DQ<7>	30
83 12	MEM_A_DQ<56>	MAKE_BASE=TRUE	=MEM_A_DQ<6>	30

CPU CHANNEL B DQS 0 -> DIMM B DQS 7

83 12	MEM_B_DQS_N<0>	MAKE_BASE=TRUE	=MEM_B_DQS_N<7>	31
83 12	MEM_B_DQS_P<0>	MAKE_BASE=TRUE	=MEM_B_DQS_P<7>	31
83 12	MEM_B_DM<0>	MAKE_BASE=TRUE	=MEM_B_DM<7>	31
83 12	MEM_B_DQ<7>	MAKE_BASE=TRUE	=MEM_B_DQ<61>	31
83 12	MEM_B_DQ<6>	MAKE_BASE=TRUE	=MEM_B_DQ<60>	31
83 12	MEM_B_DQ<5>	MAKE_BASE=TRUE	=MEM_B_DQ<62>	31
83 12	MEM_B_DQ<4>	MAKE_BASE=TRUE	=MEM_B_DQ<63>	31
83 12	MEM_B_DQ<3>	MAKE_BASE=TRUE	=MEM_B_DQ<56>	31
83 12	MEM_B_DQ<2>	MAKE_BASE=TRUE	=MEM_B_DQ<57>	31
83 12	MEM_B_DQ<1>	MAKE_BASE=TRUE	=MEM_B_DQ<58>	31
83 12	MEM_B_DQ<0>	MAKE_BASE=TRUE	=MEM_B_DQ<59>	31

CPU CHANNEL B DQS 1 -> DIMM B DQS 6

83 12	MEM_B_DQS_N<1>	MAKE_BASE=TRUE	=MEM_B_DQS_N<6>	31
83 12	MEM_B_DQS_P<1>	MAKE_BASE=TRUE	=MEM_B_DQS_P<6>	31
83 12	MEM_B_DM<1>	MAKE_BASE=TRUE	=MEM_B_DM<6>	31
83 12	MEM_B_DQ<15>	MAKE_BASE=TRUE	=MEM_B_DQ<51>	31
83 12	MEM_B_DQ<14>	MAKE_BASE=TRUE	=MEM_B_DQ<54>	31
83 12	MEM_B_DQ<13>	MAKE_BASE=TRUE	=MEM_B_DQ<53>	31
83 12	MEM_B_DQ<12>	MAKE_BASE=TRUE	=MEM_B_DQ<52>	31
83 12	MEM_B_DQ<11>	MAKE_BASE=TRUE	=MEM_B_DQ<48>	31
83 12	MEM_B_DQ<10>	MAKE_BASE=TRUE	=MEM_B_DQ<49>	31
83 12	MEM_B_DQ<9>	MAKE_BASE=TRUE	=MEM_B_DQ<50>	31
83 12	MEM_B_DQ<8>	MAKE_BASE=TRUE	=MEM_B_DQ<55>	31

CPU CHANNEL B DQS 2 -> DIMM B DQS 5

83 12	MEM_B_DQS_N<2>	MAKE_BASE=TRUE	=MEM_B_DQS_N<5>	31
83 12	MEM_B_DQS_P<2>	MAKE_BASE=TRUE	=MEM_B_DQS_P<5>	31
83 12	MEM_B_DM<2>	MAKE_BASE=TRUE	=MEM_B_DM<5>	31
83 12	MEM_B_DQ<23>	MAKE_BASE=TRUE	=MEM_B_DQ<45>	31
83 12	MEM_B_DQ<22>	MAKE_BASE=TRUE	=MEM_B_DQ<40>	31
83 12	MEM_B_DQ<21>	MAKE_BASE=TRUE	=MEM_B_DQ<42>	31
83 12	MEM_B_DQ<20>	MAKE_BASE=TRUE	=MEM_B_DQ<47>	31
83 12	MEM_B_DQ<19>	MAKE_BASE=TRUE	=MEM_B_DQ<44>	31
83 12	MEM_B_DQ<18>	MAKE_BASE=TRUE	=MEM_B_DQ<41>	31
83 12	MEM_B_DQ<17>	MAKE_BASE=TRUE	=MEM_B_DQ<46>	31
83 12	MEM_B_DQ<16>	MAKE_BASE=TRUE	=MEM_B_DQ<43>	31

CPU CHANNEL B DQS 3 -> DIMM B DQS 4

83 12	MEM_B_DQS_N<3>	MAKE_BASE=TRUE	=MEM_B_DQS_N<4>	31
83 12	MEM_B_DQS_P<3>	MAKE_BASE=TRUE	=MEM_B_DQS_P<4>	31
83 12	MEM_B_DM<3>	MAKE_BASE=TRUE	=MEM_B_DM<4>	31
83 12	MEM_B_DQ<31>	MAKE_BASE=TRUE	=MEM_B_DQ<32>	31
83 12	MEM_B_DQ<30>	MAKE_BASE=TRUE	=MEM_B_DQ<39>	31
83 12	MEM_B_DQ<29>	MAKE_BASE=TRUE	=MEM_B_DQ<37>	31
83 12	MEM_B_DQ<28>	MAKE_BASE=TRUE	=MEM_B_DQ<38>	31
83 12	MEM_B_DQ<27>	MAKE_BASE=TRUE	=MEM_B_DQ<35>	31
83 12	MEM_B_DQ<26>	MAKE_BASE=TRUE	=MEM_B_DQ<34>	31
83 12	MEM_B_DQ<25>	MAKE_BASE=TRUE	=MEM_B_DQ<36>	31
83 12	MEM_B_DQ<24>	MAKE_BASE=TRUE	=MEM_B_DQ<33>	31

CPU CHANNEL B DQS 4 -> DIMM B DQS 3

83 12	MEM_B_DQS_N<4>	MAKE_BASE=TRUE	=MEM_B_DQS_N<3>	31
83 12	MEM_B_DQS_P<4>	MAKE_BASE=TRUE	=MEM_B_DQS_P<3>	31
83 12	MEM_B_DM<4>	MAKE_BASE=TRUE	=MEM_B_DM<3>	31
83 12	MEM_B_DQ<39>	MAKE_BASE=TRUE	=MEM_B_DQ<28>	31
83 12	MEM_B_DQ<38>	MAKE_BASE=TRUE	=MEM_B_DQ<24>	31
83 12	MEM_B_DQ<37>	MAKE_BASE=TRUE	=MEM_B_DQ<31>	31
83 12	MEM_B_DQ<36>	MAKE_BASE=TRUE	=MEM_B_DQ<30>	31
83 12	MEM_B_DQ<35>	MAKE_BASE=TRUE	=MEM_B_DQ<29>	31
83 12	MEM_B_DQ<34>	MAKE_BASE=TRUE	=MEM_B_DQ<25>	31
83 12	MEM_B_DQ<33>	MAKE_BASE=TRUE	=MEM_B_DQ<27>	31
83 12	MEM_B_DQ<32>	MAKE_BASE=TRUE	=MEM_B_DQ<26>	31

CPU CHANNEL B DQS 5 -> DIMM B DQS 2

83 12	MEM_B_DQS_N<5>	MAKE_BASE=TRUE	=MEM_B_DQS_N<2>	31
83 12	MEM_B_DQS_P<5>	MAKE_BASE=TRUE	=MEM_B_DQS_P<2>	31
83 12	MEM_B_DM<5>	MAKE_BASE=TRUE	=MEM_B_DM<2>	31
83 12	MEM_B_DQ<47>	MAKE_BASE=TRUE	=MEM_B_DQ<18>	31
83 12	MEM_B_DQ<46>	MAKE_BASE=TRUE	=MEM_B_DQ<20>	31
83 12	MEM_B_DQ<45>	MAKE_BASE=TRUE	=MEM_B_DQ<23>	31
83 12	MEM_B_DQ<44>	MAKE_BASE=TRUE	=MEM_B_DQ<22>	31
83 12	MEM_B_DQ<43>	MAKE_BASE=TRUE	=MEM_B_DQ<19>	31
83 12	MEM_B_DQ<42>	MAKE_BASE=TRUE	=MEM_B_DQ<21>	31
83 12	MEM_B_DQ<41>	MAKE_BASE=TRUE	=MEM_B_DQ<17>	31
83 12	MEM_B_DQ<40>	MAKE_BASE=TRUE	=MEM_B_DQ<16>	31

CPU CHANNEL B DQS 6 -> DIMM B DQS 1

83 12	MEM_B_DQS_N<6>	MAKE_BASE=TRUE	=MEM_B_DQS_N<1>	31
83 12	MEM_B_DQS_P<6>	MAKE_BASE=TRUE	=MEM_B_DQS_P<1>	31
83 12	MEM_B_DM<6>	MAKE_BASE=TRUE	=MEM_B_DM<1>	31
83 12	MEM_B_DQ<55>	MAKE_BASE=TRUE	=MEM_B_DQ<8>	31
83 12	MEM_B_DQ<54>	MAKE_BASE=TRUE	=MEM_B_DQ<14>	31
83 12	MEM_B_DQ<53>	MAKE_BASE=TRUE	=MEM_B_DQ<10>	31
83 12	MEM_B_DQ<52>	MAKE_BASE=TRUE	=MEM_B_DQ<11>	31
83 12	MEM_B_DQ<51>	MAKE_BASE=TRUE	=MEM_B_DQ<9>	31
83 12	MEM_B_DQ<50>	MAKE_BASE=TRUE	=MEM_B_DQ<15>	31
83 12	MEM_B_DQ<49>	MAKE_BASE=TRUE	=MEM_B_DQ<12>	31
83 12	MEM_B_DQ<48>	MAKE_BASE=TRUE	=MEM_B_DQ<13>	31

CPU CHANNEL B DQS 7 -> DIMM B DQS 0

83 12	MEM_B_DQS_N<7>	MAKE_BASE=TRUE	=MEM_B_DQS_N<0>	31
83 12	MEM_B_DQS_P<7>	MAKE_BASE=TRUE	=MEM_B_DQS_P<0>	31
83 12	MEM_B_DM<7>	MAKE_BASE=TRUE	=MEM_B_DM<0>	31
83 12	MEM_B_DQ<63>	MAKE_BASE=TRUE	=MEM_B_DQ<2>	31
83 12	MEM_B_DQ<62>	MAKE_BASE=TRUE	=MEM_B_DQ<1>	31
83 12	MEM_B_DQ<61>	MAKE_BASE=TRUE	=MEM_B_DQ<3>	31
83 12	MEM_B_DQ<60>	MAKE_BASE=TRUE	=MEM_B_DQ<7>	31
83 12	MEM_B_DQ<59>	MAKE_BASE=TRUE	=MEM_B_DQ<4>	31
83 12	MEM_B_DQ<58>	MAKE_BASE=TRUE	=MEM_B_DQ<5>	31
83 12	MEM_B_DQ<57>	MAKE_BASE=TRUE	=MEM_B_DQ<6>	31
83 12	MEM_B_DQ<56>	MAKE_BASE=TRUE	=MEM_B_DQ<0>	31

MEMORY CLOCK ALIASING

83 12	MEM_A_CLK_P<0>	MAKE_BASE=TRUE	=MEM_A_CLK_P<0>	30
83 12	MEM_A_CLK_N<0>	MAKE_BASE=TRUE	=MEM_A_CLK_N<0>	30
83 12	MEM_A_CLK_P<1>	MAKE_BASE=TRUE	=MEM_A_CLK_P<1>	30
83 12	MEM_A_CLK_N<1>	MAKE_BASE=TRUE	=MEM_A_CLK_N<1>	30
83 12	MEM_A_CLK_P<2>	MAKE_BASE=TRUE	=MEM_A_CLK_P<2>	30
83 12	MEM_A_CLK_N<2>	MAKE_BASE=TRUE	=MEM_A_CLK_N<2>	30
83 12	MEM_A_CLK_P<3>	MAKE_BASE=TRUE	=MEM_A_CLK_P<3>	30
83 12	MEM_A_CLK_N<3>	MAKE_BASE=TRUE	=MEM_A_CLK_N<3>	30
83 12	MEM_B_CLK_P<0>	MAKE_BASE=TRUE	=MEM_B_CLK_P<0>	31
83 12	MEM_B_CLK_N<0>	MAKE_BASE=TRUE	=MEM_B_CLK_N<0>	31
83 12	MEM_B_CLK_P<1>	MAKE_BASE=TRUE	=MEM_B_CLK_P<1>	31
83 12	MEM_B_CLK_N<1>	MAKE_BASE=TRUE	=MEM_B_CLK_N<1>	31
83 12	MEM_B_CLK_P<2>	MAKE_BASE=TRUE	=MEM_B_CLK_P<2>	31
83 12	MEM_B_CLK_N<2>	MAKE_BASE=TRUE	=MEM_B_CLK_N<2>	31
83 12	MEM_B_CLK_P<3>	MAKE_BASE=TRUE	=MEM_B_CLK_P<3>	31
83 12	MEM_B_CLK_N<3>	MAKE_BASE=TRUE	=MEM_B_CLK_N<3>	31

SYNC_MASTER=K75F_MLB SYNC_DATE=04/14/2010

PAGE TITLE

DDR3 SUPPORT AND BITSWAPS

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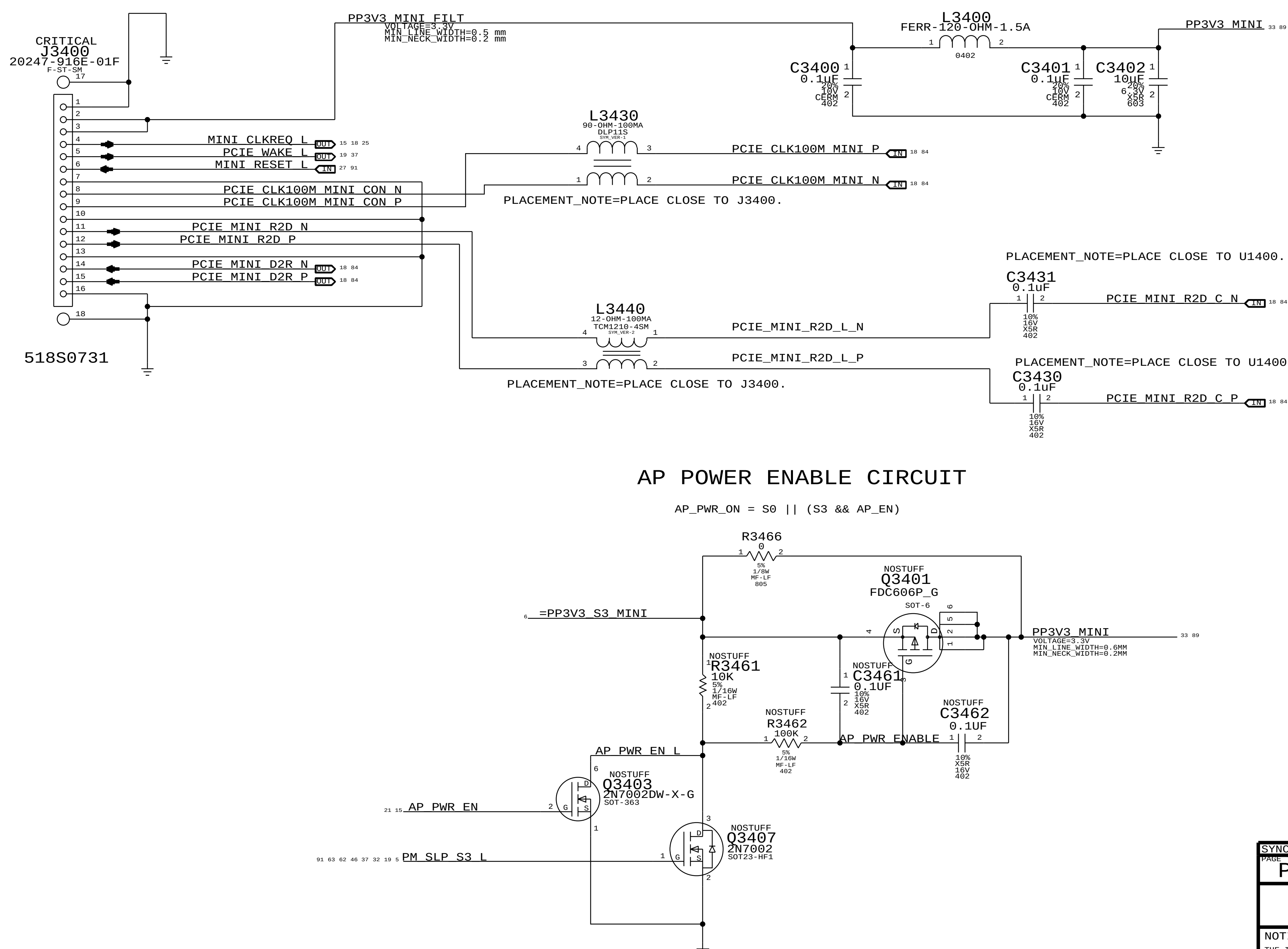

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PCI-E Wireless Connector

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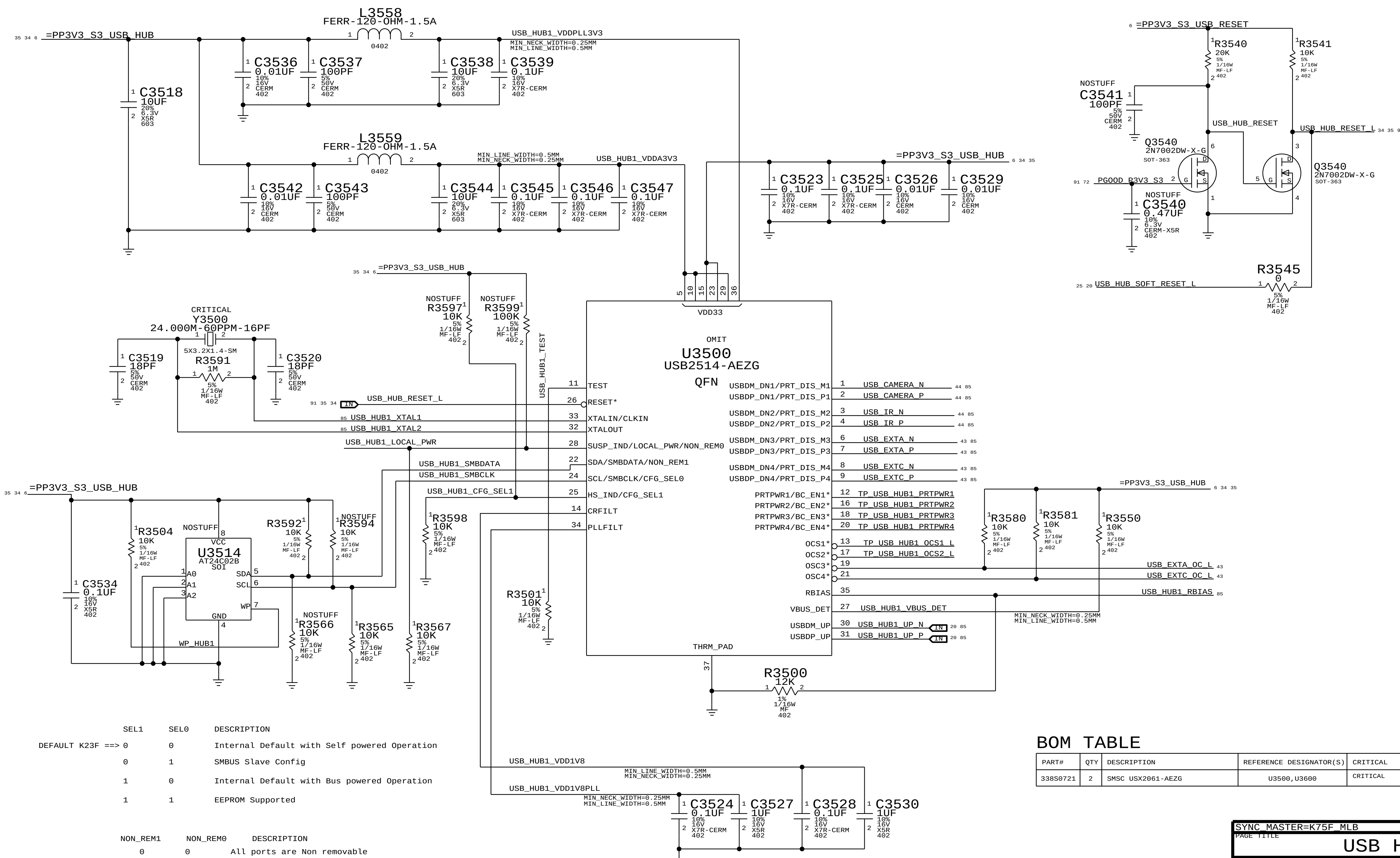
USB HUB-1



	SEL1	SEL0	DESCRIPTION
DEFAULT K23F ==>	0	0	Internal Default with Self powered Operation
	0	1	SMBUS Slave Config
	1	0	Internal Default with Bus powered Operation
	1	1	EEPROM Supported

	NON_REM1	NON_REM0	DESCRIPTION
	0	0	All ports are Non removable
	0	1	Port1 is non removable
DEFAULT K23F ==>	1	0	Port 1 and 2 are non removable
	1	1	Port1,2 and 3 are non Removable

BOM TABLE

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
338S0721	2	SMSC USX2061-AEZG	U3500,U3600	CRITICAL	

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PAGE TITLE
USB HUB 1

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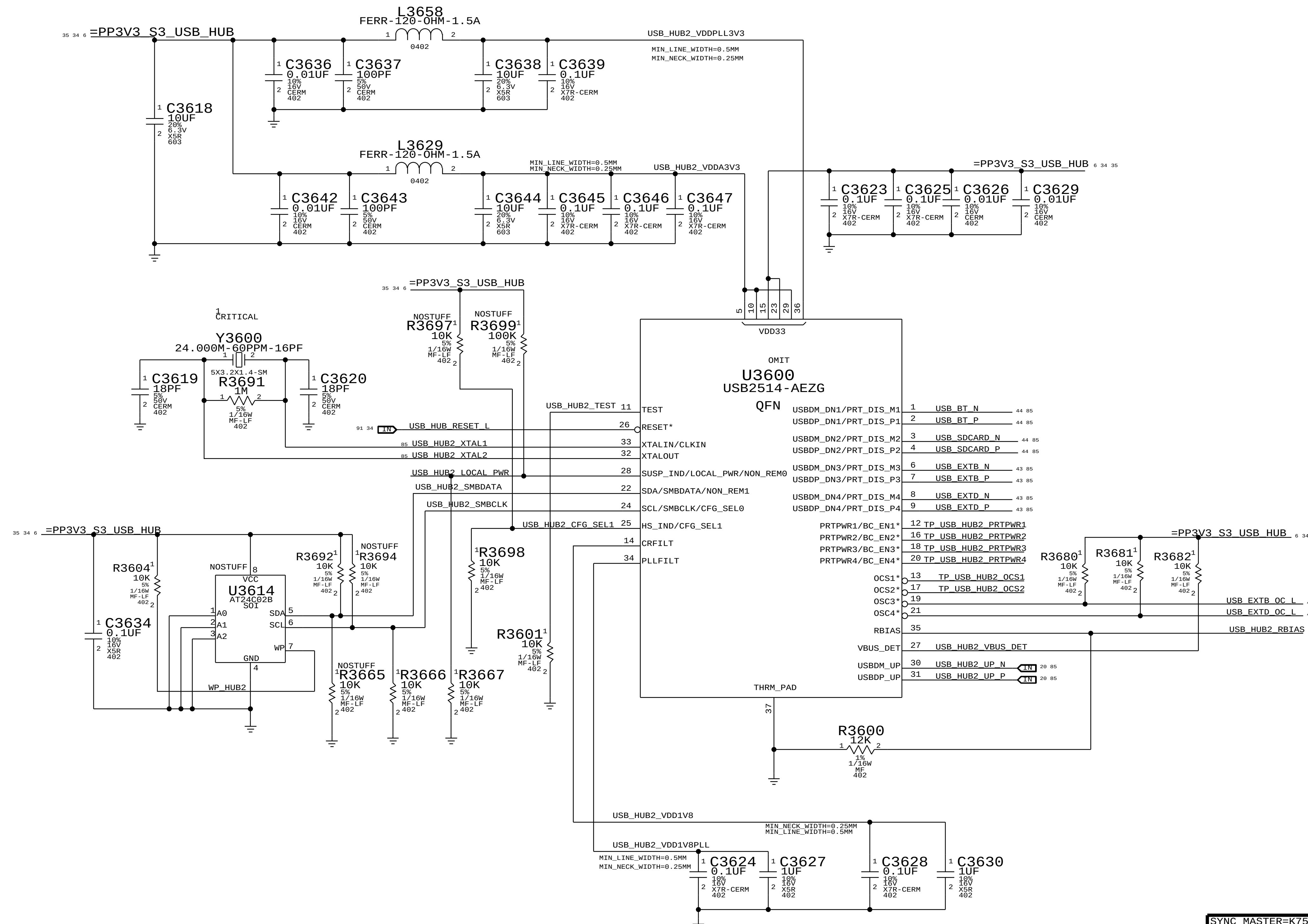
PAGE 35 OF 110

USB HUB-2



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USB HUB 2

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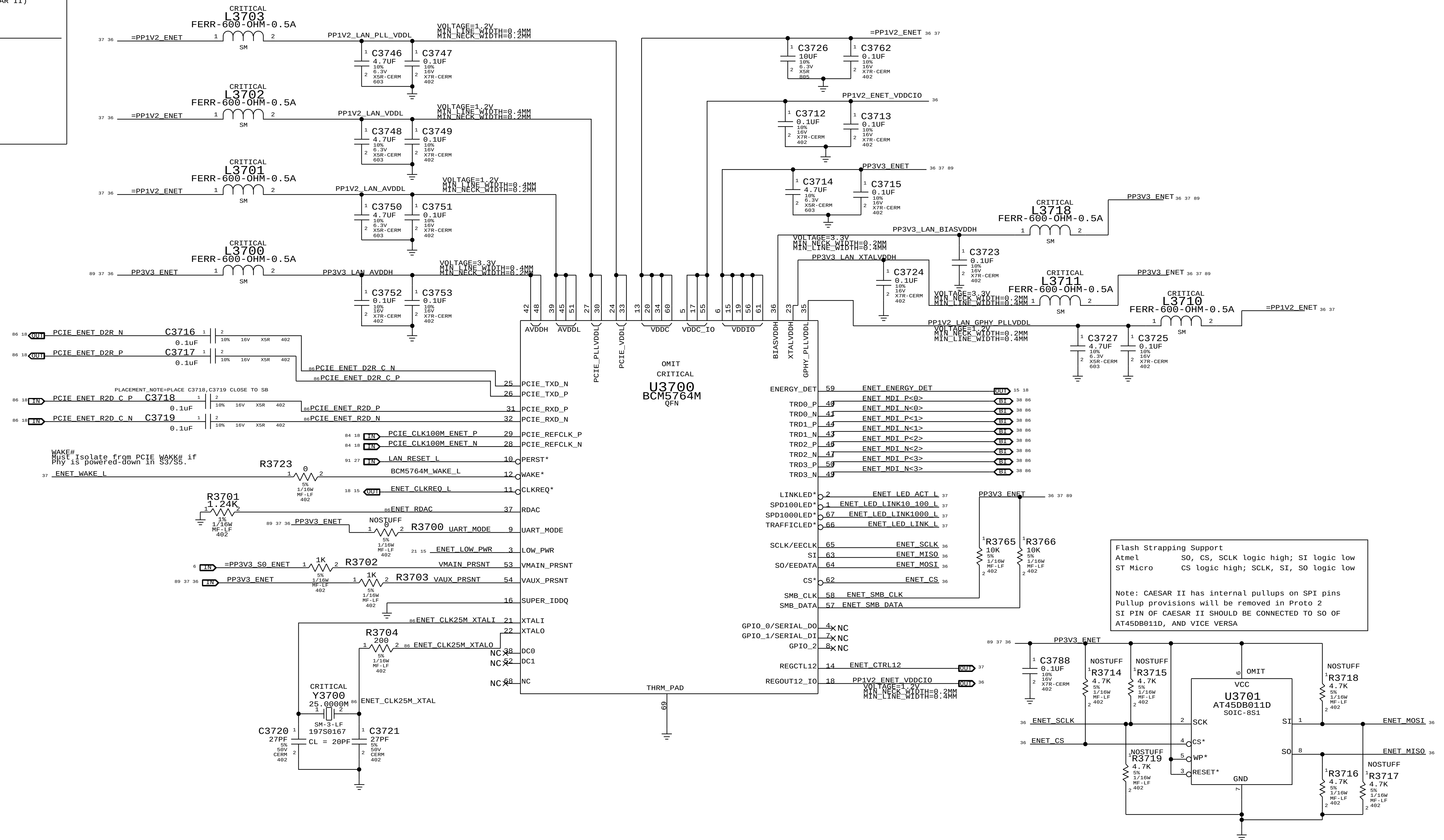
Page Notes

Power aliases required by this page:
- PP3V3_ENET (CAESAR II)
- =PP1V2_ENET



Flash Strapping Support

Atmel	SO, CS, SCLK logic high; SI logic low
ST Micro	CS logic high; SCLK, SI, SO logic low

Note: CAESAR II has internal pullups on SPI pins
Pullup provisions will be removed in Proto 2
SI PIN OF CAESAR II SHOULD BE CONNECTED TO SO OF AT45DB011D, AND VICE VERSA

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PAGE TITLE
ETHERNET (CAESAR II)

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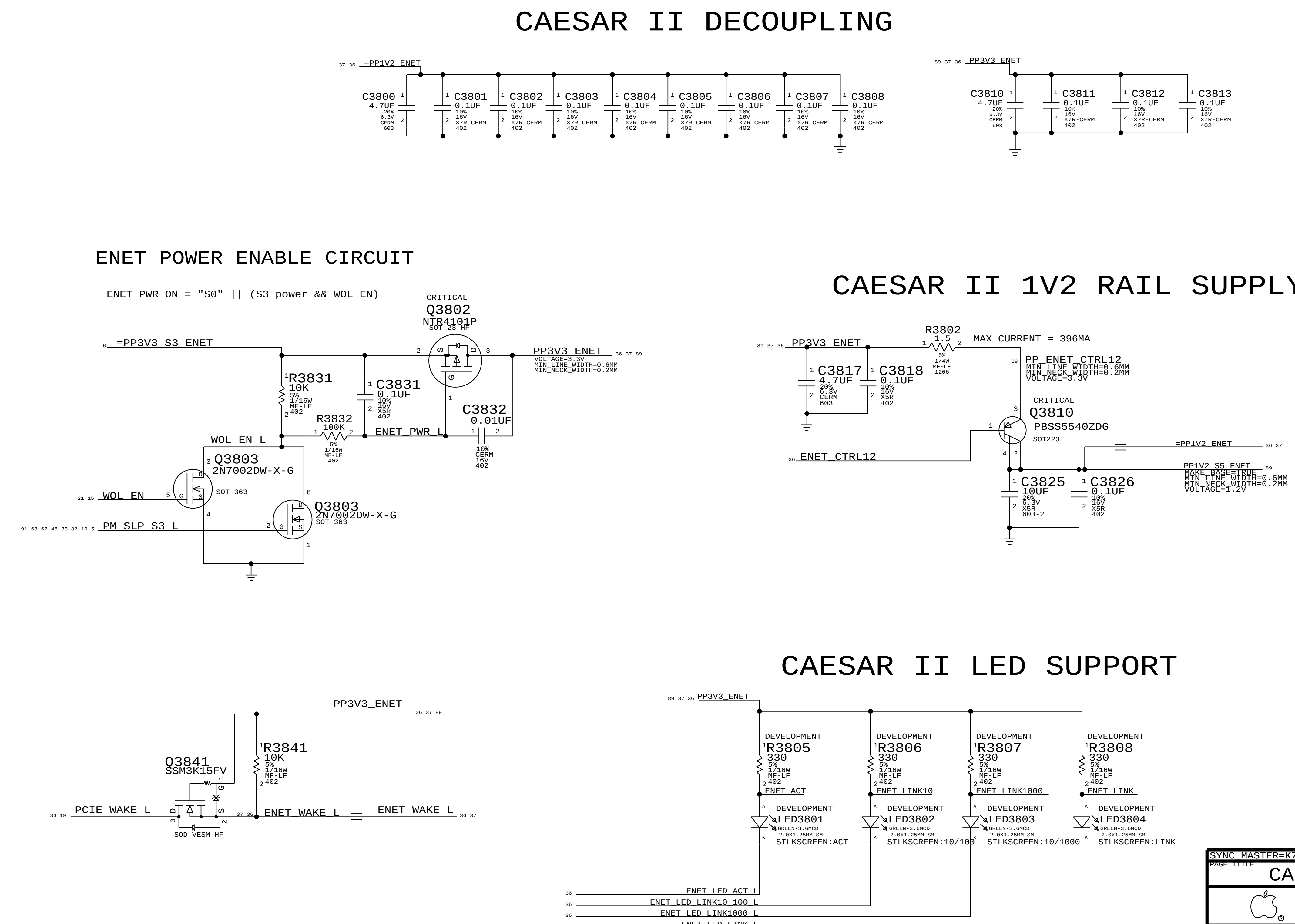

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PAGE TITLE
CAESAR II SUPPORT

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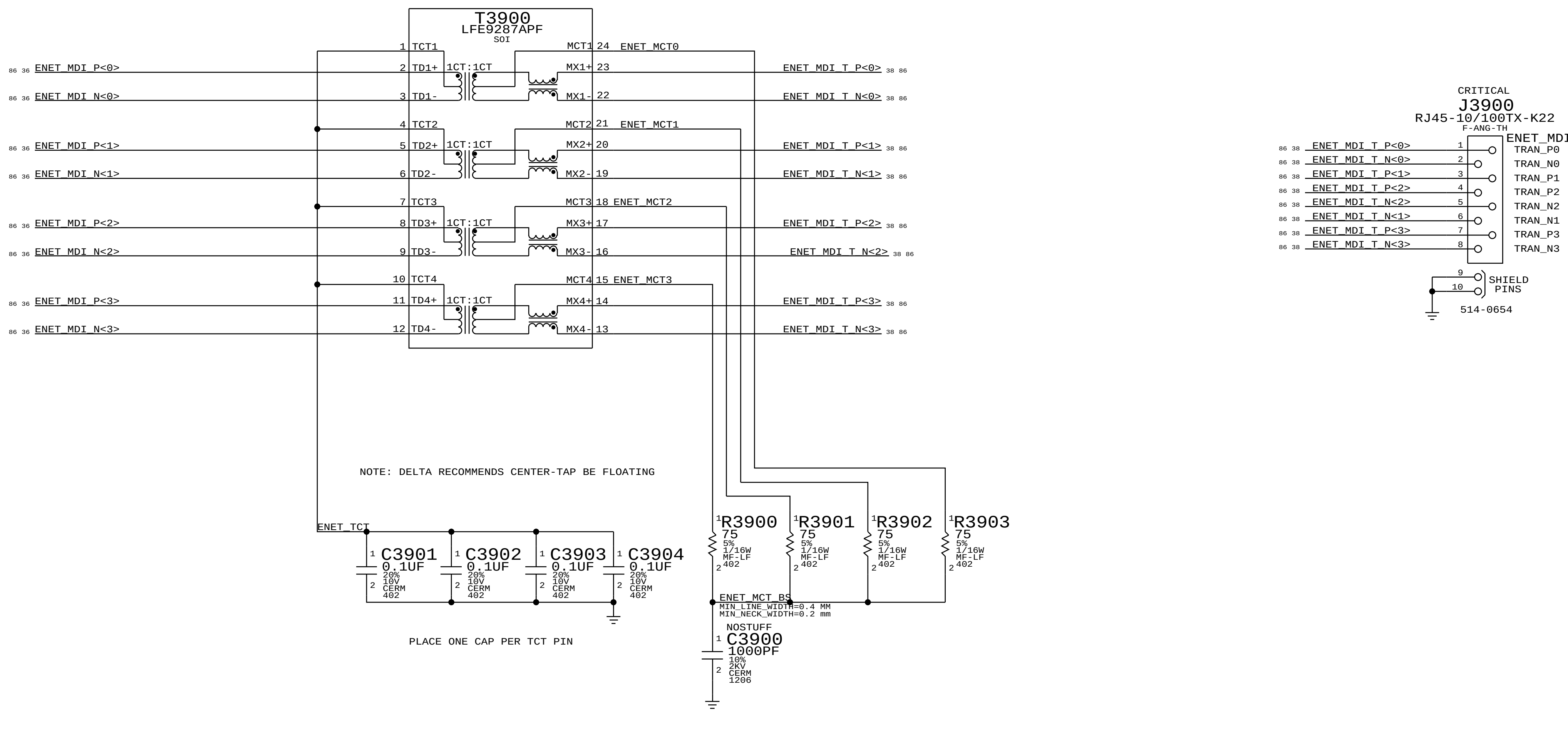

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PAGE TITLE

ETHERNET CONNECTOR

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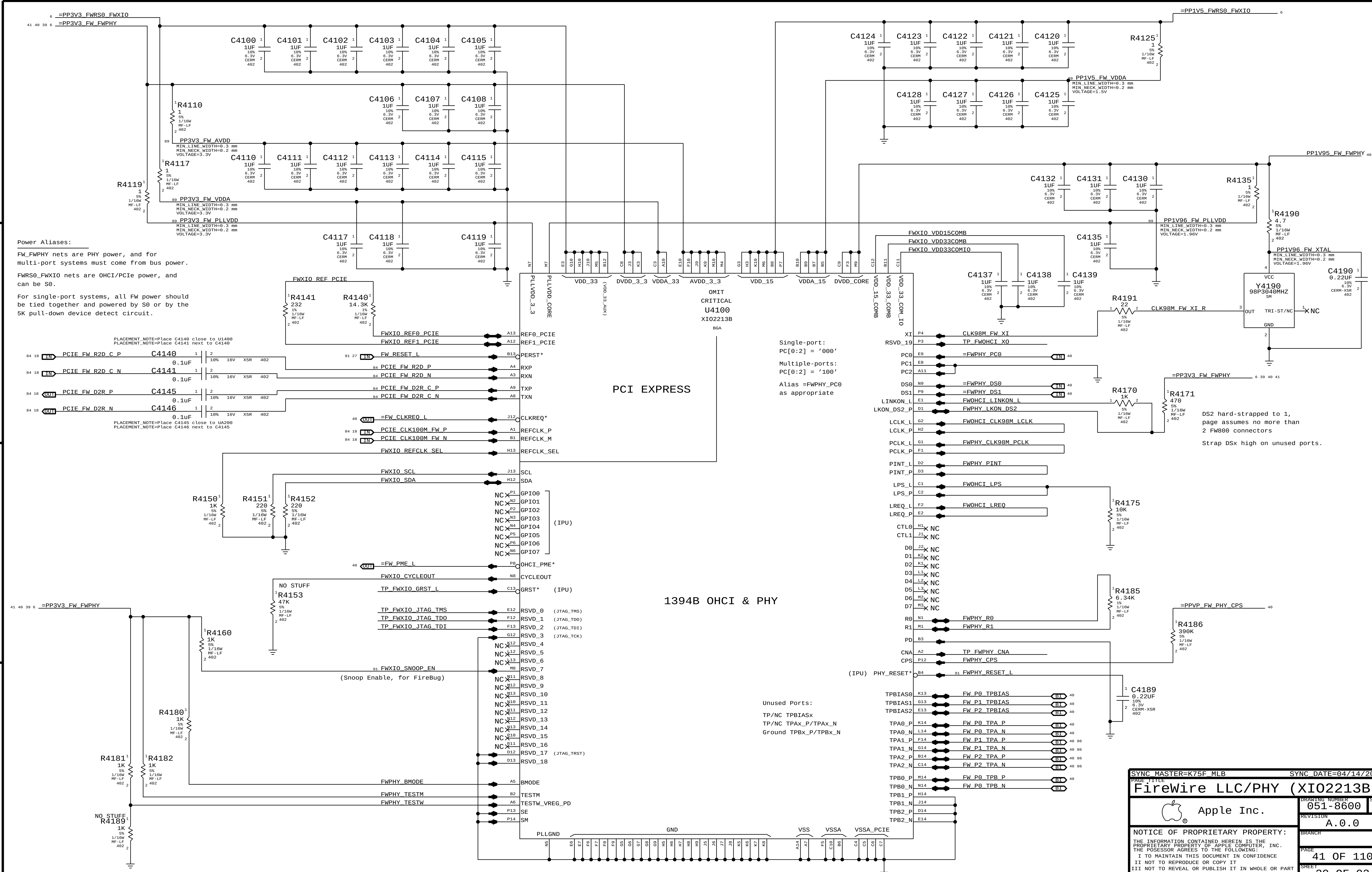

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FireWire LLC/PHY (XIO2213B)

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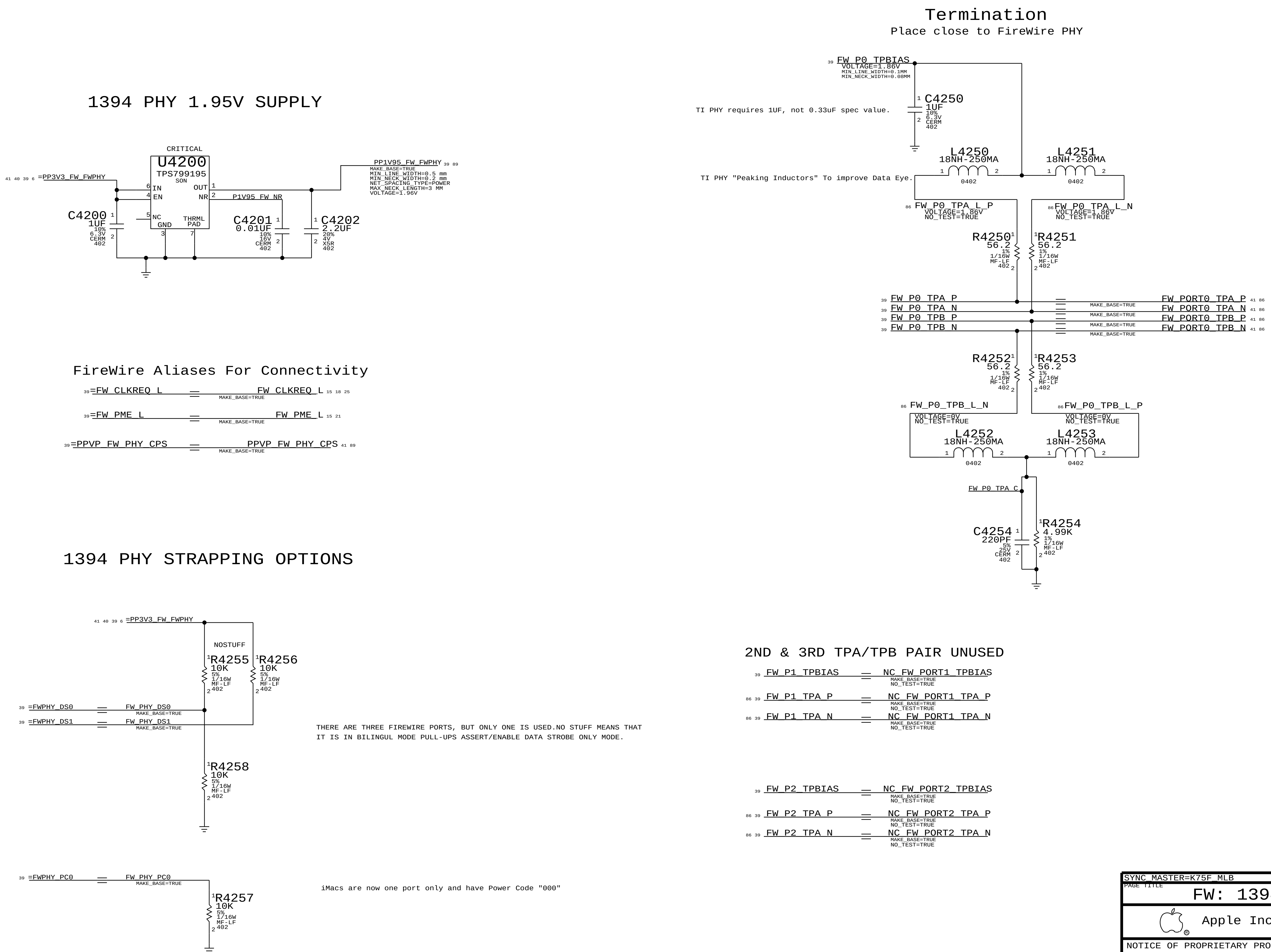

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FW: 1394B MISC

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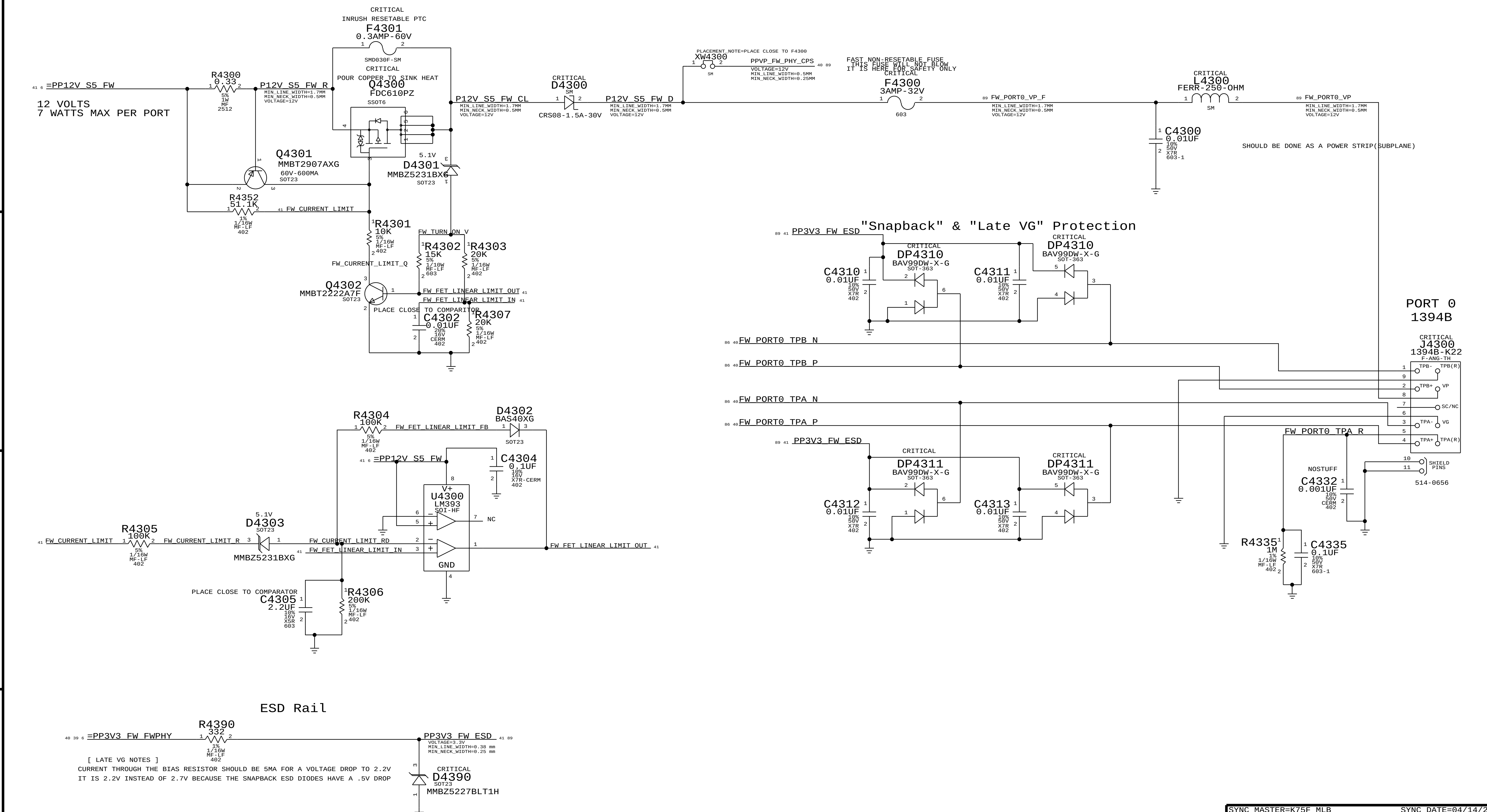

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FIREWIRE CONNECTOR

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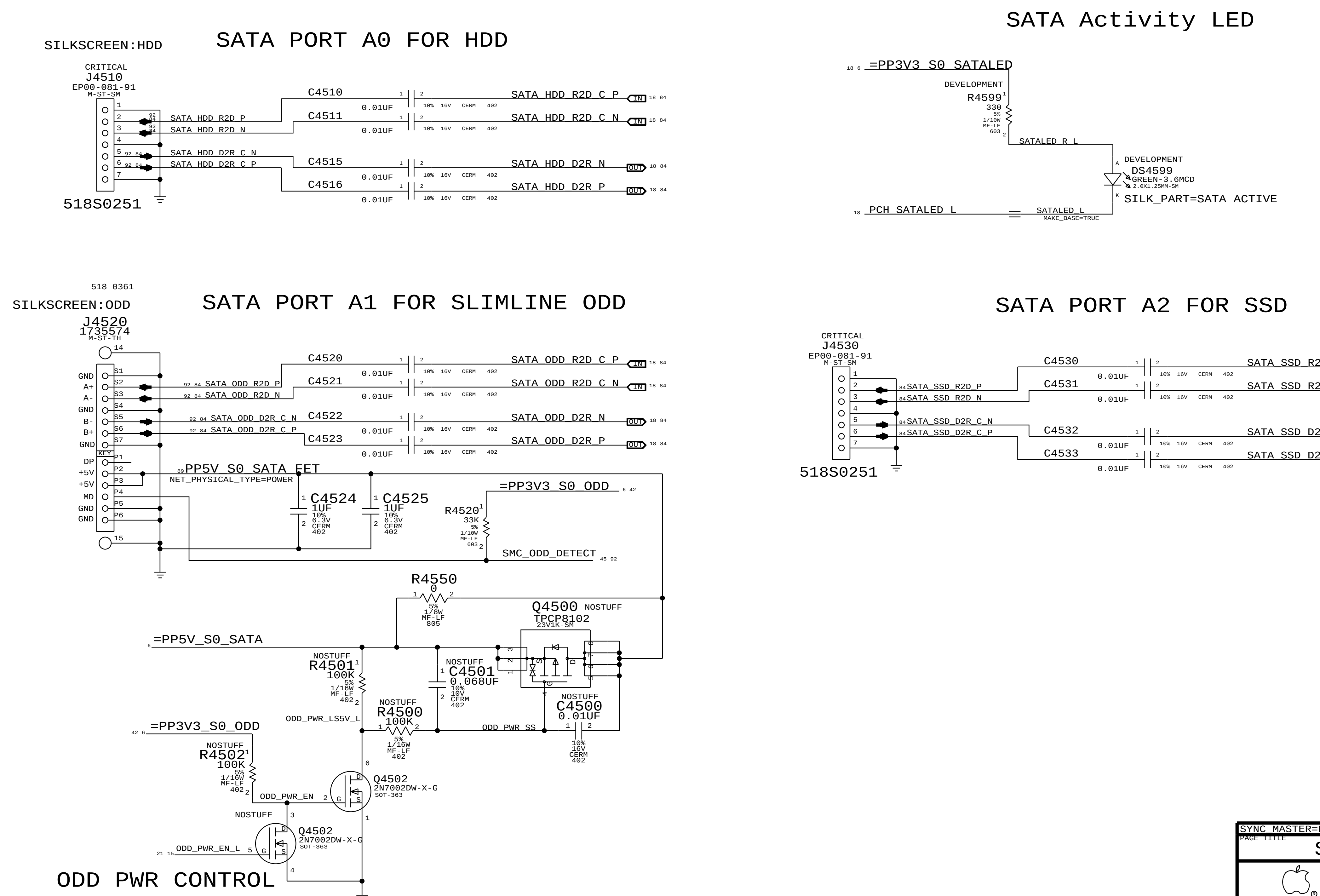

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SATA Connectors

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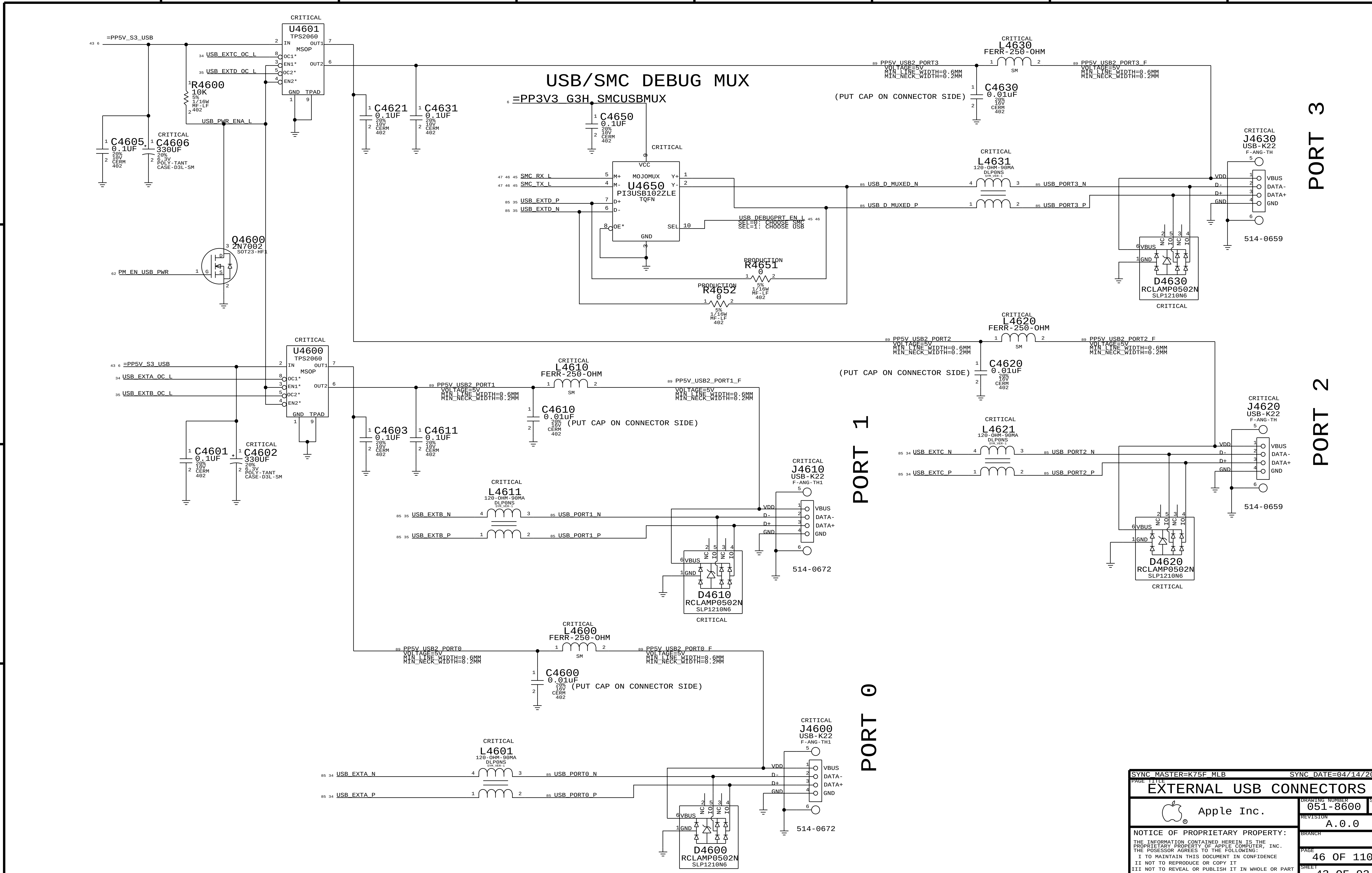

SYNC_MASTER=K75F_MLB SYNC_DATE=04/14/2010

PAGE TITLE

EXTERNAL USB CONNECTORS

Apple Inc.

DRAWING NUMBER 051-8600 SIZE D

REVISION A.0.0

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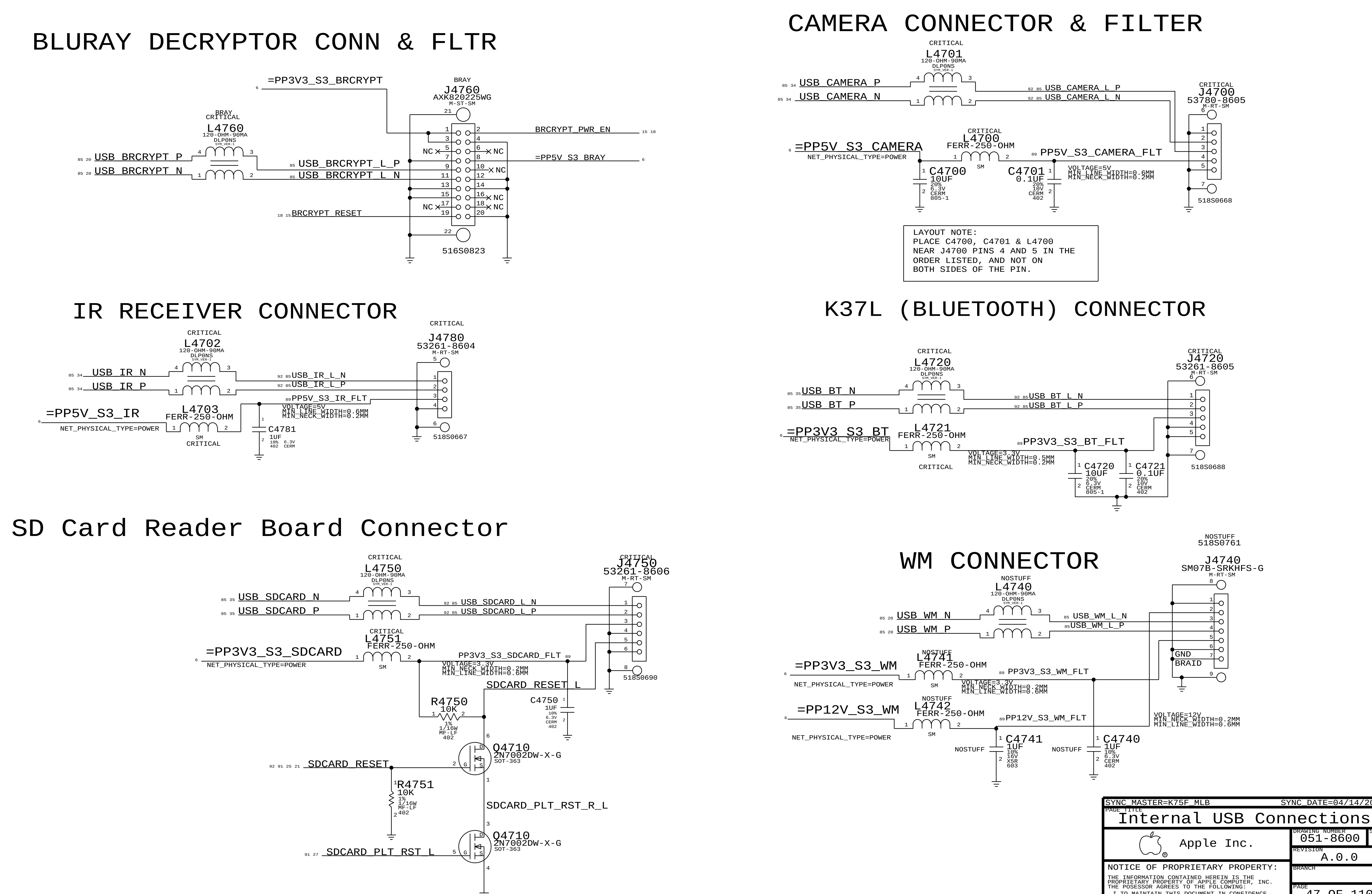

SYNC_MASTER=K75F_MLB SYNC_DATE=04/14/2010

PAGE TITLE: Internal USB Connections

Apple Inc.

DRAWING NUMBER: 051-8600 SIZE: D

REVISION: A.0.0

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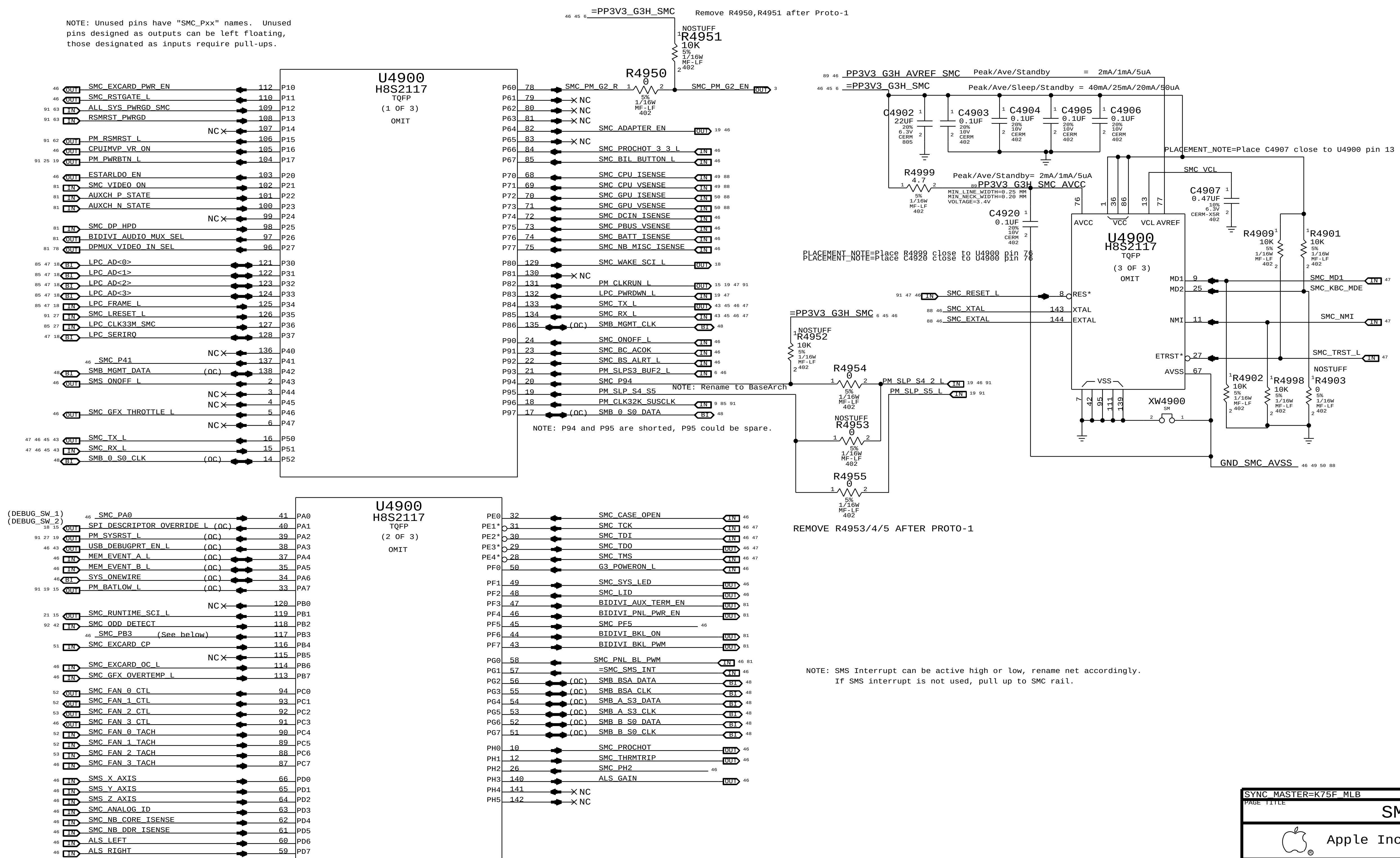
PAGE: 47 OF 110

SHEET: 44 OF 92

NOTE: Unused pins have "SMC_Pxx" names. Unused pins designed as outputs can be left floating, those designated as inputs require pull-ups.



SMC_PB3:

SMC_IG_THROTTLE_L for MG systems.
Otherwise, TP/NC okay (was ISENSE_CAL_EN)

SYNC_MASTER=K75F_MLB SYNC_DATE=04/14/2010

PAGE TITLE
SMC

Apple Inc.

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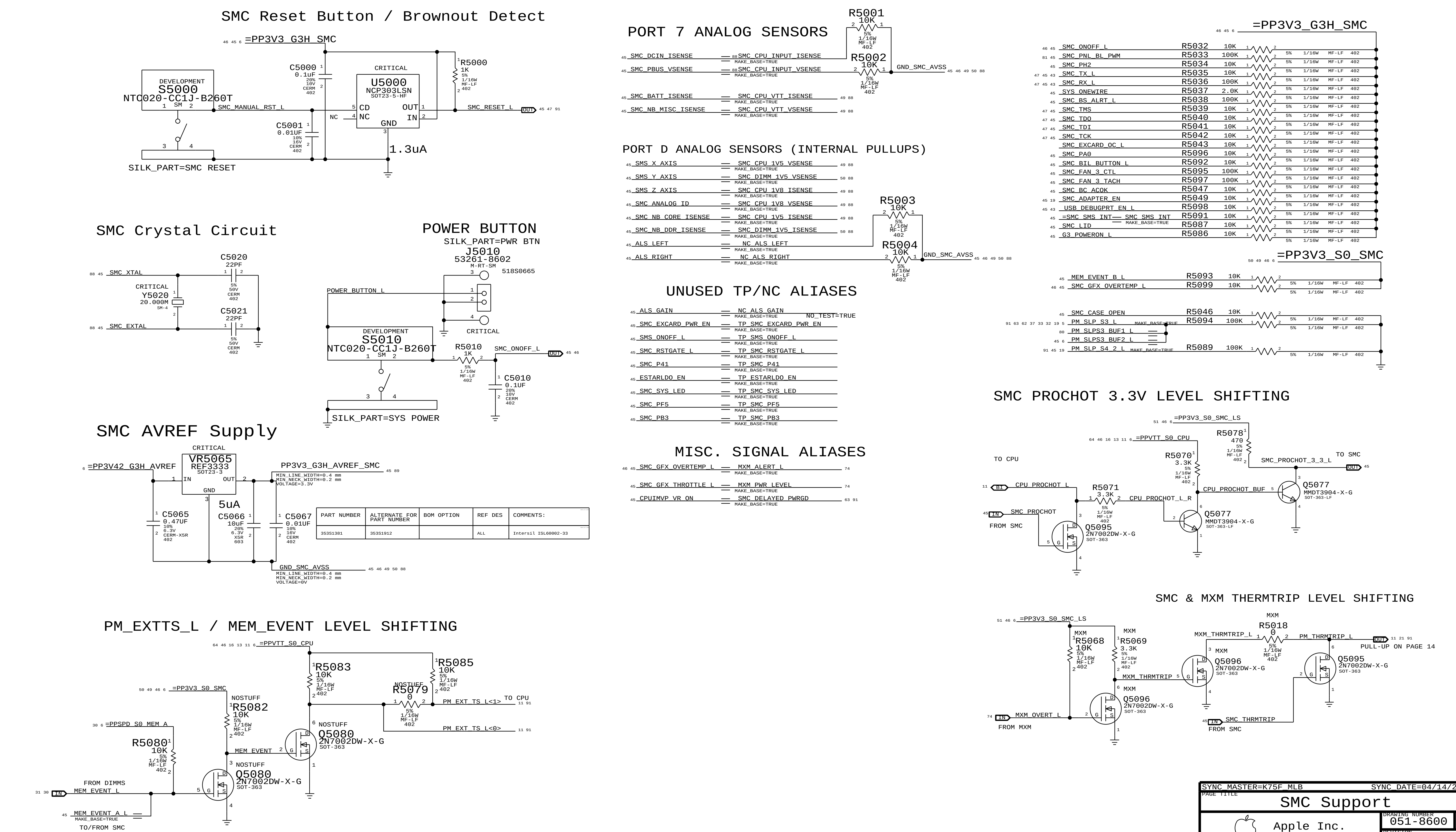

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
353S1381	353S1912		ALL	Intersil ISL60002-33

SYNC_MASTER=K75F_MLB SYNC_DATE=04/14/2010

PAGE TITLE
SMC Support

Apple Inc.

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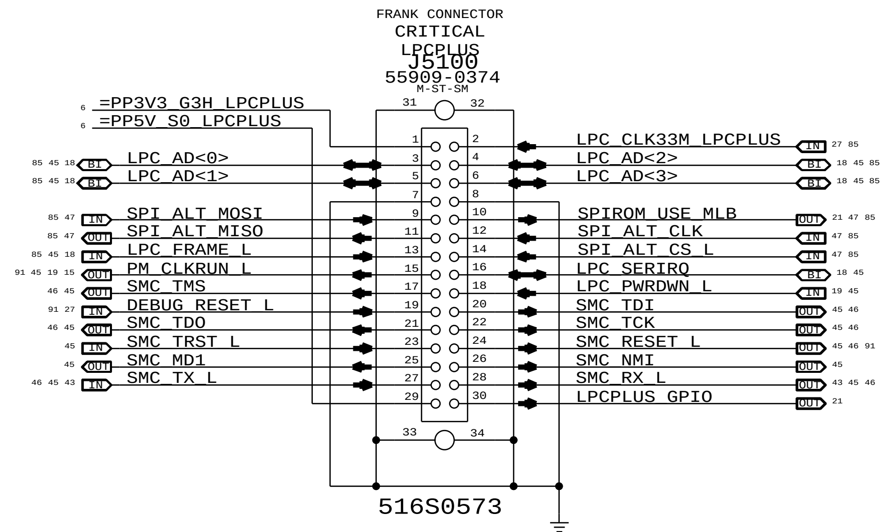
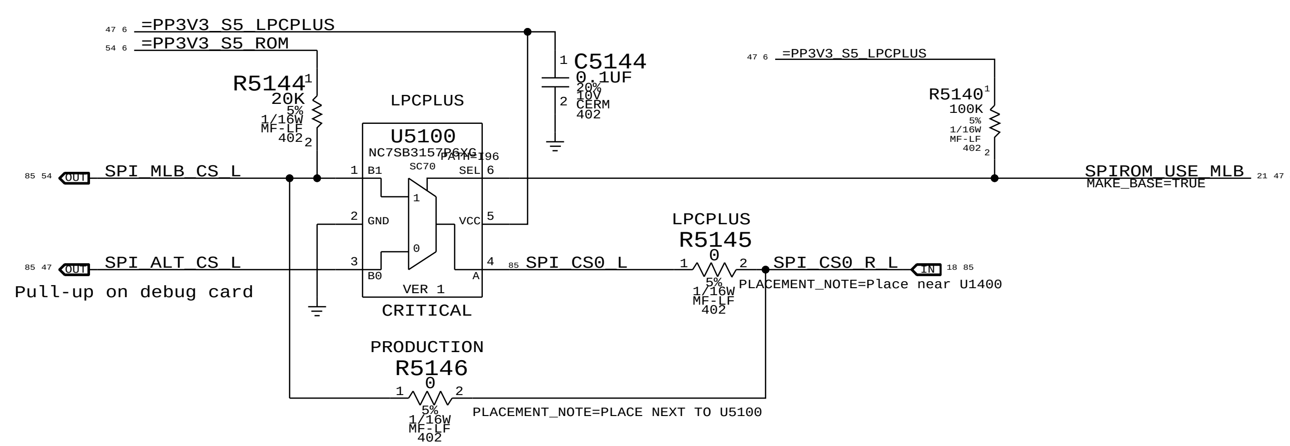
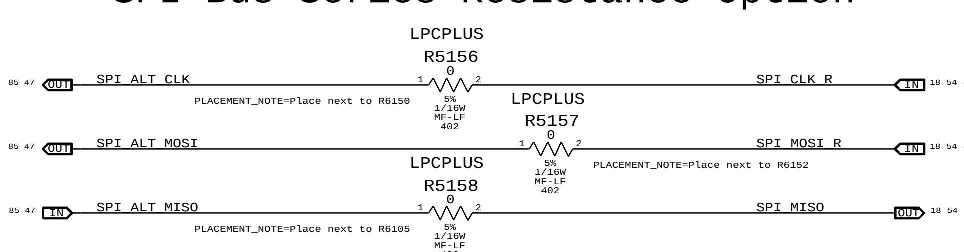
LPC+SPI Connector



Alternate SPI ROM Support



SPI Bus Series Resistance Option



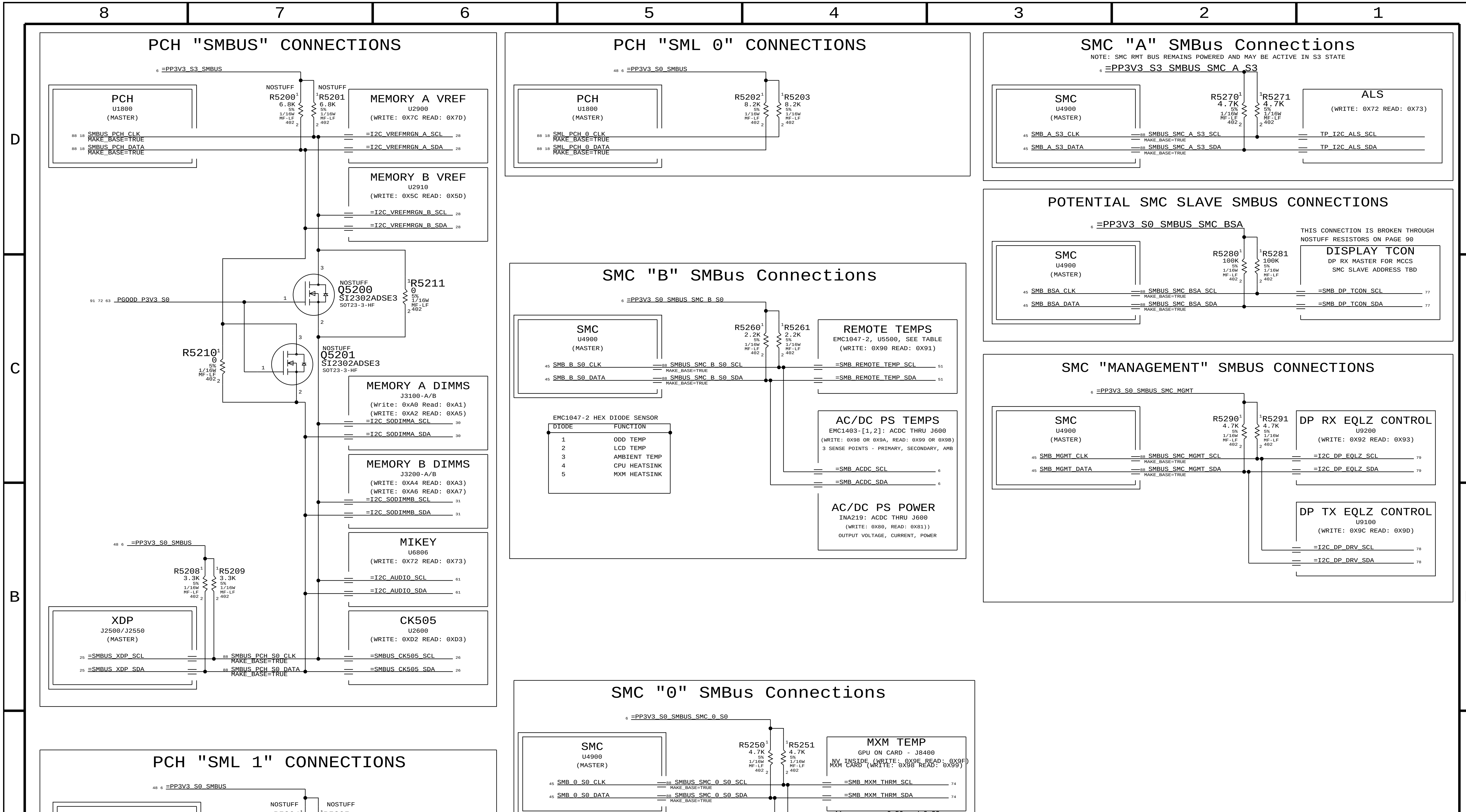
SYNC_MASTER=K75F_MLB SYNC_DATE=04/14/2010

PAGE TITLE

LPC+SPI Debug Connector

Apple Inc.

DRAWING NUMBER	051-8600
SIZE	D
REVISION	A.0.0
BRANCH	
PAGE	51 OF 110
SHEET	47 OF 92



SYNC_MASTER=K75F_MLB SYNC_DATE=04/14/2010

PAGE TITLE

SMBus Connections

Apple Inc.

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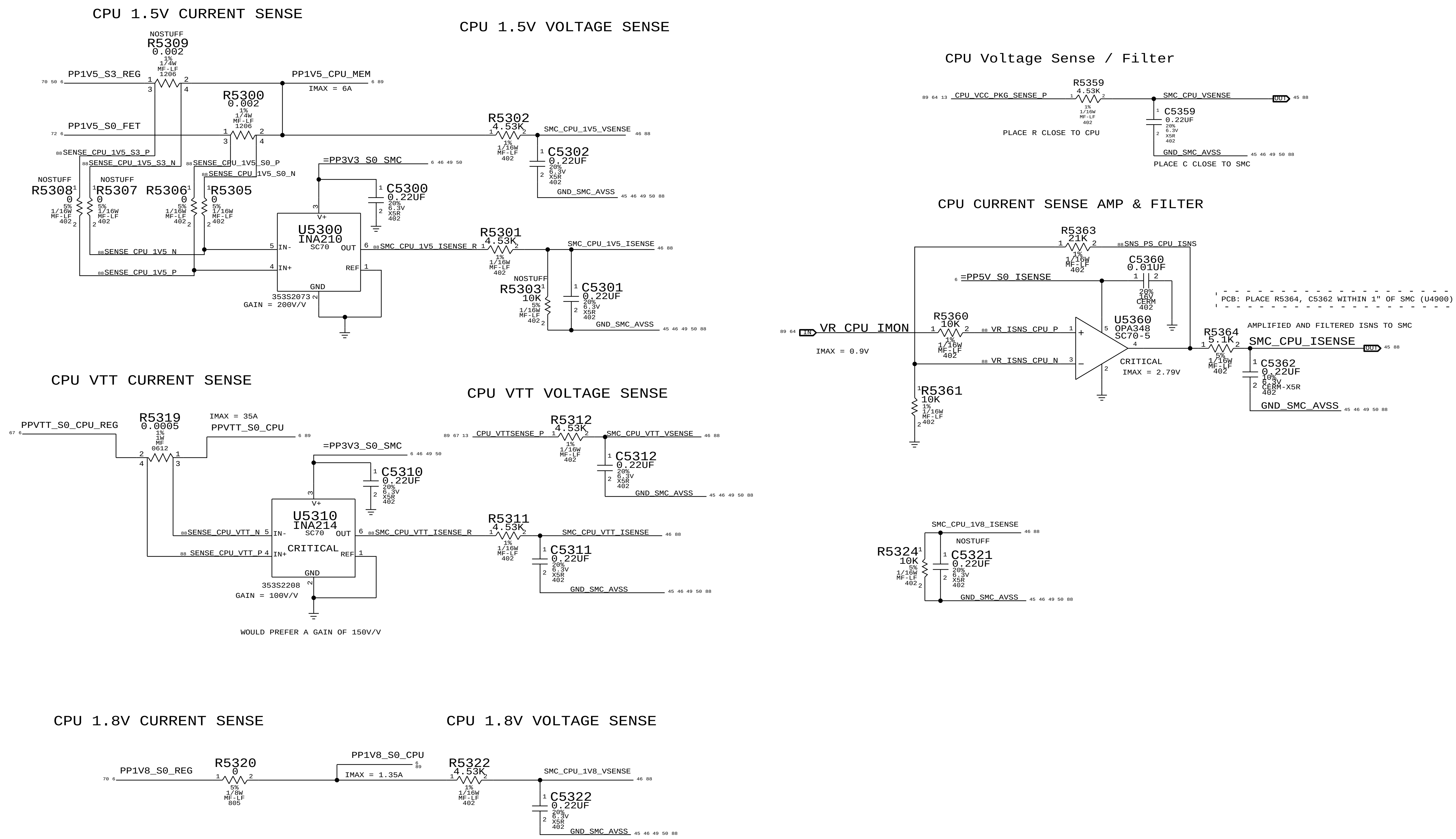

SYNC_MASTER=K75F_MLB SYNC_DATE=04/14/2010

PAGE TITLE

CPU POWER SENSE

Apple Inc.

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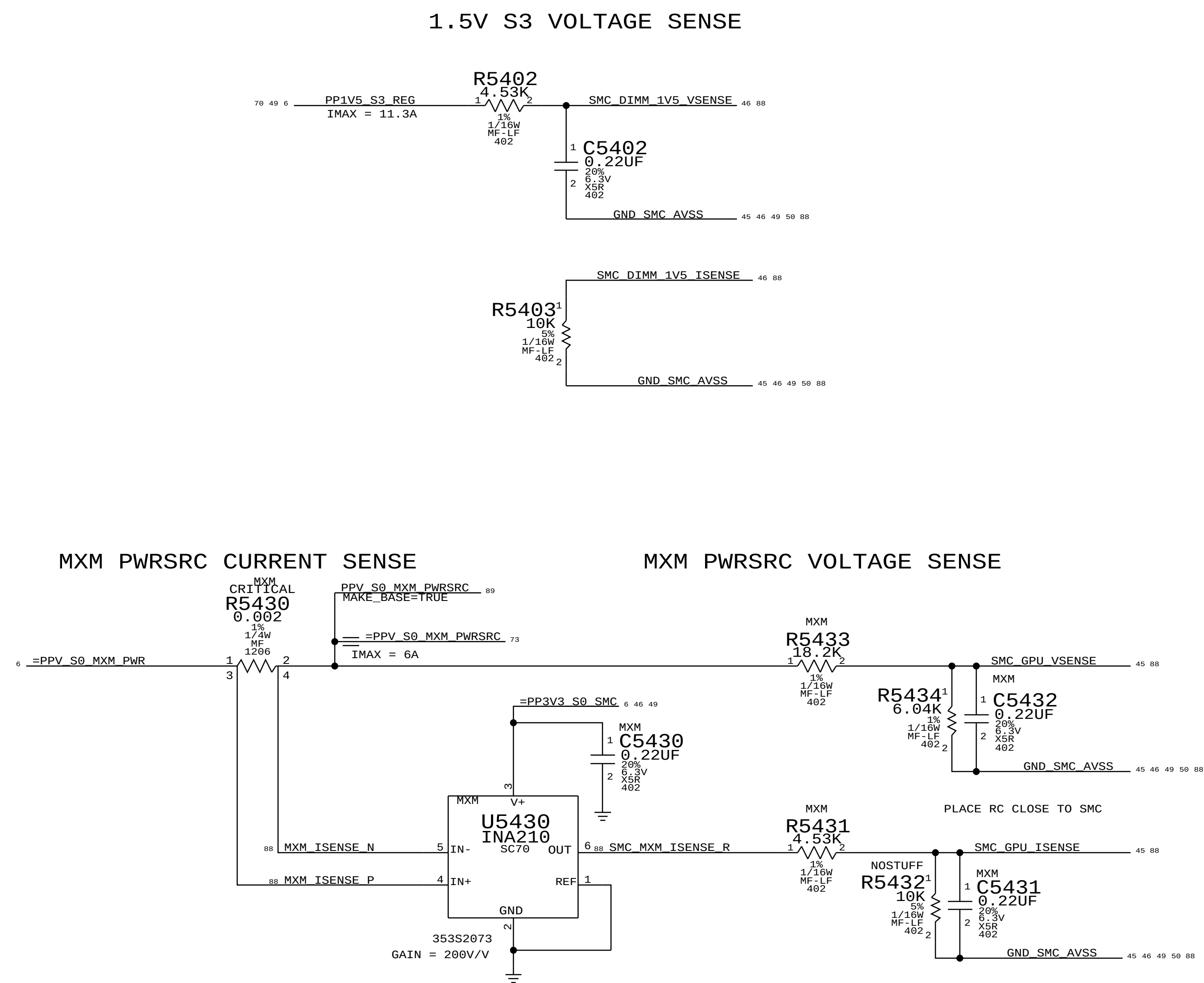
NOTICE OF PROPRIETARY PROPERTY:

PAGE 53 OF 110

SHEET 49 OF 92



SYNC_MASTER=K75F_MLB SYNC_DATE=04/14/2010

PAGE TITLE

GRAPHICS / DIMM POWER SENSE

Apple Inc.

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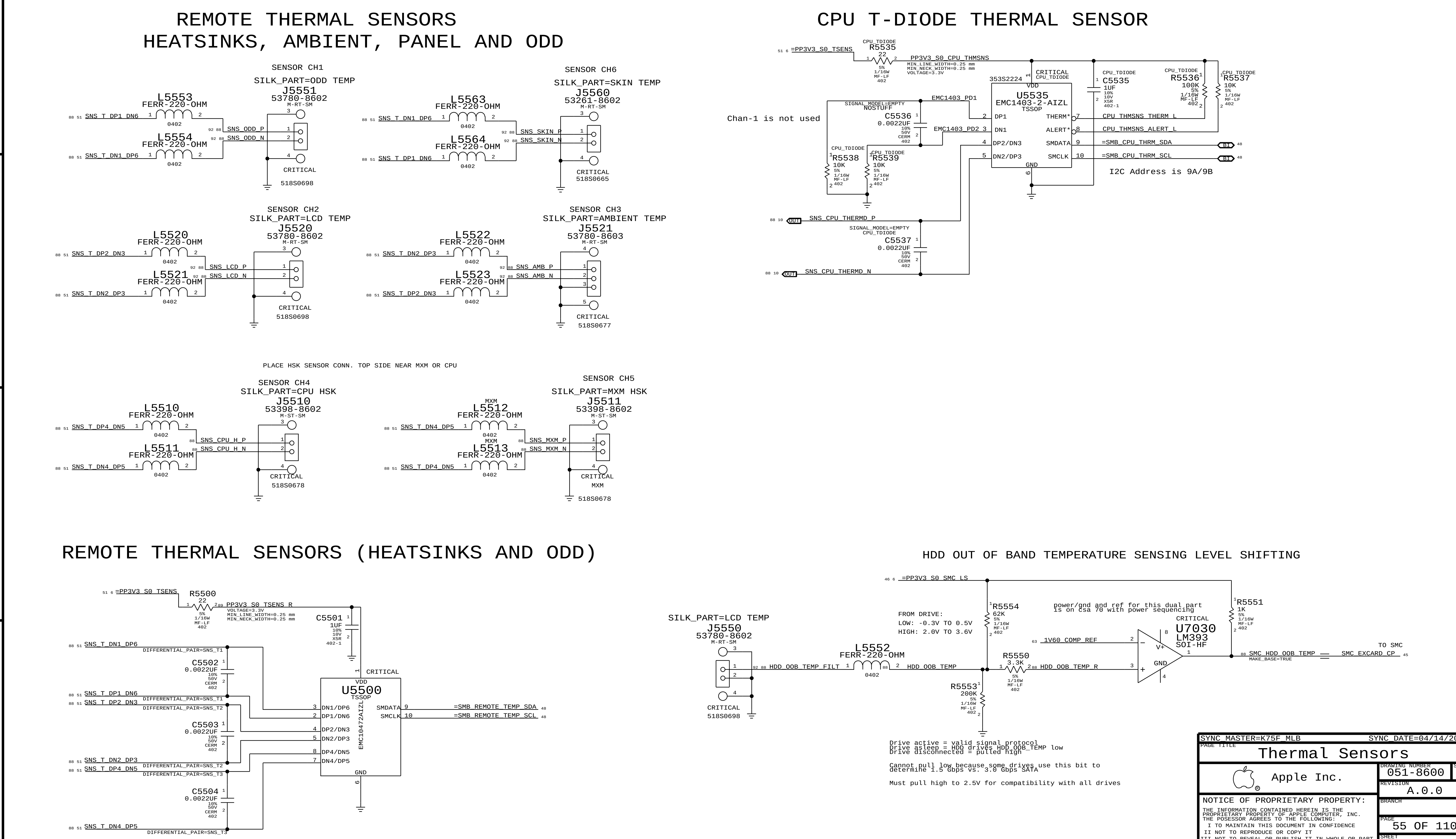

SYNC_MASTER=K75F_MLB SYNC_DATE=04/14/2010

PAGE TITLE: Thermal Sensors

Apple Inc.

DRAWING NUMBER: 051-8600 SIZE: D

REVISION: A.0.0

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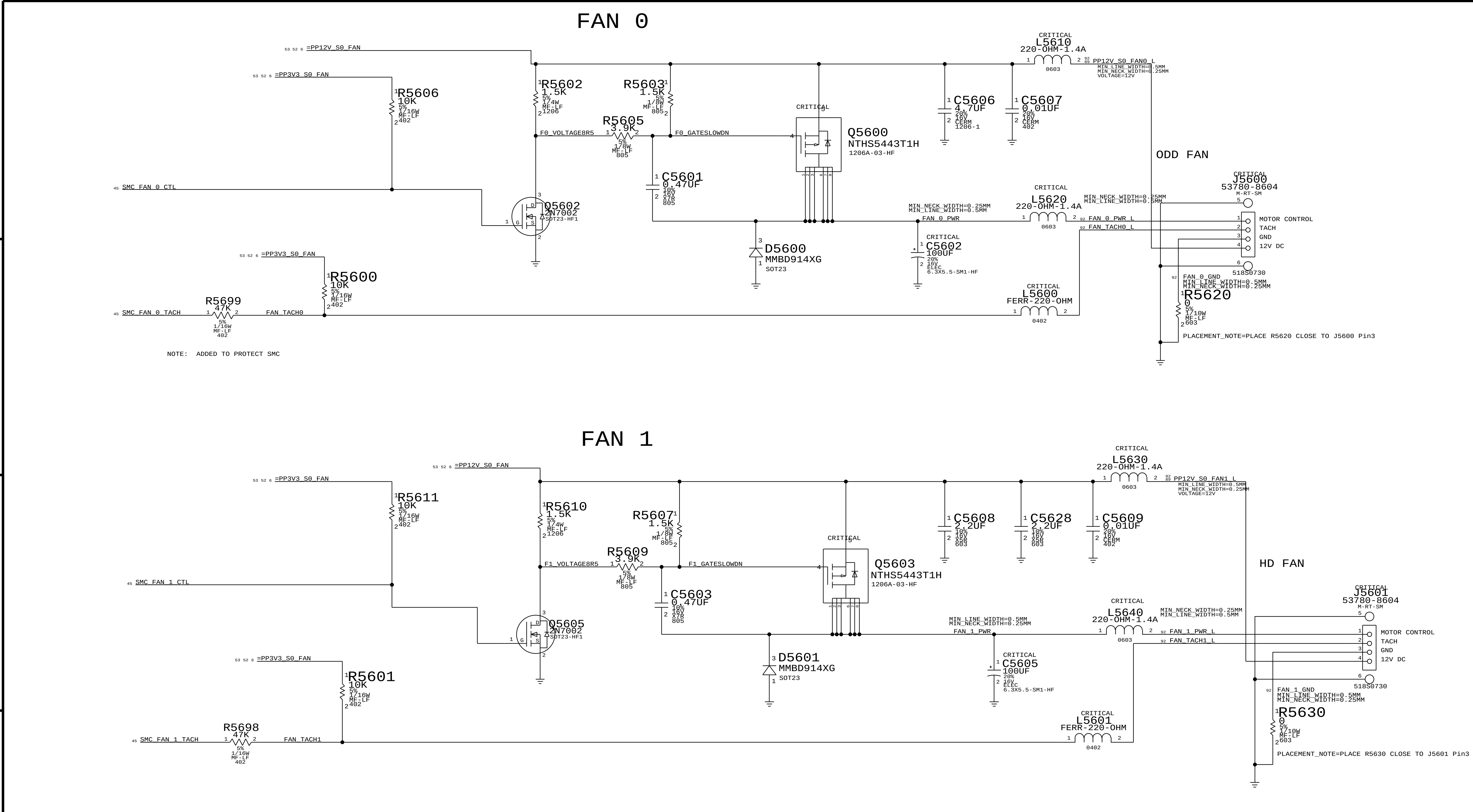
NOTICE OF PROPRIETARY PROPERTY:



SYNC_MASTER=K75F_MLB SYNC_DATE=04/14/2010

PAGE TITLE

HD AND OD FAN

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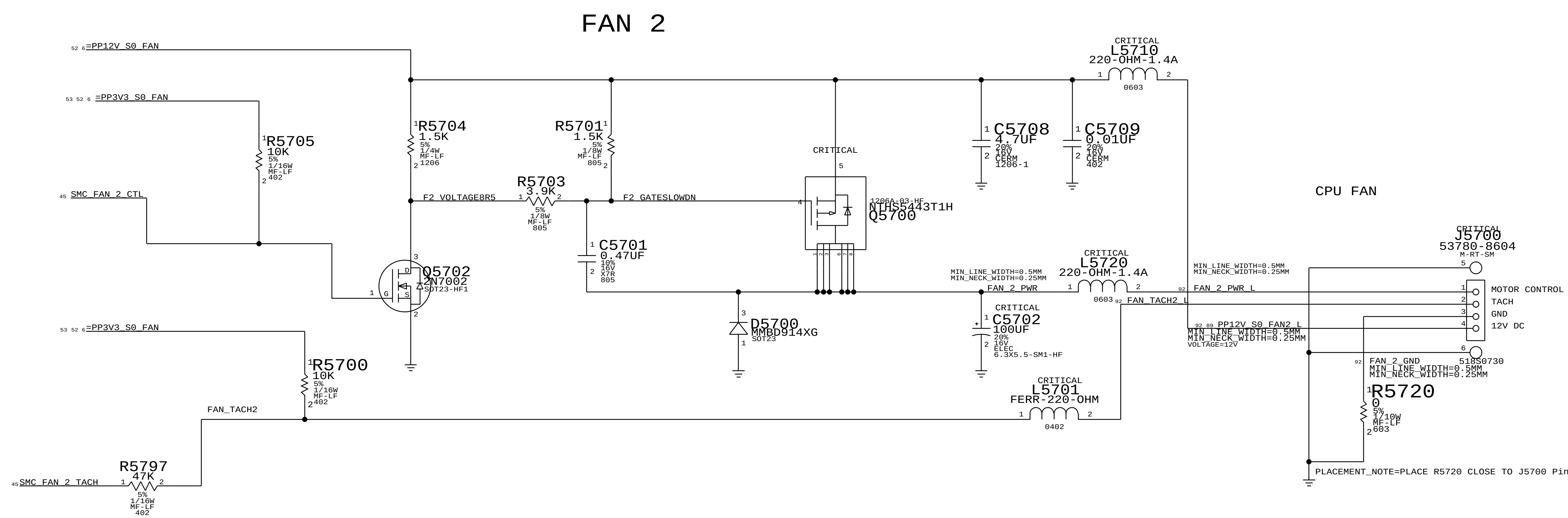
8 7 6 5 4 3 2 1

D C B A



SYNC_MASTER=K75F_MLB SYNC_DATE=04/14/2010

PAGE TITLE

CPU FAN

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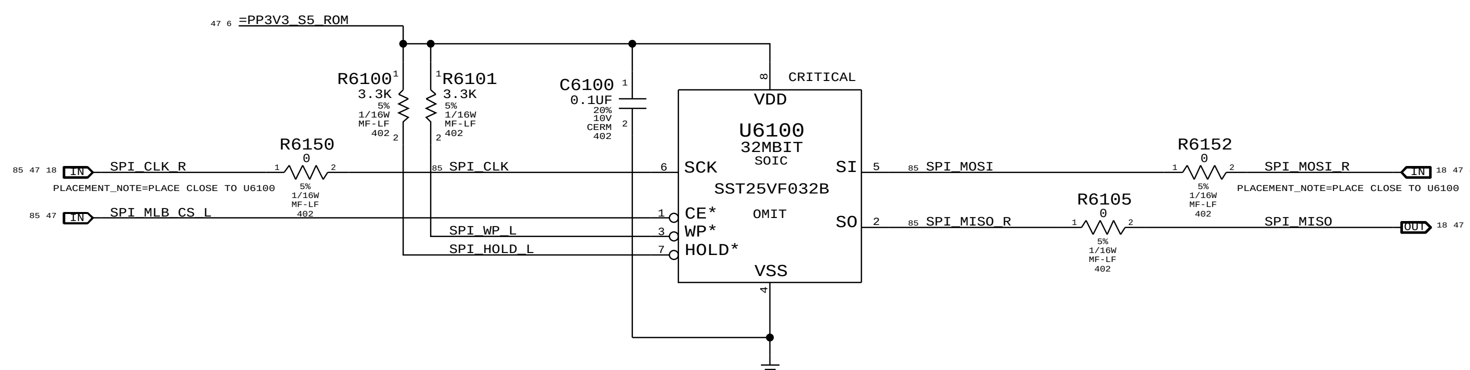
NOTICE OF PROPRIETARY PROPERTY:

8 7 6 5 4 3 2 1



SYNC_MASTER=K75F_MLB SYNC_DATE=04/14/2010

PAGE TITLE

SPI ROM

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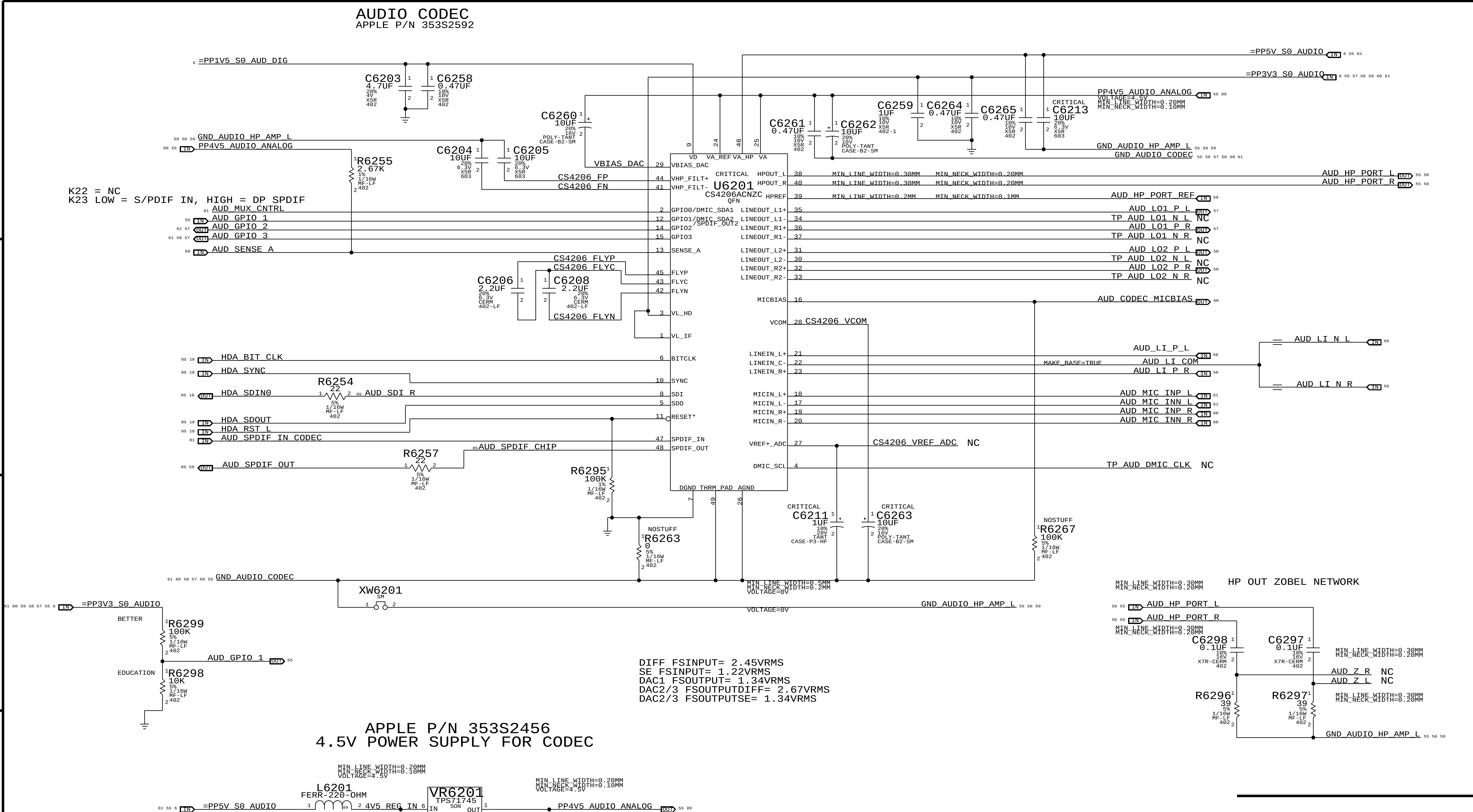

SYNC_MASTER=K75F_MLB SYNC_DATE=04/14/2010

PAGE TITLE

AUDIO: CODEC/REGULATOR

Apple Inc.

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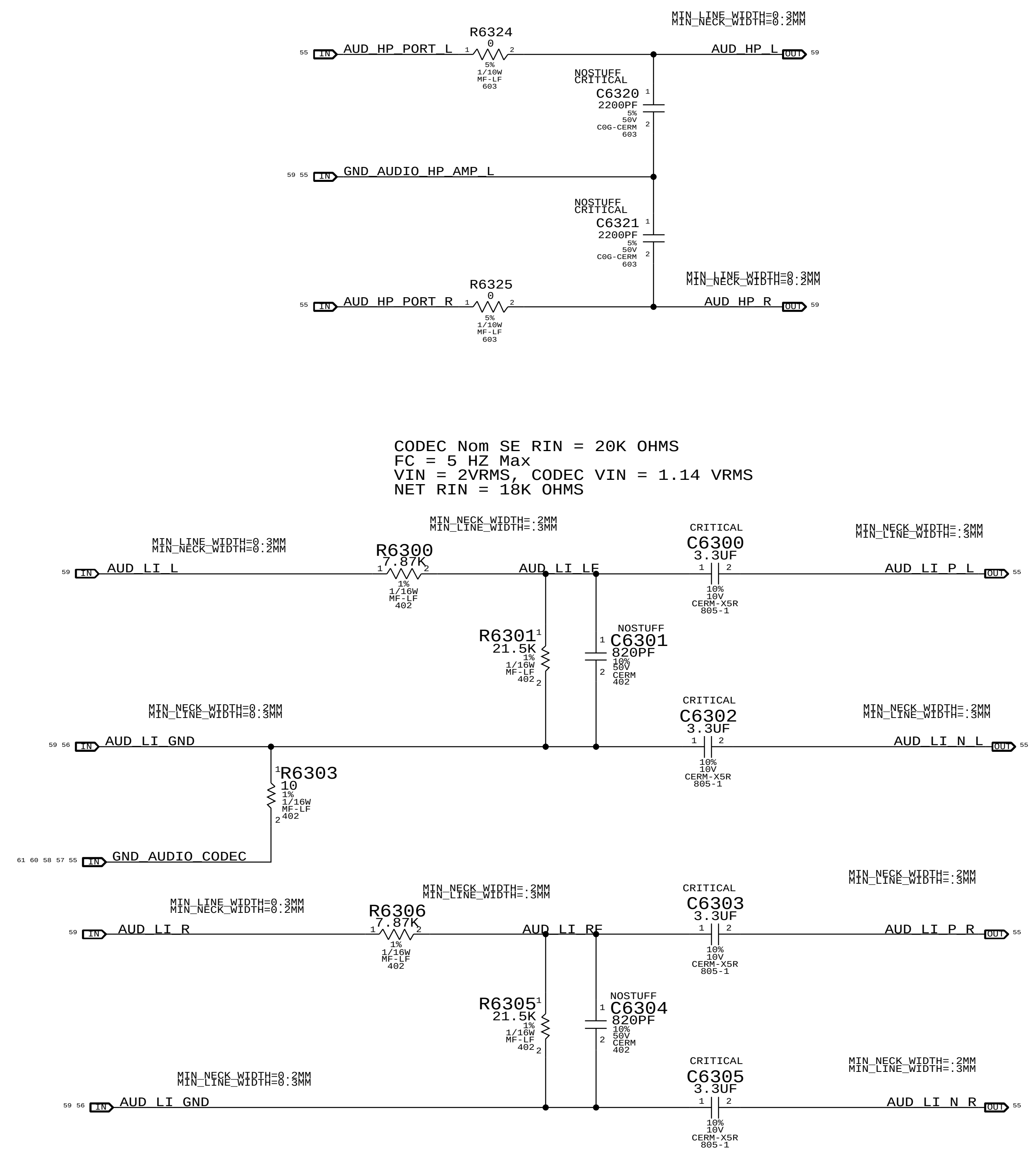
1ST ORDER DAC FILTER PLACEHOLDER



SYNC_MASTER=K75F_MLB SYNC_DATE=04/14/2010

PAGE TITLE

AUDIO: FILTER/BUFFER

Apple Inc.

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NOTICE OF PROPRIETARY PROPERTY:

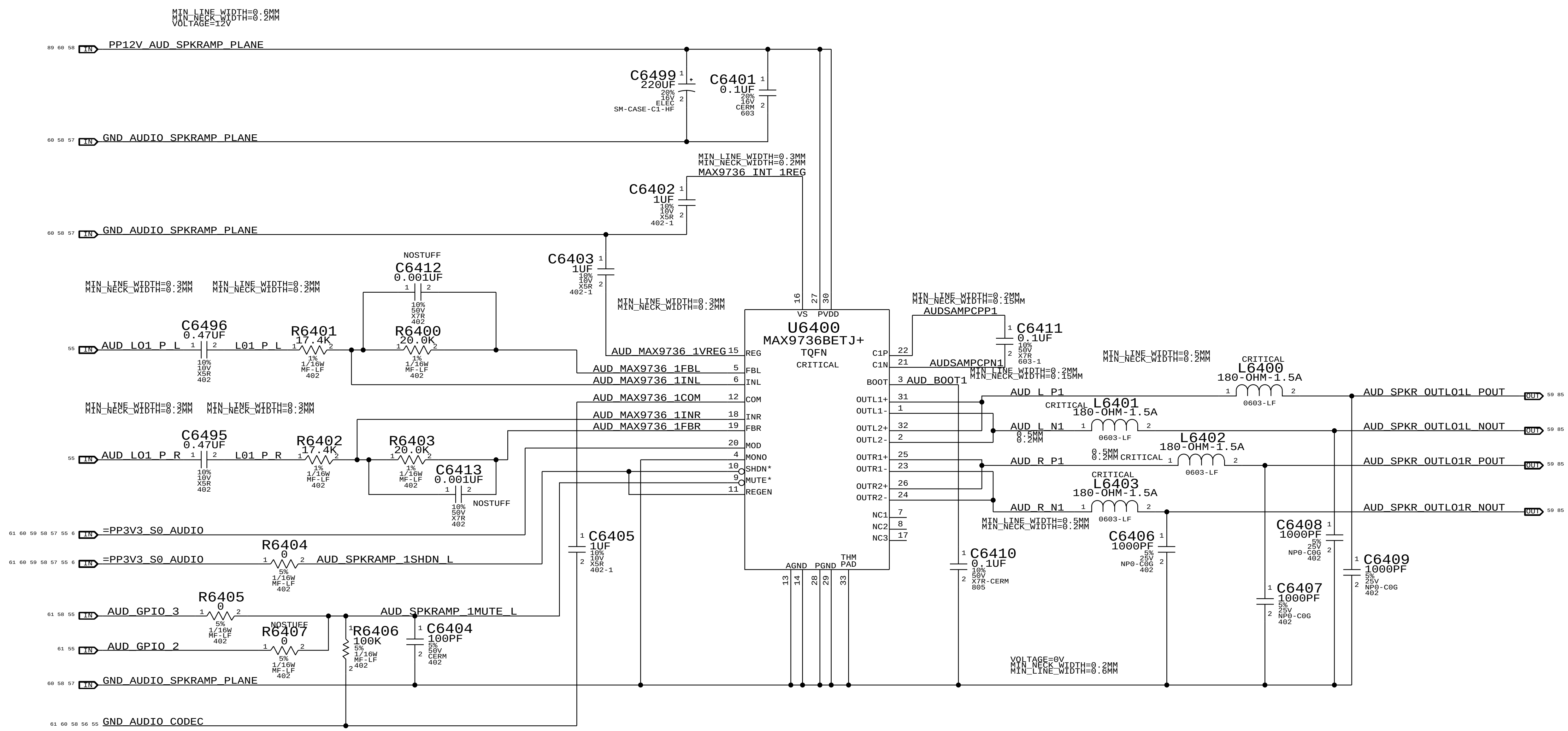
TWEETER SPEAKER AMPLIFIER

MAX9736B APN:353S2042

GAIN = -4.8(20K/17.4K)
CODEC OUT = 1.335VRMS
AMP VOUT = 7.355VRMS

TURN ON TIME: 110MS
TURN ON DELAY: 150MS
RIN = 17.4 OHMS
FC = 19.5 HZ

POUT = 6.76 W INTO 8 OHMS @ 1% THD+N



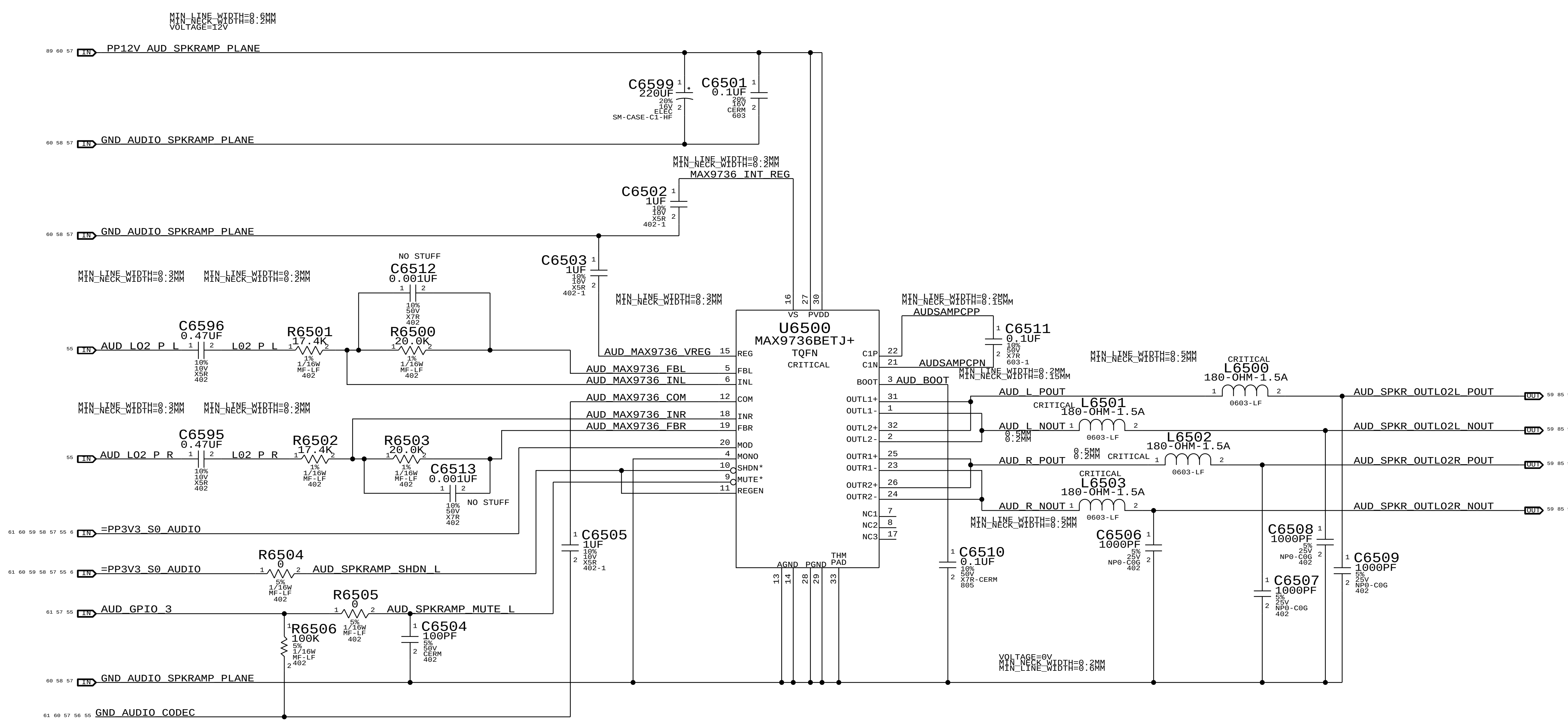
SYNC_MASTER=K75F_MLB	SYNC_DATE=04/14/2010
PAGE TITLE	AUDIO: Tweeter Amp 1
Apple Inc.	DRAWING NUMBER 051-8600 SIZE D
	REVISION A.0.0
	BRANCH
	PAGE 64 OF 110
	SHEET 57 OF 92

WOOFER SPEAKER AMPLIFIER

MAX9736B APN:353S2042

GAIN = -4.8(20K/17.4K)
CODEC OUT = 1.335VRMS
AMP VOUT = 7.355VRMS
POUT = 6.76 W INTO 8 OHMS @ 1% THD+N

TURN ON TIME: 110MS
TURN ON DELAY: 150MS
RIN = 17.4 OHMS
FC = 19.5 HZ



SYNC_MASTER=K75F_MLB SYNC_DATE=04/14/2010

PAGE TITLE: AUDIO: Woofer Amp

Apple Inc.

DRAWING NUMBER: 051-8600 SIZE: D

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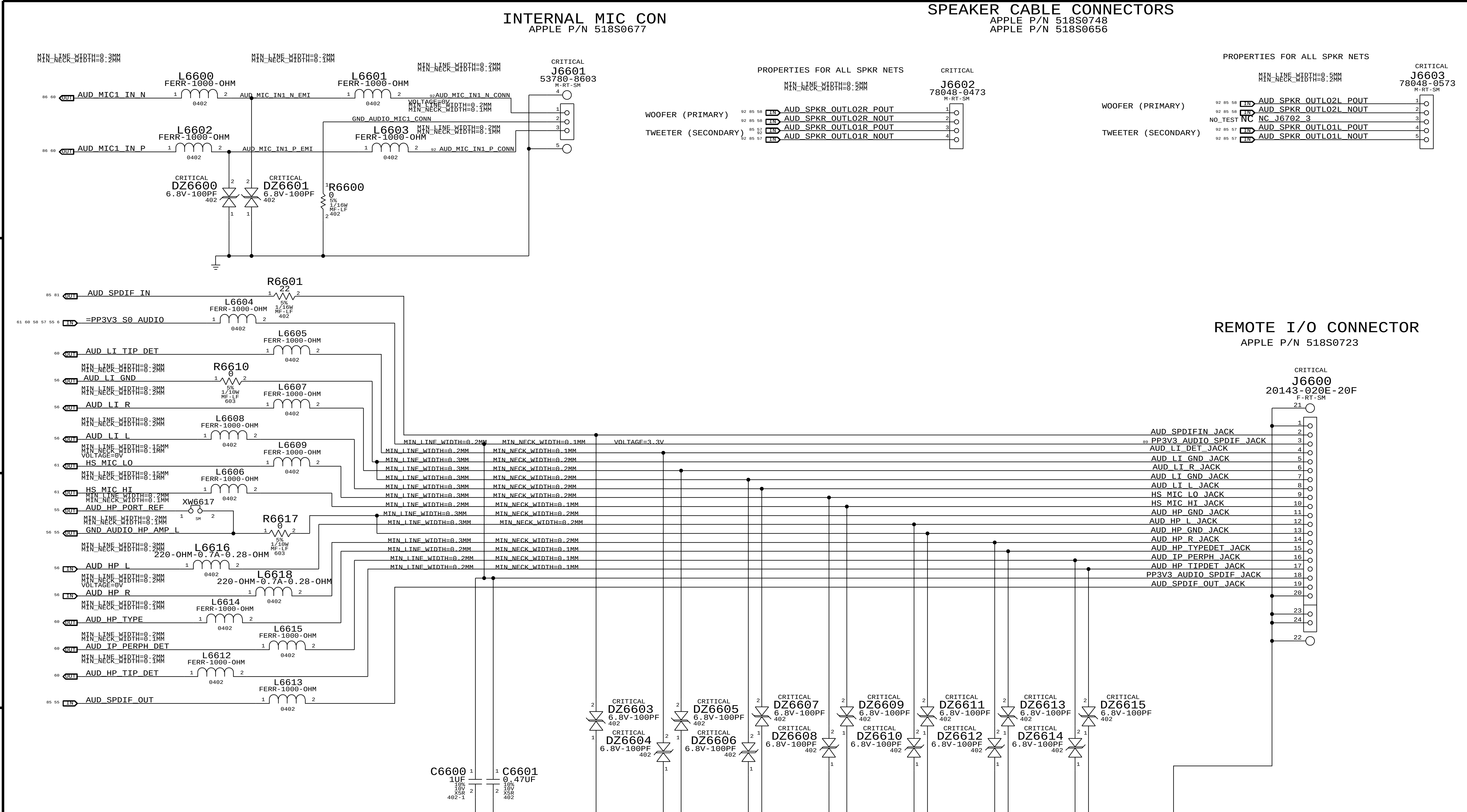

SYNC_MASTER=K75F_MLB SYNC_DATE=04/14/2010

PAGE TITLE

Audio: MLB to I/O Conn.

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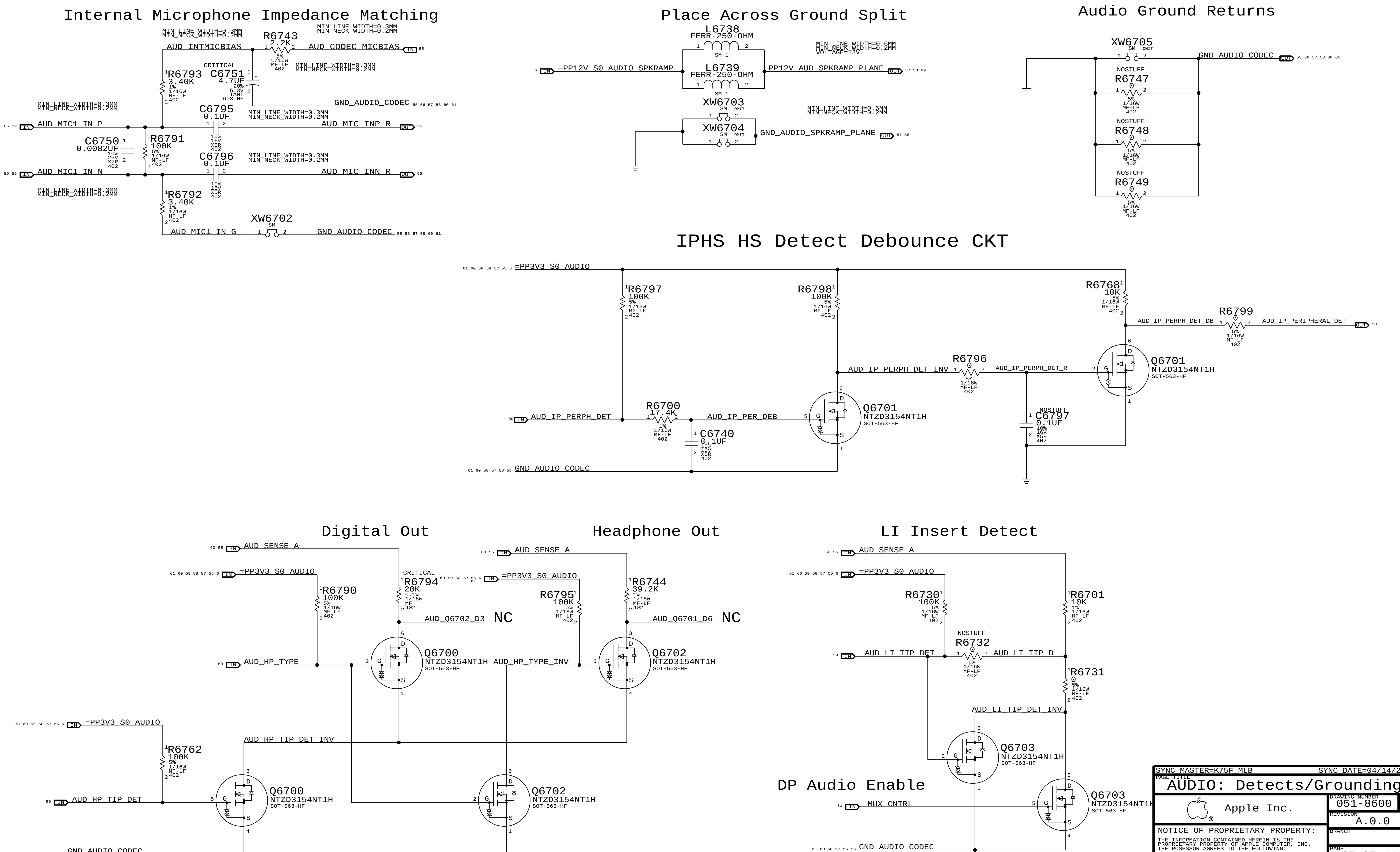
PAGE 66 OF 110

SHEET 59 OF 92



SYNC_MASTER=K75F_MLB SYNC_DATE=04/14/2010

PAGE TITLE

AUDIO: Detects/Grounding

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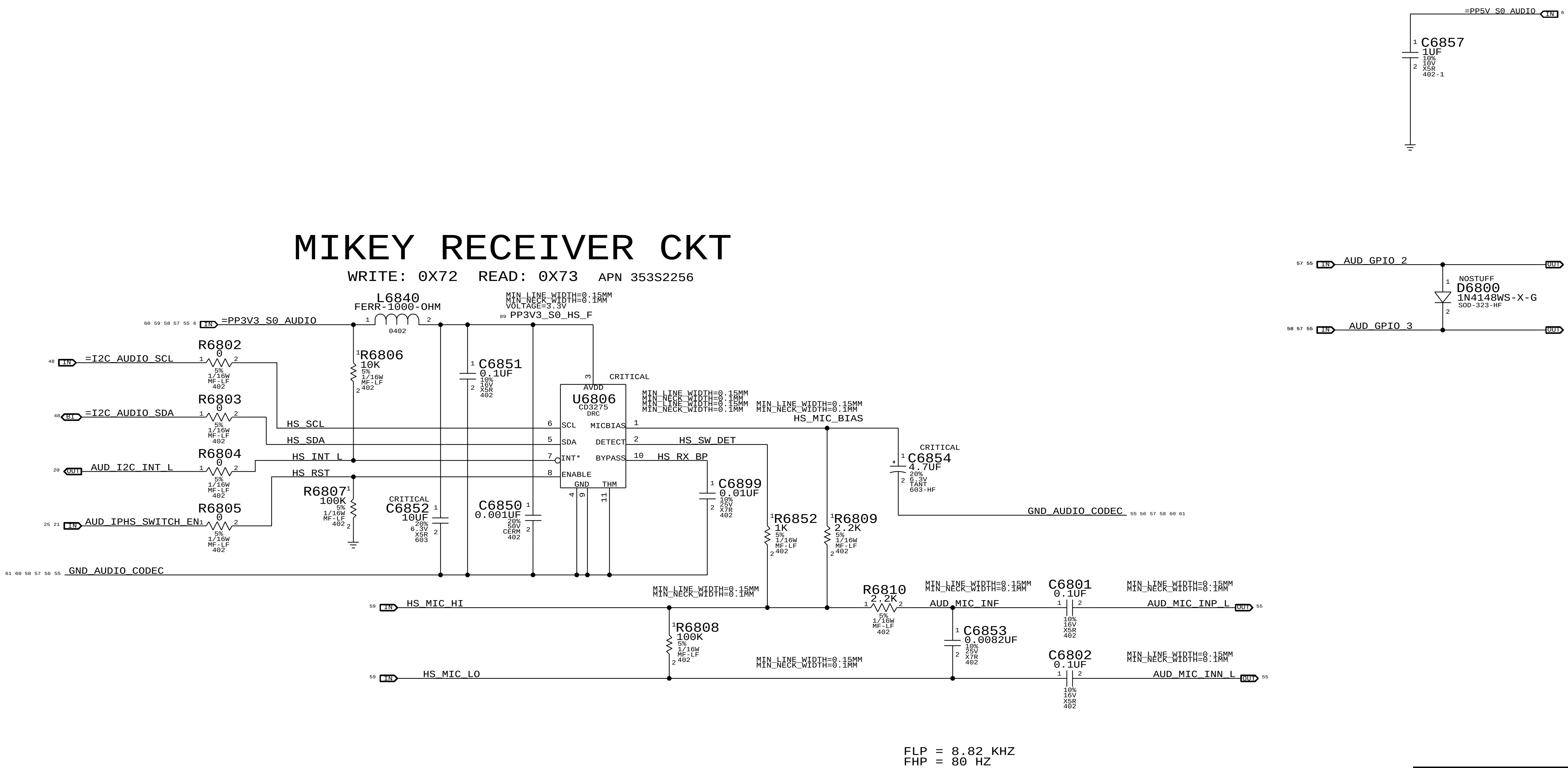
NOTICE OF PROPRIETARY PROPERTY:

FUNCTION	PIN	CONVERTER	VOLUME	ENABLE/ CNTRL TYPE	DETECT/INTERRUPT
PRIMARY	0X0B	0X04	0X04	GPIO 3	N/A
SECONDARY	0X0A	0X03	0X03	GPIO 3	N/A
HEADPHONES	0X09	0X02	0X02	N/A	0X09 (A)
LINE INPUT	0X0C	0X05	0X05	N/A	LINE IN
BUILT-IN MICROPHONE	0X0D(13,B,RIGHT)	0X06	0X06	MICBIAS 80%	N/A
HEADSET MICROPHONE	0X0D (13,V22,B,LEFT)	0X06	0X06	MIKEY	MIKEY
SPDIF OUT	0X10	0X08	N/A	N/A	0X0C (B)
SPDIF IN	0X0F	0X07	N/A	N/A	N/A
MIKEY	N/A	N/A	N/A	MCP GPIO_38	MCP GPIO_5



SYNC_MASTER=K75F_MLB SYNC_DATE=04/14/2010

PAGE TITLE: AUDIO: Mikey

Apple Inc.

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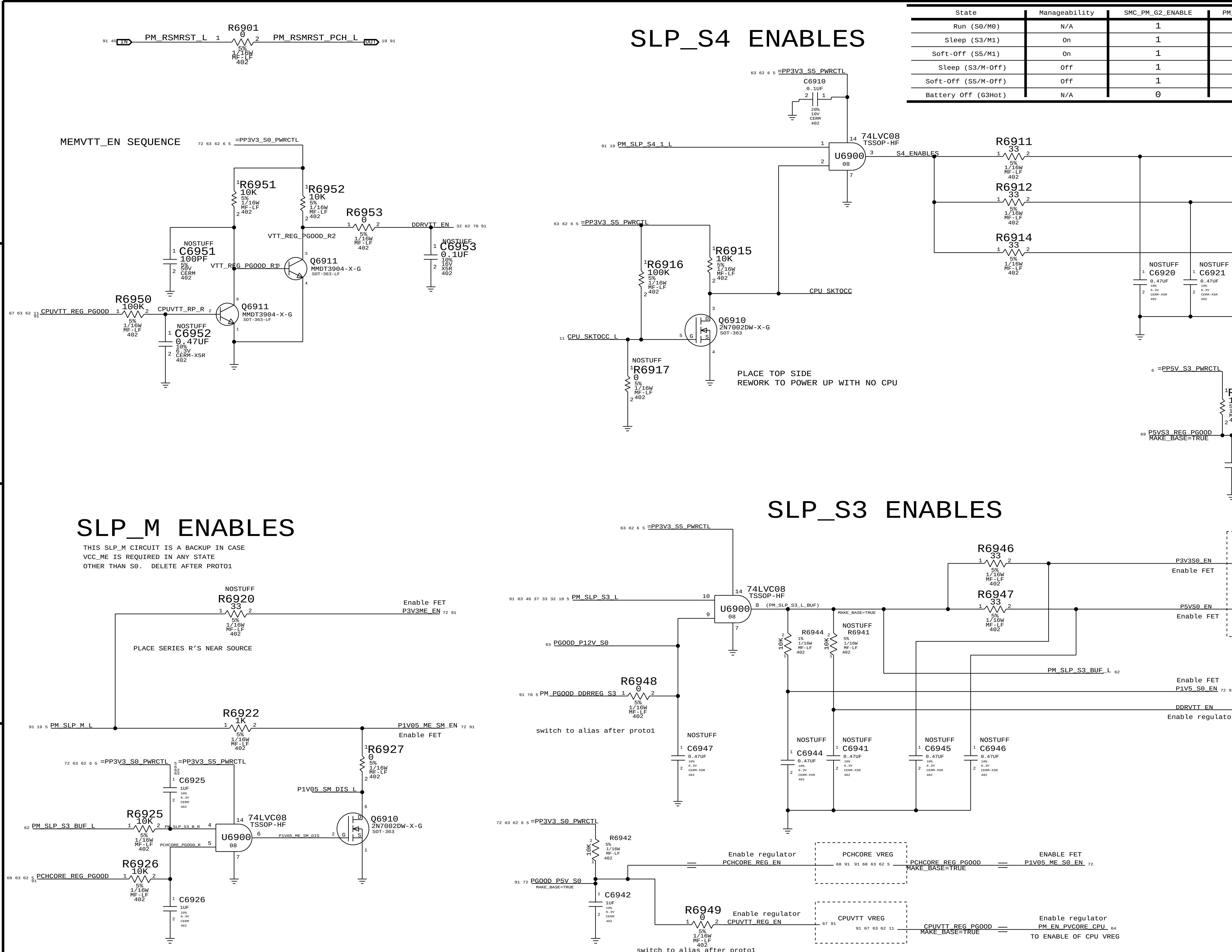

State	Manageability	SMC_PM_G2_ENABLE	PM_S4_STATE_L	PM_SLP_S3_L	PM_SLP_S4_L	PM_SLP_M_L
Run (S0/M0)	N/A	1	1	1	1	1
Sleep (S3/M1)	On	1	1	0	1	1
Soft-Off (S5/M1)	On	1	0	0	1	1
Sleep (S3/M-Off)	Off	1	1	0	1	0
Soft-Off (S5/M-Off)	Off	1	0	0	0	0
Battery Off (G3Hot)	N/A	0	0	0	0	0

SYNC_MASTER=K75F_MLB SYNC_DATE=04/14/2010

PAGE TITLE

POWER SEQUENCING ENABLES

Apple Inc.

DRAWING NUMBER 051-8600 SIZE D

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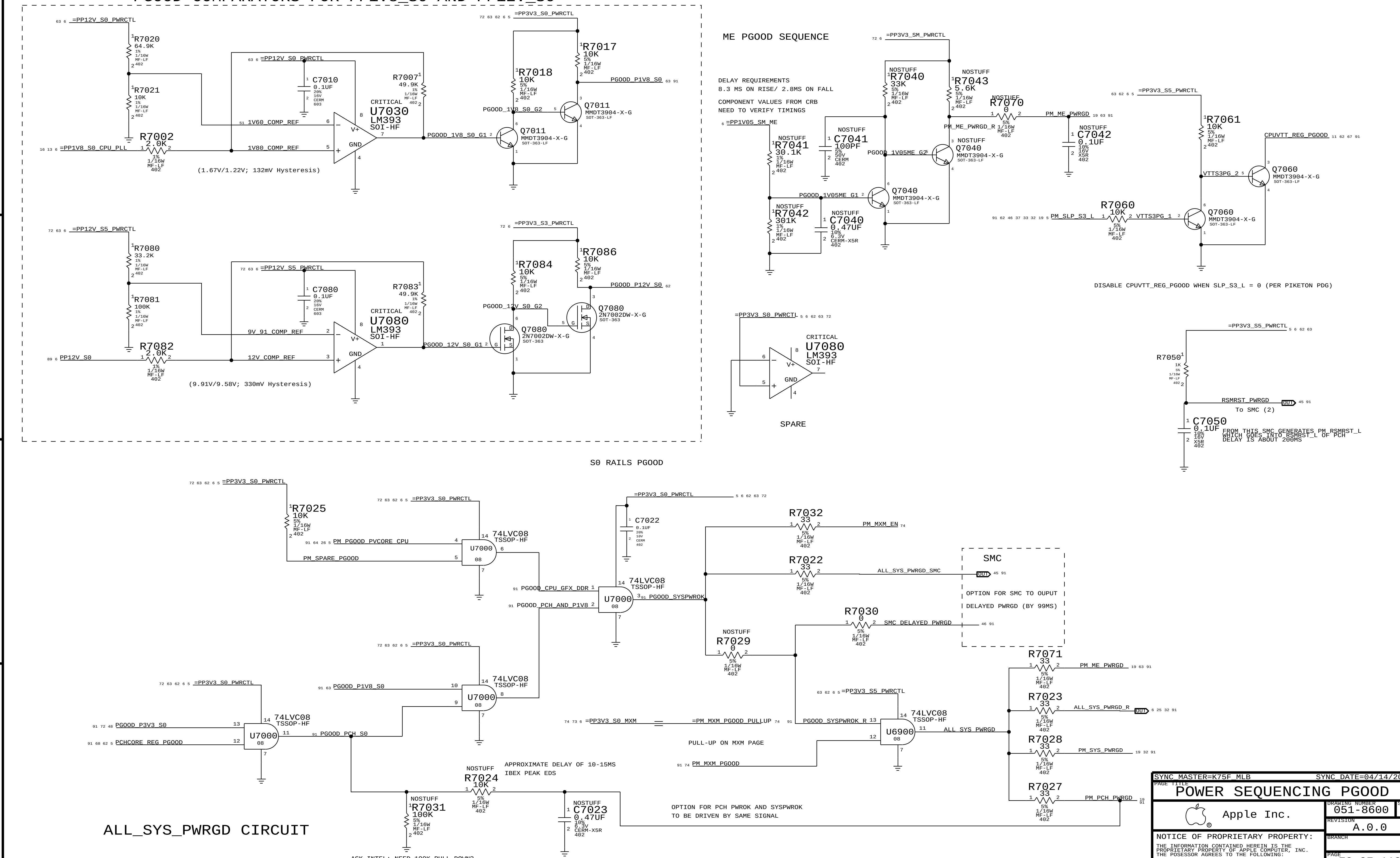
PGOOD COMPARATORS FOR PP1V8_S0 AND PP12V_S0



ALL_SYS_PWRGD CIRCUIT

SYNC_MASTER=K75F_MLB SYNC_DATE=04/14/2010

PAGE TITLE: POWER SEQUENCING PGOOD

Apple Inc.

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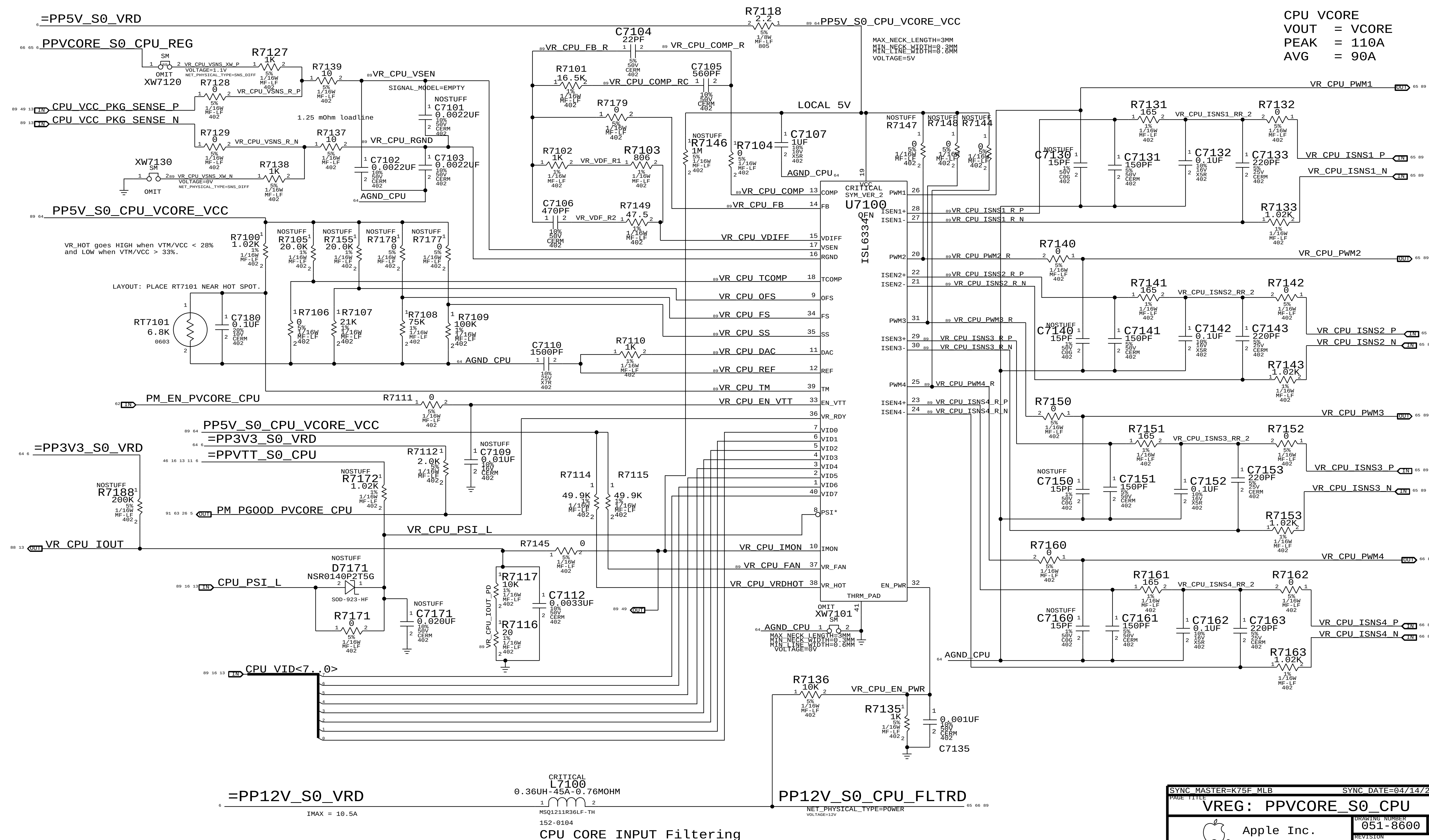
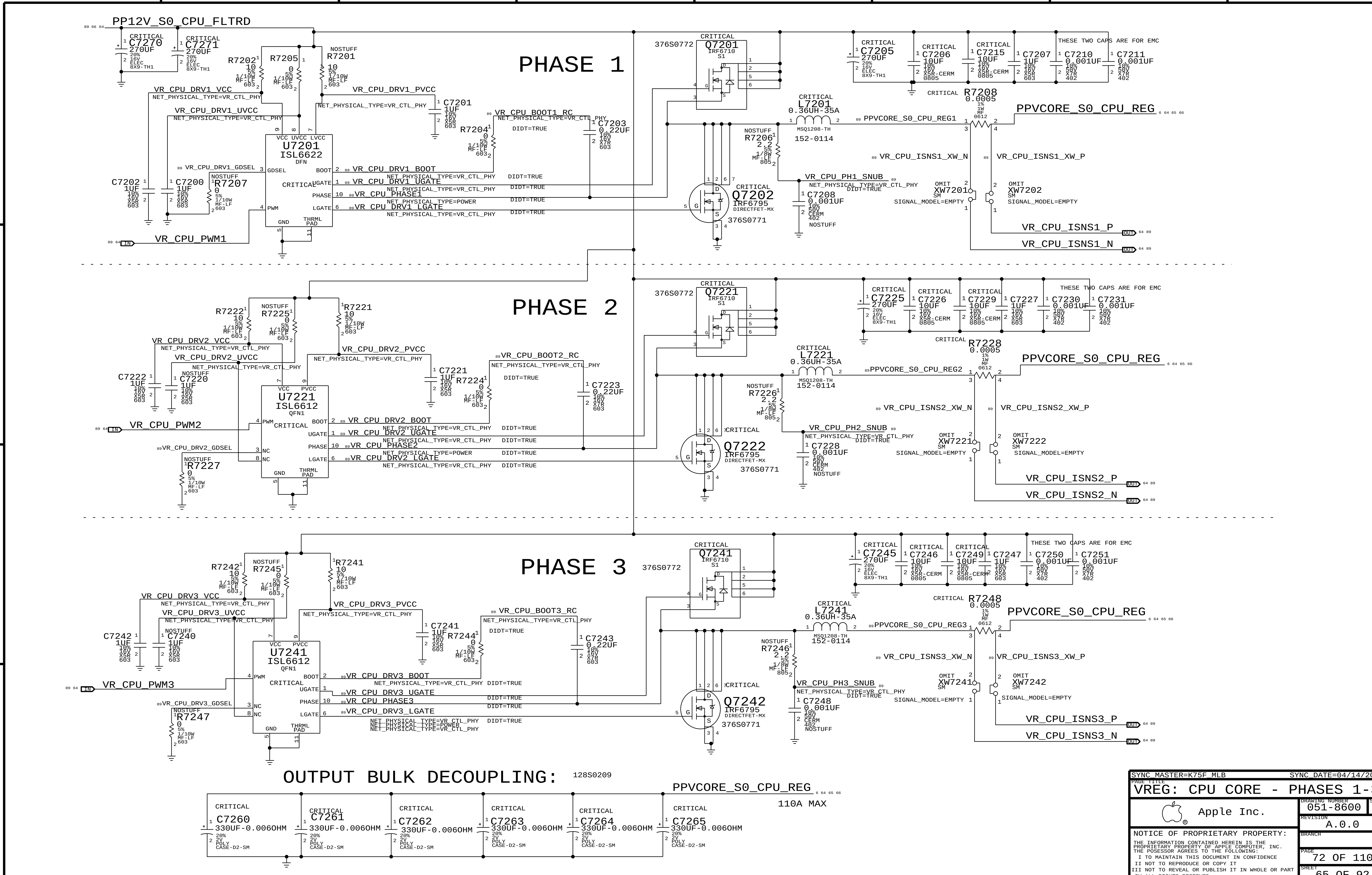
SHEET 63 OF 92

CPU CORE REG 1.1V/110A O/P= PPVCORE_S0_CPU_REG



CPU CORE INPUT Filtering

SYNC_MASTER=K75F_MLB	SYNC_DATE=04/14/2010
PAGE TITLE	VREG: PPVCORE_S0_CPU
Apple Inc.	DRAWING NUMBER 051-8600 SIZE D
	REVISION A.0.0
	BRANCH
	PAGE 71 OF 110
	SHEET 64 OF 92



SYNC_MASTER=K75F_MLB SYNC_DATE=04/14/2010

PAGE TITLE

VREG: CPU CORE - PHASES 1-3

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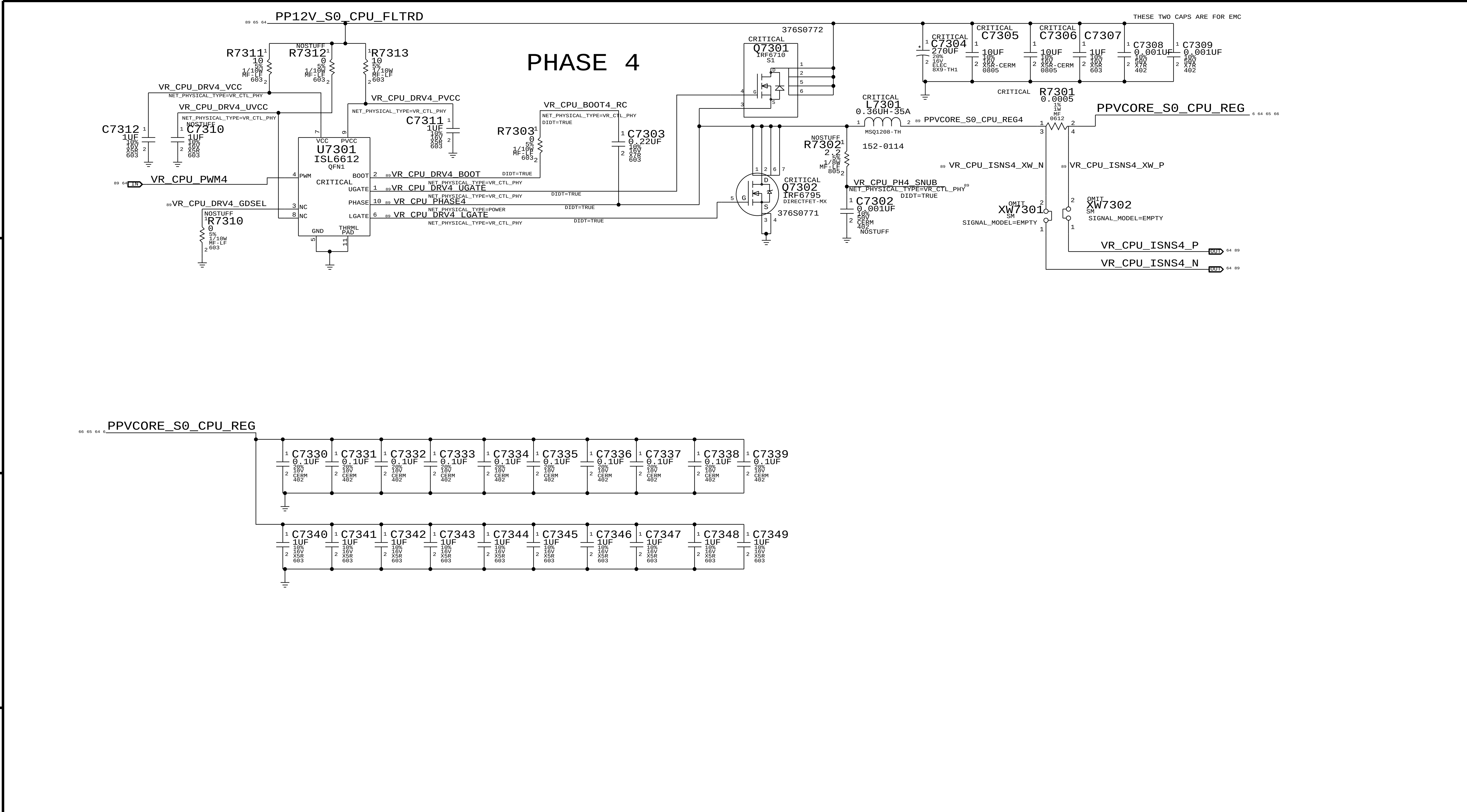
SHEET 65 OF 92



SYNC_MASTER=K75F_MLB SYNC_DATE=04/14/2010

PAGE TITLE

VREG: CPU CORE - PHASE 4

Apple Inc.

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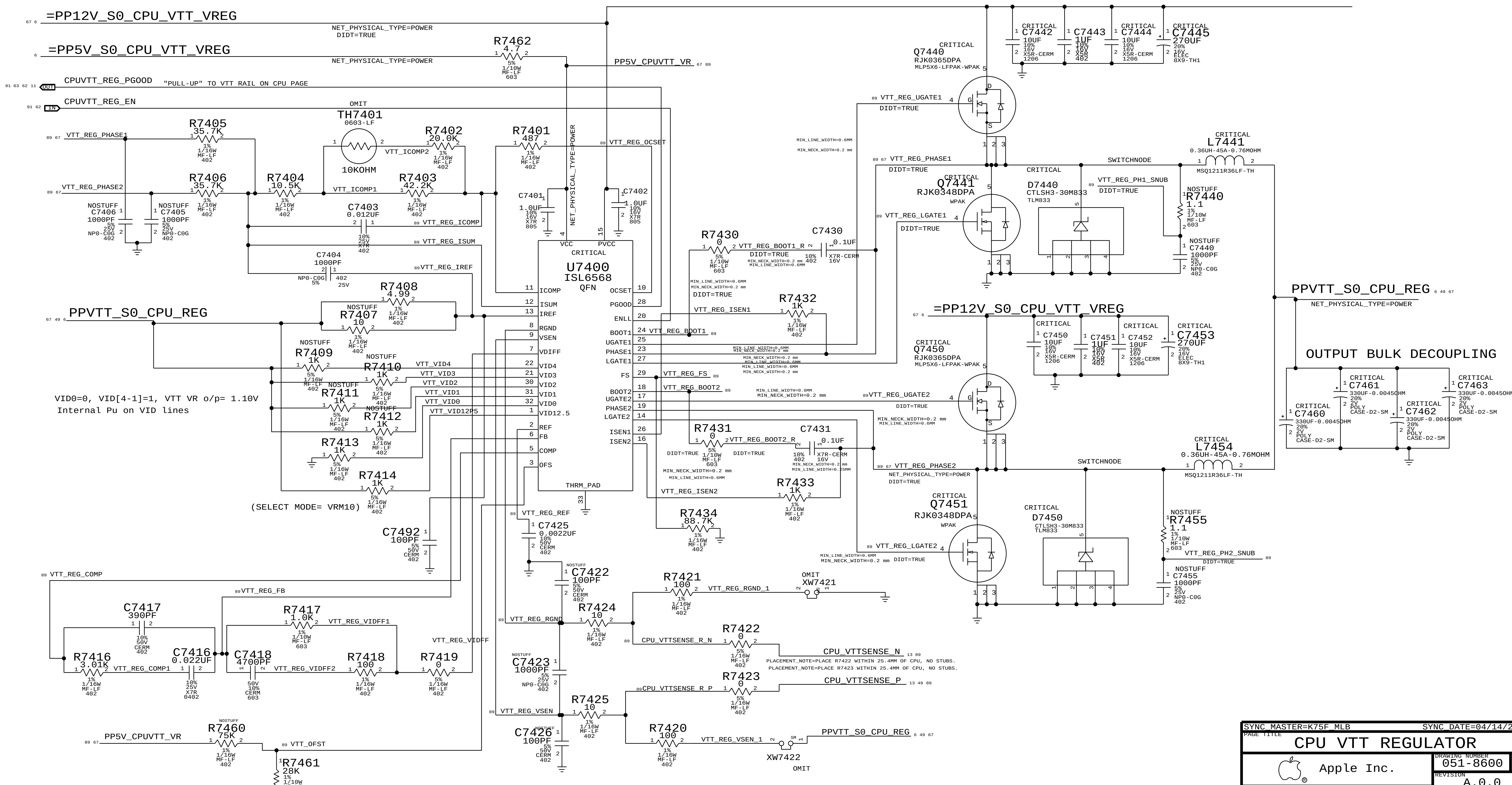
NOTICE OF PROPRIETARY PROPERTY:

CPU VTT REG 1.1V/30A O/P= PPVTT_S0_CPU_REG

CPU VTT

VOUT = 1.1V OR 1.05V
PEAK = 35A
AVG = 30A



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
113S0127	1	RES,68k,0603,5%	TH7401	

SYNC_MASTER=K75F_MLB SYNC_DATE=04/14/2010

PAGE TITLE: CPU VTT REGULATOR

Apple Inc.

DRAWING NUMBER: 051-8600 SIZE: D

REVISION: A.0.0

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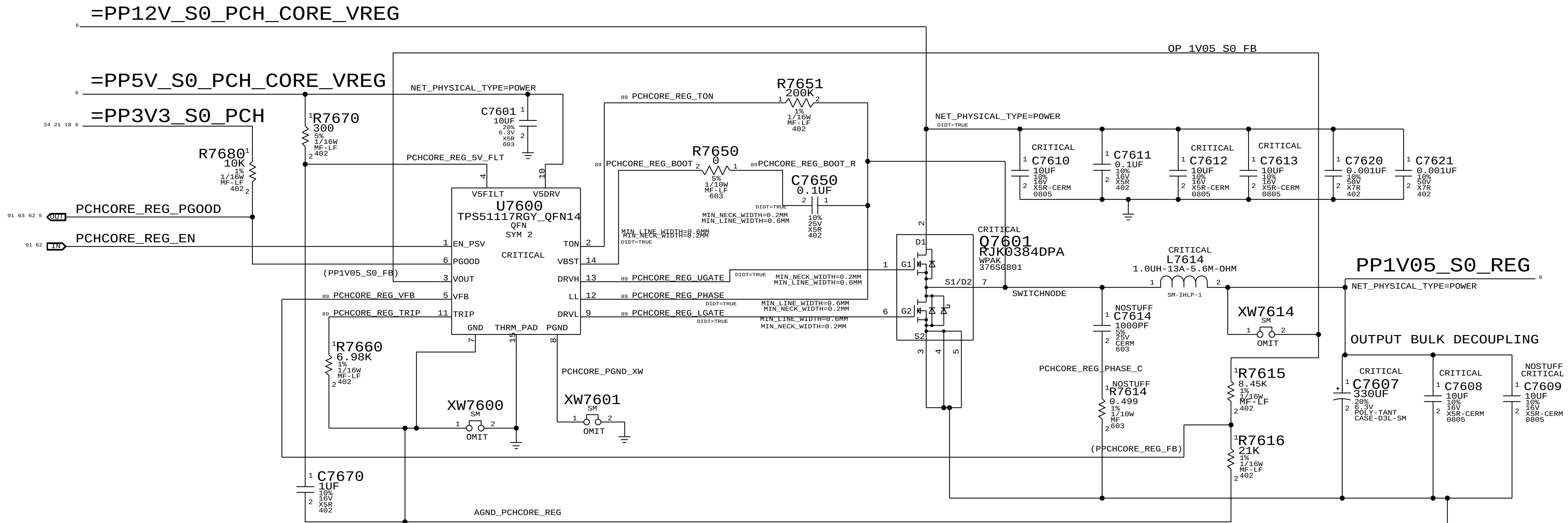
IBEX PEAK CORE REG 1.05V OUTPUT = PP1V05_S0_REG

PP1V05_S0_REG

VOUT = 1.05V
PEAK = 7.5A
AVG = 3A



Vout= 0.75*(1+R7615/R7616)
Vout= 0.75*(1 + 8.45/21) = 1.05

PAGE TITLE
IBEX PEAK CORE

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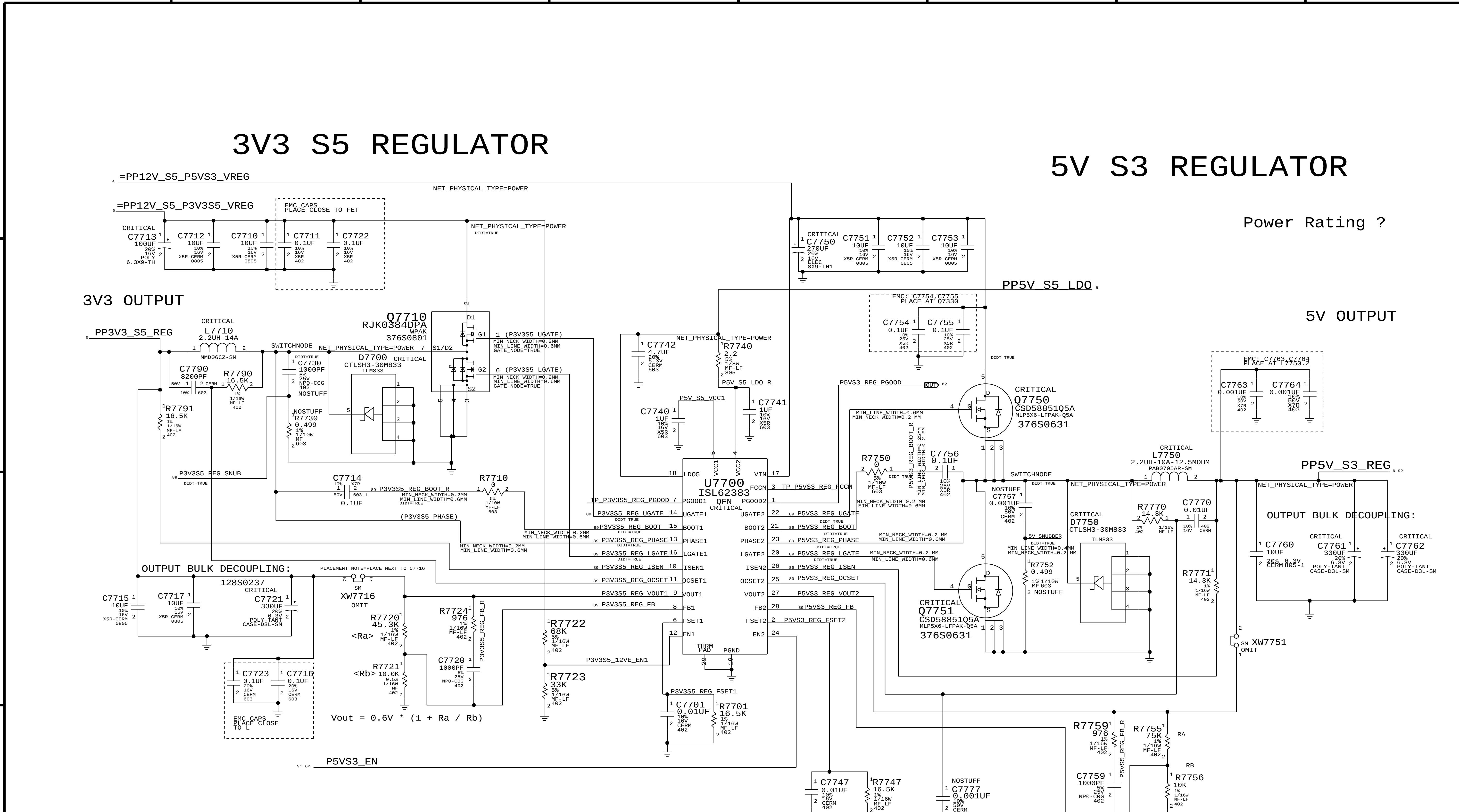
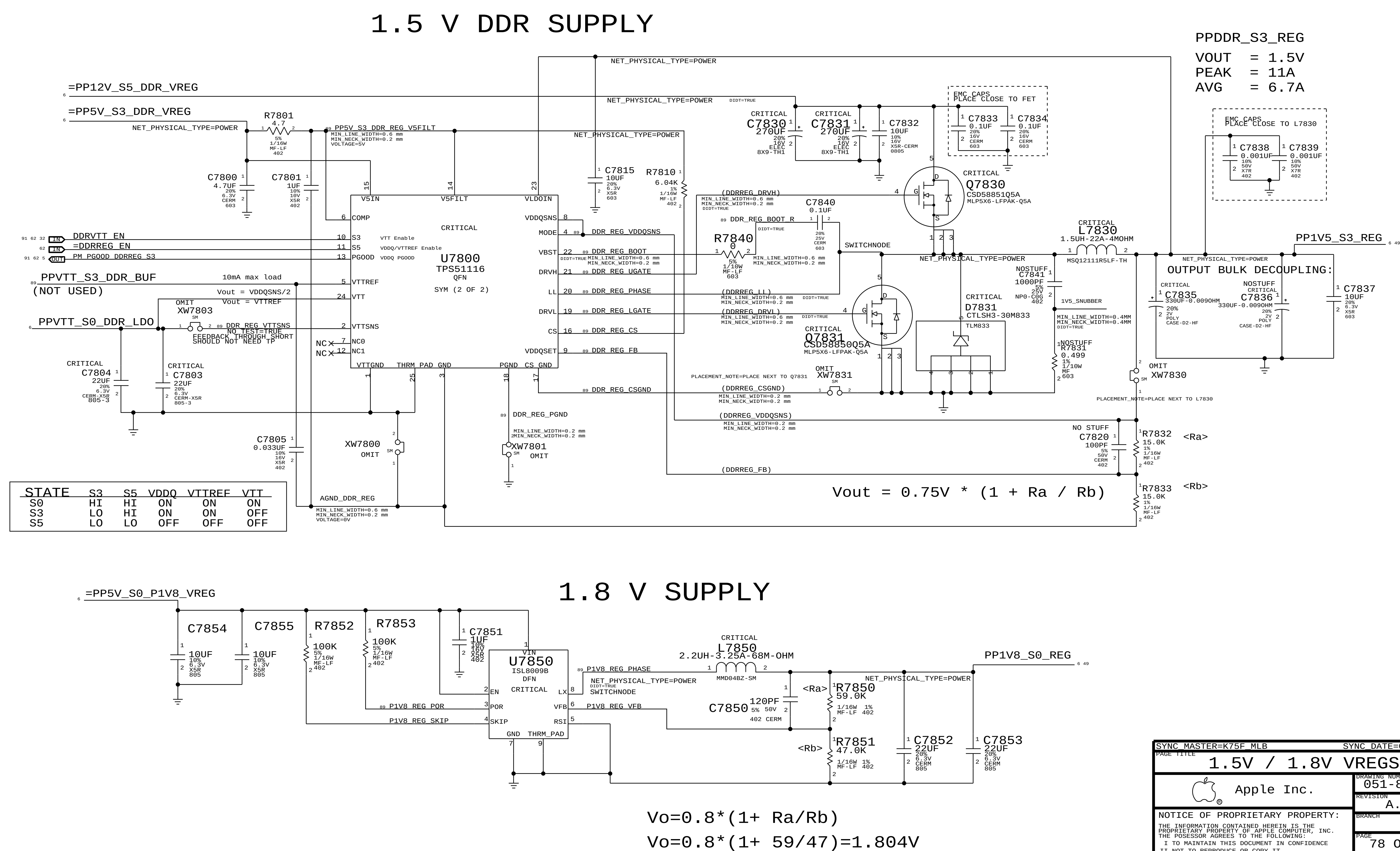
PAGE 76 OF 110

SHEET 68 OF 92



SYNC_MASTER=K75F_MLB	SYNC_DATE=04/14/2010
PAGE TITLE	5V_S3 / 3V3_S5 VREGS
Apple Inc.	DRAWING NUMBER 051-8600 SIZE D
	REVISION A.0.0
	BRANCH
	PAGE 77 OF 110
	SHEET 69 OF 92



STATE	S3	S5	VDDQ	VTTREF	VTT
S0	HI	HI	ON	ON	ON
S3	LO	HI	ON	ON	OFF
S5	LO	LO	OFF	OFF	OFF

SYNC_MASTER=K75F_MLB SYNC_DATE=04/14/2010

PAGE TITLE

1.5V / 1.8V VREGS

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DRAWING NUMBER 051-8600 SIZE D

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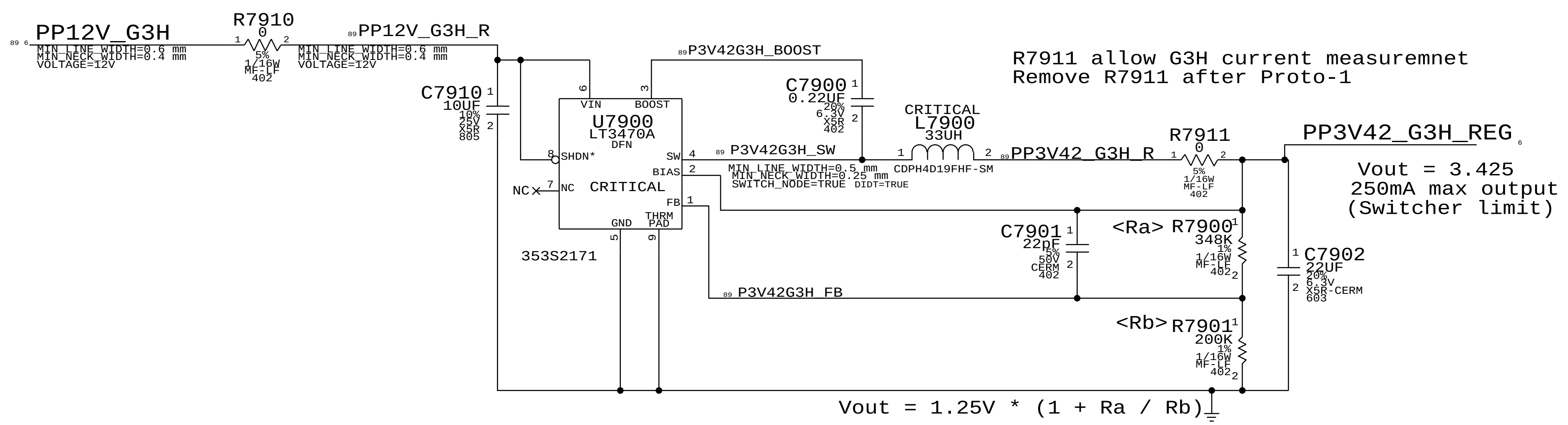
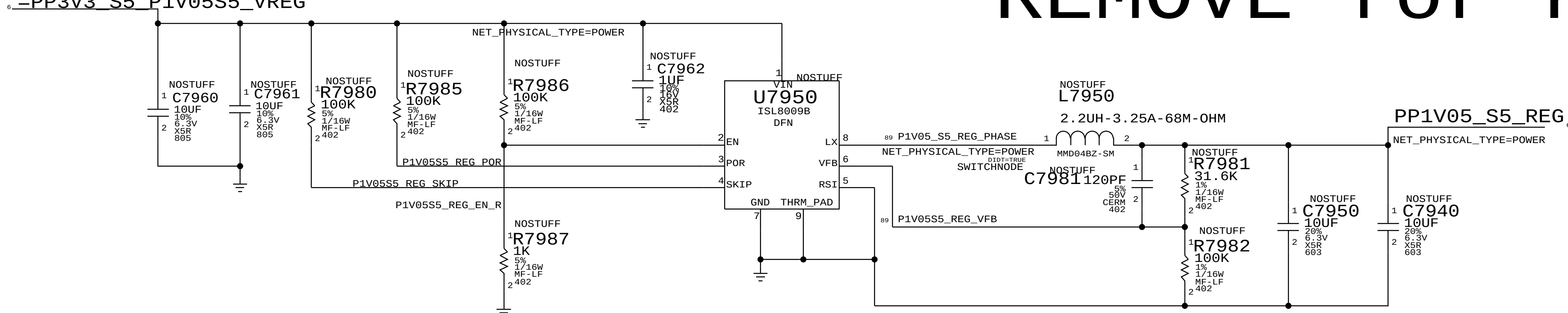
3.425V "G3Hot" Supply

Supply needs to guarantee 3.31V delivered to SMC VRef generator



1.05V S5 SUPPLY

REMOVE for K60/K61



SYNC_MASTER=K75F_MLB SYNC_DATE=04/14/2010

PAGE TITLE: 1.05 S5 SUPPLY

Apple Inc.

DRAWING NUMBER: 051-8600 SIZE: D

REVISION: A.0.0

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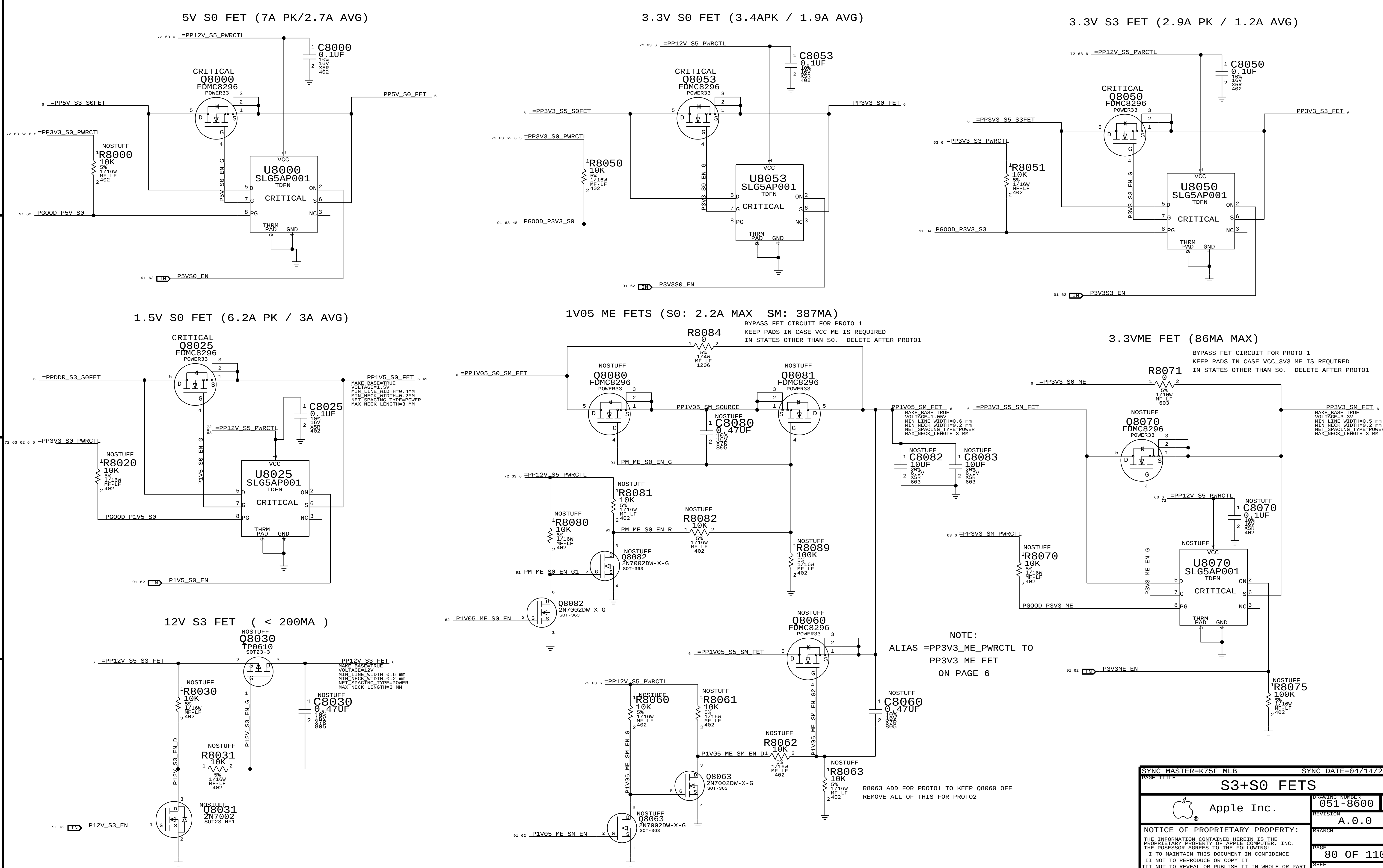
PAGE: 79 OF 110

SHEET: 71 OF 92



SYNC_MASTER=K75F_MLB SYNC_DATE=04/14/2010

PAGE TITLE

S3+S0 FETS

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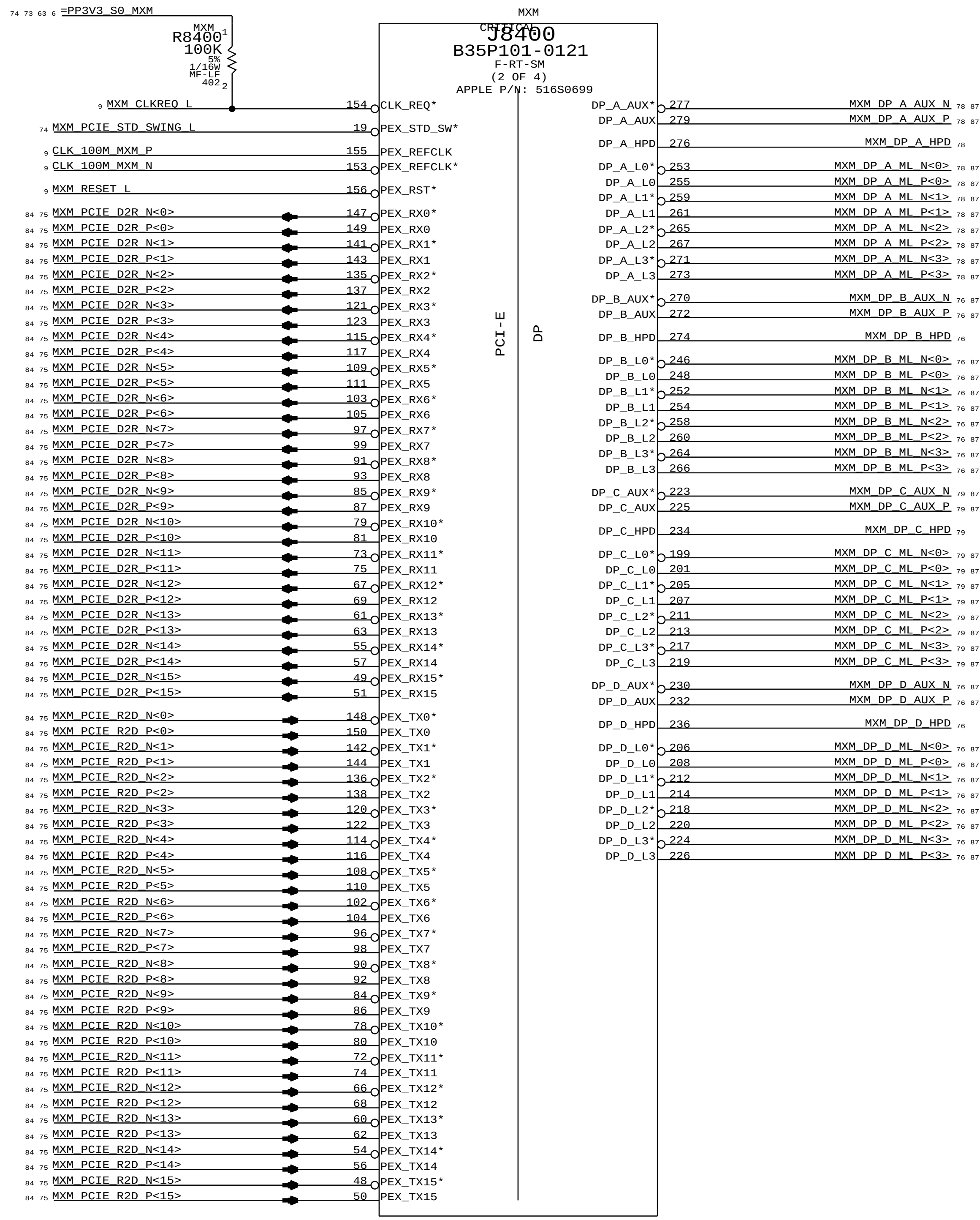
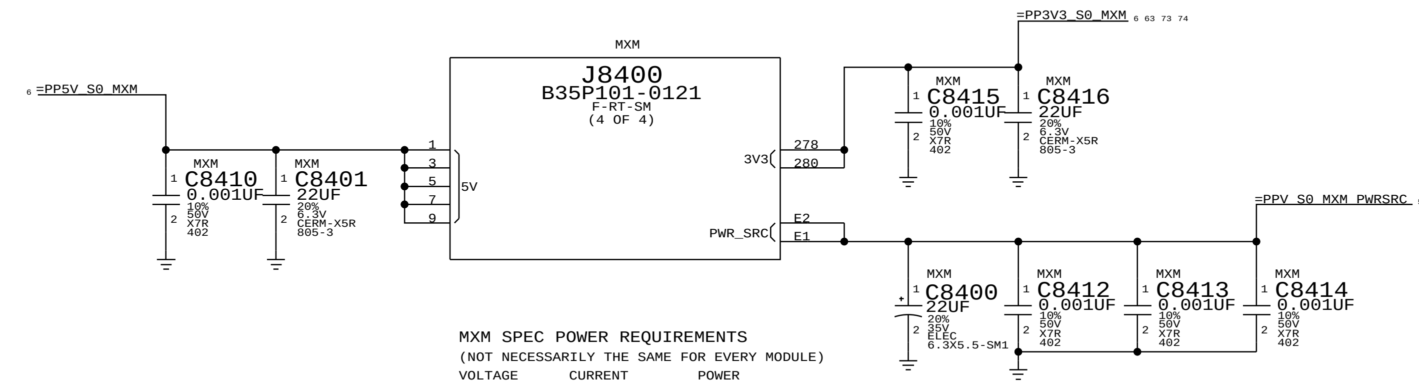
Page Notes

Power aliases required by this page:
- =PP3V3_S0_MXM
- =PP5V_S0_MXM
- =PPV_S0_MXM_PWRSRC

Signal aliases required by this page:
(NONE)

BOM options provided by this page:
- MXM





MXM SPEC POWER REQUIREMENTS
(NOT NECESSARILY THE SAME FOR EVERY MODULE)

VOLTAGE	CURRENT	POWER
3V3	1.0 A	3.3 W
5V	2.5 A	12.5 W
PWR (7-20V)	UP TO 10 A	PLATFORM DEPENDENT

SYNC_MASTER=K75F_MLB SYNC_DATE=04/14/2010

PAGE TITLE
MXM PCIe, DP & Power

Apple Inc.

DRAWING NUMBER 051-8600 SIZE D

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Page Notes

Power aliases required by this page:
- =PP3V3_S0_MXM

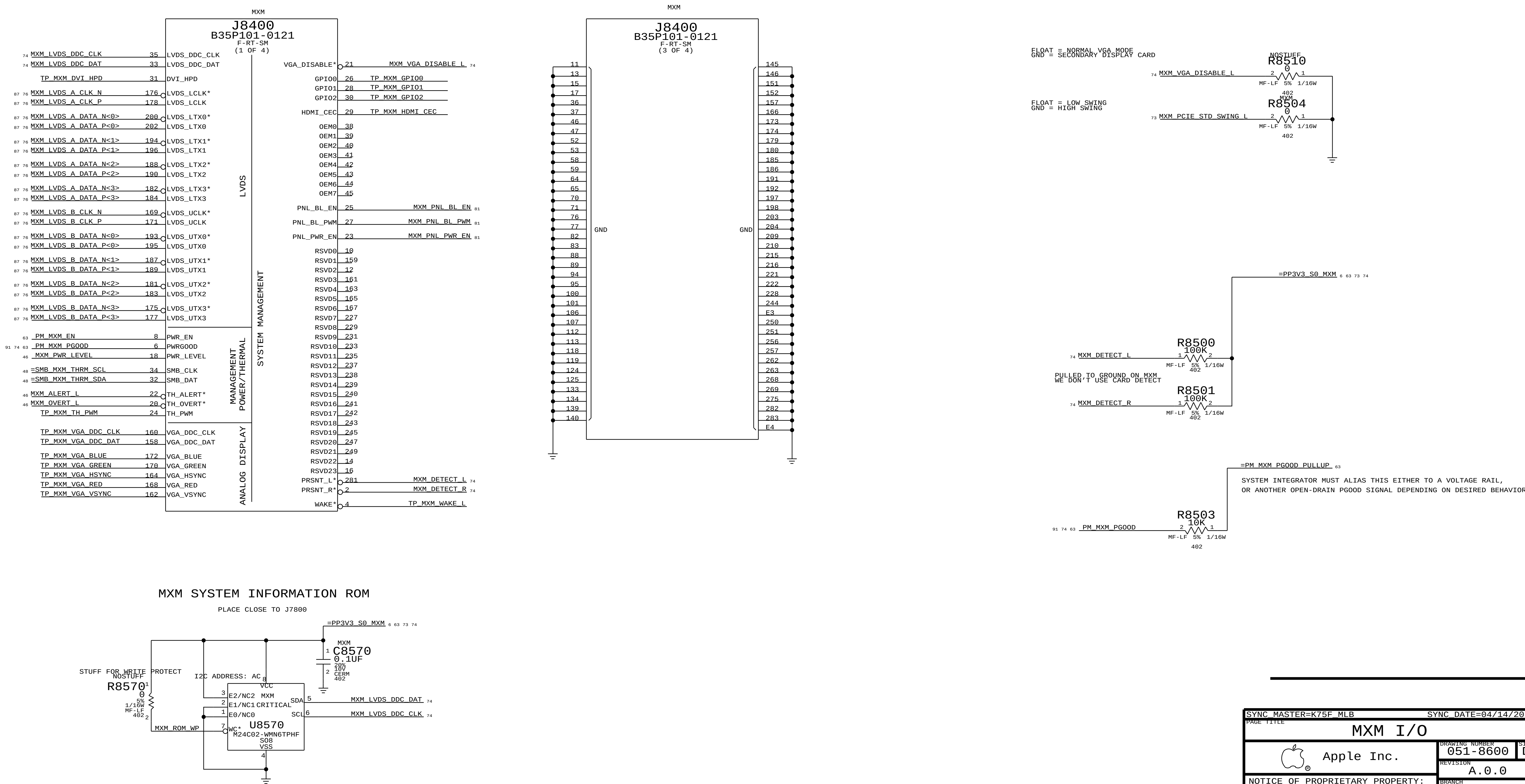
Signal aliases required by this page:
- =SMB_MXM_THRM_DATA - =PM_MXM_PGOOD_PULLUP
- =SMB_MXM_THRM_CLK

BOM options provided by this page:

PULLUPS & PULLDOWNS AT MXM CONNECTOR

MXM SYSTEM INFORMATION ROM

PLACE CLOSE TO J7800



SYNC_MASTER=K75F_MLB SYNC_DATE=04/14/2010

PAGE TITLE

MXM I/O

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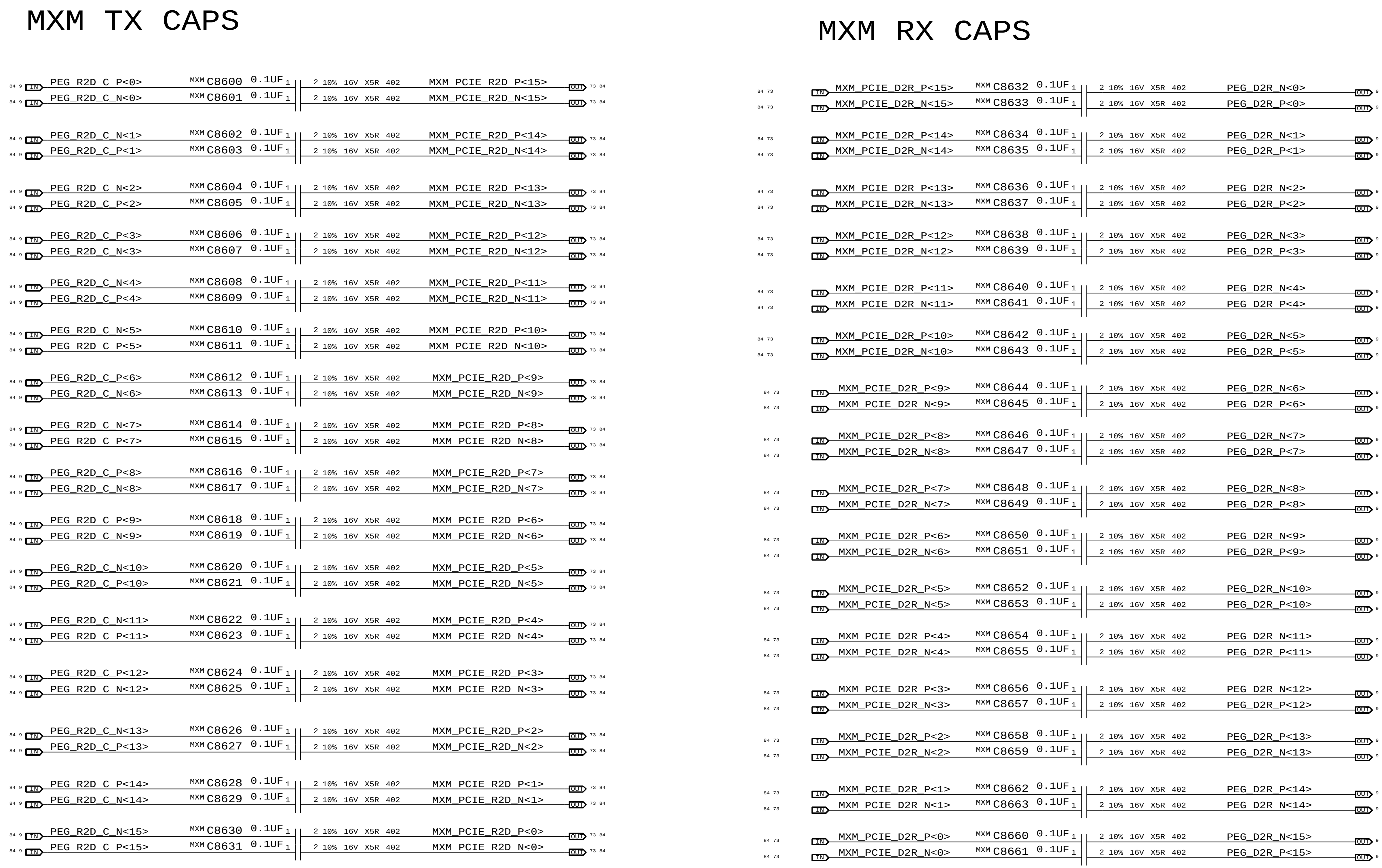
NOTICE OF PROPRIETARY PROPERTY:



SYNC_MASTER=K75F_MLB SYNC_DATE=04/14/2010

PAGE TITLE: MXM PCIE CAPS

Apple Inc.

DRAWING NUMBER: 051-8600 SIZE: D

REVISION: A.0.0

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8 7 6 5 4 3 2 1

Page Notes

Power aliases required by this page:
- =PP3V3_S0_DP

Signal aliases required by this page:
(NONE)

BOM options provided by this page:
(NONE)

Unused MXM Interfaces

87 74	MXM_LVDS_A_CLK_N	NC_MXM_LVDS_A_CLK_N	MAKE_BASE=TRUE NO_TEST=TRUE
87 74	MXM_LVDS_A_CLK_P	NC_MXM_LVDS_A_CLK_P	MAKE_BASE=TRUE NO_TEST=TRUE
87 74	MXM_LVDS_A_DATA_N<0>	NC_MXM_LVDS_A_DATA_N<0>	MAKE_BASE=TRUE NO_TEST=TRUE
87 74	MXM_LVDS_A_DATA_P<0>	NC_MXM_LVDS_A_DATA_P<0>	MAKE_BASE=TRUE NO_TEST=TRUE
87 74	MXM_LVDS_A_DATA_N<1>	NC_MXM_LVDS_A_DATA_N<1>	MAKE_BASE=TRUE NO_TEST=TRUE
87 74	MXM_LVDS_A_DATA_P<1>	NC_MXM_LVDS_A_DATA_P<1>	MAKE_BASE=TRUE NO_TEST=TRUE
87 74	MXM_LVDS_A_DATA_N<2>	NC_MXM_LVDS_A_DATA_N<2>	MAKE_BASE=TRUE NO_TEST=TRUE
87 74	MXM_LVDS_A_DATA_P<2>	NC_MXM_LVDS_A_DATA_P<2>	MAKE_BASE=TRUE NO_TEST=TRUE
87 74	MXM_LVDS_A_DATA_N<3>	NC_MXM_LVDS_A_DATA_N<3>	MAKE_BASE=TRUE NO_TEST=TRUE
87 74	MXM_LVDS_A_DATA_P<3>	NC_MXM_LVDS_A_DATA_P<3>	MAKE_BASE=TRUE NO_TEST=TRUE
87 74	MXM_LVDS_B_CLK_N	NC_MXM_LVDS_B_CLK_N	MAKE_BASE=TRUE NO_TEST=TRUE
87 74	MXM_LVDS_B_CLK_P	NC_MXM_LVDS_B_CLK_P	MAKE_BASE=TRUE NO_TEST=TRUE
87 74	MXM_LVDS_B_DATA_N<0>	NC_MXM_LVDS_B_DATA_N<0>	MAKE_BASE=TRUE NO_TEST=TRUE
87 74	MXM_LVDS_B_DATA_P<0>	NC_MXM_LVDS_B_DATA_P<0>	MAKE_BASE=TRUE NO_TEST=TRUE
87 74	MXM_LVDS_B_DATA_N<1>	NC_MXM_LVDS_B_DATA_N<1>	MAKE_BASE=TRUE NO_TEST=TRUE
87 74	MXM_LVDS_B_DATA_P<1>	NC_MXM_LVDS_B_DATA_P<1>	MAKE_BASE=TRUE NO_TEST=TRUE
87 74	MXM_LVDS_B_DATA_N<2>	NC_MXM_LVDS_B_DATA_N<2>	MAKE_BASE=TRUE NO_TEST=TRUE
87 74	MXM_LVDS_B_DATA_P<2>	NC_MXM_LVDS_B_DATA_P<2>	MAKE_BASE=TRUE NO_TEST=TRUE
87 74	MXM_LVDS_B_DATA_N<3>	NC_MXM_LVDS_B_DATA_N<3>	MAKE_BASE=TRUE NO_TEST=TRUE
87 74	MXM_LVDS_B_DATA_P<3>	NC_MXM_LVDS_B_DATA_P<3>	MAKE_BASE=TRUE NO_TEST=TRUE

Unused MXM DP Interfaces

87 73	MXM_DP_B_ML_P<0..3>	NC_MXM_DP_B_ML_P<0..3>	MAKE_BASE=TRUE NO_TEST=TRUE
87 73	MXM_DP_B_ML_N<0..3>	NC_MXM_DP_B_ML_N<0..3>	MAKE_BASE=TRUE NO_TEST=TRUE
87 73	MXM_DP_B_AUX_P	NC_MXM_DP_B_AUX_P	MAKE_BASE=TRUE NO_TEST=TRUE
87 73	MXM_DP_B_AUX_N	NC_MXM_DP_B_AUX_N	MAKE_BASE=TRUE NO_TEST=TRUE
73	MXM_DP_B_HPD	NC_MXM_DP_B_HPD	MAKE_BASE=TRUE NO_TEST=TRUE
87 73	MXM_DP_D_ML_P<0..3>	NC_MXM_DP_D_ML_P<0..3>	MAKE_BASE=TRUE NO_TEST=TRUE
87 73	MXM_DP_D_ML_N<0..3>	NC_MXM_DP_D_ML_N<0..3>	MAKE_BASE=TRUE NO_TEST=TRUE
87 73	MXM_DP_D_AUX_P	NC_MXM_DP_D_AUX_P	MAKE_BASE=TRUE NO_TEST=TRUE
87 73	MXM_DP_D_AUX_N	NC_MXM_DP_D_AUX_N	MAKE_BASE=TRUE NO_TEST=TRUE
73	MXM_DP_D_HPD	NC_MXM_DP_D_HPD	MAKE_BASE=TRUE NO_TEST=TRUE

Display: Aliases

SYNC_MASTER=K75F_MLB SYNC_DATE=04/14/2010

APPLE INC.

SIZE	DRAWING NUMBER	REV.
D	051-8600	3.0.0

SCALE	SHT	OF
NONE	76	92

8 7 6 5 4 3 2 1

Page Notes

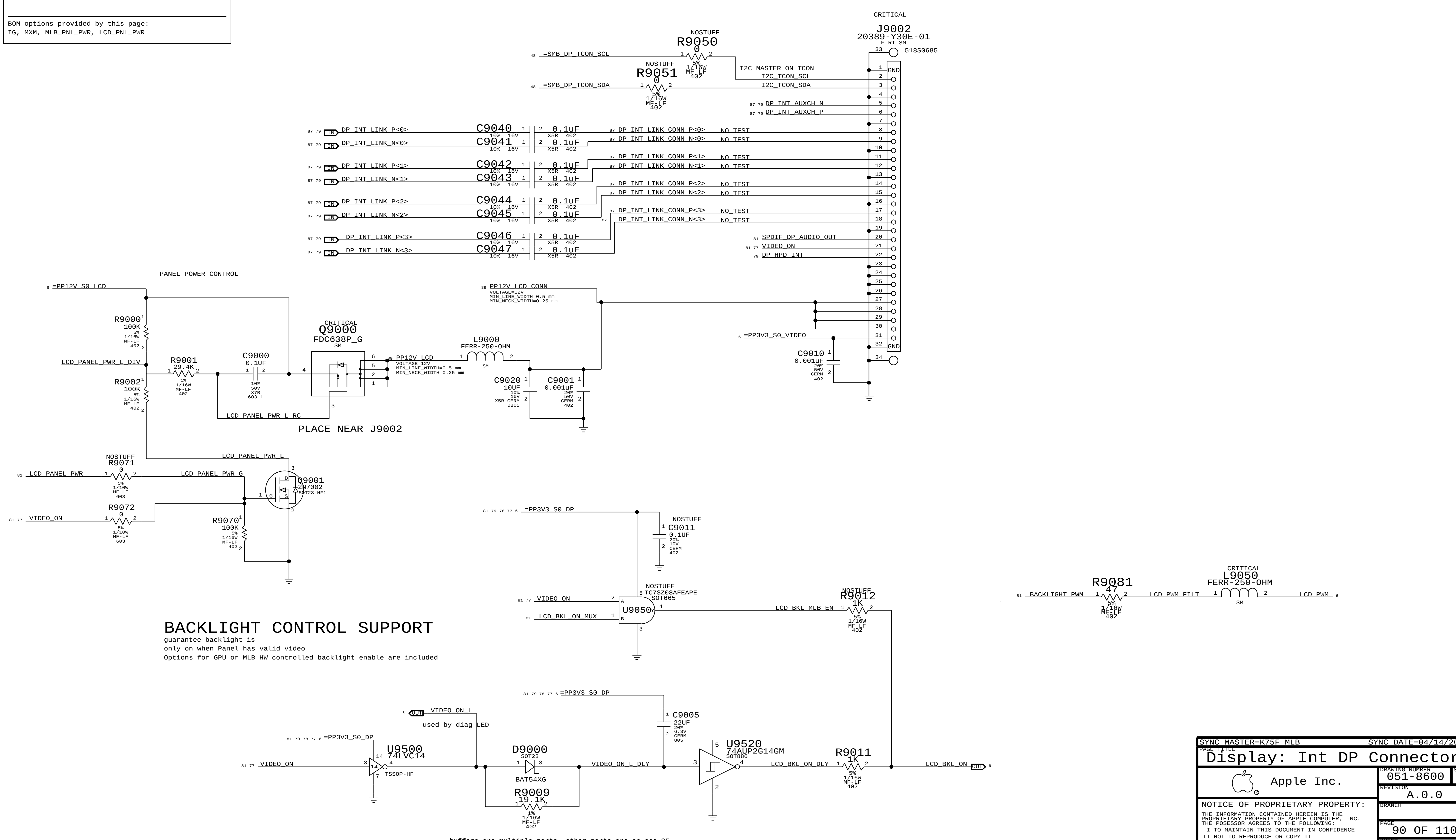
Power aliases required by this page:

- =PP12V_S0_LCD
- =PP3V3_S0_VIDEO

Signal aliases required by this page:
(NONE)

BOM options provided by this page:
IG, MXM, MLB_PNL_PWR, LCD_PNL_PWR

INTERNAL DP INTERFACE



BACKLIGHT CONTROL SUPPORT

guarantee backlight is
only on when Panel has valid video
Options for GPU or MLB HW controlled backlight enable are included

buffers are multiple parts, other parts are on csa 95

SYNC_MASTER=K75F_MLB SYNC_DATE=04/14/2010

PAGE TITLE
Display: Int DP Connector

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DRAWING NUMBER 051-8600 SIZE D

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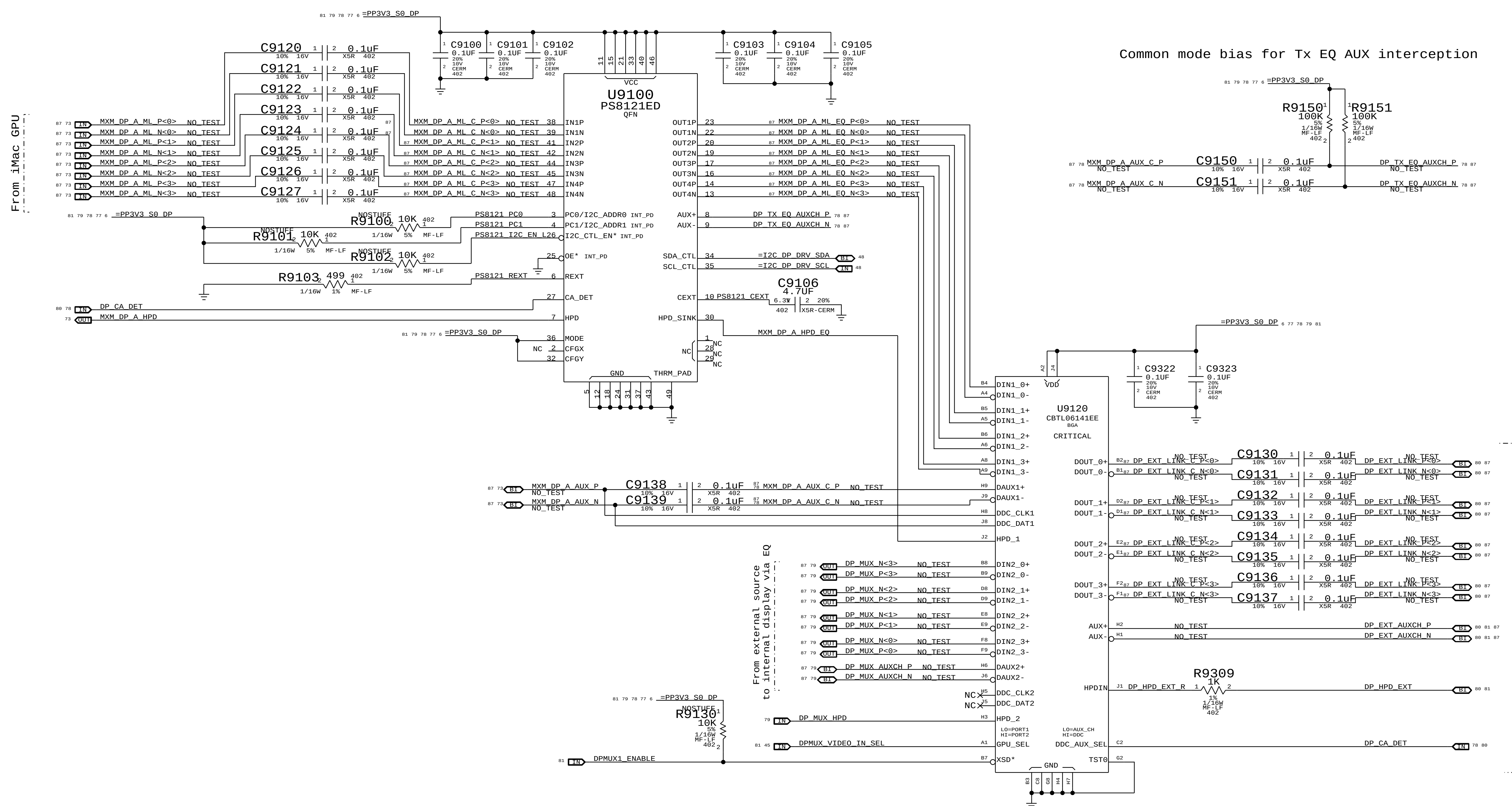
EQ & Re-Driver for DP source

Common mode bias for Tx EQ AUX interception



DisplayPort Mux 1
Analog mux at External Connector

SYNC_MASTER=K75F_MLB SYNC_DATE=04/14/2010

PAGE TITLE
Display: BiDiVi Mux1

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DRAWING NUMBER 051-8600 SIZE D

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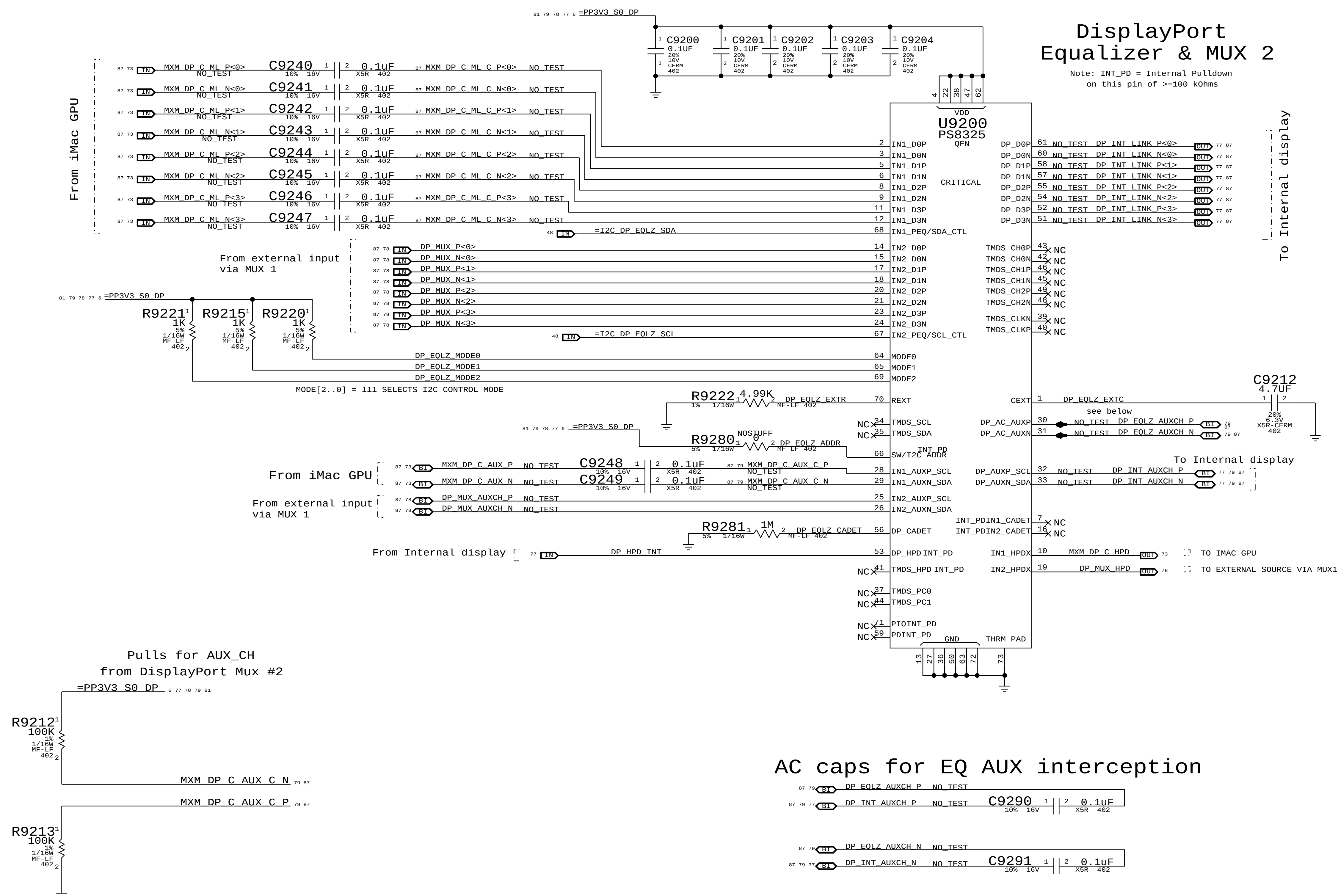

SYNC_MASTER=K75F_MLB SYNC_DATE=04/14/2010

PAGE TITLE

BIDIVI DP MUX2

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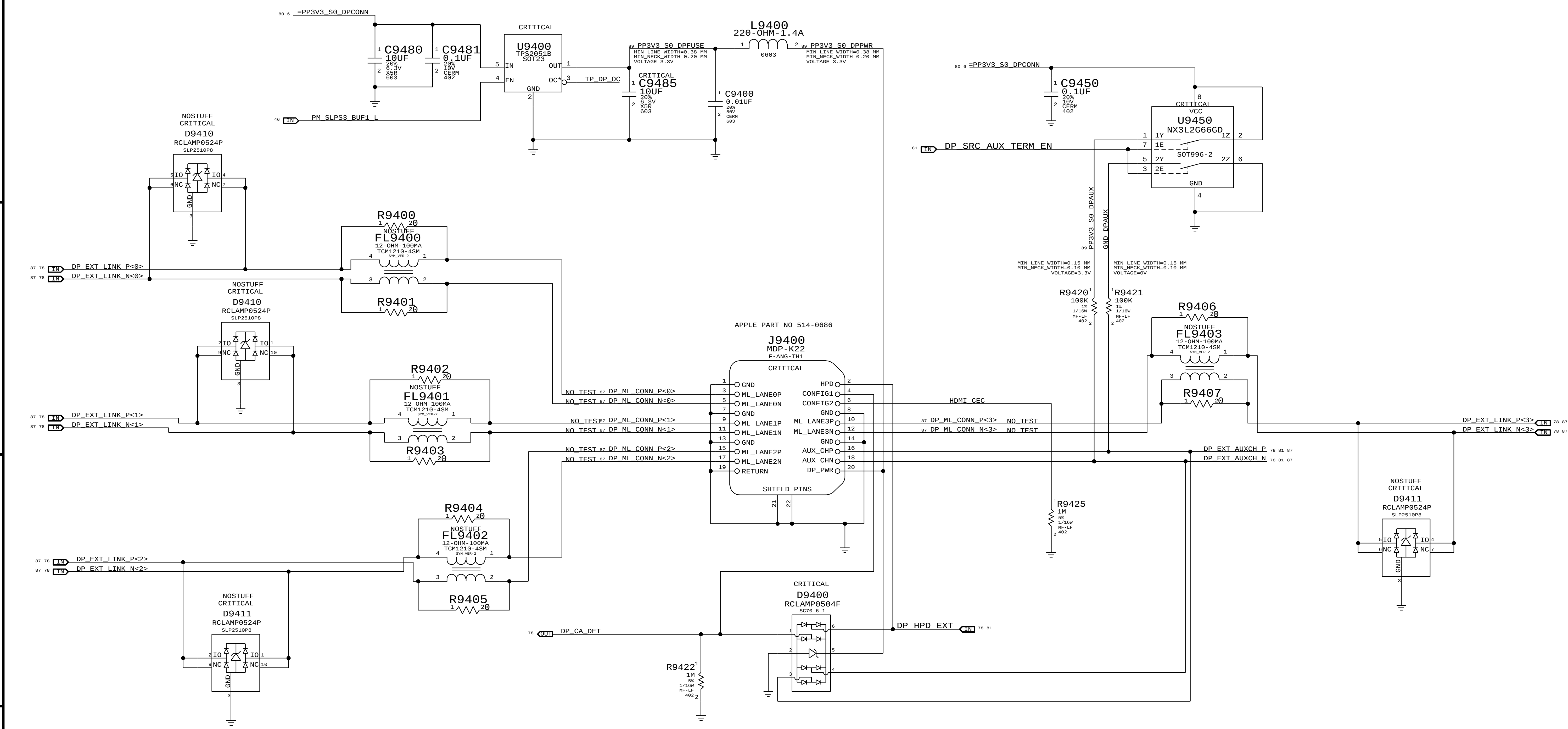

SYNC_MASTER=K75F_MLB SYNC_DATE=04/14/2010

PAGE TITLE: Display: Ext DP Connector

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DRAWING NUMBER 051-8600 SIZE D

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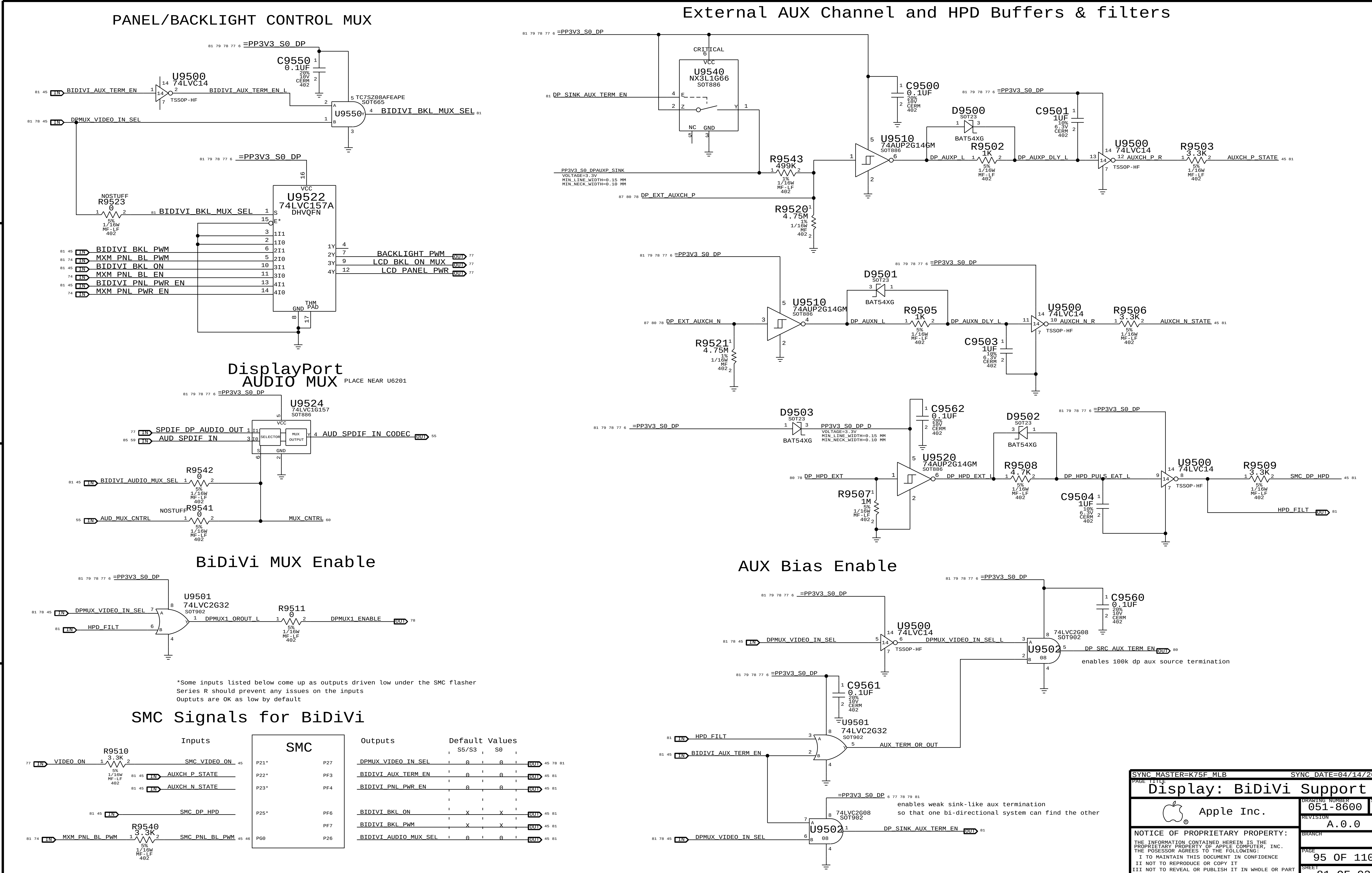

SYNC_MASTER=K75F_MLB SYNC_DATE=04/14/2010

PAGE TITLE

Display: BiDiVi Support

Apple Inc.

DRAWING NUMBER 051-8600 SIZE D

REVISION A.0.0

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K60/K61 BOARD-SPECIFIC SPACING & PHYSICAL CONSTRAINTS

BOARD LAYERS	BOARD AREAS	BOARD UNITS (MIL or MM)	ALLEGRO VERSION
TOP,ISL2,ISL3,ISL4,ISL5,ISL6,ISL7,BOTTOM	NO_TYPE,BGA_P1MM	MM	15.5.1

PHYSICAL CONSTRAINTS

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DEFAULT	*	Y	=50_OHM_SE	=50_OHM_SE	100 MM	0 MM	0 MM
STANDARD	*	Y	=DEFAULT	=DEFAULT	12.7 MM	=DEFAULT	=DEFAULT

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
35_OHM_SE	TOP,BOTTOM	Y	0.21 MM	0.085 MM	=STANDARD		
35_OHM_SE	*	Y	0.19 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
39_OHM_SE	TOP,BOTTOM	Y	0.175 MM	0.085 MM	=STANDARD		
39_OHM_SE	*	Y	0.16 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
45_OHM_SE	TOP,BOTTOM	Y	0.135 MM	0.085 MM	=STANDARD		
45_OHM_SE	*	Y	0.12 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
50_OHM_SE	TOP,BOTTOM	Y	0.1 MM	0.085 MM	15 MM		
50_OHM_SE	*	Y	0.1 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
55_OHM_SE	TOP,BOTTOM	Y	0.085 MM	0.085 MM	=STANDARD		
55_OHM_SE	*	Y	0.076 MM	0.075 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
70_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
70_OHM_DIFF	ISL3,ISL6	Y	0.155 MM	0.085 MM	=STANDARD	0.135 MM	0.1 MM
70_OHM_DIFF	TOP,BOTTOM	Y	0.165 MM	0.085 MM	=STANDARD	0.130 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
85_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
85_OHM_DIFF	ISL3,ISL6	Y	0.115 MM	0.085 MM	=STANDARD	0.2 MM	0.1 MM
85_OHM_DIFF	TOP,BOTTOM	Y	0.125 MM	0.085 MM	=STANDARD	0.2 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
90_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
90_OHM_DIFF	ISL3,ISL6	Y	0.099 MM	0.085 MM	12 MM	0.200 MM	0.1 MM
90_OHM_DIFF	TOP,BOTTOM	Y	0.110 MM	0.085 MM	=STANDARD	0.200 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
100_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
100_OHM_DIFF	ISL3,ISL6	Y	0.081 MM	0.085 MM	=STANDARD	0.25 MM	0.1 MM
100_OHM_DIFF	TOP,BOTTOM	Y	0.091 MM	0.085 MM	=STANDARD	0.25 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
110_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
110_OHM_DIFF	TOP,BOTTOM	Y	0.075 MM	0.085 MM	=STANDARD	0.320 MM	0.15 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
1:1_DIFFPAIR	*	Y	=STANDARD	=STANDARD	=STANDARD	0.1 MM	0.085 MM
POWER_WIDTH	*	Y	0.600 MM	0.200 MM	3.0 MM	=STANDARD	=STANDARD
POWER_CTL	*	Y	0.300 MM	0.200 MM	3.0 MM	=STANDARD	=STANDARD

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
POWER	BGA_P1MM	POWER_CTL
POWER	*	POWER_WIDTH
VR_CTL_PHY	BGA_P1MM	DEFAULT
VR_CTL_PHY	*	POWER_CTL

SPACING RULE SET

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DEFAULT	*	0.1 MM	?
STANDARD	*	=DEFAULT	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
1.5:1_SPACING	*	0.15 MM	?
2:1_SPACING	*	0.2 MM	?
2.5:1_SPACING	*	0.25 MM	?
3:1_SPACING	*	0.3 MM	?
4:1_SPACING	*	0.4 MM	?
5:1_SPACING	*	0.5 MM	?
6:1_SPACING	*	0.6 MM	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GND	*	=STANDARD	?
GND_P2MM	*	0.2 MM	1000
PWR_P2MM	*	0.2 MM	1000
SWITCHNODE	*	0.8 MM	1000

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
2X_DIELECTRIC	*	0.150 MM	?
2X_DIELECTRIC	TOP,BOTTOM	0.160 MM	?
3X_DIELECTRIC	*	0.220 MM	?
3X_DIELECTRIC	TOP,BOTTOM	0.240 MM	?
4X_DIELECTRIC	*	0.300 MM	?
4X_DIELECTRIC	TOP,BOTTOM	0.320 MM	?
5X_DIELECTRIC	*	0.380 MM	?
5X_DIELECTRIC	TOP,BOTTOM	0.400 MM	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GND	*	*	STANDARD

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
POWER	*	*	STANDARD

CONSTRAINTS FOR BGA AREA

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
BGA_P1MM	*	=DEFAULT	?
BGA_P2MM	*	0.2 MM	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
*	*	BGA_P1MM	BGA_P1MM
MEM_CLK	*	BGA_P1MM	BGA_P2MM
CLK_PCIE	*	BGA_P1MM	BGA_P1MM
CLK_LPC	*	BGA_P1MM	BGA_P1MM
CLK_PCI	*	BGA_P1MM	BGA_P1MM

K60/K61 RULE DEFINITIONS

SYNC_MASTER=K75F_MLB SYNC_DATE=04/14/2010

APPLE INC.

SIZE	DRAWING NUMBER	REV.
D	051-8600	3.0.0

Memory Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MEM_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD
MEM_39S	*	=39_OHM_SE	=39_OHM_SE	=39_OHM_SE	=39_OHM_SE	=STANDARD	=STANDARD
MEM_35S	*	=35_OHM_SE	=35_OHM_SE	=35_OHM_SE	=35_OHM_SE	=STANDARD	=STANDARD
MEM_70D	*	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MEM_CLK2MEM	*	=4:1_SPACING	?
MEM_CTRL2CTRL	*	=2:1_SPACING	?
MEM_CTRL2MEM	*	=2.5:1_SPACING	?
MEM_CMD2CMD	*	=1.5:1_SPACING	?
MEM_CMD2MEM	*	=3:1_SPACING	?
MEM_DQ_ODD2DQ_ODD	*	=3:1_SPACING	?
MEM_DQ_ODD2MEM	*	=3:1_SPACING	?
MEM_DQ_EVEN2DQ_EVEN	*	=3:1_SPACING	?
MEM_DQ_EVEN2MEM	*	=3:1_SPACING	?
MEM_DQ_EVEN2DQ_ODD	*	=5:1_SPACING	?
MEM_DQS2MEM	*	=3:1_SPACING	?
MEM_2OTHER	*	=3:1_SPACING	?

Memory Bus Spacing Group Assignments

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CLK	MEM_CLK	*	MEM_CLK2MEM
MEM_CLK	MEM_CTRL	*	MEM_CLK2MEM
MEM_CLK	MEM_CMD	*	MEM_CLK2MEM
MEM_CLK	MEM_DQ_ODD	*	MEM_CLK2MEM
MEM_CLK	MEM_DQS	*	MEM_CLK2MEM
MEM_CLK	MEM_DQ_EVEN	*	MEM_CLK2MEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_DQS	MEM_CLK	*	MEM_DQS2MEM
MEM_DQS	MEM_CTRL	*	MEM_DQS2MEM
MEM_DQS	MEM_CMD	*	MEM_DQS2MEM
MEM_DQS	MEM_DQ_ODD	*	MEM_DQS2MEM
MEM_DQS	MEM_DQS	*	MEM_DQS2MEM
MEM_DQS	MEM_DQ_EVEN	*	MEM_DQS2MEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_DQ_ODD	MEM_CLK	*	MEM_DQ_ODD2MEM
MEM_DQ_ODD	MEM_CTRL	*	MEM_DQ_ODD2MEM
MEM_DQ_ODD	MEM_CMD	*	MEM_DQ_ODD2MEM
MEM_DQ_ODD	MEM_DQ_ODD	*	MEM_DQ_ODD2DQ_ODD
MEM_DQ_ODD	MEM_DQS	*	MEM_DQ_ODD2MEM
MEM_DQ_ODD	MEM_DQ_EVEN	*	MEM_DQ_EVEN2DQ_ODD

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_DQ_EVEN	MEM_CLK	*	MEM_DQ_EVEN2MEM
MEM_DQ_EVEN	MEM_CTRL	*	MEM_DQ_EVEN2MEM
MEM_DQ_EVEN	MEM_CMD	*	MEM_DQ_EVEN2MEM
MEM_DQ_EVEN	MEM_DQ_EVEN	*	MEM_DQ_EVEN2DQ_EVEN
MEM_DQ_EVEN	MEM_DQS	*	MEM_DQ_EVEN2MEM
MEM_DQ_EVEN	MEM_DQ_ODD	*	MEM_DQ_EVEN2DQ_ODD

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CTRL	MEM_CLK	*	MEM_CTRL2MEM
MEM_CTRL	MEM_CTRL	*	MEM_CTRL2CTRL
MEM_CTRL	MEM_CMD	*	MEM_CTRL2MEM
MEM_CTRL	MEM_DQ_ODD	*	MEM_CTRL2MEM
MEM_CTRL	MEM_DQS	*	MEM_CTRL2MEM
MEM_CTRL	MEM_DQ_EVEN	*	MEM_CTRL2MEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CMD	MEM_CLK	*	MEM_CMD2MEM
MEM_CMD	MEM_CTRL	*	MEM_CMD2MEM
MEM_CMD	MEM_CMD	*	MEM_CMD2CMD
MEM_CMD	MEM_DQ_ODD	*	MEM_CMD2MEM
MEM_CMD	MEM_DQS	*	MEM_CMD2MEM
MEM_CMD	MEM_DQ_EVEN	*	MEM_CMD2MEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CLK	*	*	MEM_2OTHER
MEM_CTRL	*	*	MEM_2OTHER
MEM_CMD	*	*	MEM_2OTHER
MEM_DQ_ODD	*	*	MEM_2OTHER
MEM_DQS	*	*	MEM_2OTHER
MEM_DQ_EVEN	*	*	MEM_2OTHER

Need to support MEM_*-style wildcards!

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MEM_POWER_WIDTH	*	Y	0.500 MM	0.175 MM	=STANDARD	=STANDARD	=STANDARD

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
MEM_POWER_PHY	*	MEM_POWER_WIDTH

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MEM_POWER	*	0.2 MM	?

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MEM_RCOMP_PHY	*	Y	0.175 MM	0.175 MM	=STANDARD	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MEM_RCOMP	*	0.2 MM	?

Memory Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
	MEM_70D	MEM_CLK	MEM_A_CLK_P<3..0>	12 32
	MEM_70D	MEM_CLK	MEM_A_CLK_N<3..0>	12 32
	MEM_39S	MEM_CTRL	MEM_A_CKE<3..0>	12 30
	MEM_39S	MEM_CTRL	MEM_A_CS_L<3..0>	12 30
	MEM_39S	MEM_CTRL	MEM_A_ODT<3..0>	12 30
	MEM_35S	MEM_CMD	MEM_A_A<15..0>	12 30
	MEM_35S	MEM_CMD	MEM_A_BA<2..0>	12 30
	MEM_35S	MEM_CMD	MEM_A_RAS_L	12 30
	MEM_35S	MEM_CMD	MEM_A_CAS_L	12 30
	MEM_35S	MEM_CMD	MEM_A_WE_L	12 30
	MEM_45S	MEM_DQ_EVEN	MEM_A_DQ<7..0>	12 32
	MEM_45S	MEM_DQ_EVEN	MEM_A_DM<0>	12 32
	MEM_45S	MEM_DQ_ODD	MEM_A_DQ<15..8>	12 32
	MEM_45S	MEM_DQ_ODD	MEM_A_DM<1>	12 32
	MEM_45S	MEM_DQ_EVEN	MEM_A_DQ<23..16>	12 32
	MEM_45S	MEM_DQ_EVEN	MEM_A_DM<2>	12 32
	MEM_45S	MEM_DQ_ODD	MEM_A_DQ<31..24>	12 32
	MEM_45S	MEM_DQ_ODD	MEM_A_DM<3>	12 32
	MEM_45S	MEM_DQ_EVEN	MEM_A_DQ<39..32>	12 32
	MEM_45S	MEM_DQ_EVEN	MEM_A_DM<4>	12 32
	MEM_45S	MEM_DQ_ODD	MEM_A_DQ<47..40>	12 32
	MEM_45S	MEM_DQ_ODD	MEM_A_DM<5>	12 32
	MEM_45S	MEM_DQ_EVEN	MEM_A_DQ<55..48>	12 32
	MEM_45S	MEM_DQ_EVEN	MEM_A_DM<6>	12 32
	MEM_45S	MEM_DQ_ODD	MEM_A_DQ<63..56>	12 32
	MEM_45S	MEM_DQ_ODD	MEM_A_DM<7>	12 32
	MEM_70D	MEM_DQS	MEM_A_DQS_P<0>	12 32
	MEM_70D	MEM_DQS	MEM_A_DQS_N<0>	12 32
	MEM_70D	MEM_DQS	MEM_A_DQS_P<1>	12 32
	MEM_70D	MEM_DQS	MEM_A_DQS_N<1>	12 32
	MEM_70D	MEM_DQS	MEM_A_DQS_P<2>	12 32
	MEM_70D	MEM_DQS	MEM_A_DQS_N<2>	12 32
	MEM_70D	MEM_DQS	MEM_A_DQS_P<3>	12 32
	MEM_70D	MEM_DQS	MEM_A_DQS_N<3>	12 32
	MEM_70D	MEM_DQS	MEM_A_DQS_P<4>	12 32
	MEM_70D	MEM_DQS	MEM_A_DQS_N<4>	12 32
	MEM_70D	MEM_DQS	MEM_A_DQS_P<5>	12 32
	MEM_70D	MEM_DQS	MEM_A_DQS_N<5>	12 32
	MEM_70D	MEM_DQS	MEM_A_DQS_P<6>	12 32
	MEM_70D	MEM_DQS	MEM_A_DQS_N<6>	12 32
	MEM_70D	MEM_DQS	MEM_A_DQS_P<7>	12 32
	MEM_70D	MEM_DQS	MEM_A_DQS_N<7>	12 32
	MEM_70D	MEM_CLK	MEM_B_CLK_P<3..0>	12 32
	MEM_70D	MEM_CLK	MEM_B_CLK_N<3..0>	12 32
	MEM_39S	MEM_CTRL	MEM_B_CKE<3..0>	12 31
	MEM_39S	MEM_CTRL	MEM_B_CS_L<3..0>	12 31
	MEM_39S	MEM_CTRL	MEM_B_ODT<3..0>	12 31
	MEM_35S	MEM_CMD	MEM_B_A<15..0>	12 31
	MEM_35S	MEM_CMD	MEM_B_BA<2..0>	12 31
	MEM_35S	MEM_CMD	MEM_B_RAS_L	12 31
	MEM_35S	MEM_CMD	MEM_B_CAS_L	12 31
	MEM_35S	MEM_CMD	MEM_B_WE_L	12 31
	MEM_45S	MEM_DQ_EVEN	MEM_B_DQ<7..0>	12 32
	MEM_45S	MEM_DQ_EVEN	MEM_B_DM<0>	12 32
	MEM_45S	MEM_DQ_ODD	MEM_B_DQ<15..8>	12 32
	MEM_45S	MEM_DQ_ODD	MEM_B_DM<1>	12 32
	MEM_45S	MEM_DQ_EVEN	MEM_B_DQ<23..16>	12 32
	MEM_45S	MEM_DQ_EVEN	MEM_B_DM<2>	12 32
	MEM_45S	MEM_DQ_ODD	MEM_B_DQ<31..24>	12 32
	MEM_45S	MEM_DQ_ODD	MEM_B_DM<3>	12 32
	MEM_45S	MEM_DQ_EVEN	MEM_B_DQ<39..32>	12 32
	MEM_45S	MEM_DQ_EVEN	MEM_B_DM<4>	12 32
	MEM_45S	MEM_DQ_ODD	MEM_B_DQ<47..40>	12 32
	MEM_45S	MEM_DQ_ODD	MEM_B_DM<5>	12 32
	MEM_45S	MEM_DQ_EVEN	MEM_B_DQ<55..48>	12 32
	MEM_45S	MEM_DQ_EVEN	MEM_B_DM<6>	12 32
	MEM_45S	MEM_DQ_ODD	MEM_B_DQ<63..56>	12 32
	MEM_45S	MEM_DQ_ODD	MEM_B_DM<7>	12 32

MEMORY POWER PROPERTIES

VOLTAGE	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
I155	MEM_POWER_PHY	MEM_POWER	CPU_DIMM_VREF_A	12 28
I154	MEM_POWER_PHY	MEM_POWER	CPU_DIMM_VREF_B	12 28
I166	MEM_POWER_PHY	MEM_POWER	VREFMARGIN_DIMMA_DQ	28
I167	MEM_POWER_PHY	MEM_POWER	VREFMARGIN_DIMMB_DQ	28
I169	MEM_POWER_PHY	MEM_POWER	CPU_DIMM_VREF_A_SW	28
I168	MEM_POWER_PHY	MEM_POWER	CPU_DIMM_VREF_B_SW	28

Memory Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
	MEM_70D	MEM_DQS	MEM_B_DQS_P<0>	12 32
	MEM_70D	MEM_DQS	MEM_B_DQS_N<0>	12 32
	MEM_70D	MEM_DQS	MEM_B_DQS_P<1>	12 32
	MEM_70D	MEM_DQS	MEM_B_DQS_N<1>	12 32
	MEM_70D	MEM_DQS	MEM_B_DQS_P<2>	12 32
	MEM_70D	MEM_DQS	MEM_B_DQS_N<2>	12 32
	MEM_70D	MEM_DQS	MEM_B_DQS_P<3>	12 32
	MEM_70D	MEM_DQS	MEM_B_DQS_N<3>	12 32
	MEM_70D	MEM_DQS	MEM_B_DQS_P<4>	12 32
	MEM_70D	MEM_DQS	MEM_B_DQS_N<4>	12 32
	MEM_70D	MEM_DQS	MEM_B_DQS_P<5>	12 32
	MEM_70D	MEM_DQS	MEM_B_DQS_N<5>	12 32
	MEM_70D	MEM_DQS	MEM_B_DQS_P<6>	12 32
	MEM_70D	MEM_DQS	MEM_B_DQS_N<6>	12 32
	MEM_70D	MEM_DQS	MEM_B_DQS_P<7>	12 32
	MEM_70D	MEM_DQS	MEM_B_DQS_N<7>	12 32
I160	MEM_RCOMP_PHY	MEM_RCOMP	CPU_SM_RCOMP0	11
I159	MEM_RCOMP_PHY	MEM_RCOMP	CPU_SM_RCOMP1	11
I161	MEM_RCOMP_PHY	MEM_RCOMP	CPU_SM_RCOMP2	11
I162	MEM_70D	MEM_DQS	TP_MEM_B_DQS_P<8>	8 12
I163	MEM_70D	MEM_DQS	TP_MEM_B_DQS_N<8>	8 12
I164	MEM_70D	MEM_DQS	TP_MEM_A_DQS_P<8>	8 12
I165	MEM_70D	MEM_DQS	TP_MEM_A_DQS_N<8>	8 12

ADD RULES TO NC_DQS<8>
TO CLEAR CHECK_PLUS ERRORS

SYNC_MASTER=K75F_MLB SYNC_DATE=04/14/2010

PAGE TITLE
Memory Constraints

Apple Inc.

DRAWING NUMBER 051-8600 SIZE D

REVISION A.0.0

BRANCH

PCI-Express

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCIE_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF
CLK_PCIE_100D	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCIE	*	=4X_DIELECTRIC	?
CLK_PCIE	*	0.5 MM	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCIE	TOP,BOTTOM	=4X_DIELECTRIC	?

CPU

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CPU_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CPU_AGTL	*	=STANDARD	?
CPU_ITP	*	0.2 MM	?
CPU_RCOMP	*	0.2 MM	?

SATA Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SATA_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SATA	*	=5X_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SATA	TOP,BOTTOM	=5X_DIELECTRIC	?

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
FDI_MISC				
	CPU_50S	CPU_AGTL	FDI_FSYNC<1..0>	
	CPU_50S	CPU_AGTL	FDI_LSYNC<1..0>	
	CPU_50S	CPU_AGTL	FDI_INT	
SATA SSD				
	SATA_85D	SATA	SATA_SSD_R2D_C_P	18 42
	SATA_85D	SATA	SATA_SSD_R2D_C_N	18 42
	SATA_85D	SATA	SATA_SSD_R2D_P	42
	SATA_85D	SATA	SATA_SSD_R2D_N	42
	SATA_85D	SATA	SATA_SSD_D2R_P	18 42
	SATA_85D	SATA	SATA_SSD_D2R_N	18 42
	SATA_85D	SATA	SATA_SSD_D2R_C_P	42
	SATA_85D	SATA	SATA_SSD_D2R_C_N	42
CLOCKS				
	CLK_PCIE_100D	CLK_PCIE	DMI_MIDBUS_CLK100M_P	10 18
	CLK_PCIE_100D	CLK_PCIE	DMI_MIDBUS_CLK100M_N	10 18
CPU ITP				
	CPU_50S	CPU_ITP	XDP_BPM_L<7..0>	11 25
	CPU_50S	CPU_ITP	CPU_CFG<17..0>	10 15 25
	CPU_50S	CPU_ITP	XDP_OBSDATA_A<3..0>	25
CPU_MISC				
	CPU_50S	CPU_RCOMP	CPU_PEG_COMP	10
	CPU_50S	CPU_RCOMP	CPU_PEG_RBIAS	10
	CPU_50S	CPU_RCOMP	CPU_COMP3	11
	CPU_50S	CPU_RCOMP	CPU_COMP2	11
	CPU_50S	CPU_RCOMP	CPU_COMP1	11
	CPU_50S	CPU_RCOMP	CPU_COMP0	11

ANY OTHER LYNNFIELD CONSTRAINTS NOT COVERED ON PAGES 101 AND 107 SHOULD GO ON THIS PAGE TOO

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
PCIE GRAPHICS				
	PCIE_85D	PCIE	PEG_R2D_C_P<15..0>	9 75
	PCIE_85D	PCIE	PEG_R2D_C_N<15..0>	9 75
	PCIE_85D	PCIE	PEG_D2R_P<15..0>	9 75
	PCIE_85D	PCIE	PEG_D2R_N<15..0>	9 75
	PCIE_85D	PCIE	MXM_PCIE_R2D_P<15..0>	73 75
	PCIE_85D	PCIE	MXM_PCIE_R2D_N<15..0>	73 75
	PCIE_85D	PCIE	MXM_PCIE_D2R_P<15..0>	73 75
	PCIE_85D	PCIE	MXM_PCIE_D2R_N<15..0>	73 75
PCIE I/O				
	PCIE_85D	PCIE	PCIE_MINI_R2D_P	33
	PCIE_85D	PCIE	PCIE_MINI_R2D_N	33
	PCIE_85D	PCIE	PCIE_MINI_R2D_C_P	18 33
	PCIE_85D	PCIE	PCIE_MINI_R2D_C_N	18 33
	PCIE_85D	PCIE	PCIE_MINI_D2R_P	18 33
	PCIE_85D	PCIE	PCIE_MINI_D2R_N	18 33
	PCIE_85D	PCIE	PCIE_MINI_R2D_L_P	33
	PCIE_85D	PCIE	PCIE_MINI_R2D_L_N	33
	PCIE_85D	PCIE	PCIE_FW_R2D_P	39
	PCIE_85D	PCIE	PCIE_FW_R2D_N	39
	PCIE_85D	PCIE	PCIE_FW_R2D_C_P	18 39
	PCIE_85D	PCIE	PCIE_FW_R2D_C_N	18 39
	PCIE_85D	PCIE	PCIE_FW_D2R_P	18 39
	PCIE_85D	PCIE	PCIE_FW_D2R_N	18 39
	PCIE_85D	PCIE	PCIE_FW_D2R_C_P	39
	PCIE_85D	PCIE	PCIE_FW_D2R_C_N	39
DMI				
	PCIE_85D	PCIE	DMI_S2N_P<3..0>	10 19
	PCIE_85D	PCIE	DMI_S2N_N<3..0>	10 19
	PCIE_85D	PCIE	DMI_N2S_P<3..0>	10 19
	PCIE_85D	PCIE	DMI_N2S_N<3..0>	10 19
FDI				
	PCIE_85D	PCIE	FDI_DATA_N<7..0>	
	PCIE_85D	PCIE	FDI_DATA_P<15..0>	
PCIE REF CLOCKS				
	CLK_PCIE_100D	CLK_PCIE	GPU_CLK100M_PCIE_P	9
	CLK_PCIE_100D	CLK_PCIE	GPU_CLK100M_PCIE_N	9
	CLK_PCIE_100D	CLK_PCIE	PCIE_CLK100M_MINI_CON_P	33
	CLK_PCIE_100D	CLK_PCIE	PCIE_CLK100M_MINI_CON_N	33
	CLK_PCIE_100D	CLK_PCIE	PCIE_CLK100M_MINI_P	18 33
	CLK_PCIE_100D	CLK_PCIE	PCIE_CLK100M_MINI_N	18 33
	CLK_PCIE_100D	CLK_PCIE	PCIE_CLK100M_FW_P	18 39
	CLK_PCIE_100D	CLK_PCIE	PCIE_CLK100M_FW_N	18 39
	ENET_100D	ENET_MII	PCIE_CLK100M_ENET_P	18 36
	ENET_100D	ENET_MII	PCIE_CLK100M_ENET_N	18 36
SATA				
	SATA_85D	SATA	SATA_HDD_R2D_C_P	18 42
	SATA_85D	SATA	SATA_HDD_R2D_C_N	18 42
	SATA_85D	SATA	SATA_HDD_R2D_P	42 92
	SATA_85D	SATA	SATA_HDD_R2D_N	42 92
	SATA_85D	SATA	SATA_HDD_D2R_P	18 42
	SATA_85D	SATA	SATA_HDD_D2R_N	18 42
	SATA_85D	SATA	SATA_HDD_D2R_C_P	42 92
	SATA_85D	SATA	SATA_HDD_D2R_C_N	42 92
	SATA_85D	SATA	SATA_ODD_R2D_C_P	18 42
	SATA_85D	SATA	SATA_ODD_R2D_C_N	18 42
	SATA_85D	SATA	SATA_ODD_R2D_P	42 92
	SATA_85D	SATA	SATA_ODD_R2D_N	42 92
	SATA_85D	SATA	SATA_ODD_D2R_P	18 42
	SATA_85D	SATA	SATA_ODD_D2R_N	18 42
	SATA_85D	SATA	SATA_ODD_D2R_C_P	42 92
	SATA_85D	SATA	SATA_ODD_D2R_C_N	42 92
CLOCKS				
	CLK_PCIE_100D	CLK_PCIE	FSB_CLK133M_CPU_P	11 21
	CLK_PCIE_100D	CLK_PCIE	FSB_CLK133M_CPU_N	11 21
	CLK_PCIE_100D	CLK_PCIE	GFX_CLK120M_DPLLSS_P	11 18
	CLK_PCIE_100D	CLK_PCIE	GFX_CLK120M_DPLLSS_N	11 18
	CLK_PCIE_100D	CLK_PCIE	FSB_CLK133M_ITP_P	11 25
	CLK_PCIE_100D	CLK_PCIE	FSB_CLK133M_ITP_N	11 25
	CLK_PCIE_100D	CLK_PCIE	PCIE_CLK100M_CPU_P	11 18
	CLK_PCIE_100D	CLK_PCIE	PCIE_CLK100M_CPU_N	11 18
	CLK_PCIE_100D	CLK_PCIE	PCIE_CLK100M_PCH_P	18 26
	CLK_PCIE_100D	CLK_PCIE	PCIE_CLK100M_PCH_N	18 26
	CLK_PCIE_100D	CLK_PCIE	FSB_CLK133M_PCH_P	18 26
	CLK_PCIE_100D	CLK_PCIE	FSB_CLK133M_PCH_N	18 26
	CLK_PCIE_100D	CLK_PCIE	PCH_CLK96M_DOT_P	18 26
	CLK_PCIE_100D	CLK_PCIE	PCH_CLK96M_DOT_N	
	CLK_PCIE_100D	CLK_PCIE	PCH_CLK100M_SATA_P	18 26
	CLK_PCIE_100D	CLK_PCIE	PCH_CLK100M_SATA_N	18 26

SYNC_MASTER=K75F_MLB SYNC_DATE=04/14/2010

PAGE TITLE
PCIE/DMI/FDI/SATA CONSTRAINTS

Apple Inc.

DRAWING NUMBER 051-8600 SIZE D

REVISION A.0.0

BRANCH

PCH CONSTRAINTS

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCH_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
CLK_PCH_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_PCH	*	0.2 MM	?
COMP_PCH	*	0.2 MM	?

PCI Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
CLK_PCI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCI	*	=STANDARD	?
CLK_PCI	*	0.2 MM	?

LPC Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
LPC_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
CLK_LPC_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
LPC	*	0.15 MM	?
CLK_LPC	*	0.2 MM	?

USB 2.0 Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
USB_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
USB	*	=2x_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
USB	TOP,BOTTOM	=4x_DIELECTRIC	?

SMBus Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SMB_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SMB	*	=2x_DIELECTRIC	?

HD Audio Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
HDA_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
HDA	*	=2x_DIELECTRIC	?

SPI Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SPI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SPI	*	0.2 MM	?

XTAL Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CLK_XTAL	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
XTAL	*	=4X_DIELECTRIC	?

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
	PCI_55S	PCI	PCI_REQ0_L	20
	PCI_55S	PCI	PCI_REQ1_L	20
	CLK_PCI_55S	CLK_PCI	PCH_CLK33M_PCIOUT	20 27
	CLK_PCI_55S	CLK_PCI	PCH_CLK33M_PCIIN	18 27
	LPC_55S	LPC	LPC_AD<3..0>	18 45 47
	LPC_55S	LPC	LPC_FRAME_L	18 45 47
	CLK_LPC_55S	CLK_LPC	LPC_CLK33M_SMC_R	20 27
	CLK_LPC_55S	CLK_LPC	LPC_CLK33M_SMC	27 45
	CLK_LPC_55S	CLK_LPC	LPC_CLK33M_LPCPLUS	27 47
	CLK_LPC_55S	PM	PM_CLK32K_SUSCLK_R	9 19 91
	CLK_LPC_55S	PM	PM_CLK32K_SUSCLK	9 45 91
	CLK_LPC_55S	CLK_LPC	LPC_CLK33M_LPCPLUS_R	20 27
	USB_90D	USB	USB_EXTA_P	34 43
	USB_90D	USB	USB_EXTA_N	34 43
	USB_90D	USB	USB_PORT0_P	43
	USB_90D	USB	USB_PORT0_N	43
	USB_90D	USB	USB_EXTB_P	35 43
	USB_90D	USB	USB_EXTB_N	35 43
	USB_90D	USB	USB_PORT1_P	43
	USB_90D	USB	USB_PORT1_N	43
	USB_90D	USB	USB_EXTC_P	34 43
	USB_90D	USB	USB_EXTC_N	34 43
	USB_90D	USB	USB_PORT2_P	43
	USB_90D	USB	USB_PORT2_N	43
	USB_90D	USB	USB_EXTD_P	35 43
	USB_90D	USB	USB_EXTD_N	35 43
	USB_90D	USB	USB_D_MUXED_P	43
	USB_90D	USB	USB_D_MUXED_N	43
	USB_90D	USB	USB_PORT3_P	43
	USB_90D	USB	USB_PORT3_N	43
	USB_90D	USB	USB_CAMERA_P	34 44
	USB_90D	USB	USB_CAMERA_N	34 44
	USB_90D	USB	USB_CAMERA_L_P	44 92
	USB_90D	USB	USB_CAMERA_L_N	44 92
	USB_90D	USB	USB_BT_P	35 44
	USB_90D	USB	USB_BT_N	35 44
	USB_90D	USB	USB_BT_L_P	44 92
	USB_90D	USB	USB_BT_L_N	44 92
	USB_90D	USB	USB_IR_P	34 44
	USB_90D	USB	USB_IR_N	34 44
	USB_90D	USB	USB_IR_L_P	44 92
	USB_90D	USB	USB_IR_L_N	44 92
	USB_90D	USB	USB_SDCARD_P	35 44
	USB_90D	USB	USB_SDCARD_N	35 44
	USB_90D	USB	USB_SDCARD_L_P	44 92
	USB_90D	USB	USB_SDCARD_L_N	44 92
	USB_90D	USB	USB_WM_P	20 44
	USB_90D	USB	USB_WM_N	20 44
	USB_90D	USB	USB_WM_L_P	44
	USB_90D	USB	USB_WM_L_N	44
	USB_90D	USB	USB_MINI_P	
	USB_90D	USB	USB_MINI_N	
	USB_90D	USB	USB_BRCRYPT_P	20 44
	USB_90D	USB	USB_BRCRYPT_N	20 44
	CLK_XTAL	XTAL	PCH_CLK32K_RTCX1	18 27
	CLK_XTAL	XTAL	PCH_CLK32K_RTCX2	18 27
	CLK_XTAL	XTAL	CK505_XTAL_IN	26
	CLK_XTAL	XTAL	CK505_XTAL_OUT	26
	CLK_PCH_55S	CLK_PCH	PCH_CLK14P3M_REFCLK	18 26
	USB_90D	USB	USB_BRCRYPT_L_P	44
	USB_90D	USB	USB_BRCRYPT_L_N	44
	USB_90D	USB	USB_HUB1_UP_P	20 34
	USB_90D	USB	USB_HUB1_UP_N	20 34
	USB_90D	USB	USB_HUB2_UP_P	20 35
	USB_90D	USB	USB_HUB2_UP_N	20 35

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
	SPI_55S	SPI	SPI_CLK_R	18 47 54
	SPI_55S	SPI	SPI_CLK	54
	SPI_55S	SPI	SPI_MOSI_R	18 47 54
	SPI_55S	SPI	SPI_MOSI	54
	SPI_55S	SPI	SPI_MISO	18 47 54
	SPI_55S	SPI	SPI_MISO_R	54
	SPI_55S	SPI	SPI_CS0_R_L	18 47
	SPI_55S	SPI	SPI_CS0_L	47
	SPI_55S	SPI	SPI_MLB_CS_L	47 54
	SPI_55S	SPI	SPI_ALT_CS_L	47
	SPI_55S	SPI	SPIROM_USE_MLB	21 47
	SPI_55S	SPI	SPI_ALT_MOSI	47
	SPI_55S	SPI	SPI_ALT_MISO	47
	SPI_55S	SPI	SPI_ALT_CLK	47
	HDA_55S	HDA	HDA_BIT_CLK	18 55
	HDA_55S	HDA	HDA_BIT_CLK_R	18
	HDA_55S	HDA	HDA_RST_L	18 55
	HDA_55S	HDA	HDA_RST_R_L	18
	HDA_55S	HDA	HDA_SDOUT	18 55
	HDA_55S	HDA	HDA_SDOUT_R	18
	HDA_55S	HDA	HDA_SYNC	18 55
	HDA_55S	HDA	HDA_SYNC_R	18
	HDA_55S	HDA	HDA_SDIN0	18 55
	HDA_55S	HDA	AUD_SDI_R	55
		HDA	AUD_SPDIF_IN	59 81
		HDA	AUD_SPDIF_OUT	55 59
		HDA	AUD_SPDIF_CHIP	55
	HDA_55S	HDA	AUD_SPKR_OUTLO1L_NOUT	57 59 92
	HDA_55S	HDA	AUD_SPKR_OUTLO1L_POUT	57 59 92
	HDA_55S	HDA	AUD_SPKR_OUTLO1R_NOUT	57 59 92
	HDA_55S	HDA	AUD_SPKR_OUTLO1R_POUT	57 59 92
	HDA_55S	HDA	AUD_SPKR_OUTLO2L_NOUT	58 59 92
	HDA_55S	HDA	AUD_SPKR_OUTLO2L_POUT	58 59 92
	HDA_55S	HDA	AUD_SPKR_OUTLO2R_NOUT	58 59 92
	HDA_55S	HDA	AUD_SPKR_OUTLO2R_POUT	58 59 92
	CLK_XTAL	XTAL	PCH_CLK25M_XTALOUT	18 27
	CLK_XTAL	XTAL	PCH_CLK25M_XTALIN	18 27
	PCH_55S	COMP_PCH	PCH_USB_RBIAS	20
	PCH_55S	COMP_PCH	PCH_SATAICOMP	18
	PCH_55S	COMP_PCH	PCH_XCLK_RCOMP	18
	PCH_55S	COMP_PCH	PCH_DMI_COMP	19
	CLK_XTAL	XTAL	USB_HUB1_XTAL1	34
	CLK_XTAL	XTAL	USB_HUB1_XTAL2	34
	PCH_55S	COMP_PCH	USB_HUB1_RBIAS	34
	CLK_XTAL	XTAL	USB_HUB2_XTAL1	35
	CLK_XTAL	XTAL	USB_HUB2_XTAL2	35
	PCH_55S	COMP_PCH	USB_HUB2_RBIAS	35

SYNC_MASTER=K75F_MLB SYNC_DATE=04/14/2010

PAGE TITLE

IBEX PEAK CONSTRAINTS

Apple Inc.

DRAWING NUMBER 051-8600 SIZE D

REVISION A.0.0

BRANCH

CAESAR II (ETHERNET) CONSTRAINTS

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
ENET_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
BUF0_CLK	*	=3:1_SPACING	?
ENET_MII	*	0.3 MM	?
ENET_SE	*	=STANDARD	?

SOURCE:BROADCOM 5764M-DS04-RDS. PAGE 38

CAESAR II (ETHERNET) CONSTRAINTS

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
ENET_100D	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
ENET_DIFF	*	0.6 MM	?

SOURCE: BROADCOM 5764-DS04-RDS. PAGE 38

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
	ENET_50S	ENET_SE	ENET_RDAC	36
	ENET_50S	BUF0_CLK	ENET_CLK25M_XTALI	36
	ENET_50S	BUF0_CLK	ENET_CLK25M_XTALO	36
	ENET_50S	BUF0_CLK	ENET_CLK25M_XTAL	36
	ENET_100D	ENET_DIFF	ENET_MDI_P<3..0>	36 38
	ENET_100D	ENET_DIFF	ENET_MDI_N<3..0>	36 38
	ENET_100D	ENET_DIFF	ENET_MDI_T_P<3..0>	38
	ENET_100D	ENET_DIFF	ENET_MDI_T_N<3..0>	38
	ENET_100D	ENET_MII	PCIE_ENET_R2D_P	36
	ENET_100D	ENET_MII	PCIE_ENET_R2D_N	36
	ENET_100D	ENET_MII	PCIE_ENET_D2R_P	18 36
	ENET_100D	ENET_MII	PCIE_ENET_D2R_N	18 36
	ENET_100D	ENET_MII	PCIE_ENET_R2D_C_P	18 36
	ENET_100D	ENET_MII	PCIE_ENET_R2D_C_N	18 36
	ENET_100D	ENET_MII	PCIE_ENET_D2R_C_P	36
	ENET_100D	ENET_MII	PCIE_ENET_D2R_C_N	36

FireWire Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
FW_110D	*	=110_OHM_DIFF	=110_OHM_DIFF	=110_OHM_DIFF	=110_OHM_DIFF	=110_OHM_DIFF	=110_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
FW_TP	*	=3:1_SPACING	?

FireWire Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
AUDIO_PHY	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
AUDIO	*	=STANDARD	?

FireWire Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
	FW_110D	FW_TP	FW_PORT0_TPA_P	40 41
	FW_110D	FW_TP	FW_PORT0_TPA_N	40 41
	FW_110D	FW_TP	FW_PORT0_TPB_P	40 41
	FW_110D	FW_TP	FW_PORT0_TPB_N	40 41

PORT 1 & 2 NOT USED

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
	FW_110D	FW_TP	FW_P0_TPA_L_P	40
	FW_110D	FW_TP	FW_P0_TPA_L_N	40
	FW_110D	FW_TP	FW_P0_TPB_L_P	40
	FW_110D	FW_TP	FW_P0_TPB_L_N	40

UNUSED FW NETS PHYSICAL PROPERTIES

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
	FW_110D	FW_TP	FW_P1_TPA_P	39 40
	FW_110D	FW_TP	FW_P1_TPA_N	39 40
	FW_110D	FW_TP	FW_P2_TPA_P	39 40
	FW_110D	FW_TP	FW_P2_TPA_N	39 40

AUDIO MIC PHYSICAL PROPERTIES

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
	AUDIO_PHY	AUDIO	AUD_MIC1_IN_N	59 60
	AUDIO_PHY	AUDIO	AUD_MIC1_IN_P	59 60

SYNC_MASTER=K75F_MLB SYNC_DATE=04/14/2010

PAGE TITLE
ENET/FIREWIRE CONSTRAINTS

Apple Inc.

DRAWING NUMBER 051-8600 SIZE D

REVISION A.0.0

BRANCH

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DP_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	0.08MM	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DISPLAYPORT	*	=3:1_SPACING	?

PAIRS SHOULD BE WITHIN 100 MILS OF CLOCK LENGTH.
DisplayPort/TMDS intra-pair matching should be 5 ps. Inter-pair matching should be within 150 ps.
DIsplayPort AUX CH intra-pair matching should be 5 ps. No relationship to other signals.
Max length of LVDS/DisplayPort/TMDS traces: 12 inches.

ASSINGED IN CONT. MGR.

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
	DP_85D	DISPLAYPORT	DP_ML_CONN_P<3..0>	80
	DP_85D	DISPLAYPORT	DP_ML_CONN_N<3..0>	80
	DP_85D	DISPLAYPORT	DP_INT_LINK_CONN_P<3..0>	77
	DP_85D	DISPLAYPORT	DP_INT_LINK_CONN_N<3..0>	77
	DP_85D	DISPLAYPORT	DP_INT_LINK_P<3..0>	77 79
	DP_85D	DISPLAYPORT	DP_INT_LINK_N<3..0>	77 79
	DP_85D	DISPLAYPORT	DP_INT_AUXCH_P	77 79
	DP_85D	DISPLAYPORT	DP_INT_AUXCH_N	77 79
	DP_85D	DISPLAYPORT	DP_EXT_LINK_P<3..0>	78 80
	DP_85D	DISPLAYPORT	DP_EXT_LINK_N<3..0>	78 80
	DP_85D	DISPLAYPORT	DP_EXT_AUXCH_P	78 80 81
	DP_85D	DISPLAYPORT	DP_EXT_AUXCH_N	78 80 81
	DP_85D	DISPLAYPORT	DP_EXT_LINK_C_P<3..0>	78
	DP_85D	DISPLAYPORT	DP_EXT_LINK_C_N<3..0>	78
	DP_85D	DISPLAYPORT	MXM_DP_A_ML_P<3..0>	73 78
	DP_85D	DISPLAYPORT	MXM_DP_A_ML_N<3..0>	73 78
	DP_85D	DISPLAYPORT	MXM_DP_A_AUX_C_P	78
	DP_85D	DISPLAYPORT	MXM_DP_A_AUX_C_N	78
	DP_85D	DISPLAYPORT	MXM_DP_A_AUX_P	73 78
	DP_85D	DISPLAYPORT	MXM_DP_A_AUX_N	73 78
	DP_85D	DISPLAYPORT	MXM_DP_C_ML_P<3..0>	73 79
	DP_85D	DISPLAYPORT	MXM_DP_C_ML_N<3..0>	73 79
	DP_85D	DISPLAYPORT	MXM_DP_C_AUX_P	73 79
	DP_85D	DISPLAYPORT	MXM_DP_C_AUX_N	73 79
	DP_85D	DISPLAYPORT	MXM_DP_C_AUX_C_P	79
	DP_85D	DISPLAYPORT	MXM_DP_C_AUX_C_N	79
	DP_85D	DISPLAYPORT	DP_MUX_P<3..0>	78 79
	DP_85D	DISPLAYPORT	DP_MUX_N<3..0>	78 79
	DP_85D	DISPLAYPORT	DP_MUX_AUXCH_P	78 79
	DP_85D	DISPLAYPORT	DP_MUX_AUXCH_N	78 79
	DP_85D	DISPLAYPORT	DP_EQLZ_AUXCH_P	79
	DP_85D	DISPLAYPORT	DP_EQLZ_AUXCH_N	79
	DP_85D	DISPLAYPORT	MXM_DP_A_ML_C_P<3..0>	78
	DP_85D	DISPLAYPORT	MXM_DP_A_ML_C_N<3..0>	78
	DP_85D	DISPLAYPORT	MXM_DP_C_ML_C_P<3..0>	79
	DP_85D	DISPLAYPORT	MXM_DP_C_ML_C_N<3..0>	79
	DP_85D	DISPLAYPORT	DP_TX_EQ_AUXCH_P	78
	DP_85D	DISPLAYPORT	DP_TX_EQ_AUXCH_N	78
	DP_85D	DISPLAYPORT	MXM_DP_A_ML_EQ_P<3..0>	78
	DP_85D	DISPLAYPORT	MXM_DP_A_ML_EQ_N<3..0>	78

UNUSED VIDEO NET PHYSICAL CONSTRAINTS

	PHYSICAL	SPACING	NET	PAGES
	DP_85D	DISPLAYPORT	MXM_DP_B_AUX_P	73 76
	DP_85D	DISPLAYPORT	MXM_DP_B_AUX_N	73 76
	DP_85D	DISPLAYPORT	MXM_DP_D_AUX_P	73 76
	DP_85D	DISPLAYPORT	MXM_DP_D_AUX_N	73 76
	DP_85D	DISPLAYPORT	MXM_LVDS_A_CLK_P	74 76
	DP_85D	DISPLAYPORT	MXM_LVDS_A_CLK_N	74 76
	DP_85D	DISPLAYPORT	MXM_LVDS_B_CLK_P	74 76
	DP_85D	DISPLAYPORT	MXM_LVDS_B_CLK_N	74 76
	DP_85D	DISPLAYPORT	MXM_DP_B_ML_P<3..0>	73 76
	DP_85D	DISPLAYPORT	MXM_DP_B_ML_N<3..0>	73 76
	DP_85D	DISPLAYPORT	MXM_DP_D_ML_P<3..0>	73 76
	DP_85D	DISPLAYPORT	MXM_DP_D_ML_N<3..0>	73 76
	DP_85D	DISPLAYPORT	MXM_LVDS_A_DATA_P<3..0>	74 76
	DP_85D	DISPLAYPORT	MXM_LVDS_A_DATA_N<3..0>	74 76
	DP_85D	DISPLAYPORT	MXM_LVDS_B_DATA_P<3..0>	74 76
	DP_85D	DISPLAYPORT	MXM_LVDS_B_DATA_N<3..0>	74 76

8 | 7 | 6 | 5 | 4 | 3 | 2 | 1

SMBus Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SMB_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SMB	*	=2x_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
THERMAL	*	*	4:1_SPACING
THERMAL	POWER	*	PWR_P2MM
THERMAL	GND	*	GND_P2MM

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
THERM_DIFF	*	1:1_DIFFPAIR
SNS_DIFF	*	1:1_DIFFPAIR

SMC SMBus Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
	SMB_55S	SMB	SMBUS_SMC_A_S3_SCL	48
	SMB_55S	SMB	SMBUS_SMC_A_S3_SDA	48
	SMB_55S	SMB	SMBUS_SMC_B_S0_SCL	48
	SMB_55S	SMB	SMBUS_SMC_B_S0_SDA	48
	SMB_55S	SMB	SMBUS_SMC_0_S0_SCL	48
	SMB_55S	SMB	SMBUS_SMC_0_S0_SDA	48
	SMB_55S	SMB	SMBUS_SMC_BSA_SCL	48
	SMB_55S	SMB	SMBUS_SMC_BSA_SDA	48
	SMB_55S	SMB	SMBUS_SMC_MGMT_SCL	48 88
	SMB_55S	SMB	SMBUS_SMC_MGMT_SDA	48 88
	SMB_55S	SMB	SMBUS_SMC_MGMT_SCL	48 88
	SMB_55S	SMB	SMBUS_SMC_MGMT_SDA	48 88
I136	SMB_55S	SMB	SMBUS_PCH_S0_CLK	48
I135	SMB_55S	SMB	SMBUS_PCH_S0_DATA	48
I160	SMB_55S	SMB	SMBUS_PCH_CLK	18 48
I161	SMB_55S	SMB	SMBUS_PCH_DATA	18 48
I137	SMB_55S	SMB	SML_PCH_0_CLK	18 48
I139	SMB_55S	SMB	SML_PCH_0_DATA	18 48
I138	SMB_55S	SMB	SML_PCH_1_CLK	18 48
I140	SMB_55S	SMB	SML_PCH_1_DATA	18 48
I197	CLK_XTAL	XTAL	SMC_EXTAL	45 46
I196	CLK_XTAL	XTAL	SMC_XTAL	45 46

SMC THERMAL NET PROPERTIES

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
I87	THERM_DIFF	THERMAL	SNS_T_DP1_DN6	51 88
I88	THERM_DIFF	THERMAL	SNS_T_DN1_DP6	51 88
I89	THERM_DIFF	THERMAL	SNS_T_DP2_DN3	51
I90	THERM_DIFF	THERMAL	SNS_T_DN2_DP3	51
I200	THERM_DIFF	THERMAL	SNS_T_DN1_DP6	51 88
I201	THERM_DIFF	THERMAL	SNS_T_DP1_DN6	51 88
I93	THERM_DIFF	THERMAL	SNS_T_DP4_DN5	51
I94	THERM_DIFF	THERMAL	SNS_T_DN4_DP5	51
I101	THERM_DIFF	THERMAL	SNS_LCD_P	51 92
I102	THERM_DIFF	THERMAL	SNS_LCD_N	51 92
I103	THERM_DIFF	THERMAL	SNS_ODD_P	51 92
I105	THERM_DIFF	THERMAL	SNS_ODD_N	51 92
I104	THERM_DIFF	THERMAL	SNS_CPU_H_P	51
I106	THERM_DIFF	THERMAL	SNS_CPU_H_N	51
I198	THERM_DIFF	THERMAL	SNS_SKIN_P	51 92
I199	THERM_DIFF	THERMAL	SNS_SKIN_N	51 92
I109	THERM_DIFF	THERMAL	SNS_AMB_P	51 92
I110	THERM_DIFF	THERMAL	SNS_AMB_N	51 92
I111	THERM_DIFF	THERMAL	SNS_MXM_P	51
I112	THERM_DIFF	THERMAL	SNS_MXM_N	51
I194	THERM_DIFF	THERMAL	SNS_CPU_THERMD_P	10 51
I195	THERM_DIFF	THERMAL	SNS_CPU_THERMD_N	10 51
I115		THERMAL	HDD_OOB_TEMP_FILT	51 92
I133		THERMAL	HDD_OOB_TEMP	51
I134		THERMAL	HDD_OOB_TEMP_R	51
I132		THERMAL	SMC_HDD_OOB_TEMP	51

SMC VOLTAGE/CURRENT NET PROPERTIES

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
I97	THERM_DIFF	THERMAL	MXM_ISENSE_P	50
I98	THERM_DIFF	THERMAL	MXM_ISENSE_N	50
I99	THERM_DIFF	THERMAL	SENSE_CPU_1V5_S3_P	49
I100	THERM_DIFF	THERMAL	SENSE_CPU_1V5_S3_N	49
I148	THERM_DIFF	THERMAL	SENSE_CPU_1V5_S0_P	49
I149	THERM_DIFF	THERMAL	SENSE_CPU_1V5_S0_N	49
I150	THERM_DIFF	THERMAL	SENSE_CPU_1V5_P	49
I151	THERM_DIFF	THERMAL	SENSE_CPU_1V5_N	49
I152	THERM_DIFF	THERMAL	SENSE_CPU_VTT_P	49
I153	THERM_DIFF	THERMAL	SENSE_CPU_VTT_N	49
I181	THERM_DIFF	THERMAL	SENSE_CPU_VTT1_P	
I182	THERM_DIFF	THERMAL	SENSE_CPU_VTT1_N	
I183	THERM_DIFF	THERMAL	SENSE_CPU_VTT2_P	
I184	THERM_DIFF	THERMAL	SENSE_CPU_VTT2_N	
I162		THERMAL	GND_SMC_AVSS	45 46 49 50
I124		THERMAL	SMC_CPU_1V5_ISENSE	46 49
I123		THERMAL	SMC_CPU_1V5_ISENSE_R	49
I122		THERMAL	SMC_CPU_1V5_VSENSE	46 49
I163		THERMAL	SMC_CPU_VTT_ISENSE	46 49
I164		THERMAL	SMC_CPU_VTT_ISENSE_R	49
I165		THERMAL	SMC_CPU_VTT_VSENSE	46 49
I166		THERMAL	SMC_CPU_1V8_ISENSE	46 49
I167		THERMAL	SMC_CPU_1V8_ISENSE_R	
I168		THERMAL	SMC_CPU_1V8_VSENSE	46 49
I128		THERMAL	SMC_CPU_VSENSE	45 49
I113	VID_PHY	VR_CTL	VR_CPU_IOUT	13 64
I131	THERM_DIFF	THERMAL	VR_ISNS_CPU_P	49
I130	THERM_DIFF	THERMAL	VR_ISNS_CPU_N	49
I129		THERMAL	SNS_PS_CPU_ISNS	49
I114		THERMAL	SMC_CPU_ISENSE	45 49
I169		THERMAL	SMC_CPU_INPUT_ISENSE	46
I171		THERMAL	SMC_CPU_INPUT_VSENSE	46
I172		THERMAL	SMC_DIMM_1V5_ISENSE	46 50
I173		THERMAL	SMC_1V5_S3_ISENSE_R	
I174		THERMAL	SMC_DIMM_1V5_VSENSE	46 50
I178		THERMAL	SMC_GPU_ISENSE	45 50
I179		THERMAL	SMC_MXM_ISENSE_R	50
I180		THERMAL	SMC_GPU_VSENSE	45 50

SYNC_MASTER=K75F_MLB SYNC_DATE=04/14/2010

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SMC Constraints

Apple Inc.

DRAWING NUMBER 051-8600 SIZE D

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NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
SWITCHNODE	SWITCHNODE	BGA_P1MM	BGA_P2MM
SWITCHNODE	POWER	BGA_P1MM	BGA_P2MM
SWITCHNODE	GND	BGA_P1MM	BGA_P2MM
SWITCHNODE	*	BGA_P1MM	BGA_P2MM
SWITCHNODE	POWER	*	6:1_SPACING
SWITCHNODE	GND	*	6:1_SPACING
SWITCHNODE	*	*	SWITCHNODE

POWER NET PROPERTIES

	NET_TYPE PHYSICAL	SPACING	VOLTAGE	NET	
I128	POWER	SWITCHNODE	1.5V	VR_CPU_PHASE1	65
I129	POWER	SWITCHNODE	1.5V	VR_CPU_PHASE2	65
I130	POWER	SWITCHNODE	1.5V	VR_CPU_PHASE3	65
I131	POWER	SWITCHNODE	1.5V	VR_CPU_PHASE4	66
I132	POWER	SWITCHNODE	3.3V	P3V3S5_REG_PHASE	69
I133	POWER	SWITCHNODE	5V	P5VS3_REG_PHASE	69
I134	POWER	SWITCHNODE	1.1V	VTT_REG_PHASE1	67
I135	POWER	SWITCHNODE	1.1V	VTT_REG_PHASE2	67
I136	POWER	SWITCHNODE	3.4V	P3V42G3H_SW	71
I137	POWER	SWITCHNODE	1.05V	PCHCORE_REG_PHASE	68
I138	POWER	SWITCHNODE	1.05V	P1V05_S5_REG_PHASE	71
I139	POWER	SWITCHNODE	1.5V	DDR_REG_PHASE	70
I140	POWER	SWITCHNODE	1.8V	P1V8_REG_PHASE	70
I141	POWER	POWER	1.5V	PP0V75_S3_MEM_VREFCA_A	28 30
I142	POWER	POWER	1.5V	PP0V75_S3_MEM_VREFCA_B	28 31
I143	POWER	POWER	1.5V	PP0V75_S3_MEM_VREFDQ_A	28 30
I144	POWER	POWER	1.5V	PP0V75_S3_MEM_VREFDQ_B	28 31
I585	POWER	POWER	12V	PP12V_S0_CPUVTT_FLTD	
I145	POWER	POWER	12V	PP12V_AUD_SPKRAMP_PLANE	57 58 60
I146	POWER	POWER	12V	PP12V_LCD	77
I147	POWER	POWER	12V	PP12V_LCD_CONN	77
I148	POWER	POWER	12V	PP12V_S0	6 63
I149	POWER	POWER	12V	PP12V_S0_CPU_FLTRD	64 65 66
I175	POWER	POWER	12V	PP12V_S0_FAN0_L	52 92
I174	POWER	POWER	12V	PP12V_S0_FAN1_L	52 92
I173	POWER	POWER	12V	PP12V_S0_FAN2_L	53 92
I604	POWER	POWER	12V	PP12V_G3H	6 71
I605	POWER	POWER	12V	PP12V_G3H_R	71
I178	POWER	POWER	12V	PP12V_S5	6
I150	POWER	POWER	12V	P12V_S0_FW	
I276	POWER	POWER	12V	P12V_S0_FW_CL	
I277	POWER	POWER	12V	P12V_S0_FW_D	
I264	POWER	POWER	12V	P12V_S0_FW_R	
I267	POWER	POWER	12V	FW_PORT0_VP	41
I270	POWER	POWER	12V	FW_PORT0_VP_F	41
I544	POWER	POWER	12V	PPVP_FW_PHY_CPS	40 41
I549	POWER	POWER	12V	PP12V_S3	6
I551	POWER	POWER	12V	PP12V_S3_WM_FLT	44
I259	POWER	POWER	1.1V	PPVCORE_S0_CPU	6
I258	POWER	POWER	1.1V	PPVCORE_S0_CPU_REG1	65
I563	POWER	POWER	1.1V	PPVCORE_S0_CPU_REG2	65
I564	POWER	POWER	1.1V	PPVCORE_S0_CPU_REG3	65
I565	POWER	POWER	1.1V	PPVCORE_S0_CPU_REG4	66
I254	POWER	POWER	1.05V	PP1V05_SM_SOURCE	72
I177	POWER	POWER	1.05V	PP1V05_S0	6
I179	POWER	POWER	1.05V	PP1V05_S0_CK505_F	26
I182	POWER	POWER	1.05V	PP1V05_S0_PCH_VCCADPLLA	17 22
I181	POWER	POWER	1.05V	PP1V05_S0_PCH_VCCADPLLA_F	
I180	POWER	POWER	1.05V	PP1V05_S0_PCH_VCCADPLLB	17 22
I185	POWER	POWER	1.05V	PP1V05_S0_PCH_VCCADPLLB_F	
I184	POWER	POWER	1.05V	PP1V05_S0_PCH_VCCAPLL_EXP	22 24
I183	POWER	POWER	1.05V	PP1V05_S0_PCH_VCCAPLL_FDI	22 24
I186	POWER	POWER	1.05V	PP1V05_S0_PCH_VCCAPLL_SATA	22 24
I188	POWER	POWER	1.05V	PP1V05_S0_PCH_VCCA_CLK	22 24
I187	POWER	POWER	1.05V	PP1V05_S0_PCH_VCCA_CLK_F	24
I599	POWER	POWER	1.05V	PP1V05_S0_CIO_VDD1P0_DP	
I598	POWER	POWER	1.05V	PP1V05_S0_CIO_VDD1P0_DP_PLL	
I614	POWER	POWER	1.05V	PP1V05_SM_PCH_LAN	6
I192	POWER	POWER	1.05V	PP1V05_S5	6
I574	POWER	POWER	1.05V	PP1V05_SM	6
I269	POWER	POWER	1.1V	PPVTT_S0	6
I555	POWER	POWER	1.1V	PPVTT_S0_CPU	6 49
I268	POWER	POWER	0.75V	PPVTT_S0_DDR	6
I613	POWER	POWER	0.75V	PP0V75_S0	6
I190	POWER	POWER	1.2V	PP1V2_S5_ENET	37
I194	POWER	POWER	1.5V	PP1V5_S0	6
I193	POWER	POWER	1.5V	PP1V5_S0_CK505_F	26
I587	POWER	POWER	1.5V	PP1V5_S0_CK505_R	26
I198	POWER	POWER	1.5V	PP1V5_S3	6
I575	POWER	POWER	1.5V	PP1V5_CPU_MEM	6 49
I205	POWER	POWER	1.5V	PP1V8R1V5_S0_PCH_VCCVRM	22 24
I195	POWER	POWER	1.5V	PP1V5_FW_VDDA	39
I203	POWER	POWER	1.5V	PP1V8_S0	6
I204	POWER	POWER	1.8V	PP1V8_S0_CPU	6 49
I206	POWER	POWER	1.96V	PP1V96_FW_PLLVDD	39
I208	POWER	POWER	1.96V	PP1V95_FW_FWPHY	39 40

POWER NET PROPERTIES

NET_TYPE PHYSICAL	SPACING	VOLTAGE	NET	
POWER	POWER	3.3V	PP3V3_S0	6
POWER	POWER	3.3V	PP3V3_S0_CK505_F	26
POWER	POWER	3.3V	PP3V3_S0_DPAUX	80
POWER	POWER	3.3V	PP3V3_S0_DPFUSE	80
POWER	POWER	3.3V	PP3V3_S0_DPPWR	80
POWER	POWER	3.3V	PP3V3_S0_HS_F	61
POWER	POWER	3.3V	PP3V3_S0M	6
POWER	POWER	3.3V	PP3V3_MINI	33
POWER	POWER	3.3V	PP3V3_ENET	36 37
POWER	POWER	3.3V	PP3V3_S0_PCH_VCCA_DAC	17 22
POWER	POWER	3.3V	PP3V3_S0_PCH_VCCA_DAC_F	
POWER	POWER	3.3V	PP3V3_S0_TSENS_R	51
POWER	POWER	3.3V	PP3V3_S3	6 92
POWER	POWER	3.3V	PP3V3_S3_BT_FLT	44
POWER	POWER	3.3V	PP3V3_S3_SDCARD_FLT	44
POWER	POWER	3.3V	PP3V3_S3_WM_FLT	44
POWER	POWER	3.3V	PP3V3_S5	6
POWER	POWER	3.3V	PPVTT_S3_DDR_BUF	70
POWER	POWER	3.3V	PPV_S0_MXM_PWRSRC	50
POWER	POWER	3.3V	PPVOUT_S0_PCH_DCPSST	22
POWER	POWER	3.3V	PPVOUT_S5_PCH_DCPSUS	22
POWER	POWER	3.3V	PPVOUT_S5_PCH_DCPSUSBYP	22
POWER	POWER	3.3V	PPVOUT_G3_PCH_DCPRTC	22
POWER	POWER	3.3V	PPVOUT_S0_PCH_VCCRTC_NCTF	
POWER	POWER	3.3V	PPVBATT_G3_RTC	27
POWER	POWER	3.3V	PPVBATT_G3_RTC_R	27
POWER	POWER	3.3V	PP3V3_AUDIO_SPDIF_JACK	59
POWER	POWER	3.3V	PP3V3_FW_AVDD	39
POWER	POWER	3.3V	PP3V3_FW_ESD	41
POWER	POWER	3.3V	PP3V3_FW_PLLVDD	39
POWER	POWER	3.3V	PP3V3_FW_VDDA	39
POWER	POWER	3.3V	PP3V3_G3_RTC	18 22 24 27
POWER	POWER	3.3V	PP_ENET_CTRL12	37
POWER	POWER	3.4V	PP3V3_G3H_SMC_AVCC	45
POWER	POWER	3.3V	PP3V3_G3H_AVREF_SMC	45 46
POWER	POWER	3.42V	PP3V42_G3H	6
POWER	POWER	3.42V	PP3V42_G3H_R	71
POWER	POWER	4.5V	4V5_REG_IN	55
POWER	POWER	4.5V	PP4V5_AUDIO_ANALOG	55
POWER	POWER	5V	PP5V_S0	6 92
POWER	POWER	5V	PP5V_S0_CPU_VCORE_VCC	64
POWER	POWER	5V	PP5V_S0_PCH_V5REF	22 24
POWER	POWER	5V	PP5V_S0_SATA_FET	42
POWER	POWER	5V	PP5V_S3	6
POWER	POWER	5V	PP5V_S3_DDR_REG_V5FILT	70
POWER	POWER	5V	PP5V_S3_CAMERA_FLT	44
POWER	POWER	5V	PP5V_S3_IR_FLT	44
POWER	POWER	5V	VREFMARGIN_DIMMA_P5V	28
POWER	POWER	5V	VREFMARGIN_DIMMB_P5V	28
POWER	POWER	5V	PP5V_S5	6
POWER	POWER	5V	PP5V_S5_PCH_V5REFSUS	22 24
POWER	POWER	5V	PP5V_CPUVTT_VR	67
POWER	POWER	5V	PP5V_USB2_PORT0	43
POWER	POWER	5V	PP5V_USB2_PORT0_F	43
POWER	POWER	5V	PP5V_USB2_PORT1	43
POWER	POWER	5V	PP5V_USB2_PORT1_F	43
POWER	POWER	5V	PP5V_USB2_PORT2	43
POWER	POWER	5V	PP5V_USB2_PORT2_F	43
POWER	POWER	5V	PP5V_USB2_PORT3	43
POWER	POWER	5V	PP5V_USB2_PORT3_F	43
POWER	POWER		DDR_REG_PGND	70
POWER	POWER		DDR_REG_CSGND	70

SENSING NET PROPERTIES

	NET_TYPE PHYSICAL	SPACING	NET	
I155	SNS_DIFF	THERMAL	VR_CPU_ISNS1_P	64 65
I156	SNS_DIFF	THERMAL	VR_CPU_ISNS1_N	64 65
I157	SNS_DIFF	THERMAL	VR_CPU_ISNS1_R_P	64
I158	SNS_DIFF	THERMAL	VR_CPU_ISNS1_R_N	64
I160	SNS_DIFF	THERMAL	VR_CPU_ISNS2_P	64 65
I159	SNS_DIFF	THERMAL	VR_CPU_ISNS2_N	64 65
I161	SNS_DIFF	THERMAL	VR_CPU_ISNS2_R_P	64
I163	SNS_DIFF	THERMAL	VR_CPU_ISNS2_R_N	64
I162	SNS_DIFF	THERMAL	VR_CPU_ISNS3_P	64 65
I164	SNS_DIFF	THERMAL	VR_CPU_ISNS3_N	64 65
I165	SNS_DIFF	THERMAL	VR_CPU_ISNS3_R_P	64
I166	SNS_DIFF	THERMAL	VR_CPU_ISNS3_R_N	64
I167	SNS_DIFF	THERMAL	VR_CPU_ISNS4_P	64 66
I168	SNS_DIFF	THERMAL	VR_CPU_ISNS4_N	64 66
I169	SNS_DIFF	THERMAL	VR_CPU_ISNS4_R_P	64
I170	SNS_DIFF	THERMAL	VR_CPU_ISNS4_R_N	64
I566	SNS_DIFF	THERMAL	VR_CPU_ISNS1_XW_P	65
I567	SNS_DIFF	THERMAL	VR_CPU_ISNS1_XW_N	65
I568	SNS_DIFF	THERMAL	VR_CPU_ISNS2_XW_P	65
I569	SNS_DIFF	THERMAL	VR_CPU_ISNS2_XW_N	65
I570	SNS_DIFF	THERMAL	VR_CPU_ISNS3_XW_P	65
I571	SNS_DIFF	THERMAL	VR_CPU_ISNS3_XW_N	65
I572	SNS_DIFF	THERMAL	VR_CPU_ISNS4_XW_P	66
I573	SNS_DIFF	THERMAL	VR_CPU_ISNS4_XW_N	66
I578	SNS_DIFF		CPU_VCC_PKG_SENSE_P	13 49 64
I579	SNS_DIFF		CPU_VCC_PKG_SENSE_N	13 64
I56	SNS_DIFF		CPU_VTTSENSE_P	13 49 67
I57	SNS_DIFF		CPU_VTTSENSE_N	13 67
I172	SNS_DIFF		CPU_VTTSENSE_R_P	67
I171	SNS_DIFF		CPU_VTTSENSE_R_N	67
I58	SNS_DIFF		VR_CPU_VSEN	64
I59	SNS_DIFF		VR_CPU_RGND	64
I151	SNS_DIFF		VR_CPU_VSNS_R_N	64
I152	SNS_DIFF		VR_CPU_VSNS_R_P	64
I153	SNS_DIFF		VR_CPU_VSNS_XW_P	64
I154	SNS_DIFF		VR_CPU_VSNS_XW_N	64

VR CTRL NET PROPERTIES

	NET_TYPE PHYSICAL	SPACING	NET	
I278	VR_CTL_PHY	VR_CTL	VR_CPU_PH1_SNUB	65
I338	VR_CTL_PHY	VR_CTL	VR_CPU_PH2_SNUB	65
I339	VR_CTL_PHY	VR_CTL	VR_CPU_PH3_SNUB	65
I340	VR_CTL_PHY	VR_CTL	VR_CPU_PH4_SNUB	66
I341	VR_CTL_PHY	VR_CTL	VR_CPU_PWM1	64 65
I342	VR_CTL_PHY	VR_CTL	VR_CPU_PWM2	64 65
I343	VR_CTL_PHY	VR_CTL	VR_CPU_PWM2_R	64
I344	VR_CTL_PHY	VR_CTL	VR_CPU_PWM3	64 65
I345	VR_CTL_PHY	VR_CTL	VR_CPU_PWM3_R	64
I346	VR_CTL_PHY	VR_CTL	VR_CPU_PWM4	64 66
I347	VR_CTL_PHY	VR_CTL	VR_CPU_PWM4_R	64
I348	VR_CTL_PHY	VR_CTL	VR_CPU_REF	64
I349	VR_CTL_PHY	VR_CTL	VR_CPU_SS	64
I350	VR_CTL_PHY	VR_CTL	VR_CPU_TCOMP	64
I353	VR_CTL_PHY	VR_CTL	VR_CPU_TM	64
I352	VR_CTL_PHY	VR_CTL	VR_CPU_BOOT1_RC	65
I351	VR_CTL_PHY	VR_CTL	VR_CPU_BOOT2_RC	65
I354	VR_CTL_PHY	VR_CTL	VR_CPU_BOOT3_RC	65
I356	VR_CTL_PHY	VR_CTL	VR_CPU_COMP	64
I355	VR_CTL_PHY	VR_CTL	VR_CPU_COMP_R	64
I357	VR_CTL_PHY	VR_CTL	VR_CPU_COMP_RC	64
I358	VR_CTL_PHY	VR_CTL	VR_CPU_DAC	64
I359	VR_CTL_PHY	VR_CTL	VR_CPU_DRV1_BOOT	65
I360	VR_CTL_PHY	VR_CTL	VR_CPU_DRV1_GDSEL	65
I363	VR_CTL_PHY	SWITCHNODE	VR_CPU_DRV1_LGATE	65
I361	VR_CTL_PHY	SWITCHNODE	VR_CPU_DRV1_UGATE	65
I365	VR_CTL_PHY	VR_CTL	VR_CPU_DRV2_BOOT	65
I367	VR_CTL_PHY	VR_CTL	VR_CPU_DRV2_GDSEL	65
I368	VR_CTL_PHY	SWITCHNODE	VR_CPU_DRV2_LGATE	65
I370	VR_CTL_PHY	SWITCHNODE	VR_CPU_DRV2_UGATE	65
I371	VR_CTL_PHY	VR_CTL	VR_CPU_DRV3_BOOT	65
I374	VR_CTL_PHY	VR_CTL	VR_CPU_DRV3_GDSEL	65
I376	VR_CTL_PHY	SWITCHNODE	VR_CPU_DRV3_LGATE	65
I377	VR_CTL_PHY	SWITCHNODE	VR_CPU_DRV3_UGATE	65
I380	VR_CTL_PHY	VR_CTL	VR_CPU_DRV4_BOOT	66
I383	VR_CTL_PHY	VR_CTL	VR_CPU_DRV4_GDSEL	66
I382	VR_CTL_PHY	SWITCHNODE	VR_CPU_DRV4_LGATE	66
I384	VR_CTL_PHY	SWITCHNODE	VR_CPU_DRV4_UGATE	66
I387	VR_CTL_PHY	VR_CTL	VR_CPU_FAN	64
I388	VR_CTL_PHY	VR_CTL	VR_CPU_FB	64
I389	VR_CTL_PHY	VR_CTL	VR_CPU_FB_R	64
I390	VR_CTL_PHY	VR_CTL	VR_CPU_FS	64
I393	VR_CTL_PHY	VR_CTL	VR_CPU_IMON	49 64
I391	VR_CTL_PHY	VR_CTL	VR_CPU_IOUT_PD	64
I394	VR_CTL_PHY	SWITCHNODE	PCHCORE_REG_UGATE	68
I396	VR_CTL_PHY	SWITCHNODE	PCHCORE_REG_LGATE	68
I395	VR_CTL_PHY	VR_CTL	PCHCORE_REG_VFB	68
I397	VR_CTL_PHY	VR_CTL	PCHCORE_REG_TON	68
I398	VR_CTL_PHY	VR_CTL	PCHCORE_REG_TRIP	68
I399	VR_CTL_PHY	VR_CTL	PCHCORE_REG_BOOT	68
I594	VR_CTL_PHY	VR_CTL	PCHCORE_REG_BOOT_R	68
I400	VR_CTL_PHY	VR_CTL	VTT_REG_BOOT1	67
I403	VR_CTL_PHY	VR_CTL	VTT_REG_BOOT2	67
I402	VR_CTL_PHY	VR_CTL	VTT_REG_COMP	67
I401	VR_CTL_PHY	VR_CTL	VTT_REG_FB	67
I404	VR_CTL_PHY	VR_CTL	VTT_REG_FS	67
I406	VR_CTL_PHY	VR_CTL	VTT_REG_ICOMP	67
I405	VR_CTL_PHY	VR_CTL	VTT_REG_IREF	67
I409	VR_CTL_PHY	VR_CTL	VTT_REG_ISUM	67
I408	VR_CTL_PHY	SWITCHNODE	VTT_REG_LGATE1	67
I407	VR_CTL_PHY	SWITCHNODE	VTT_REG_LGATE2	67
I410	VR_CTL_PHY	VR_CTL	VTT_REG_OCSET	67
I411	VR_CTL_PHY	VR_CTL	VTT_OFST	67
I412	VR_CTL_PHY	VR_CTL	VTT_REG_REF	67
I413	VR_CTL_PHY	VR_CTL	VTT_REG_RGND	67
I414	VR_CTL_PHY	VR_CTL	VTT_SEL	
I415	VR_CTL_PHY	VR_CTL	VTT_REG_UGATE2	67
I416	VR_CTL_PHY	VR_CTL	VTT_REG_UGATE1	67
I418	VR_CTL_PHY	VR_CTL	VTT_REG_PH1_SNUB	67
I417	VR_CTL_PHY	VR_CTL	VTT_REG_PH2_SNUB	67
I419	VR_CTL_PHY	VR_CTL	VTT_REG_VSEN	67
I597	VR_CTL_PHY	VR_CTL	P1V05S5_REG_VFB	71
I607	VR_CTL_PHY	VR_CTL	P3V42G3H_BOOST	71
I608	VR_CTL_PHY	VR_CTL	P3V42G3H_FB	71

VR CTRL NET PROPERTIES

	NET_TYPE PHYSICAL	SPACING	NET	
I437	VR_CTL_PHY	VR_CTL	DDR_REG_CS	70
I595	VR_CTL_PHY	VR_CTL	DDR_REG_FB	70
I439	VR_CTL_PHY	SWITCHNODE	DDR_REG_LGATE	70
I596	VR_CTL_PHY	SWITCHNODE	DDR_REG_UGATE	70
I443	VR_CTL_PHY	VR_CTL	DDR_REG_BOOT	70
I442	VR_CTL_PHY	VR_CTL	DDR_REG_BOOT_R	70
I441	VR_CTL_PHY	VR_CTL	DDR_REG_VDDQSNS	70
I444	VR_CTL_PHY	VR_CTL	DDR_REG_VTTSNS	70
I449	VR_CTL_PHY	VR_CTL	P1V8_REG_POR	70
I450	VR_CTL_PHY	VR_CTL	P3V3S5_REG_BOOT	69
I452	VR_CTL_PHY	VR_CTL	P3V3S5_REG_BOOT_R	69
I451	VR_CTL_PHY	VR_CTL	P3V3S5_REG_FB	69
I453	VR_CTL_PHY	VR_CTL	P3V3S5_REG_ISEN	69
I455	VR_CTL_PHY	SWITCHNODE	P3V3S5_REG_LGATE	69
I454	VR_CTL_PHY	VR_CTL	P3V3S5_REG_OCSET	69
I458	VR_CTL_PHY	SWITCHNODE	P3V3S5_REG_UGATE	69
I457	VR_CTL_PHY	VR_CTL	P3V3S5_REG_SNUB	69
I459	VR_CTL_PHY	VR_CTL	P5VS3_REG_BOOT	69
I461	VR_CTL_PHY	VR_CTL	P5VS3_REG_FB	69
I462	VR_CTL_PHY	VR_CTL	P5VS3_REG_ISEN	69
I463	VR_CTL_PHY	SWITCHNODE	P5VS3_REG_LGATE	69
I464	VR_CTL_PHY	VR_CTL	P5VS3_REG_OCSET	69
I466	VR_CTL_PHY	SWITCHNODE	P5VS3_REG_UGATE	69

VR VID NET PROPERTIES

PULL-UP STUB < 1-INCH
VID LENGTH SKEW < 1-INCH
VID LENGTH RANGE< 1 TO 15-INCH

	NET_TYPE PHYSICAL	SPACING	NET	
I491	VID_PHY	VR_CTL	CPU_VID<0>	13 16 64
I492	VID_PHY	VR_CTL	CPU_VID<1>	13 16 64
I521	VID_PHY	VR_CTL	CPU_VID<2>	13 16 64
I522	VID_PHY	VR_CTL	CPU_VID<3>	13 16 64
I523	VID_PHY	VR_CTL	CPU_VID<4>	13 16 64
I524	VID_PHY	VR_CTL	CPU_VID<5>	13 16 64
I525	VID_PHY	VR_CTL	CPU_VID<6>	13 16 64
I526	VID_PHY	VR_CTL	CPU_VID<7>	13 16 64
I554	VID_PHY	VR_CTL	CPU_PSI_L	13 16 64

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
VID_PHY	*	39_OHM_SE

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
VR_CTL	*	0.2MM	?

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POWER CONSTRAINTS

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PM NET PROPERTIES
(PM, RESET, EN, PGOOD)

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
PM	*	*	2:1_SPACING
PM_VTT	PM_VTT	*	2:1_SPACING
PM_VTT	*	*	3:1_SPACING
PM_VTT	GND	*	DEFAULT
PM	GND	*	DEFAULT

Instance	NET_TYPE PHYSICAL	NET_TYPE SPACING	Net	Pages
I2		PM	PLT_RESET_L	20 27
I1		PM_VTT	PLT_RESET_LS1V1_L	11
I4		PM	PM_ACDC_PS_ON	6
I3		PM	PM_BATLOW_L	15 19 45
I5		PM	PM_CLK32K_SUSCLK	9 45 85
I7		PM	PM_CLK32K_SUSCLK_R	9 19 85
I6		PM	PM_CLKRUN_L	15 19 45 47
I10		PM	PM_EXT_TS_L<0>	11 46
I9		PM	PM_EXT_TS_L<1>	11 46
I12		PM	PM_LAN_PWRGD	15 19
I117		PM_VTT	FSB_CPURSTOUT_L	11 25
I123		PM	USB_HUB_RESET_L	34 35
I25		PM_VTT	PM_MEM_PWRGD	11 19
I24		PM	PM_ME_PWRGD	19 63
I27		PM	PM_ME_S0_EN_G	72
I26		PM	PM_ME_S0_EN_G1	72
I28		PM	PM_ME_S0_EN_R	72
I30		PM	PM_MXM_PGOOD	63 74
I29		PM	PM_PCH_PWRGD	19 63
I31		PM	PM_PGOOD_DDRREG_S3	5 62 70
I33		PM	PM_PGOOD_PVCORE_CPU	5 26 63 64
I32		PM	PM_PWRBTN_L	19 25 45
I35		PM	PM_RSMRST_L	45 62
I34		PM	PM_RSMRST_PCH_L	19 62
I36		PM	PM_SLP_M_L	5 19 62
I38		PM	PM_SLP_M_R	
I37		PM	PM_SLP_S3_L	5 19 32 33 37 46 62 63
I40		PM	PM_SLP_S3_L_AND_S0_RDY	
I120		PM	PM_SLP_S4_1_L	19 62
I119		PM	PM_SLP_S4_2_L	19 45 46
I118		PM	PM_SLP_S4_3_L	5 19
I39		PM	PM_SLP_S4_L	19 32
I41		PM	PM_SLP_S5_L	19 45
I43		PM	PM_SUS_PWR_ACK	19
I42		PM_VTT	PM_SYNC	11 19
I122		PM	SDCARD_PLT_RST_L	27 44
I45		PM	PM_SYSRST_L	19 27 45
I44		PM	PM_SYS_PWRGD	19 32 63
I46		PM_VTT	PM_THRMTRIP_L	11 21 46
I48		PM	RSMRST_PWRGD	45 63
I47		PM	RTC_RESET_L	18 91
I50		PM_VTT	CPU_PWRGD	11 21 25
I49		PM	CPU_RESET_L	11 27
I51		PM	PGOOD_1V05ME_G1	63
I53		PM	PGOOD_1V05ME_G2	63
I52		PM	PGOOD_1V8_S0_G1	63
I54		PM	PGOOD_1V8_S0_G2	63
I56		PM	PGOOD_CPU_GFX_DDR	63
I55		PM	PGOOD_P12V_S3	
I58		PM	PGOOD_P1V05_ME_S5	
I57		PM	PGOOD_P1V5_S0	72
I59		PM	PGOOD_P1V8_S0	63
I61		PM	PGOOD_P3V3_ME	72
I60		PM	PGOOD_P3V3_S0	48 63 72
I63		PM	PGOOD_P3V3_S3	34 72
I62		PM	PGOOD_P5V_S0	62 72
I64		PM	PGOOD_PCH_AND_P1V8	63
I66		PM	PGOOD_PCH_S0	63
I65		PM	PGOOD_SYSPWROK	63
I67		PM	PGOOD_SYSPWROK_R	63
I68		PM	RTC_RESET_L	18 91
I69		PM	P12V_S3_EN	62 72
I71		PM	P1V05_ME_SM_EN	62 72
I70		PM	P1V5_S0_EN	62 72
I72		PM	P3V3ME_EN	62 72
I74		PM	P3V3S0_EN	62 72
I73		PM	P3V3S3_EN	62 72
I76		PM	P5VS0_EN	62 72
I75		PM	P5VS3_EN	62 69
I77		PM	PCHCORE_REG_EN	62 68
I79		PM	PCHCORE_REG_PGOOD	5 62 63 68
I78		PM	PEG_RESET_L	9 27
I80		PM	SDCARD_RESET	21 25 44 92

Instance	NET_TYPE PHYSICAL	NET_TYPE SPACING	Net	Pages
I82		PM	4V5_REG_EN	55
I81		PM	ALL_SYS_PWRGD_R	6 25 32 63
I84		PM	ALL_SYS_PWRGD_SMC	45 63
I83		PM	CK505_27MHZ_EN	26
I85		PM	CPUVTT_REG_EN	62 67
I86		PM_VTT	CPUVTT_REG_PGOOD	11 62 63 67
I87		PM	CPU_MEM_RESET_L	11 32
I88		PM	DDRVTT_EN	32 62 70
I89		PM	DEBUG_RESET_L	27 47
I90		PM	FWPHY_RESET_L	39
I91		PM	FWXIO_SNOOP_EN	39
I92		PM	FW_RESET_L	27 39
I93		PM	GFX_VR_EN	
I94		PM	GFX_VR_PGOOD	
I95		PM	LAN_RESET_L	27 36
I97		PM	MEM_RESET_L	30 31 32
I96		PM	MINI_RESET_L	27 33
I98		PM	SMC_DELAYED_PWRGD	46 63
I100		PM	SMC_LRESET_L	27 45
I99		PM	SMC_RESET_L	45 46 47
I102		PM	T28_RESET_L	
I101		PM_VTT	XDP_CPUPWRGD	11 25
I103		PM_VTT	XDP_DBRESET_L	11 25 27
I104		PM_VTT	XDP_PWRGD	25

NET PHYSICAL FOR NC NETS
REMOVE WHEN CHECKPLUS IS FIXED

Instance	ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	Net
	PCIE GRAPHICS			
I105		PCIE_85D	PCIE	NC_PCIE_CLK100M_EXCARD_N
I106		PCIE_85D	PCIE	NC_PCIE_CLK100M_EXCARD_P
I107		PCIE_85D	PCIE	NC_PCIE_EXCARD_D2R_N
I108		PCIE_85D	PCIE	NC_PCIE_EXCARD_D2R_P
I109		PCIE_85D	PCIE	NC_PCIE_EXCARD_R2D_C_N
I110		PCIE_85D	PCIE	NC_PCIE_EXCARD_R2D_C_P
I112	NO_TEST=TRUE	USB_90D	USB	NC_USB_EXCARD_N
I111	NO_TEST=TRUE	USB_90D	USB	NC_USB_EXCARD_P
I114	NO_TEST=TRUE	USB_90D	USB	NC_USB_EXTE_N
I113	NO_TEST=TRUE	USB_90D	USB	NC_USB_EXTE_P
I116	NO_TEST=TRUE	USB_90D	USB	NC_USB_TPAD_N
I115	NO_TEST=TRUE	USB_90D	USB	NC_USB_TPAD_P

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FUNCTIONAL TESTPOINTS FOR MAC-1 & ICT

J4700 USB CAMERA

- IN PP5V_S3_CAMERA FUNC_TEST=TRUE MIN_ALLOWED_TPS=1
- 85 44 IN USB_CAMERA_L_P FUNC_TEST=TRUE
- 85 44 IN USB_CAMERA_L_N FUNC_TEST=TRUE

1 PP5V_S3_REG Testpoint near J4700
2 Ground Testpoints near J4700

J4750 USB CARD READER

- 85 44 IN USB_SDCARD_L_P FUNC_TEST=TRUE
- 85 44 IN USB_SDCARD_L_N FUNC_TEST=TRUE
- 91 44 25 21 IN SDCARD_RESET FUNC_TEST=TRUE

1 PP3V3_S3 Testpoint near J4750
2 Ground Testpoints near J4750

J4720 USB BLUETOOTH

- 85 44 IN USB_BT_L_P FUNC_TEST=TRUE
- 85 44 IN USB_BT_L_N FUNC_TEST=TRUE

1 PP3V3_S3 Testpoint near J4720
2 Ground Testpoints near J4720

J4780 IR BOARD

- 85 44 IN USB_IR_L_P FUNC_TEST=TRUE
- 85 44 IN USB_IR_L_N FUNC_TEST=TRUE

1 PP5V_S3_REG Testpoint near J4780
2 Ground Testpoints near J4780

J4520 SATA ODD (HIGH SPEED)

- 84 42 IN SATA_ODD_R2D_P FUNC_TEST=TRUE
- 84 42 IN SATA_ODD_R2D_N FUNC_TEST=TRUE
- 84 42 IN SATA_ODD_D2R_C_N FUNC_TEST=TRUE
- 84 42 IN SATA_ODD_D2R_C_P FUNC_TEST=TRUE
- 45 42 IN SMC_ODD_DETECT FUNC_TEST=TRUE

1 PP5V_S0 Testpoint near J4520
5 Ground Testpoints near J4520

J4510 SATA HDD (HIGH SPEED)

- 84 42 IN SATA_HDD_R2D_P FUNC_TEST=TRUE
- 84 42 IN SATA_HDD_R2D_N FUNC_TEST=TRUE
- 84 42 IN SATA_HDD_D2R_C_N FUNC_TEST=TRUE
- 84 42 IN SATA_HDD_D2R_C_P FUNC_TEST=TRUE

3 Ground Testpoints near J4510

J5520 ANALOG LCD TEMP SENSOR

- 88 51 IN SNS_LCD_P FUNC_TEST=TRUE
- 88 51 IN SNS_LCD_N FUNC_TEST=TRUE

J5521 AMBIENT TEMP SENSOR

- 88 51 IN SNS_AMB_P FUNC_TEST=TRUE
- 88 51 IN SNS_AMB_N FUNC_TEST=TRUE

J5551 ODD TEMP SENSOR

- 88 51 IN SNS_ODD_P FUNC_TEST=TRUE
- 88 51 IN SNS_ODD_N FUNC_TEST=TRUE

J5600 ODD FAN

- 52 IN FAN_0_PWR_L FUNC_TEST=TRUE
- 52 IN FAN_TACH0_L FUNC_TEST=TRUE
- 89 52 IN PP12V_S0_FAN0_L FUNC_TEST=TRUE
- 52 IN FAN_0_GND FUNC_TEST=TRUE

J5700 CPU FAN

- 53 IN FAN_2_PWR_L FUNC_TEST=TRUE
- 53 IN FAN_TACH2_L FUNC_TEST=TRUE
- 89 53 IN PP12V_S0_FAN2_L FUNC_TEST=TRUE
- 53 IN FAN_2_GND FUNC_TEST=TRUE

J5601 HD FAN

- 52 IN FAN_1_PWR_L FUNC_TEST=TRUE
- 52 IN FAN_TACH1_L FUNC_TEST=TRUE
- 89 52 IN PP12V_S0_FAN1_L FUNC_TEST=TRUE
- 52 IN FAN_1_GND FUNC_TEST=TRUE

J5550 HDD TEMP SENSOR

- 88 51 IN HDD_OOB_TEMP_FILT FUNC_TEST=TRUE

J5560 SKIN TEMP SENSOR

- 88 51 IN SNS_SKIN_P FUNC_TEST=TRUE
- 88 51 IN SNS_SKIN_N FUNC_TEST=TRUE

J6601 AUDIO MICROPHONE

- 59 IN AUD_MIC_IN1_N_CONN FUNC_TEST=TRUE
- 59 IN GND_AUDIO_MIC1_CONN FUNC_TEST=TRUE
- 59 IN AUD_MIC_IN1_P_CONN FUNC_TEST=TRUE

1 Ground Testpoint near J6601

J6602 AUDIO RIGHT SPEAKER

- 85 59 58 IN AUD_SPKR_OUTL02R_POUT FUNC_TEST=TRUE
- 85 59 58 IN AUD_SPKR_OUTL02R_NOUT FUNC_TEST=TRUE
- 85 59 57 IN AUD_SPKR_OUTL01R_POUT FUNC_TEST=TRUE
- 85 59 57 IN AUD_SPKR_OUTL01R_NOUT FUNC_TEST=TRUE

J6603 AUDIO LEFT SPEAKER

- 85 59 58 IN AUD_SPKR_OUTL02L_POUT FUNC_TEST=TRUE
- 85 59 58 IN AUD_SPKR_OUTL02L_NOUT FUNC_TEST=TRUE
- 85 59 57 IN AUD_SPKR_OUTL01L_POUT FUNC_TEST=TRUE
- 85 59 57 IN AUD_SPKR_OUTL01L_NOUT FUNC_TEST=TRUE

Power / Ground

- GND 16 TP'S FUNC_TEST=TRUE MIN_ALLOWED_TPS=16
- 89 6 IN PP3V3_S3 2 TP'S FUNC_TEST=TRUE MIN_ALLOWED_TPS=2
- 60 6 IN PP5V_S3_REG 2 TP'S FUNC_TEST=TRUE MIN_ALLOWED_TPS=1
- 89 6 IN PP5V_S0 FUNC_TEST=TRUE MIN_ALLOWED_TPS=1

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